Universidad Autónoma de Madrid | Escuela de Doctorado

Tesis Doctoral

Stability analysis and special solutions in fluid mechanics models

Miguel Martínez González

Programa de Doctorado en Matemáticas

Centro de realización de la tesis
Instituto de Ciencias Matemáticas

Dirección:
Ángel Castro Martínez

Madrid,
2026

This PhD thesis was written by

Miguel Martínez González

under the supervision of

Dr. Ángel Castro Martínez
Instituto de Ciencias Matemáticas (ICMAT),
CSIC-UAM-UC3M-UCM.

Tutor:

Prof. Daniel Faraco Hurtado
Departamento de Matemáticas
Universidad Autónoma de Madrid

within the PhD Programme in Mathematics at
Universidad Autónoma de Madrid.

This thesis was carried out at the
Instituto de Ciencias Matemáticas (ICMAT),
CSIC-UAM-UC3M-UCM.

In the publications related to this thesis,
the author has used the name
Miguel M.G. Pascual-Caballo.

A mi familia.

Acknowledgments

Aún recuerdo aquel correo del 13 de septiembre que me encontré al despertar. Gracias, Ángel, por darme la oportunidad de hacer esta tesis, por tu paciencia y por enseñarme a escribir. Gracias, Faraco, ha sido genial conocerte e importunarte para que firmes burocracia.

Gracias también a Javi por acogerme durante mi estancia, por nuestras sesiones clandestinas de matemáticas y por sacar tiempo para mí aun no teniéndolo.

I would also like to thank the referees of this thesis, Michele Dolce and Zineb Hassainia, for agreeing to review it.

Several people have shared the office with me, and I would like to thank them all. Especialmente a Jesús, por recibirme el primer día y enseñarme a trabajar. Ojalá hubiésemos coincidido más tiempo. Thanks to everyone I met at Brown for the good company and all the days we spent climbing, en especial a Álvaro y Fernando por abrirme las puertas de su casa. Gracias a los compañeros del ICMAT, de la UAM y de la Complu, a los beodos y a los sanos, a los matemáticos y a los algebristas, a los que empiezan y a los que ya terminaron.

Gracias a Esther, Marta y Natalia por su ayuda y paciencia con la burocracia. Gracias también a Mar por su ayuda con los trámites del depósito y a Lucía por resolver mis dudas con el clúster.

Gracias a Conde y a Quirós por hacerme ver el análisis con otros ojos. También a Otto y a Marco por dirigirme por aquellos derroteros. Y, antes aún, a Elisa y a Lada por formarme en mis primeras etapas. Todavía recuerdo la explicación de qué es una función mediante un exprimidor. Gracias a todos los que saltaron a la palestra para transmitirme sus enseñanzas.

Todavía me quedan más colegas a quienes dar las gracias. Gracias por estar ahí, por las partidas de dardos, por las expediciones micológicas, por las tardes en la Ferro, por las crónicas del gym, por las noches de karaoke y, por supuesto, por esas quedadas para degustar el etanol en todas sus formas.

Ainda preciso aprender mais português pra conseguir te agradecer direito por ter entrado na minha vida na reta final da tese. Obrigado, Giulia, por me acompanhar e me apoiar quando tudo parece dar errado. Queria ter te conhecido antes. Obrigado por ser quem você é e por dividir o mesmo neurônio comigo.

Por último, gracias a mi familia, por escucharme hablar de cosas que no entiendo ni yo. A mis abuelos, por esos billetes que enfadaban a mi madre, por descubrirme el flamenco y por las historias de vuestra juventud. Y a Arturo e Isabel, por aguantarme estos cuatro años cuando nada salía y parecía que la tesis no iba a terminar bien.

Funding

This work was supported by grant PRE2021-099600, funded by MCIN/AEI/10.13039/501100011033 and by the European Social Fund Plus (ESF+), and by research project PID2020-114703GB-I00, funded by MCIN/AEI/10.13039/501100011033 (*Dinámica de fluidos incompresibles*). The author also acknowledges financial support from the Severo Ochoa Programme for Centres of Excellence, Grants CEX2019-000904-S and CEX2023-001347-S, funded by MCIN/AEI/10.1303 9/501100011033, and from grants PID2024-158418NB-I00 and RED2024-153842-T, funded by

MICIU/AEI/10.13039/501100011033. Parts of this thesis were carried out during a stay at the Department of Mathematics of Brown University, whose hospitality is gratefully acknowledged.

Generative AI statement

During the preparation of this thesis, the author used Claude (Anthropic) to assist with copyediting and language revision. In Chapters II–IV, it was also used to review the mathematical arguments and the logical structure of the proofs for internal consistency and possible gaps. In Chapter II, it was further used to refine the discussion of regularity. In Chapter III, it was additionally used to review and debug the code involved in the computer-assisted parts of the proof. In Chapter IV, it was likewise used to review and debug the computer-assisted code, and parts of this code were generated with its assistance and subsequently reviewed, tested, and verified by the author. The author reviewed and verified the content of this thesis and takes full responsibility for it.

Abstract

The incompressible Euler equations admit simpler nonlocal models that retain part of their structure. In this thesis, we study special solutions of three such models: homogeneous stationary solutions of the inviscid surface quasi-geostrophic (SQG) equation, stationary states of the De Gregorio equation, and traveling waves of the Burgers–Hilbert equation.

We first prove, by a fixed-point argument, the existence of nontrivial homogeneous stationary solutions of the two-dimensional SQG equation. These solutions have infinite energy and are unbounded at infinity. The same argument also yields nontrivial stationary solutions of the generalized De Gregorio equation on the torus when $\alpha > \frac{1}{2}$.

Next, for the Burgers–Hilbert equation on the torus, we prove the existence of a traveling wave and show that it is spectrally unstable by exhibiting an unstable eigenvalue of the linearized equation.

Finally, for the De Gregorio equation on the torus, we prove that every excited stationary state is linearly unstable by showing that the linearized equation has an unstable eigenvalue.

In the instability analyses for the Burgers–Hilbert and De Gregorio equations, we reduce the corresponding spectral problems to a finite set of explicit inequalities, which are rigorously verified by a computer-assisted argument based on interval arithmetic.

Keywords: fluid dynamics, surface quasi-geostrophic equation, De Gregorio equation, Burgers–Hilbert equation, computer-assisted proof.

Resumen

Las ecuaciones de Euler incompresibles admiten modelos no locales más sencillos que conservan parte de su estructura. En esta tesis estudiamos soluciones especiales de tres de estos modelos: soluciones estacionarias homogéneas de la ecuación cuasigeostrófica superficial inviscida (SQG), estados estacionarios de la ecuación de De Gregorio y ondas viajeras de la ecuación de Burgers–Hilbert.

En primer lugar, demostramos, mediante un argumento de punto fijo, la existencia de soluciones estacionarias homogéneas no triviales de la ecuación SQG bidimensional. Estas soluciones tienen energía infinita y no están acotadas en el infinito. El mismo argumento proporciona también soluciones estacionarias no triviales de la ecuación de De Gregorio generalizada en el toro cuando $\alpha > \frac{1}{2}$.

A continuación, para la ecuación de Burgers–Hilbert en el toro, demostramos la existencia de una onda viajera y probamos que es espectralmente inestable mediante la exhibición de un valor propio inestable de la ecuación linealizada.

Finalmente, para la ecuación de De Gregorio en el toro, demostramos que todo estado estacionario excitado es linealmente inestable mostrando que la ecuación linealizada tiene un valor propio inestable.

En los análisis de inestabilidad de las ecuaciones de Burgers–Hilbert y de De Gregorio, reducimos los correspondientes problemas espectrales a un conjunto finito de desigualdades explícitas, que verificamos rigurosamente mediante un argumento asistido por ordenador basado en aritmética de intervalos.

Palabras clave: dinámica de fluidos, ecuación cuasigeostrófica de superficie, ecuación de De Gregorio, ecuación de Burgers–Hilbert, demostración asistida por ordenador.

Contents

Acknowledgments **v**

List of Symbols and Notation **1**

I Introduction **3**
I.1 Incompressible fluids . . . 3
I.1.1 The Hilbert transform . . . 5
I.2 Organization of the thesis . . . 5
I.2.1 Chapter II: SQG steady states . . . 6
I.2.2 Chapter III: instability of a Burgers–Hilbert traveling wave . . . 6
I.2.3 Chapter IV: instability of the De Gregorio excited states . . . 7

II SQG steady states **9**
II.1 Introduction . . . 9
II.2 Existence of stationary solutions . . . 11
II.2.1 Existence argument . . . 15
II.2.2 Regularity of the solution . . . 17
II.2.3 Verifying Schauder's hypotheses for $-\mathcal{S}_n$ . . . 17
II.3 Main theorem . . . 25
II.A Auxiliary lemmas . . . 26

III Burgers–Hilbert traveling wave **31**
III.1 Introduction . . . 31
III.1.1 Computational aspects of the proof . . . 32
III.1.2 The vortex patch problem . . . 33
III.1.3 Organization of the chapter . . . 34
III.1.4 Functional setting and notation . . . 34
III.2 Traveling wave existence . . . 36
III.3 Traveling wave stability . . . 40
III.3.1 Fixed point argument to prove linear instability . . . 44
III.3.2 Conclusion of the proof of Theorem 3.2 . . . 46
III.4 Proof of Proposition 2.2 . . . 46
III.4.1 The linear operator $\mathcal{L}_{\text{exis}}^{\text{ap, hol}}$ . . . 49
III.4.1.1 Fuchsian theory . . . 50
III.4.1.2 Invertibility of $\mathcal{L}_{\text{exis}}^{\text{ap, hol}}$ . . . 58
III.4.1.3 Estimates of the explicit functions and operators . . . 63
III.4.2 Conclusion . . . 66
III.5 Proof of Proposition 3.11: Lemmas 3.12–3.14 . . . 68
III.5.1 The operator T_{stab}^{θ} . . . 68
III.5.2 The linear operator $\mathcal{L}_{\theta}^{\text{ap}}$ . . . 68
III.5.2.1 Invertibility of $\mathcal{L}_{\theta,\lambda}^{\text{ap, hol}}$ . . . 73

III.5.2.2 Restricted invertibility of $\mathcal{L}^{\mathrm{ap,\,hol}}_{\theta,\lambda^{\mathrm{ap}}}$ . . . 78
III.5.2.3 Estimates of the explicit functions and operators . . . 83
III.5.3 The choice for η . . . 88
III.5.4 Proofs of Lemmas 3.12–3.14 . . . 90
III.A Classical inequalities . . . 91
III.B Commutator of Hilbert transform and cosine . . . 92
III.C Palindromic polynomial . . . 93
III.D Treatment of the solutions of the differential equations . . . 95
III.D.1 Existence argument . . . 95
III.D.2 Stability argument . . . 100
III.D.2.1 The functions $J^+F_v^{\mathrm{ap}+}|_{\mathbb{T}}$ and $J^-F_v^{\mathrm{ap}-}|_{\mathbb{T}}$ . . . 104
III.E Computer-assisted proofs . . . 106
III.E.1 Supplementary data . . . 106
III.E.2 Objects, methods and routines . . . 108
III.E.3 Details of the proofs . . . 111

IV de Gregorio steady states **121**
IV.1 Introduction . . . 121
IV.1.1 Computer-assisted methods . . . 122
IV.1.2 Functional setting and notation . . . 123
IV.2 The linearized equation . . . 124
IV.2.1 Decoupling the modes . . . 126
IV.2.2 Conclusion of the proof of Theorem 2.1 . . . 128
IV.3 Proof of Proposition 2.7 . . . 129
IV.3.1 Reduction to finite dimension . . . 129
IV.3.2 Second-order ODE system . . . 132
IV.3.2.1 Local analysis . . . 132
IV.3.2.2 Global analysis . . . 135
IV.3.3 Conclusion of the proof . . . 139
IV.A Fuchsian theory . . . 140
IV.A.1 Homogeneous solutions . . . 140
IV.A.2 Analytic particular solutions . . . 144
IV.A.2.1 A holomorphic solution from the Frobenius recurrence . . . 144
IV.A.3 A priori estimate at the boundary singular points . . . 146
IV.A.3.1 Weighted Hardy operators . . . 146
IV.B Error bounds for Taylor approximations . . . 149
IV.B.1 A residual estimate for inhomogeneous Fuchsian equations . . . 149
IV.B.2 Application to the entries of $\mathbf{M}_{\lambda,\theta}$ . . . 151
IV.B.3 Rigorous control of the tails . . . 154
IV.B.4 Certified bounds for the matrix entries . . . 156
IV.C Computer-assisted implementation . . . 157
IV.C.1 Supplementary code and computational data . . . 157
IV.C.2 Objects, methods and routines . . . 158
IV.C.3 Details of the proof . . . 159

V Conclusions and future work **161**

List of Symbols and Notation

This list collects notation used throughout the thesis. Notation specific to a single chapter is introduced where it is needed.

Sets and Domains

$\mathbb{R}$	the real line.
$\mathbb{C}$	the complex plane.
$\mathbb{Z}$	the set of integers.
$\mathbb{N}$	the set of positive integers.
$\mathbb{T}$	the one-dimensional torus $\mathbb{R}/2\pi\mathbb{Z}$, often identified with $\{z \in \mathbb{C} : \lvert z\rvert = 1\}$.
$\mathbb{D}$	the unit disk, $\{z \in \mathbb{C} : \lvert z\rvert < 1\}$.

Operators

∂_x	differentiation with respect to x.
H	the Hilbert transform.
Λ	the Fourier multiplier $(-\Delta)^{1/2}$.
∂_x^{-1}	the zero-mean primitive, defined on mean-zero functions.

Fourier and Norm Conventions

Fourier coefficients on $\mathbb{T}$ are normalized by

$$\widehat{f}_k := \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\, e^{-ikx}\, dx.$$

With this convention,

$$\|f\|_{L^2(\mathbb{T})}^2 := \sum_{k\in\mathbb{Z}} |\widehat{f}_k|^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} |f(x)|^2\, dx.$$

I. Introduction

The Euler and Navier–Stokes equations are far from fully understood. Their study has motivated the development of simplified models that encode part of their structure. This thesis concerns three nonlocal equations of this kind: the surface quasi-geostrophic equation in two dimensions, and the De Gregorio and Burgers–Hilbert equations in one dimension.

We study the existence of steady states for the surface quasi-geostrophic equation and of traveling waves for the Burgers–Hilbert equation, as well as the stability of the De Gregorio excited states and of the Burgers–Hilbert traveling waves. The SQG existence argument extends to the generalized De Gregorio equation for $\alpha > \frac{1}{2}$.

I.1 Incompressible fluids

The Euler equations for an incompressible fluid in dimension $d \in \{2, 3\}$ are

Incompressible Euler

$$\begin{cases} \partial_t u + (u \cdot \nabla) u = -\nabla p, \quad (x,t) \in \mathbb{R}^d \times [0,\infty), \\ \nabla \cdot u = 0, \\ u(x,0) = u_0(x), \end{cases} \tag{1.1}$$

where $u : \mathbb{R}^d \times [0,\infty) \to \mathbb{R}^d$ is the velocity field and $p : \mathbb{R}^d \times [0,\infty) \to \mathbb{R}$ is the pressure. Adding a viscous term $\nu\Delta u$ to the right member of the first equation of (1.1) gives the Navier–Stokes equations. Throughout this thesis, we focus on inviscid models.

Taking the curl of (1.1) removes the pressure and gives the equation for the vorticity

$$\omega = \operatorname{curl} u.$$

In three dimensions,

3D Euler, vorticity form

$$\begin{cases} \partial_t \omega + (u \cdot \nabla)\omega = (\omega \cdot \nabla) u, \quad (x,t) \in \mathbb{R}^3 \times [0,\infty), \\ u = \operatorname{curl} (-\Delta)^{-1} \omega, \\ \omega(x,0) = \omega_0(x). \end{cases} \tag{1.2}$$

The left-hand side of the first equation in (1.2) transports the vorticity along the flow, while the right-hand side stretches it. The second equation recovers the velocity by inverting the Laplacian. This is the Biot–Savart law, which can be written explicitly as

$$u(x,t) = \frac{1}{4\pi} \int_{\mathbb{R}^3} \frac{(x-y) \times \omega(y,t)}{|x-y|^3}\, dy. \tag{1.3}$$

In the plane, the vorticity formulation reads as follows

2D Euler, vorticity form

$$\begin{cases} \partial_t \omega + (u \cdot \nabla)\omega = 0, \quad (x,t) \in \mathbb{R}^2 \times [0,\infty), \\ u = \nabla^\perp (-\Delta)^{-1}\omega, \\ \omega(x,0) = \omega_0(x). \end{cases} \tag{1.4}$$

Here $\nabla^\perp = (-\partial_{x_2}, \partial_{x_1})$ and the vorticity ω is a scalar function. In contrast with (1.2) the stretching term vanishes and we have a different Biot–Savart law, namely:

$$u(x,t) = \frac{1}{2\pi} \int_{\mathbb{R}^2} \frac{(x-y)^\perp}{|x-y|^2}\, \omega(y,t)\, dy, \tag{1.5}$$

In two dimensions, the vorticity equation is a pure transport equation by an incompressible velocity field. As a consequence, all L^p norms of the vorticity are conserved, in particular

$$\|\omega(t)\|_{L^\infty} = \|\omega_0\|_{L^\infty}.$$

The Beale–Kato–Majda criterion [MB02, Theorem 4.3] then rules out finite-time blow-up for smooth solutions of two-dimensional Euler.

The situation is different in three dimensions. The vorticity equation contains the stretching term $(\omega \cdot \nabla)u$, so the vorticity is no longer merely transported and its L^∞ norm is not conserved by the preceding argument. Whether this stretching can produce a finite-time singularity from smooth, finite-energy data remains a central open problem.

For the classical theory of incompressible Euler flows, including the vorticity formulation and the Biot–Savart law, we refer to Majda and Bertozzi [MB02]. For less regular data, finite-time singularities have recently been proved, both for $C^{1,\alpha}$ velocities in $\mathbb{R}^3$ [Elg21] and in the presence of a boundary [CH21].

One approach to the interplay between transport and stretching in (1.2) is to study lower-dimensional models that retain an analogue of this mechanism. The family introduced by Okamoto, Sakajo, and Wunsch [OSW08] is one such example:

Generalized De Gregorio

$$\partial_t \omega + \alpha u\, \partial_x \omega = \omega\, \partial_x u, \qquad u = -\Lambda^{-1}\omega, \quad \alpha \in \mathbb{R}, \tag{1.6}$$

where $\Lambda = (-\Delta)^{1/2}$ and both ω and u are scalar functions. In this thesis, we consider it on the torus. The parameter α controls the relative weight of the transport term. At $\alpha = 0$ one recovers the Constantin–Lax–Majda model [CLM85], which retains only the stretching term and admits finite-time blow-up from smooth data. At $\alpha = 1$ one obtains the De Gregorio equation [De 90; De 96].

The inviscid surface quasi-geostrophic (SQG) equation is

Inviscid SQG equation

$$\partial_t \theta + (u \cdot \nabla)\theta = 0, \qquad u = \nabla^\perp (-\Delta)^{-1/2}\theta. \tag{1.7}$$

Here θ is a scalar and u is a two-dimensional divergence-free velocity field. In Chapter II, we use the whole-plane formulation on $\mathbb{R}^2$.

The SQG model arises in geophysical fluid dynamics as a model for the temperature at the surface of a rapidly rotating stratified fluid. Mathematically, it is notable for a structural analogy with three-dimensional Euler: in (1.2), the velocity gradient ∇u has the same order of regularity as the vorticity, and the same holds in SQG for ∇u and $q := \nabla^\perp \theta$. Indeed, taking $\nabla^\perp$ in the first equation of (1.7) and using the incompressibility condition on yields

$$\partial_t q + (u \cdot \nabla)q = (q \cdot \nabla)u.$$

This analogy was emphasized by Constantin, Majda, and Tabak [CMT94a] and is one of the main reasons why SQG is often used as a model problem for aspects of three-dimensional Euler dynamics.

The Burgers–Hilbert equation

$$\begin{gathered}\textit{Burgers–Hilbert}\\ \partial_t u + u\,\partial_x u + Hu = 0,\end{gathered} \tag{1.8}$$

arises as a model for the motion of a vorticity front. Here, u is a scalar function and H denotes the Hilbert transform, a linear Fourier multiplier defined in Section I.1.1. In Chapter III, we work on the torus $\mathbb{T}$.

More precisely, when two regions of constant but different vorticity are separated by a nearly flat interface, the interface displacement satisfies (1.8) at quadratic order in the slope. The equation was introduced by Marsden and Weinstein [MW83] and derived rigorously for small-slope fronts by Biello and Hunter [BH10]. The approximation is valid on time scales of order ε^{-1} [Hun+22].

Remark 1.1 *The Burgers–Hilbert equation appears in the literature with different sign conventions. These conventions are equivalent up to the elementary transformations*

$$u(x,t) \mapsto u(-x,t), \qquad u(x,t) \mapsto -u(x,t),$$

and their composition $u(x,t) \mapsto -u(-x,t)$.

I.1.1 The Hilbert transform

The Hilbert transform is the common nonlocal operator behind the one-dimensional formulations used in this thesis. On the torus $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, it is defined by the Fourier multiplier

$$(\widehat{Hf})_k = -i\mathrm{sgn}(k)\,\widehat{f}_k, \qquad k \in \mathbb{Z}, \tag{1.9}$$

with $\mathrm{sgn}(0) = 0$.

Equivalently, for sufficiently regular functions,

$$Hf(x) = \frac{1}{2\pi}\,\mathrm{P.V.}\int_{-\pi}^{\pi} \cot\!\left(\frac{x-y}{2}\right) f(y)\,dy, \tag{1.10}$$

where P.V. denotes the Cauchy principal value.

It satisfies

$$H\partial_x f = \partial_x H f = \Lambda f, \tag{1.11}$$

where $\Lambda = (-\Delta)^{1/2}$.

In the Burgers–Hilbert equation, H appears explicitly as the linear term in (1.8). In the generalized De Gregorio equation, the relation $\partial_x u = H\omega$ rewrites the nonlocal coupling in terms of H. In the SQG chapter, the homogeneous ansatz reduces the problem to an equation for an angular profile on $\mathbb{T}$, again involving a nonlocal operator on the circle. The splitting of H on positive and negative Fourier modes is also used in the holomorphic reformulations of Chapters III and IV.

I.2 Organization of the thesis

The three main chapters are based on independent works and are presented in chronological order. They can be read separately, although they share a common interest in special solutions of nonlocal fluid models. Each chapter includes its own introduction and develops the notation needed for its main result.

The computer-assisted arguments in Chapters III and IV are described in the corresponding appendices. The accompanying code is available at https://github.com/MiguelMGPascualCaballo/. The thesis ends with a short chapter on conclusions and future directions.

We now summarize the main contributions of each chapter.

I.2.1 Chapter II: SQG steady states

Chapter II concerns the existence of nontrivial stationary solutions of the inviscid SQG equation (1.7). These steady states are additionally 1-homogeneous:

$$\theta(x_1, x_2) = rf(x),$$

where $(r, x) \in [0, \infty) \times \mathbb{T}$ are polar coordinates. Hence they have infinite energy and are unbounded at infinity.

The proof is analytic rather than constructive, since it relies on Schauder's fixed point theorem, and same argument also applies when the SQG angular operator is replaced by $-\Lambda^{-1}$. This gives stationary states for the generalized De Gregorio family (1.6) in the range $\alpha > \frac{1}{2}$.

I.2.2 Chapter III: instability of a Burgers–Hilbert traveling wave

Chapter III constructs a traveling wave for the Burgers–Hilbert equation (1.8) on $\mathbb{T}$ and proves its linear instability. A traveling wave of speed c is a solution of the form $u(x,t) = v(x + ct)$. Substituting this ansatz into (1.8) gives the profile equation

$$c\partial_x v + Hv + v\,\partial_x v = 0.$$

We show that there exists a traveling wave v with minimal period 2π.

For every $n \geq 2$,

$$v^n(x) := \frac{1}{n} v(nx)$$

defines another traveling wave with speed $\frac{c}{n}$ and period $\frac{2\pi}{n}$.

We then prove linear instability of the rescaled wave v^3. At the level of the integrated perturbation f, the linearized equation has the form

$$\partial_t f + \mathcal{L}^{c/3, v^3} f = 0,$$

where

$$\mathcal{L}^{c/3, v^3} f := \frac{c}{3}\partial_x f + v^3\,\partial_x f + Hf - \frac{1}{2\pi}\int_{-\pi}^{\pi} v^3(x)\,\partial_x f(x)\,dx.$$

We prove that $\mathcal{L}^{c/3, v^3}$ has an eigenvalue λ with $\mathrm{Re}\lambda < 0$ and a smooth eigenfunction. Thus $e^{-\lambda t} f$ is an exponentially growing solution of the linearized equation, and hence v^3 is linearly unstable.

In [CCZ23], small-amplitude $\frac{2\pi}{m}$-periodic traveling-wave branches bifurcate from zero at speed $c_m = \frac{1}{m}$, and the spectrum of the corresponding linearized operators remains purely imaginary near the beginning of these branches. The wave v constructed in Chapter III has minimal period 2π and speed $c \approx 1.1$, close to the speed of the cusped waves constructed in [DG23].

A V-state is a rotating vortex patch. Near zero, small-amplitude $\frac{2\pi}{m}$-periodic Burgers–Hilbert waves correspond to m-fold V-states close to the disk. The instability found here may therefore be viewed as a step toward instability questions for n-fold V-states.

The proof has two parts: constructing v and finding the unstable eigenpair. Both reduce to fixed-point arguments around explicit numerical approximations. In each case, the central analytic step is to invert a linear operator with an explicit bound on the inverse norm. After a holomorphic change of variables on the unit disk, this inversion problem becomes an ODE with regular singular points. The estimates needed to close the fixed point arguments are verified by interval arithmetic.

I.2.3 Chapter IV: instability of the De Gregorio excited states

Chapter IV concerns the De Gregorio equation on $\mathbb{T}$, whose stationary states include

$$x \mapsto -\sin(nx) \qquad \text{for all } n \geq 1.$$

We call $-\sin x$ the ground state and the higher modes $-\sin(nx)$, $n \geq 2$, excited states.

The ground state $-\sin x$ is known to be nonlinearly stable in a C^2 neighborhood [JSS19]: solutions starting close to it exist globally and converge to nearby equilibrium. At the linear level, the distinction is reflected in the local structure of the eigenfunctions. Near the singular point, the two Frobenius branches have orders $|x|^{1-\mathrm{Re}\lambda}$ and $|x|^{1+\mathrm{Re}\lambda}$. For the eigenvalues considered here, with $\mathrm{Re}\lambda > \frac{1}{2}$, the first branch is too singular to belong to $H^1(\mathbb{T})$, while the second belongs to $H^2(\mathbb{T})$. This also separates our eigenfunctions from the formal unstable branches in the ground-state analysis of [JSS19]: for $\mathrm{Re}\lambda > \frac{1}{2}$, the branch of order $|x|^{1-\mathrm{Re}\lambda}$ is not in $H^1(\mathbb{T})$.

To state the linearized operator, fix $n \geq 2$ and write

$$S_n(x) := \sin(nx) \qquad \text{and} \qquad C_n(x) := \cos(nx).$$

For a perturbation of the form

$$\omega = -S_n + \eta,$$

the linearized equation is

$$\partial_t \eta = L_n \eta,$$

where

$$L_n\eta := -\frac{1}{n} S_n\, \partial_x \eta + n C_n\, H \partial_x^{-1} \eta - S_n\, H\eta + C_n\, \eta.$$

Here ∂_x^{-1} is defined on mean-zero functions by

$$\widehat{(\partial_x^{-1} f)}_k = \frac{1}{ik} \widehat{f}_k \qquad \text{for } k \neq 0.$$

We show that L_n has an eigenvalue $\lambda_n \in (0.56, 0.6)$, with an associated eigenfunction $\eta_n \in H^2(\mathbb{T})$. Hence

$$\eta(x,t) = e^{\lambda_n t} \eta_n(x)$$

solves the linearized equation and grows exponentially.

A useful feature of the operator L_n is that it respects the decomposition of Fourier modes into congruence classes modulo n. The class $l = 0$, which behaves like the ground-state problem, is not needed for the instability argument. The remaining classes lead to a family of operators L_θ, with $\theta = \frac{l}{n}$ and $1 \leq l \leq n-1$. For every $n \geq 2$, one can choose such a fraction $\theta \in [\frac{1}{3}, \frac{1}{2}]$, so the instability of $-\sin(nx)$ reduces to finding an unstable eigenvalue of L_θ uniformly for θ in this range.

The eigenvalue equation for L_θ is converted, via a holomorphic change of variables on the unit disk, into a system of ODEs with regular singular points. Matching the local Frobenius solutions near each singularity reduces the problem to the vanishing of the determinant of an explicit 4×4 matrix $\mathbf{M}_{\lambda,\theta}$. The small size of this matrix makes the certification simpler than in Chapter III. A finite computer-assisted verification of the sign change of $\det \mathbf{M}_{\lambda,\theta}$, using interval arithmetic, yields an unstable eigenvalue for every excited state.

II. Stationary homogeneous solutions for the inviscid SQG equation

This chapter is based on the preprint [Pas25] and corresponds to its first version posted on arXiv, with corrections incorporated. The text has been adapted to the format and notation of this thesis.

II.1 Introduction

The inviscid Surface Quasi-geostrophic equation (SQG) in $\mathbb{R}^2$ is given by the following transport problem

$$\begin{cases} \partial_t \theta + \nabla^\perp u \cdot \nabla \theta = 0, \quad (x,t) \in \mathbb{R}^2 \times [0,\infty), \\ u = -\Lambda^{-1}\theta, \\ \theta(x,0) = \theta_0(x), \end{cases} \tag{1.1}$$

where $\Lambda := (-\Delta)^{\frac{1}{2}}$. No decay condition is imposed on θ at spatial infinity, since the homogeneous solutions considered in this chapter are unbounded as $r \to \infty$. For such θ, Λ^{-1} is understood as in [CCZ21; EJ20c].

This equation describes the temperature evolution in a general quasi-geostrophic framework, applicable to atmospheric and oceanic flows, assuming small Rossby and Ekman numbers, along with constant potential vorticity. We refer to [CMT94a; Hel+95; MB02; Ped79] for more details.

The mathematical study of this equation received a great boost with the works of Constantin, Majda, and Tabak, [CMT94a] and [CMT94b]. In them, the authors revealed the similarities between SQG and 3D Euler, and proposed it as an interesting model in which to search for singularities.

There are several known results concerning the existence of solutions to the SQG model. In [Res95], Resnick proved global existence in L^2, and Marchand later extended this result in [Mar08] to L^p, $p > \frac{4}{3}$. Proofs for local well-posedness in H^s, $s > 2$ can be found in [Cha03; CMT94a]. Wu also proved local well-posedness in [Wu05] for $C^\gamma \cap L^p$, $\gamma > 1$ and $p > 1$. In [CIN25; CN18a; CN18b], the authors studied SQG on bounded domains.

Regarding non-uniqueness, Buckmaster, Shkoller, and Vicol [BV19] constructed weak solutions satisfying $\Lambda^{-1}\theta \in C^\gamma$, $\frac{1}{2} < \gamma < \frac{4}{5}$. This result was extended to $\Lambda^{-1}\theta \in C^\gamma$ for every $\gamma < 1$ by Dai, Giri, and Radu [DGR24], as well as by Isett and Looi [LI24] (see also [Zha25]).

In the forcing case, there are similar non-uniqueness results by Bulut, Huynh, and Palasek [BHP23] and by Dai and Peng [DP23]. Furthermore, Castro, Faraco, Mengual, and Solera [Cas+25a; Cas+25b] constructed non-unique solutions for $\theta, f \in L^1 \cap L^p$, $1 < p < \infty$ and $\theta, f \in W^{s,p} \cap \dot{H}^{-\frac{1}{2}}$, $0 \le s < 1 + \frac{2}{p}$, $1 \le p \le \infty$.

A central question in fluid mechanics is whether SQG develops finite-time singularities. Kiselev and Nazarov [KN12] showed the existence of arbitrarily small initial data in H^s, $s \ge 11$, whose corresponding solution's H^s-norm exceeds any prescribed bound within a finite time interval. Moreover, Friedlander and Shvydkoy [FS05] found unstable eigenvalues for the linearization around a steady state. Additionally, He and Kiselev [HK21] showed that solutions exist for which either

the C^2-norm grows exponentially for all time or the solution blows up in finite time. Several examples of global solutions with nontrivial dynamics have also been constructed. We refer to [Ao+21; CCG20b; God21; GS19].

Córdoba and Martínez-Zoroa [CM22] showed ill-posedness in C^k for $k \geq 2$ and in H^s with $\frac{3}{2} < s < 2$. See also the works of Jeong and Kim [JK24], as well as Elgindi and Masmoudi [EM20]. In addition, Córdoba, Martínez-Zoroa, and Ożański [CMO24] showed that this loss of regularity is continuous in time, with the solution's Sobolev index decreasing steadily from s rather than dropping at a single instant.

In this chapter, we will consider homogeneous solutions, which are of the form

$$\theta(x_1, x_2, t) = rf(x, t) \tag{1.2}$$

where $(r, x) \in \mathbb{R}^+ \times \mathbb{T}$ are the usual polar coordinates, and $\mathbb{T} := \mathbb{R}/2\pi\mathbb{Z}$ denotes the one-dimensional torus, identified with $(-\pi, \pi]$.

Introducing the ansatz (1.2) into (1.1), one obtains the following equation for a function $f : \mathbb{T} \times [0, \infty) \to \mathbb{R}$ whose Fourier coefficients satisfy $\widehat{f}_k = 0$ for $|k| \leq 2$,

$$\partial_t f + 2\mathcal{S}f\, \partial_x f - \partial_x \mathcal{S}f\, f = 0, \tag{1.3}$$

where $\mathcal{S}$ is a Fourier multiplier defined by

$$(\widehat{\mathcal{S}f})_k := \widehat{\mathcal{S}}_k \widehat{f}_k := -\frac{1}{|k|}\frac{k^2 - 1}{k^2 - 4}\widehat{f}_k, \quad |k| \geq 3. \tag{1.4}$$

Detailed computations leading to this expression can be found in [CCZ21] and [EJ20c].

The goal of this chapter is to prove the existence of nontrivial stationary solutions of (1.3), which is a first step toward studying the long-time dynamics of nearby solutions or constructing non-uniqueness examples.

Theorem 1.1 *For each $n \geq 3$, there exists an odd $\frac{2\pi}{n}$-periodic function f, belonging to $C^{\frac{1}{2}}(\mathbb{T})$, such that $\widehat{f}_n \neq 0$, solving* (1.3)*. In addition, the n-fold symmetric function $\Theta : \mathbb{R}^2 \to \mathbb{R}$ given by*

$$\Theta(x_1, x_2) := rf(x)$$

is a stationary solution of the SQG equation (1.1)*.*

Remark 1.2 *The solutions constructed here do not depend on time, so* (1.1) *reduces to*

$$\nabla^\perp u \cdot \nabla\Theta = 0, \qquad u = -\Lambda^{-1}\Theta.$$

This is understood in the distributional sense, that is,

$$\int_{\mathbb{R}^2} \Theta\, \nabla^\perp u \cdot \nabla\psi\, dx = 0 \qquad \text{for every } \psi \in C_c^\infty(\mathbb{R}^2),$$

which comes from $\nabla^\perp u \cdot \nabla\Theta = \nabla \cdot (\Theta \nabla^\perp u)$, valid because $\nabla^\perp u$ is divergence free. Since $\nabla\psi$ is bounded and supported in a compact set, the integral is absolutely convergent as soon as $\Theta\nabla^\perp u$ is locally integrable.

Remark 1.3 *Uniqueness, for fixed n, is not proven with our strategy.*

Once this project was completed, we became aware of the paper [AGJ25] by Abe, Gómez-Serrano, and Jeong. Their results for SQG also include solutions of the form $\theta = rf(x)$, obtained through variational methods, whereas ours uses a Schauder fixed-point argument. Our results additionally cover the generalized De Gregorio equation (1.7) below on $\mathbb{T}$, not addressed in [AGJ25]. We agreed to synchronize the upload of both works to arXiv.

In the process of proving Theorem 1.1, we prove a more general result. Specifically, we consider the equation

$$\alpha \mathcal{S}f\, \partial_x f = \partial_x \mathcal{S}f\, f, \tag{1.5}$$

with $\alpha > \frac{1}{2}$. We obtain nontrivial, odd, and $\frac{2\pi}{n}$-periodic solutions for each $n \geq 3$.

The family of equations

$$\partial_t f + \alpha \mathcal{S}f\, \partial_x f - \partial_x \mathcal{S}f\, f = 0, \tag{1.6}$$

with $\alpha \in \mathbb{R}$, is intimately related to the generalized De Gregorio equation,

$$\begin{cases} \partial_t w + \alpha u\, \partial_x w = w\, \partial_x u, \quad (x,t) \in \mathbb{R} \times [0,\infty),\ \alpha \in \mathbb{R}, \\ u = -\Lambda^{-1} w, \\ w(x,0) = w_0(x). \end{cases} \tag{1.7}$$

Here $\Lambda := (-\Delta)^{\frac{1}{2}}$. Equation (1.7) with $\alpha = 1$ was proposed by De Gregorio [De 90; De 96] as a one-dimensional model that captures the competition between advection and vortex stretching in the Euler equation. Okamoto, Sakajo, and Wunsch [OSW08] introduced the parameter $\alpha \in \mathbb{R}$ into the De Gregorio equation, obtaining the generalized model, also known in the literature as the generalized Constantin–Lax–Majda (gCLM) equation.

If we fix $\alpha = -1$ in (1.7), we obtain the Córdoba–Córdoba–Fontelos model [CCF05]. Furthermore, the case $\alpha = 0$ corresponds to the Constantin–Lax–Majda equation, see [CLM85].

Equations (1.6) and (1.7) are closely related because

$$\mathcal{S} = -\Lambda^{-1} + \text{smoothing operator}.$$

Okamoto, Sakajo, and Wunsch [OSW14] studied the existence of nontrivial steady states of (1.7) in $C^\gamma(\mathbb{T})$, with $0 < \alpha \leq 1$. In this chapter, we prove the existence of nontrivial solutions to (1.5) for every $\alpha > \frac{1}{2}$, in $C^{\frac{1}{\alpha}}(\mathbb{T})$ when $\alpha > 1$, and in $C^{1,\frac{1}{\alpha}-1}(\mathbb{T})$ when $\alpha < 1$. Our argument, inspired by [OSW14], is also valid for the generalized De Gregorio equation (1.7).

II.2 Existence of stationary solutions

Here and below, ∂_x^{-1} denotes the zero-mean primitive on $\mathbb{T}$ and H the Hilbert transform. On mean-zero functions they are given by the Fourier multipliers

$$\widehat{(\partial_x^{-1} f)}_k := \frac{1}{ik} \widehat{f}_k \qquad \text{and} \qquad \widehat{(Hf)}_k := -i \operatorname{sgn}(k)\, \widehat{f}_k \qquad \text{for } k \neq 0.$$

Notice that the operator Λ^{-1} in (1.7) can be written as $\Lambda^{-1} = -\partial_x^{-1} H = -H \partial_x^{-1}$. We also write $A \lesssim B$ to mean $A \leq CB$ for a constant C that may change from one line to the next.

We seek nontrivial solutions to (1.5). We will assume that $f \in L^2(\mathbb{T})$ is odd and $\frac{2\pi}{n}$-periodic, $n \geq 3$, so we can expand

$$f(x) = \sum_{k \geq 1} \widehat{f}^{\,\mathrm{s}}_{nk} \sin(nkx),$$

where the sine coefficients of f are given by

$$\widehat{f}^{\,\mathrm{s}}_k := \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(kx)\, dx = \frac{2}{\pi} \int_0^{\pi} f(x) \sin(kx)\, dx.$$

Remark 2.1 *Suppose that there is an odd stationary solution of* (1.7)*, with nonzero first Fourier coefficient. From such a solution one can construct new ones by rescaling the period. This follows*

from the fact that $-\Lambda^{-1} = \partial_x^{-1} H$ *and that, for any mean-zero function* $g \in C(\mathbb{T})$*, if we set* $g_n := g(nx)$*, then*

$$\partial_x^{-1} H g_n(x) = \frac{1}{n} \partial_x^{-1} H g(nx), \quad n \geq 1.$$

Notice that this is not the case for $\mathcal{S}$*, since* $\mathcal{S} g_n(x) \neq \frac{1}{n} \mathcal{S} g(nx)$*, and this is what makes our analysis different. We are nevertheless able to prove the theorem for every* $n \geq 3$*, thereby generating this family of solutions.*

For $n \geq 3$, we denote by $\mathcal{S}_n$ the Fourier multiplier given by

$$(\widehat{\mathcal{S}_n w})_k := (\widehat{\mathcal{S}_n})_k \widehat{w}_k := -\frac{1}{|k|} \frac{n^2 k^2 - 1}{n^2 k^2 - 4} \widehat{w}_k, \qquad k \neq 0. \tag{2.1}$$

The following lemma collects the properties of $\mathcal{S}_n$ and $\mathcal{S}$ that we will use, and the relationship between them.

Lemma 2.2 *Let* $n \geq 3$*. Both* $\mathcal{S}_n$ *and* $\mathcal{S}$ *take odd functions to odd functions. Moreover, if* $f \in L^2(\mathbb{T})$ *is odd and* $\frac{2\pi}{n}$*-periodic, then, for* $g(x) := f(\frac{x}{n})$*, both* $\mathcal{S}_n g$ *and* $\mathcal{S} f$ *belong to* $H^1(\mathbb{T}) \subset C(\mathbb{T})$*, and*

$$\mathcal{S}_n g(x) = n \, \mathcal{S} f(\tfrac{x}{n}) \qquad \text{for every } x \in \mathbb{T}.$$

Proof: If w is odd then $\widehat{w}_{-k} = -\widehat{w}_k$, and since $(\widehat{\mathcal{S}_n})_k$ depends on k only through $|k|$,

$$(\widehat{\mathcal{S}_n w})_{-k} = (\widehat{\mathcal{S}_n})_{-k} \widehat{w}_{-k} = -(\widehat{\mathcal{S}_n})_k \widehat{w}_k = -(\widehat{\mathcal{S}_n w})_k,$$

so $\mathcal{S}_n w$ is odd. The same applies to $\mathcal{S}$, whose symbol is defined for $|k| \geq 3$.

Let now $f \in L^2(\mathbb{T})$ be odd and $\frac{2\pi}{n}$-periodic. It can be written as the sine Fourier series

$$f(x) = \sum_{k \geq 1} \widehat{f}^{\mathrm{s}}_{nk} \sin(knx),$$

converging in $L^2(\mathbb{T})$, and hence

$$g(x) := f(\tfrac{x}{n}) = \sum_{k \geq 1} \widehat{f}^{\mathrm{s}}_{nk} \sin(kx) \tag{2.2}$$

in $L^2(\mathbb{T})$ as well.

Both symbols are defined and bounded, see (1.4) and (2.1), so that $\mathcal{S}_n g$ and $\mathcal{S} f$ are the $L^2(\mathbb{T})$ limits of the corresponding partial sums. Now, observe that

$$\widehat{\mathcal{S}}_{kn} = -\frac{1}{n|k|} \frac{n^2 k^2 - 1}{n^2 k^2 - 4} = \frac{1}{n} (\widehat{\mathcal{S}_n})_k. \tag{2.3}$$

Since f is $\frac{2\pi}{n}$-periodic, its only nonzero Fourier coefficients occur at multiples of n, so, using (2.2) and (2.3),

$$\mathcal{S}_n g(x) = \sum_{k \geq 1} (\widehat{\mathcal{S}_n})_k \widehat{g}^{\mathrm{s}}_k \sin(kx) = \sum_{k \geq 1} (\widehat{\mathcal{S}_n})_k \widehat{f}^{\mathrm{s}}_{nk} \sin(kx) = n \sum_{k \geq 1} \widehat{\mathcal{S}}_{kn} \widehat{f}^{\mathrm{s}}_{nk} \sin(kx) = n \, \mathcal{S} f(\tfrac{x}{n})$$

as an identity in $L^2(\mathbb{T})$. Moreover, $(\widehat{\mathcal{S}_n})_k = O(\frac{1}{k})$ and $\widehat{\mathcal{S}}_k = O(\frac{1}{k})$, so both sides above in fact belong to $H^1(\mathbb{T}) \subset C(\mathbb{T})$, and the identity holds for every $x \in \mathbb{T}$. □

Then, from Lemma 2.2, equation (1.5) becomes

$$\alpha \mathcal{S}_n g \, \partial_x g = g \, \partial_x \mathcal{S}_n g, \tag{2.4}$$

for a 2π-periodic, mean zero and odd function $g \in L^2(\mathbb{T})$.

As $\mathcal{S}_n$ maps sine series to sine series, it is natural to let it act on functions on $[0,\pi]$, by applying it to the odd extension and restricting the result back to $[0,\pi]$. Following the strategy outlined in [OSW14], instead of (2.4), we will work with the equation

$$v(x) = -\mathcal{S}_n(v^{\frac{1}{\alpha}})(x) \quad \text{for every } x \in [0,\pi], \tag{2.5}$$

where v is a nonnegative function. Once we have a solution v of (2.5), we also have a solution g of (2.4), as the following lemma states.

Lemma 2.3 *Let $\alpha > \frac{1}{2}$ with $\alpha \neq 1$, let $n \geq 3$, and let $v \in C^{1,\gamma}([0,\pi])$, $\gamma \in (0,1)$ with $\gamma \geq \max\{0, \frac{1}{\alpha} - 1\}$, satisfying $v(x) > 0$ for every $x \in (0,\pi)$ and $v(0) = v(\pi) = 0$, be a solution of (2.5). Let $g \in C^{\gamma}([0,\pi])$ be such that $\mathcal{S}_n g = v$. Then, the odd extension of g is in $C^{\frac{1}{\alpha}}(\mathbb{T})$ if $\alpha > 1$, and in $C^{1,\frac{1}{\alpha}-1}(\mathbb{T})$ if $\alpha < 1$, and in both cases solves (2.4).*

Remark 2.4 *When we say that g solves (2.4), it is understood in the divergence form*

$$\alpha \partial_x (g \, \mathcal{S}_n g) = (\alpha + 1) \, g \, \partial_x \mathcal{S}_n g,$$

in which g is never differentiated, that is,

$$-\alpha \int_{-\pi}^{\pi} g(x) \, \mathcal{S}_n g(x) \, \partial_x \psi(x) \, dx = (\alpha+1) \int_{-\pi}^{\pi} g(x) \, \partial_x \mathcal{S}_n g(x) \, \psi(x) \, dx \qquad \text{for every } \psi \in C^{\infty}(\mathbb{T}).$$

Since $(\widehat{\mathcal{S}_n})_k = O(\frac{1}{k})$, the operator $\mathcal{S}_n$ maps $L^2(\mathbb{T})$ into $H^1(\mathbb{T}) \subset L^2(\mathbb{T})$, so $\mathcal{S}_n g, \partial_x \mathcal{S}_n g, g \in L^2(\mathbb{T})$. By Cauchy-Schwarz, both $g \, \mathcal{S}_n g$ and $g \, \partial_x \mathcal{S}_n g$, each a product of two $L^2(\mathbb{T})$ factors, belong to $L^1(\mathbb{T})$ and the two integrals converge. Equation (1.5) is understood in the same sense.

Proof: Both g and $v^{\frac{1}{\alpha}}$ are continuous on $[0,\pi]$, so their odd extensions belong to $L^2(\mathbb{T})$ and are given by sine series. Since $(\widehat{\mathcal{S}_n})_k \neq 0$ for every $k \geq 1$, the operator $\mathcal{S}_n$ is injective on sine series, so (2.5) gives

$$g = -v^{\frac{1}{\alpha}} \quad \text{on } [0,\pi]. \tag{2.6}$$

Suppose first that $\alpha > 1$, so that $\frac{1}{\alpha} \in (0,1)$. For $a \geq b \geq 0$ and $t \in (0,1]$, the function $\phi(x) := (b+x)^t - x^t - b^t$ satisfies $\phi(0) = 0$ and, for $x > 0$,

$$\partial_x \phi(x) = t\big[(b+x)^{t-1} - x^{t-1}\big] \leq 0,$$

since $t-1 \leq 0$ and $b+x \geq x$, so ϕ is non-increasing and $\phi(a-b) \leq \phi(0) = 0$, that is, $a^t - b^t \leq (a-b)^t$, and the case $b > a$ follows by symmetry, so

$$\big|a^t - b^t\big| \leq |a-b|^t \qquad \text{for every } a, b \geq 0 \text{ and every } t \in (0,1]. \tag{2.7}$$

Since $v \in C^{1,\gamma}([0,\pi])$, we obtain from (2.6) and (2.7), for every $x, y \in [0,\pi]$,

$$|g(x) - g(y)| = \big|v(x)^{\frac{1}{\alpha}} - v(y)^{\frac{1}{\alpha}}\big| \leq |v(x) - v(y)|^{\frac{1}{\alpha}} \leq \|\partial_x v\|^{\frac{1}{\alpha}}_{C([0,\pi])} |x-y|^{\frac{1}{\alpha}},$$

so $g \in C^{\frac{1}{\alpha}}([0,\pi])$.

Since $g = -v^{\frac{1}{\alpha}}$ by (2.6) and $v(0) = v(\pi) = 0$, we have $g(0) = g(\pi) = 0$, so the odd extension $\widetilde{g}$ of g to the torus vanishes at 0 and at π. On $[0,\pi]$, $\widetilde{g} = g \in C^{\frac{1}{\alpha}}$ as shown above, and on $[-\pi, 0]$,

$\widetilde{g}(x) = -g(-x)$ is Hölder with the same constant. It is enough to bound the distance between points separated by 0 or by π. For $y < 0 < x$,

$$\begin{aligned}|\widetilde{g}(x) - \widetilde{g}(y)| &\le |\widetilde{g}(x) - \widetilde{g}(0)| + |\widetilde{g}(0) - \widetilde{g}(y)| = |\widetilde{g}(x)| + |\widetilde{g}(y)| \\ &\lesssim |x|^{\frac{1}{\alpha}} + |y|^{\frac{1}{\alpha}} \le 2|x-y|^{\frac{1}{\alpha}}.\end{aligned}$$

Here, in the last step, we have used that $|x-y| \ge |x|, |y|$. The same argument applies at the crossing at π. This gives the statement for $\alpha > 1$.

Suppose now that $\alpha \in (\frac{1}{2}, 1)$, so that $\frac{1}{\alpha} - 1 \in (0,1)$. By (2.6) and the chain rule,

$$\partial_x g = -\frac{1}{\alpha}\, v^{\frac{1}{\alpha}-1}\, \partial_x v,$$

which is continuous on $[0,\pi]$ as a product of two continuous functions and vanishes at $0, \pi$. Applying (2.7) with $t = \frac{1}{\alpha} - 1 \in (0,1)$, we obtain

$$|v(x)^{\frac{1}{\alpha}-1} - v(y)^{\frac{1}{\alpha}-1}| \le |v(x) - v(y)|^{\frac{1}{\alpha}-1} \le \|\partial_x v\|^{\frac{1}{\alpha}-1}_{C([0,\pi])} |x-y|^{\frac{1}{\alpha}-1},$$

where the last inequality is the mean value theorem. Hence $v^{\frac{1}{\alpha}-1} \in C^{\frac{1}{\alpha}-1}([0,\pi])$.

Both $v^{\frac{1}{\alpha}-1}$ and $\partial_x v$ are bounded on $[0,\pi]$, and

$$|\partial_x v(x) - \partial_x v(y)| \le \|\partial_x v\|_{C^\gamma([0,\pi])} |x-y|^\gamma \le \|\partial_x v\|_{C^\gamma([0,\pi])}\, \pi^{\gamma - \frac{1}{\alpha} + 1} |x-y|^{\frac{1}{\alpha}-1},$$

so $\partial_x v \in C^{\frac{1}{\alpha}-1}([0,\pi])$, and the decomposition

$$|\partial_x g(x) - \partial_x g(y)| \le \tfrac{1}{\alpha}\, v(x)^{\frac{1}{\alpha}-1} |\partial_x v(x) - \partial_x v(y)| + \tfrac{1}{\alpha}\, |\partial_x v(y)|\, |v(x)^{\frac{1}{\alpha}-1} - v(y)^{\frac{1}{\alpha}-1}| \lesssim |x-y|^{\frac{1}{\alpha}-1}$$

gives $\partial_x g \in C^{\frac{1}{\alpha}-1}([0,\pi])$, that is, $g \in C^{1,\frac{1}{\alpha}-1}([0,\pi])$.

For the odd extension, $g(0) = g(\pi) = 0$ implies that $\widetilde{g}$ is continuous, and $\partial_x \widetilde{g}$ is even about 0 and about π, hence continuous on $\mathbb{T}$. For $y < 0 < x$, $\partial_x \widetilde{g}(x) = \partial_x g(x)$ and $\partial_x \widetilde{g}(y) = \partial_x g(-y)$, so

$$\begin{aligned}|\partial_x \widetilde{g}(x) - \partial_x \widetilde{g}(y)| &= |\partial_x g(x) - \partial_x g(-y)| \\ &\le \|\partial_x g\|_{C^{\frac{1}{\alpha}-1}([0,\pi])} |x+y|^{\frac{1}{\alpha}-1} \le \|\partial_x g\|_{C^{\frac{1}{\alpha}-1}([0,\pi])} |x-y|^{\frac{1}{\alpha}-1}.\end{aligned}$$

Since the same argument applies at $x = \pi$, we have $\widetilde{g} \in C^{1,\frac{1}{\alpha}-1}(\mathbb{T})$.

We now prove that $\widetilde{g}$ solves (2.4) on $\mathbb{T}$, in the sense of Remark 2.4.

From (2.6) and $\mathcal{S}_n g = v$, we obtain

$$|g|^\alpha = |\mathcal{S}_n g|. \tag{2.8}$$

Since $\mathcal{S}_n g = v > 0$ on $(0,\pi)$, and both g and $\mathcal{S}_n g$ are odd, it follows from (2.8) that $g \neq 0$ and $\mathcal{S}_n g \neq 0$ on $\mathbb{T} - \{0,\pi\}$. Differentiating (2.8) gives

$$\alpha |g|^{\alpha-1} \mathrm{sgn}(g)\, \partial_x g = \mathrm{sgn}(\mathcal{S}_n g)\, \partial_x \mathcal{S}_n g.$$

Multiplying by g, we find

$$\alpha |g|^\alpha \partial_x g = g\, \mathrm{sgn}(\mathcal{S}_n g)\, \partial_x \mathcal{S}_n g. \tag{2.9}$$

Then, using $|g|^\alpha = |\mathcal{S}_n g|$ in (2.9), we get

$$\alpha |\mathcal{S}_n g|\, \partial_x g = g\, \mathrm{sgn}(\mathcal{S}_n g)\, \partial_x \mathcal{S}_n g.$$

Multiplying by $\mathrm{sgn}(\mathcal{S}_n g)$, this yields, on $\mathbb{T} - \{0,\pi\}$,

$$\alpha \mathcal{S}_n g\, \partial_x g = g\, \partial_x \mathcal{S}_n g. \tag{2.10}$$

It remains to control the two excluded points, 0 and π. Since $\mathcal{S}_n g = v \in C^{1,\gamma}([0,\pi])$ vanishes at 0 and at π, $\widetilde{\mathcal{S}_n g} \in C^{1,\gamma}(\mathbb{T})$.

For $\alpha < 1$ nothing more is needed, since $\widetilde{g} \in C^{1,\frac{1}{\alpha}-1}(\mathbb{T})$ too, so both sides of (2.10) are continuous on $\mathbb{T}$. For $\alpha > 1$, note that $\widetilde{g}\,\widetilde{\mathcal{S}_n g}$ is continuous on $\mathbb{T}$ and that, away from 0 and π, (2.10) gives

$$\alpha \partial_x (\widetilde{g}\,\widetilde{\mathcal{S}_n g}) = (\alpha+1)\,\widetilde{g}\,\partial_x \widetilde{\mathcal{S}_n g},$$

classically on $(0,\pi)$ and on $(-\pi,0)$. Since $\partial_x(g\,\mathcal{S}_n g) = (1+\frac{1}{\alpha}) g\,\partial_x \mathcal{S}_n g$ is bounded on $(0,\pi)$ and on $(-\pi,0)$, $\widetilde{g}\,\widetilde{\mathcal{S}_n g}$ is Lipschitz on each closed arc, hence absolutely continuous, so the identity extends across 0 and π and integration by parts against $\psi \in C^\infty(\mathbb{T})$ is valid on all of $\mathbb{T}$, giving exactly the divergence form of (2.4) in the sense of Remark 2.4. □

II.2.1 Existence argument

We adapt the existence argument of [OSW14] to our setting. The strategy is still based on Schauder's fixed point theorem [Sch30], applied to the set of functions normalized so that $\int_0^\pi v(x)\sin x\,dx = 1$. We first state the result in a general form, valid for other operators. It then suffices to check that $-\mathcal{S}_n$ satisfies the required assumptions.

Proposition 2.5 *Let $M > 0$ be such that the set V_M given by*

$$V_M := \left\{ v \in C([0,\pi]) :\ 0 \le v \le M,\ \int_0^\pi v(x)\sin x\,dx = 1 \right\},$$

is non-empty.

Let $\alpha > 0$, with $\alpha \neq 1$, and let

$$S : C([0,\pi]) \to C([0,\pi])$$

be a linear compact operator. Define

$$Av := S(v^{\frac{1}{\alpha}}).$$

Assume that A satisfies the following properties

1. *There exists $\rho > 0$ such that, for every $v \in V_M$,*

$$\int_0^\pi Av(x)\sin x\,dx = \rho \int_0^\pi v(x)^{\frac{1}{\alpha}} \sin x\,dx.$$

2. *For every $v \in V_M$ and every $x \in [0,\pi]$,*

$$0 \le Av(x) \le M \int_0^\pi Av(y)\sin y\,dy.$$

Then, A has a nonzero fixed point in $C([0,\pi])$, which is a multiple of an element of V_M.

Remark 2.6 *To ensure that V_M is bounded, we have introduced the constant M, which does not appear in [OSW14]. This seems to be necessary to apply Schauder's fixed point theorem.*

Proof: Let F be the operator given by

$$Fv := \frac{Av}{\int_0^\pi Av(x)\sin x\,dx}.$$

Now, we observe that V_M is a non-empty, closed, bounded, and convex subset of $C([0,\pi])$. We also note that $Fv \in V_M$ since $0 \le Fv \le M$ and

$$\int_0^\pi Fv(x)\sin x\,dx = 1.$$

To apply Schauder's fixed point theorem, we need to show that F is compact. Notice that S is compact, so is A. Therefore, it remains to check that

$$\inf_{v\in V_M}\int_0^\pi Av(x)\sin x\,dx = \rho\inf_{v\in V_M}\int_0^\pi v(x)^{\frac{1}{\alpha}}\sin x\,dx > 0.$$

For $\alpha < 1$ we can use the same Hölder estimate as in [OSW14]:

$$1 = \int_0^\pi v(x)\sin x\,dx \le \underbrace{\left(\int_0^\pi \sin x\,dx\right)^{1-\alpha}}_{=2^{1-\alpha}}\left(\int_0^\pi v(x)^{\frac{1}{\alpha}}\sin x\,dx\right)^{\alpha}. \tag{2.11}$$

For the case $\alpha > 1$, we note that $\frac{1}{\alpha} - 1 < 0$, so

$$v^{\frac{1}{\alpha}-1} \ge M^{\frac{1}{\alpha}-1}.$$

Hence,

$$\int_0^\pi v(x)^{\frac{1}{\alpha}}\sin x\,dx \ge M^{\frac{1}{\alpha}-1}\int_0^\pi v(x)\sin x\,dx = M^{\frac{1}{\alpha}-1}. \tag{2.12}$$

By Schauder's fixed point theorem, there exists $v \in V_M$ which is a fixed point of F. Since $\alpha \neq 1$, we can construct a nonzero multiple Γv so that Γv is a fixed point of A. Indeed, set $c := \int_0^\pi Av(x)\sin x\,dx$, so that the identity $Fv = v$ reads $Av = c\,v$. Since $A(\Gamma v) = \Gamma^{\frac{1}{\alpha}}Av$, the choice $\Gamma := c^{\frac{\alpha}{\alpha-1}}$ gives $A(\Gamma v) = \Gamma v$. □

Remark 2.7 *The proposition applies for every $\alpha > 0$ with $\alpha \neq 1$, but in this chapter we verify its hypotheses only for $\alpha > \frac{1}{2}$, $\alpha \neq 1$ (see Section II.2.3).*

Remark 2.8 *In Section II.2.3 we will take $S = -\mathcal{S}_n$. For $w \in C([0,\pi])$, the function $\mathcal{S}_n w$ is the restriction to $[0,\pi]$ of an odd function of $H^1(\mathbb{T}) \subset C(\mathbb{T})$, since $(\widehat{\mathcal{S}_n})_k = O(\frac{1}{|k|})$, so it vanishes at 0 and at π. Any nonnegative fixed point $v \in C([0,\pi])$ of A satisfies*

$$v = S(v^{\frac{1}{\alpha}}),$$

and since v is continuous and nonnegative, $v^{\frac{1}{\alpha}} \in C([0,\pi])$ too, so v is likewise the restriction of an odd $H^1(\mathbb{T})$ function, hence vanishes at 0 and π and its odd extension is continuous on $\mathbb{T}$.

Remark 2.9 *Finally, we observe that if there exists a non-empty, closed, and convex subset $\widetilde{V}_M \subset V_M$ such that $F(\widetilde{V}_M) \subset \widetilde{V}_M$, then Schauder's theorem can be applied to this smaller set, thereby yielding additional properties of the fixed point. For instance, in our case we can intersect V_M with the space of functions satisfying*

$$v(\tfrac{\pi}{2} + x) = v(\tfrac{\pi}{2} - x)\,.$$

This subset is preserved by F whenever S commutes with the reflection $Rv(x) := v(\pi - x)$, since then

$$ARv = S((Rv)^{\frac{1}{\alpha}}) = SR(v^{\frac{1}{\alpha}}) = RS(v^{\frac{1}{\alpha}}) = RAv,$$

and, since $\sin(\pi - y) = \sin y$, the normalizing integral $\int_0^\pi Av(y)\sin y\,dy$ is unchanged as well, so $F(Rv) = R(Fv)$. Hence, v admits a Fourier representation of the form

$$v(x) = \sum_{k\ge 0}\widehat{v}^{\mathrm{s}}_{2k+1}\sin((2k+1)x).$$

II.2.2 Regularity of the solution

This section collects the regularity results for the solutions constructed in this chapter. Lemma 2.10 deals with a solution v_α of (2.5) on $[0,\pi]$, and Corollary 2.11 transfers the result to the corresponding solution f_α of (1.5) on $\mathbb{T}$, which is the form used in Theorem 3.1 below.

Lemma 2.10 *Let $\alpha > 0$ and let $v_\alpha \in C([0,\pi])$ be a nonnegative solution of (2.5) with $n \geq 3$, and denote $g_\alpha := -v_\alpha^{1/\alpha}$. Then*

$$v_\alpha \in C^{1,\gamma}([0,\pi]) \quad \text{and} \quad g_\alpha \in C^{\gamma}([0,\pi]) \qquad \text{for every } \gamma < \min\{1, \tfrac{1}{\alpha}\}.$$

Proof: The proof follows the argument in [OSW14, Proposition 3], which relies on the boundedness of the Hilbert transform H on $L^p([0,\pi]),\ 1 < p < \infty$, and on $C^\gamma([0,\pi]),\ 0 < \gamma < 1$, so we will not repeat the full argument here.

Nevertheless, we work with a different operator, which differs from $\partial_x^{-1}H$ by a term that decays sufficiently fast:

$$\mathcal{S}_n = \partial_x^{-1} H(I + \mathcal{R}_n).$$

Here $\mathcal{R}_n$ is the smoothing Fourier multiplier defined by

$$\widehat{(\mathcal{R}_n)}_k := \frac{3}{n^2k^2 - 4},\ k \neq 0.$$

Since $\widehat{(\mathcal{R}_n)}_k = O(k^{-2})$, the operator $\mathcal{R}_n$ is bounded on $W^{s,p}([0,\pi]),\ 1 < p < \infty$, and on $C^\gamma([0,\pi]),\ 0 < \gamma < 1$. Hence the same holds for $\partial_x \mathcal{S}_n = H(I + \mathcal{R}_n)$, and the bootstrap applies without any change. □

Corollary 2.11 *Let $\alpha > \frac{1}{2}$ with $\alpha \neq 1$, let $n \geq 3$, and let $v_\alpha \in C([0,\pi])$ be a solution of (2.5) that is positive on $(0,\pi)$ and vanishes at 0 and π. Let g_α be as in Lemma 2.10 and let $\widetilde{g}_\alpha$ be its odd extension to $\mathbb{T}$, and set $f_\alpha(x) := \widetilde{g}_\alpha(nx)$. Then f_α is odd and $\frac{2\pi}{n}$-periodic, it solves (1.5), and $\widehat{(f_\alpha)}_n \neq 0$. Moreover, f_α belongs to $C^{\frac{1}{\alpha}}(\mathbb{T})$ if $\alpha > 1$, and to $C^{1,\frac{1}{\alpha}-1}(\mathbb{T})$ if $\alpha < 1$.*

Proof: By Lemma 2.10, $v_\alpha \in C^{1,\gamma}([0,\pi])$ and $g_\alpha \in C^\gamma([0,\pi])$ for every $\gamma < \min\{1, \frac{1}{\alpha}\}$. When $\alpha < 1$ we may in addition take $\gamma \geq \frac{1}{\alpha} - 1$, since $\alpha > \frac{1}{2}$ gives $\frac{1}{\alpha} - 1 < 1$.

Since $\mathcal{S}_n g_\alpha = v_\alpha$ by (2.5), these are the hypotheses of Lemma 2.3, which therefore gives that $\widetilde{g}_\alpha$ belongs to $C^{\frac{1}{\alpha}}(\mathbb{T})$ if $\alpha > 1$ and to $C^{1,\frac{1}{\alpha}-1}(\mathbb{T})$ if $\alpha < 1$, and that it solves (2.4).

The function $f_\alpha(x) := \widetilde{g}_\alpha(nx)$ is odd and $\frac{2\pi}{n}$-periodic, and it belongs to the same space as $\widetilde{g}_\alpha$, since rescaling preserves the Hölder exponent. By Lemma 2.2, f_α solves (1.5) and $\widehat{(f_\alpha)}_n$ is a nonzero multiple of the first sine coefficient of g_α. Since $v_\alpha = \mathcal{S}_n g_\alpha$ and $\widehat{(\mathcal{S}_n)}_1 \neq 0$, that coefficient is a nonzero multiple of $\int_0^\pi v_\alpha(x) \sin x\, dx$. □

II.2.3 Verification of the hypotheses of Proposition 2.5 for the operator $-\mathcal{S}_n$

In this section, we prove that $-\mathcal{S}_n$, defined in (2.1), satisfies all the assumptions of Proposition 2.5 with $S = -\mathcal{S}_n$, where, as above, $\mathcal{S}_n$ acts on a function on $[0,\pi]$ by applying it to the odd extension and restricting back to $[0,\pi]$.

A direct computation gives

$$\begin{aligned}\int_0^\pi -\mathcal{S}_n\big(v^{\frac{1}{\alpha}}\big)(x) \sin x\, dx &= \frac{2}{\pi}\int_0^\pi \Big(\int_0^\pi \sum_{k\geq 1} \frac{1}{k} \frac{n^2k^2 - 1}{n^2k^2 - 4} v^{\frac{1}{\alpha}}(y) \sin(kx)\sin(ky)\, dy \Big) \sin x\, dx \\ &= \frac{n^2 - 1}{n^2 - 4} \int_0^\pi v^{\frac{1}{\alpha}}(y) \sin y\, dy. \end{aligned} \tag{2.13}$$

Identity (2.13) corresponds to property 1 in Proposition 2.5 with $\rho = \frac{n^2-1}{n^2-4}$.

Moreover, since $(\widehat{\mathcal{S}_n})_k$ behaves like $-\frac{1}{|k|}$ as k goes to infinity, $\mathcal{S}_n$ is compact on $C([0,\pi])$.

To verify hypothesis 2 in Proposition 2.5, we next show, following the strategy of [OSW14], that $-\mathcal{S}_n(v^{\frac{1}{\alpha}})$ admits an integral representation on $[0,\pi]$ against a certain singular kernel.

Lemma 2.12 *Let $v \in C([0,\pi])$ be nonnegative and $n \geq 3$. Then, for any $x \in [0,\pi]$ we have that*

$$-\mathcal{S}_n(v^{\frac{1}{\alpha}})(x) = \int_0^\pi v(y)^{\frac{1}{\alpha}} K_n(x,y)\, dy,$$

where

$$K_n(x,y) := \frac{1}{2\pi}\log\Big(\frac{1-\cos(x+y)}{1-\cos(x-y)}\Big) + \frac{6}{\pi}\sum_{k\geq 1}\frac{1}{n^2k^3-4k}\sin(kx)\sin(ky).$$

Proof: We write $\mathcal{S}_n = \partial_x^{-1}H(I+\mathcal{R}_n)$, where the multiplier of $\mathcal{R}_n$ is given by

$$(\widehat{\mathcal{R}_n})_k := \frac{3}{n^2k^2-4}.$$

We can write $\partial_x^{-1}H(v^{\frac{1}{\alpha}})$ as

$$\partial_x^{-1}H(v^{\frac{1}{\alpha}})(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} v(y)^{\frac{1}{\alpha}}\log(1-\cos(x-y))\, dy.$$

Assuming now that v is odd, we obtain

$$\partial_x^{-1}H\mathcal{R}_n(v^{\frac{1}{\alpha}})(x) = -\frac{6}{\pi}\sum_{k\geq 1}\int_0^\pi v(y)^{\frac{1}{\alpha}}\frac{1}{n^2k^3-4k}\sin(kx)\sin(ky)\, dy,$$

$$\partial_x^{-1}H(v^{\frac{1}{\alpha}})(x) = \frac{1}{2\pi}\int_0^\pi v(y)^{\frac{1}{\alpha}}\log\Big(\frac{1-\cos(x-y)}{1-\cos(x+y)}\Big)\, dy.$$

Thus,

$$\begin{aligned}
-\mathcal{S}_n(v^{\frac{1}{\alpha}})(x) &= -\partial_x^{-1}H(v^{\frac{1}{\alpha}})(x) - \partial_x^{-1}H\mathcal{R}_n(v^{\frac{1}{\alpha}})(x)\\
&= \frac{1}{2\pi}\int_0^\pi v(y)^{\frac{1}{\alpha}}\log\Big(\frac{1-\cos(x+y)}{1-\cos(x-y)}\Big)\, dy + \frac{6}{\pi}\sum_{k\geq 1}\int_0^\pi v(y)^{\frac{1}{\alpha}}\frac{\sin(kx)\sin(ky)}{n^2k^3-4k}\, dy\\
&= \int_0^\pi v(y)^{\frac{1}{\alpha}} K_n(x,y)\, dy.
\end{aligned}$$

□

We will use the following lemma to prove that $-\mathcal{S}_n(v^{\frac{1}{\alpha}}) \geq 0$ for every nonnegative $v \in C([0,\pi])$.

Lemma 2.13 *Let ϕ_n, with $n \geq 3$, be the function given by*

$$\phi_n(x) := \log(1-\cos x) - 6\sum_{k\geq 1}\frac{1}{n^2k^3-4k}\cos(kx).$$

Then, $\partial_x^2\phi_n(x) < 0$ for all $x \in (0,\pi)$.

Proof: We begin by computing $\partial_x^2\phi_n$

$$\begin{aligned}
\partial_x^2\phi_n(x) &= -\frac{1}{2}\frac{1}{\sin^2(x/2)} + \frac{6}{n^2-4}\cos x + 6\sum_{k\geq 2}\frac{k}{n^2k^2-4}\cos(kx)\\
&= -\frac{1}{2}\frac{1}{\sin^2(x/2)} + \frac{6}{n^2-4}\cos x + \frac{6}{n^2}\sum_{k\geq 2}\frac{1}{k}\cos(kx) + \frac{24}{n^2}\sum_{k\geq 2}\frac{1}{n^2k^3-4k}\cos(kx).
\end{aligned}$$

We use the identity

$$\begin{aligned}\log(2-2\cos x) &= \log(1-e^{ix}) + \log(1-e^{-ix}) \\ &= -\sum_{k\geq 1}\frac{e^{ikx}}{k} - \sum_{k\geq 1}\frac{e^{-ikx}}{k} = -2\sum_{k\geq 1}\frac{1}{k}\cos(kx),\end{aligned} \tag{2.14}$$

which yields

$$\begin{aligned}\frac{6}{n^2-4}\cos x + \frac{6}{n^2}\sum_{k\geq 2}\frac{1}{k}\cos(kx) &= \frac{6}{n^2-4}\cos x - \frac{3}{n^2}\log(2-2\cos x) - \frac{6}{n^2}\cos x \\ &= \frac{24}{n^2(n^2-4)}\cos x - \frac{3}{n^2}\log(2-2\cos x).\end{aligned} \tag{2.15}$$

From (2.15), we can decompose $\partial_x^2\phi_n$ as

$$\partial_x^2\phi_n = I_{\text{main}} + I_{\text{tail}},$$

where

$$\begin{aligned}I_{\text{main}}(x) &:= -\frac{1}{2}\frac{1}{\sin^2(x/2)} + \frac{24}{n^4-4n^2}\cos x - \frac{3}{n^2}\log(2-2\cos x), \\ I_{\text{tail}}(x) &:= \frac{24}{n^2}\sum_{k\geq 2}\frac{1}{n^2k^3-4k}\cos(kx).\end{aligned}$$

We first estimate I_{tail}. Define τ by

$$\tau := \sum_{k\geq 2}\frac{1}{k^2}\frac{1}{k-1}. \tag{2.16}$$

It follows that

$$\|I_{\text{tail}}\|_{L^\infty([0,\pi])} \leq \frac{24}{n^2}\sum_{k\geq 2}\frac{1}{n^2k^3-4k} \leq \frac{24}{n^4}\tau. \tag{2.17}$$

We next estimate I_{main}. First, observe that

$$\begin{aligned}-\log(2-2\cos x) &= -\log\left(\frac{1-\cos x}{2}\right) - 2\log 2 = \int_x^\pi \cot(y/2)\,dy - 2\log 2 \\ &\leq \int_x^\pi \frac{\cos(y/2)}{\sin^2(y/2)}\,dy - 2\log 2 = \frac{2}{\sin(x/2)} - 2 - 2\log 2.\end{aligned} \tag{2.18}$$

Set

$$\vartheta := \sin(x/2) \in (0,1), \quad \text{for all } x \in (0,\pi).$$

Combining (2.17) and (2.18):

$$\begin{aligned}\partial_x^2\phi_n(x) &= I_{\text{main}}(x) + I_{\text{tail}}(x) \\ &\leq -\frac{1}{2}\frac{1}{\sin^2(x/2)} + \frac{24}{n^4-4n^2}\cos x + \frac{3}{n^2}\left(\frac{2}{\sin(x/2)} - 2 - 2\log 2\right) + \frac{24}{n^4}\tau \\ &\leq -\frac{1}{2\vartheta^2} + \frac{24(1-2\vartheta^2)}{n^4-4n^2} + \frac{6}{n^2}\left(\frac{1}{\vartheta} - 1 - \log 2\right) + \frac{24}{n^4}\tau.\end{aligned}$$

Multiplying this last expression by $\vartheta^2 > 0$ preserves the inequality, and we obtain the polynomial P_n given by

$$\vartheta^2\partial_x^2\phi_n(x) \leq P_n(\vartheta) := -\frac{1}{2} + \frac{24}{n^4-4n^2}\vartheta^2(1-2\vartheta^2) + \frac{6}{n^2}\left(\vartheta - (1+\log 2)\vartheta^2\right) + \frac{24}{n^4}\tau\vartheta^2.$$

Thus, it suffices to prove that $P_n(\vartheta) < 0$ for every $\vartheta \in (0,1)$.

We first look at the quadratic coefficient of P_n, given by

$$\frac{1}{2}\partial_\vartheta^2 P_n(0) = \frac{24}{n^4-4n^2} - \frac{6}{n^2}(1+\log 2) + \frac{24}{n^4}\tau. \tag{2.19}$$

Define

$$\xi(n) := n^2\left(\frac{1}{4} - \frac{1}{n^2-4}\right),\ n \geq 3,$$

which is positive and increasing since

$$\partial_n\left(\frac{1}{4} - \frac{1}{n^2-4}\right) = \frac{2n}{(n^2-4)^2} > 0, \quad \text{for all } n \geq 3.$$

Since $\pi < \frac{22}{7}$, we have $\pi^2 < \frac{484}{49} < \frac{99}{10}$. Using also $\zeta(2) = \frac{\pi^2}{6}$, where ζ denotes the Riemann zeta function, we obtain

$$\tau < \frac{1}{4} + \frac{1}{2}\sum_{k\geq 3}\frac{1}{k^2} = \frac{\pi^2}{12} - \frac{3}{8} < \frac{33}{40} - \frac{3}{8} = \xi(3) \leq \xi(n) \quad \text{for all } n \geq 3. \tag{2.20}$$

Substituting (2.20) into (2.19) yields

$$\frac{1}{2}\partial_\vartheta^2 P_n(0) < \frac{6}{n^2}\left(\frac{4}{n^2-4} - (1+\log 2) + \frac{4}{n^2}\xi(n)\right) = -\frac{6}{n^2}\log 2 < 0.$$

Therefore, the quadratic coefficient of P_n is negative, and hence

$$\vartheta^2\partial_x^2\phi_n(x) \leq P_n(\vartheta) \leq -\frac{48}{n^4-4n^2}\vartheta^4 + \frac{6}{n^2}\vartheta - \frac{1}{2} =: Q_n(\vartheta).$$

It remains to show that $Q_n < 0$ on $(0,1)$. Its leading coefficient is negative and its derivative only has one real root, so Q_n has a unique critical point on the real line, which is therefore its maximum there. Setting $\partial_\vartheta Q_n(\vartheta) = 0$ gives $\vartheta_0^3 = \frac{n^2-4}{32}$.

Evaluating Q_n at ϑ_0, we find that

$$Q_n(\vartheta_0) = -\frac{48}{n^4-4n^2}\vartheta_0^4 + \frac{6}{n^2}\vartheta_0 - \frac{1}{2} = -\frac{3}{2n^2}\vartheta_0 + \frac{6}{n^2}\vartheta_0 - \frac{1}{2} = \frac{9}{2n^2}\left(\frac{n^2-4}{32}\right)^{1/3} - \frac{1}{2}.$$

Finally, notice that this last quantity is negative if and only if

$$32n^6 > 729(n^2-4).$$

Since $n \geq 3$,

$$32n^6 \geq 2592\, n^2 > 729(n^2-4).$$

Hence $Q_n(\vartheta_0) < 0$, and therefore $Q_n(\vartheta) < 0$ for every $\vartheta \in (0,1)$. Consequently, $P_n(\vartheta) < 0$, and therefore $\partial_x^2\phi_n(x) < 0$, as claimed. □

The following lemma shows that the kernel K_n is nonnegative, which gives the first inequality of hypothesis 2 in Proposition 2.5.

Lemma 2.14 *Let $n \geq 3$ and let $v \in C([0,\pi])$ be nonnegative and not identically zero. Then,*

$$K_n(x,y) \geq 0 \quad \textit{for a.e. } (x,y) \in [0,\pi]^2,$$

and

$$-\mathcal{S}_n(v^{\frac{1}{\alpha}}) \in C([0,\pi])$$

is nonnegative. Moreover,

$$-\mathcal{S}_n(v^{\frac{1}{\alpha}})(x) > 0 \quad \textit{for every } x \in (0,\pi),$$

while

$$-\mathcal{S}_n(v^{\frac{1}{\alpha}})(0) = -\mathcal{S}_n(v^{\frac{1}{\alpha}})(\pi) = 0.$$

Proof: Since the kernel K_n is integrable, $-\mathcal{S}_n(v^{\frac{1}{\alpha}})$ is continuous on $[0,\pi]$ for all $v \in C([0,\pi])$. In addition, K_n satisfies

$$K_n(0,y) = K_n(\pi,y) = 0 \quad \text{for all } y \in [0,\pi].$$

Hence, $-\mathcal{S}_n(v^{\frac{1}{\alpha}})(0) = -\mathcal{S}_n(v^{\frac{1}{\alpha}})(\pi) = 0$.

The nontrivial part of the statement is the nonnegativity. Recall from Lemma 2.12 that

$$-\mathcal{S}_n(v^{\frac{1}{\alpha}})(x) = \int_0^\pi v(y)^{\frac{1}{\alpha}} K_n(x,y)\,dy.$$

The kernel $K_n(x,y)$ is not defined on the diagonal $x = y$, but this set has measure zero and the singularity is integrable. Hence, it is enough to prove that $K_n(x,y) \geq 0$ for a.e. $(x,y) \in [0,\pi]^2$. We work with the variables $u := x + y$ and $z := x - y$, which lie in

$$R_{\text{squ}} := \{(u,z) :\ 0 < u + z < 2\pi,\ 0 < u - z < 2\pi\}.$$

Then, we obtain

$$\begin{aligned}\widetilde{K}_n(u,z) = K_n\Big(\frac{u+z}{2}, \frac{u-z}{2}\Big) &= \frac{1}{2\pi}\log\Big(\frac{1-\cos u}{1-\cos z}\Big) + \frac{3}{\pi}\sum_{k\geq 1}\frac{1}{n^2k^3 - 4k}\left(\cos(kz) - \cos(ku)\right)\\ &= \frac{1}{2\pi}\phi_n(u) - \frac{1}{2\pi}\phi_n(z).\end{aligned}$$

Here, ϕ_n is the function introduced in Lemma 2.13.

Differentiating gives

$$\partial_u\widetilde{K}_n(u,z) = \frac{1}{2\pi}\partial_x\phi_n(u), \qquad \partial_z\widetilde{K}_n(u,z) = -\frac{1}{2\pi}\partial_x\phi_n(z).$$

By Lemma 2.13, $\partial_x^2\phi_n(x) < 0$ for all $x \in (0,\pi)$, so $\partial_x\phi_n$ is strictly decreasing there. Since $\partial_x\phi_n$ vanishes at π, we conclude that $\partial_x\phi_n(x) > 0$ for $x \in (0,\pi)$.

Now fix $z_0 \in (0,\pi)$ and consider $\widetilde{K}_n(u + z_0, z_0)$ for $u \in (0, \pi - z_0)$. With this parametrization, we cover the triangle

$$R_{\text{tri}} := \{(u,z) :\ 0 < z, u < \pi,\ 0 < u - z < \pi\}.$$

By the mean value theorem,

$$\phi_n(u + z_0) - \phi_n(z_0) = u\partial_x\phi_n(s)$$

for some $s \in (z_0, u + z_0) \subset (0,\pi)$. Since $\partial_x\phi_n(s) > 0$ for $s \in (0,\pi)$, we conclude that

$$\widetilde{K}_n(u + z_0, z_0) = \frac{1}{2\pi}\big(\phi_n(u + z_0) - \phi_n(z_0)\big) > 0.$$

We have proved that $\widetilde{K_n}$ is positive on R_{tri}. We now extend this to all of R_{squ}, using the fact that ϕ_n satisfies

$$\phi_n(2\pi - x) = \phi_n(-x) = \phi_n(x).$$

As $\widetilde{K_n}(u,z) = \frac{1}{2\pi}\phi_n(u) - \frac{1}{2\pi}\phi_n(z)$ for every $(u,z) \in R_{\text{squ}}$, we have

$$\widetilde{K_n}(u,z) = \widetilde{K_n}(u,-z) = \widetilde{K_n}(2\pi - u, z) = \widetilde{K_n}(2\pi - u, -z). \tag{2.21}$$

Hence, every value of $\widetilde{K_n}$ on $R_{\text{squ}} - \{(u,z) \in \mathbb{R}^2 : \ z(u-\pi) = 0\}$ coincides with a value taken on R_{tri}, which is positive. This is illustrated in Figure II.1.

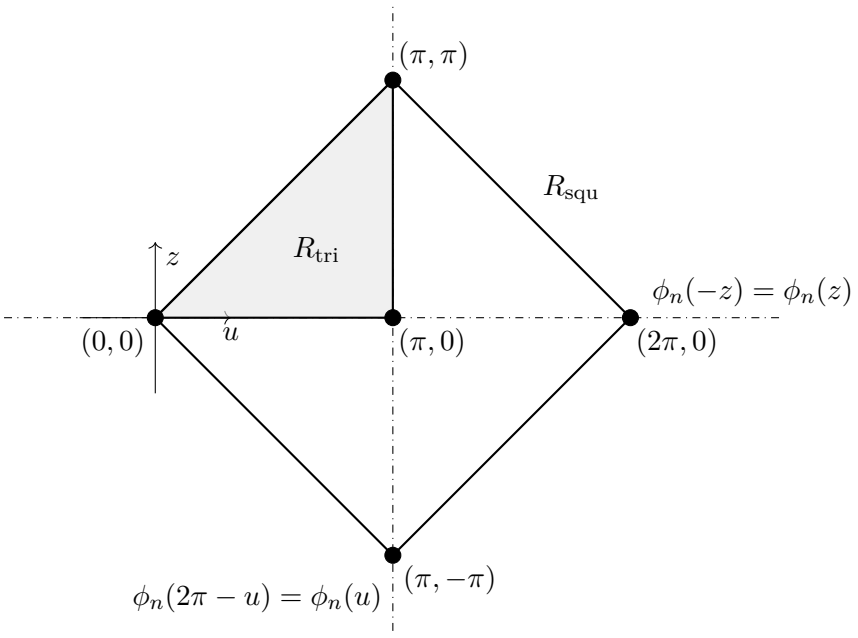


Figure II.1: Representation of R_{squ} and R_{tri}.

Finally, fix $x \in (0,\pi)$. Then $K_n(x,y) > 0$ for every $y \in (0,\pi)$ with $y \neq x$ and $y \neq \pi - x$, and v is nonnegative and does not vanish identically, so

$$-\mathcal{S}_n(v^{\frac{1}{\alpha}})(x) = \int_0^\pi v(y)^{\frac{1}{\alpha}} K_n(x,y)\, dy > 0.$$

□

The following lemma proves the second inequality of hypothesis 2 in Proposition 2.5.

Lemma 2.15 *Let $\alpha > \frac{1}{2}$ with $\alpha \neq 1$. Then, there exists $M > 0$ such that $V_M \neq \varnothing$ and, for every $v \in V_M$, the following estimate holds,*

$$\| -\mathcal{S}_n(v^{\frac{1}{\alpha}})\|_{C([0,\pi])} \leq M \int_0^\pi -\mathcal{S}_n(v^{\frac{1}{\alpha}})(y) \sin y\, dy.$$

Remark 2.16 *We cannot choose M arbitrarily small. Indeed, if $M < \frac{1}{2}$ and $v \in V_M$, then*

$$1 = \int_0^\pi v(x) \sin x\, dx \leq M \int_0^\pi \sin x\, dx = 2M < 1,$$

a contradiction. Hence, $V_M = \varnothing$ whenever $M < \frac{1}{2}$.

The proof of Lemma 2.15 relies on the following four auxiliary lemmas, whose proofs are deferred to Appendix II.A.

Lemma 2.17 *For every $(x,y)\in[0,\pi]^2$, it holds that*

$$K_n(x,y)\lesssim \sin y\,\cot\left(\frac{|x-y|}{2}\right)+\sin y.$$

Lemma 2.18 *Let $A_1,A_2>0$, $\alpha\in(0,1)$ and let $\beta:(0,\infty)\to\mathbb{R}$ be the function given by*

$$\beta(M):=A_1M^{\frac{1}{\alpha}}+A_2-M.$$

Then, β attains a negative minimum at $M_0:=(A_1\alpha^{-1})^{\frac{1}{1-\alpha^{-1}}}$ if and only if

$$A_1^{\alpha}A_2^{1-\alpha}<(1-\alpha)^{1-\alpha}\alpha^{\alpha}.$$

Lemma 2.19 *Let $\alpha\in(\frac{1}{2},1)$ and let $\mathcal{I}:\mathbb{R}^+\to\mathbb{R}$ be given by*

$$\mathcal{I}(\delta):=\sup_{x\in[0,\pi]}\int_0^{\pi}K_n(x,y)\chi_{|x-y|<\delta}(x,y)\,dy.$$

Then, we have that

$$\lim_{\delta\to0^+}\ \mathcal{I}(\delta)^{\alpha}\left(1+\cot(\delta/2)\right)^{1-\alpha}=0.$$

Lemma 2.20 *Let $\mathcal{I}:\mathbb{R}^+\to\mathbb{R}$ be the function defined in Lemma 2.19. Then, for every $\delta\in(0,\frac{\pi}{3})$, we have that*

$$\mathcal{I}(\delta)\lesssim\delta(1-\log\delta).$$

Proof of Lemma 2.15: By Lemma 2.14, $-\mathcal{S}_n(v^{\frac{1}{\alpha}})$ is continuous, so its $C([0,\pi])$ norm is well defined. Using identity (2.13), the statement of this lemma is equivalent to the estimate

$$\|-\mathcal{S}_n(v^{\frac{1}{\alpha}})\|_{C([0,\pi])}\leq M\frac{n^2-1}{n^2-4}\int_0^{\pi}v(y)^{\frac{1}{\alpha}}\sin y\,dy. \tag{2.22}$$

We split the integration domain into two parts

$$\begin{aligned}-\mathcal{S}_n(v^{\frac{1}{\alpha}})(x)&=\int_0^{\pi}v(y)^{\frac{1}{\alpha}}K_n(x,y)\,dy\\&=\int_0^{\pi}v(y)^{\frac{1}{\alpha}}K_n(x,y)\chi_{|x-y|<\delta}(x,y)\,dy+\int_0^{\pi}v(y)^{\frac{1}{\alpha}}K_n(x,y)\chi_{|x-y|>\delta}(x,y)\,dy.\end{aligned} \tag{2.23}$$

Here, $\delta>0$ will be chosen later.

Recall $\mathcal{I}(\delta)$ defined in Lemma 2.19. By Lemma 2.20, for every $\varepsilon>0$ there exists $\delta>0$ such that $\mathcal{I}(\delta)<\varepsilon$. Now, for any $v\in V_M$, we have that

$$\int_0^{\pi}v(y)^{\frac{1}{\alpha}}K_n(x,y)\chi_{|x-y|<\delta}(x,y)\,dy\leq M^{\frac{1}{\alpha}}\int_0^{\pi}K_n(x,y)\chi_{|x-y|<\delta}(x,y)\,dy\leq M^{\frac{1}{\alpha}}\mathcal{I}(\delta). \tag{2.24}$$

Denote now by $\mathcal{B}(\delta,x,v)$ the quantity given by

$$\mathcal{B}(\delta,x,v):=\int_0^{\pi}v(y)^{\frac{1}{\alpha}}K_n(x,y)\chi_{|x-y|>\delta}(x,y)\,dy.$$

By Lemma 2.17, we have

$$\mathcal{B}(\delta, x, v) \lesssim (\cot(\delta/2) + 1) \int_0^\pi v(y)^{\frac{1}{\alpha}} \sin y \, dy. \tag{2.25}$$

Now, we distinguish two cases depending on the range of α.

Case $\alpha > 1$. For $v \in V_M$, we saw in (2.12) that

$$M^{\frac{1}{\alpha}-1} \leq \int_0^\pi v(x)^{\frac{1}{\alpha}} \sin x \, dx. \tag{2.26}$$

Substituting (2.26) into (2.24) yields

$$\int_0^\pi v(y)^{\frac{1}{\alpha}} K_n(x,y) \chi_{|x-y|<\delta}(x,y) \, dy \leq M\mathcal{I}(\delta) \int_0^\pi v(x)^{\frac{1}{\alpha}} \sin x \, dx. \tag{2.27}$$

Therefore, substituting estimates (2.25) and (2.27) into (2.23), we find that

$$-\mathcal{S}_n(v^{\frac{1}{\alpha}})(x) \lesssim (M\mathcal{I}(\delta) + \cot(\delta/2) + 1) \int_0^\pi v(y)^{\frac{1}{\alpha}} \sin y \, dy.$$

Consequently, to prove inequality (2.22), it is enough to check that for any $C > 0$ we can find δ and M such that

$$M\mathcal{I}(\delta) + \cot(\delta/2) + 1 \leq CM.$$

Fix $\varepsilon < C$ and choose δ such that $\mathcal{I}(\delta) < \varepsilon$. Then, choose $M \geq \frac{1}{C-\varepsilon} [\cot(\delta/2) + 1]$ large enough that $V_M \neq \varnothing$. With these choices,

$$M\mathcal{I}(\delta) + \cot(\delta/2) + 1 \leq \ \varepsilon M + (C - \varepsilon)M = CM.$$

Case $\alpha \in (\frac{1}{2}, 1)$. For $v \in V_M$, we recall inequality (2.11),

$$2^{1-\frac{1}{\alpha}} \leq \int_0^\pi v(x)^{\frac{1}{\alpha}} \sin x \, dx. \tag{2.28}$$

Substituting (2.28) into (2.24) yields

$$\int_0^\pi v(y)^{\frac{1}{\alpha}} K_n(x,y) \chi_{|x-y|<\delta}(x,y) \, dy \leq 2^{\frac{1}{\alpha}-1} M^{\frac{1}{\alpha}} \mathcal{I}(\delta) \int_0^\pi v(x)^{\frac{1}{\alpha}} \sin x \, dx. \tag{2.29}$$

Consequently, substituting estimates (2.25) and (2.29) into (2.23) yields

$$-\mathcal{S}_n(v^{\frac{1}{\alpha}})(x) \lesssim \left(M^{\frac{1}{\alpha}} \mathcal{I}(\delta) + \cot(\delta/2) + 1\right) \int_0^\pi v(y)^{\frac{1}{\alpha}} \sin y \, dy. \tag{2.30}$$

Thus, it is enough to prove that, for any $\varepsilon > 0$, there exist δ and M such that

$$\mathcal{I}(\delta) M^{\frac{1}{\alpha}} + 1 + \cot(\delta/2) \leq \varepsilon M. \tag{2.31}$$

Notice that (2.31) is equivalent to showing that, for any $\varepsilon > 0$, there exist δ and M such that

$$\underbrace{\frac{\mathcal{I}(\delta)}{\varepsilon}}_{A_1} M^{\frac{1}{\alpha}} + \underbrace{\frac{1 + \cot(\delta/2)}{\varepsilon}}_{A_2} - M \leq 0.$$

By Lemma 2.18, this is equivalent to proving that, for any $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\left(\frac{\mathcal{I}(\delta)}{\varepsilon}\right)^\alpha \left(\frac{1 + \cot(\delta/2)}{\varepsilon}\right)^{1-\alpha} = A_1^\alpha A_2^{1-\alpha} < (1-\alpha)^{1-\alpha} \alpha^\alpha.$$

This is possible if we can make the product

$$\mathcal{I}(\delta)^\alpha \left(1+\cot(\delta/2)\right)^{1-\alpha}$$

arbitrarily small by taking δ small enough, which is true since Lemma 2.19 proves that, for $\alpha \in (\frac{1}{2}, 1)$,

$$\lim_{\delta \to 0^+} \mathcal{I}(\delta)^\alpha \left(1+\cot(\delta/2)\right)^{1-\alpha} = 0.$$

By Lemma 2.18, we take M to be

$$M_0 := \left(\frac{\mathcal{I}(\delta)}{\varepsilon}\alpha^{-1}\right)^{\frac{1}{1-\alpha^{-1}}}.$$

Notice that taking δ smaller gives a larger M_0, so that V_M is never empty. □

II.3 Main theorem

We are now ready to prove our main result.

Theorem 3.1 *For any $\alpha > \frac{1}{2}$ and any $n \geq 3$, there exists an odd $\frac{2\pi}{n}$-periodic function f_α, belonging to $C^{\frac{1}{\alpha}}(\mathbb{T})$ if $\alpha > 1$, and to $C^{1,\frac{1}{\alpha}-1}(\mathbb{T})$ if $\alpha < 1$, such that its n-th Fourier coefficient does not vanish, i.e. $(\widehat{f_\alpha})_n \neq 0$, and f_α solves equation (1.5), namely*

$$\alpha \mathcal{S} f_\alpha \, \partial_x f_\alpha = f_\alpha \, \partial_x \mathcal{S} f_\alpha.$$

Proof: For $\alpha = 1$, any multiple of $\sin(nx)$ is a solution.

For $\alpha \neq 1$, we know by Proposition 2.5, applied with $S = -\mathcal{S}_n$ and with hypotheses checked in Section II.2.3 for $\alpha > \frac{1}{2}$, $\alpha \neq 1$, that there exists a nonzero fixed point v_α of $v \mapsto -\mathcal{S}_n(v^{\frac{1}{\alpha}})$ that is a multiple of an element of V_M.

By Lemma 2.14, v_α is positive on $(0,\pi)$ and vanishes at 0 and π, so Corollary 2.11 applies. □

Proof of Theorem 1.1: This follows from the particular case $\alpha = 2$ of Theorem 3.1. Let $f := f_2$ be the function given by the theorem. It remains to verify that Θ solves (1.1) in the sense of Remark 1.2.

Write

$$u(x_1,x_2) = -\Lambda^{-1}\Theta(x_1,x_2) = r^2 \mathcal{S} f(x).$$

Since $\mathcal{S}$ differs from $\partial_x^{-1} H$ by a smoothing operator, we have $\mathcal{S} f \in C^{1,\frac{1}{2}}(\mathbb{T})$, so $u \in C^1(\mathbb{R}^2)$. Since f is continuous, $\Theta \nabla^\perp u$ is then continuous on $\mathbb{R}^2$, so it is locally integrable and the integral of Remark 1.2 is absolutely convergent.

In polar coordinates, for every $\phi(x_1,x_2) = \phi(r,x)$, in the radial-angular frame,

$$\nabla\phi = \Big(\partial_r\phi,\ \tfrac{1}{r}\partial_x\phi\Big) \qquad \text{and} \qquad \nabla^\perp\phi = \Big(-\tfrac{1}{r}\partial_x\phi,\ \partial_r\phi\Big).$$

Hence,

$$\int_{\mathbb{R}^2} \Big(\Theta\, \nabla^\perp u \cdot \nabla\psi\Big)(x_1,x_2)\, dx = \int_0^\infty \int_{-\pi}^{\pi} r^2 f(x) \Big(2\mathcal{S}f(x)\, \partial_x\psi(r,x) - r\, \partial_x \mathcal{S} f(x)\, \partial_r\psi(r,x)\Big)\, dx\, dr.$$

Integrating by parts the first term in x and the second term in r, we obtain

$$\int_{\mathbb{R}^2} \Big(\Theta\, \nabla^\perp u \cdot \nabla\psi\Big)(x_1,x_2)\, dx = \int_0^\infty \int_{-\pi}^{\pi} r^2 \Big(f(x)\, \partial_x \mathcal{S} f(x) - 2\mathcal{S}f(x)\, \partial_x f(x)\Big)\psi(r,x)\, dx\, dr, \quad (3.1)$$

with no boundary contribution in either integration by parts, in x because $f\, \mathcal{S} f\, \psi$ is continuous on $\mathbb{T}$, and in r because ψ has compact support and the integrand vanishes at $r = 0$. Finally, by (1.5) with $\alpha = 2$, understood in the sense of Remark 2.4, the right-hand side of (3.1) vanishes. □

For a 2π-periodic function f, the stationary version of (1.7) reads as

$$\alpha u\,\partial_x f = f\,\partial_x u, \qquad \partial_x u = Hf. \tag{3.2}$$

Theorem 3.2 *For any $\alpha > \frac{1}{2}$ and any $n \geq 1$, there exists an odd $\frac{2\pi}{n}$-periodic function f_α, belonging to $C^{\frac{1}{\alpha}}(\mathbb{T})$ if $\alpha > 1$, and to $C^{1,\frac{1}{\alpha}-1}(\mathbb{T})$ if $\alpha < 1$, such that its n-th Fourier coefficient does not vanish, i.e. $(\widehat{f_\alpha})_n \neq 0$, and f_α solves the stationary generalized De Gregorio equation* (3.2).

Proof: It suffices to treat $n = 1$, since the case $n \geq 2$ follows from constructing the case $n = 1$ and rescaling by $x \mapsto nx$, as in Remark 2.1. For $\alpha = 1$, any multiple of $\sin(nx)$ is a solution.

For $\alpha \neq 1$, the proof follows exactly as the proof of Theorem 3.1, applied with $S = -\partial_x^{-1}H$, whose kernel is the logarithmic term of K_n alone.

Hypothesis 1 of Proposition 2.5 holds with $\rho = 1$, by the same computation as (2.13).

For hypothesis 2, the nonnegativity of the kernel follows from Lemma 2.14's proof with $\phi(x) = \log(1 - \cos x)$, for which

$$\partial_x^2\phi(x) = -\tfrac{1}{2\sin^2(x/2)} < 0 \qquad \text{for all } x \in (0, \pi),$$

directly, so Lemma 2.13 is not needed. The remaining inequality of hypothesis 2 follows as in the proof of Lemma 2.15, where the ε-δ condition of Lemma 2.20 is now obtained in the same way as in the proof of that lemma, without having to bound the smoothing part, which there only contributes an $O(\delta)$.

Proposition 2.5 then gives a nonzero fixed point v_α of $v \mapsto -\partial_x^{-1}H\big(v^{\frac{1}{\alpha}}\big)$ that is a multiple of an element of V_M. The argument of Corollary 2.11 applies with $\mathcal{S}_n$ replaced by $\partial_x^{-1}H$, so v_α gives a function f_α that solves (3.2) and has the stated regularity. □

II.A Auxiliary lemmas

This appendix contains the proofs of the auxiliary lemmas used in the proof of Lemma 2.15, as well as a further intermediate lemma used in their proofs.

Proof of Lemma 2.17: The kernel K_n has a logarithmic part and a Fourier series part (see Lemma 2.12). To bound the latter one, we apply the symmetry condition (2.21) to $\widetilde{K_n}(u, z) = K_n\big(\frac{u+z}{2}, \frac{u-z}{2}\big)$,

$$K_n(\pi - x, \pi - y) = \widetilde{K_n}(2\pi - x - y, y - x) = \widetilde{K_n}(x + y, x - y) = K_n(x, y).$$

Hence, it suffices to prove the bound for $y \in [0, \frac{\pi}{2}]$, since the case $y \in [\frac{\pi}{2}, \pi]$ follows by symmetry.

For $y \in [0, \frac{\pi}{2}]$, we have $y \leq \pi \sin y$. Consequently, the Fourier series part of K_n can be estimated as

$$\sum_{k\geq 1} \frac{1}{n^2k^3 - 4k} \sin(kx)\sin(ky) = \int_0^y \sum_{k\geq 1} \frac{1}{n^2k^2 - 4} \sin(kx)\cos(ks)\,ds \lesssim \int_0^y ds \leq y \leq \pi \sin y. \tag{A.1}$$

It remains to control the logarithmic term. We claim that

$$\log\Big(\frac{1 - \cos(x + y)}{1 - \cos(x - y)}\Big) \leq 2 \sin y\, \cot\Big(\frac{|x - y|}{2}\Big). \tag{A.2}$$

The kernel $K_n(x, y)$ is not defined at $x = y$, as noted in the proof of Lemma 2.14. For $x \neq y$, $\theta := x - y$ satisfies $\theta \in [-\pi, \pi] - \{0\}$, so $1 - \cos\theta > 0$. Then, inequality (A.2) becomes

$$\log\Big(\frac{1 - \cos\theta\, \cos(2y) + \sin\theta\, \sin(2y)}{1 - \cos\theta}\Big) \leq 2 \sin y\, \frac{\sin|\theta|}{1 - \cos\theta} =: \Gamma. \tag{A.3}$$

Since $\Gamma \geq 0$, we have

$$1 + \Gamma + \frac{1}{2}\Gamma^2 \leq \exp(\Gamma).$$

Therefore, to prove inequality (A.3), it suffices to check that

$$\frac{1 - \cos\theta\, \cos(2y) + \sin\theta\, \sin(2y)}{1 - \cos\theta} \leq 1 + \Gamma + \frac{1}{2}\Gamma^2. \tag{A.4}$$

We now simplify (A.4) through a chain of equivalent inequalities. Multiplying both sides by $1 - \cos\theta > 0$, we obtain

$$1 - \cos\theta\, \cos(2y) + \sin\theta\, \sin(2y) \leq 1 - \cos\theta + 2\sin y\, \sin|\theta| + 2\sin^2 y \frac{1 - \cos^2\theta}{1 - \cos\theta}.$$

Now, we subtract 1 and use the identity $\frac{1-\cos^2\theta}{1-\cos\theta} = 1 + \cos\theta$, so (A.4) is equivalent to

$$-\cos\theta\, \cos(2y) + \sin\theta\, \sin(2y) \leq -\cos\theta + 2\sin y\, \sin|\theta| + 2\sin^2 y\,(1 + \cos\theta). \tag{A.5}$$

Observe that

$$-\cos\theta\, \cos(2y) = -\cos\theta + 2\sin^2 y\, \cos\theta. \tag{A.6}$$

Substituting (A.6) into (A.5), we obtain

$$\sin\theta\, \sin(2y) \leq 2\sin y\, \sin|\theta| + 2\sin^2 y. \tag{A.7}$$

Inequality (A.7) now follows from

$$\sin\theta\, \cos y \leq \sin|\theta| + \sin y,$$

after multiplying by $2\sin y \geq 0$. This last inequality holds because $\sin\theta\, \cos y \leq |\sin\theta| = \sin|\theta|$ for $\theta \in [-\pi, \pi]$, and $\sin y \geq 0$ for $y \in [0, \pi]$. The statement then follows from (A.1) and (A.2). □

Proof of Lemma 2.18: The derivative of β is given by

$$\partial_M \beta(M) = \alpha^{-1} A_1 M^{\frac{1}{\alpha}-1} - 1.$$

Hence, β has a unique critical point M_0, determined by $M_0^{1-\frac{1}{\alpha}} = A_1\alpha^{-1}$.

The second derivative satisfies

$$\partial_M^2 \beta(M) = A_1\alpha^{-1}(\alpha^{-1} - 1)M^{\frac{1}{\alpha}-2} > 0.$$

Hence, M_0 is a minimum of β.

Evaluating β at M_0, we obtain

$$\beta(M_0) = M_0(A_1 M_0^{\frac{1}{\alpha}-1} - 1) + A_2 = M_0(\alpha - 1) + A_2.$$

Hence, the minimum is negative if and only if $A_2 < (1-\alpha)M_0$. Substituting $M_0 = (A_1\alpha^{-1})^{\frac{1}{1-\alpha^{-1}}}$ and raising both sides to the power $1 - \alpha > 0$ and multiplying by A_1^α, this is equivalent to

$$A_1^\alpha A_2^{1-\alpha} < (1 - \alpha)^{1-\alpha}\alpha^\alpha.$$

□

Lemma A.1 *Given any $\delta \in (0,\pi)$, let $\mathrm{Cl}_2 : \mathbb{R} \to \mathbb{R}$ be the Clausen function, given by*

$$\mathrm{Cl}_2(x) := \sum_{k\geq 1} \frac{1}{k^2} \sin(kx),$$

and let $\rho_\delta : [0,\pi] \to \mathbb{R}$ be the function given by

$$\rho_\delta(x) := \mathrm{Cl}_2(2x-\delta) - \mathrm{Cl}_2(2x+\delta).$$

Then, ρ_δ is increasing on $[0,\frac{\pi}{2}]$ and attains its maximum at $\frac{\pi}{2}$.

Proof: First, observe that ρ_δ is symmetric with respect to $\frac{\pi}{2}$. Indeed,

$$\begin{aligned}\rho_\delta(\pi - x) &= \mathrm{Cl}_2(2(\pi-x)-\delta) - \mathrm{Cl}_2(2(\pi-x)+\delta)\\ &= \mathrm{Cl}_2(-2x-\delta) - \mathrm{Cl}_2(-2x+\delta) = \mathrm{Cl}_2(2x-\delta) - \mathrm{Cl}_2(2x+\delta) = \rho_\delta(x),\end{aligned}$$

where we have used that Cl_2 is odd and 2π-periodic. Hence, it is enough to study ρ_δ on $[0,\frac{\pi}{2}]$.

For $x \neq \frac{\delta}{2}$, neither $2x-\delta$ nor $2x+\delta$ is an integer multiple of 2π. By (2.14), whose right hand side converges uniformly on compact subsets of $(0,2\pi)$, differentiating the series of Cl_2 term by term is therefore legitimate at both points, and gives

$$\partial_x \mathrm{Cl}_2(x) = -\frac{1}{2}\log(2-2\cos x), \tag{A.8}$$

Differentiating ρ_δ and using (A.8), we obtain

$$\partial_x \rho_\delta(x) = 2\partial_x \mathrm{Cl}_2(2x-\delta) - 2\partial_x \mathrm{Cl}_2(2x+\delta) = \log\left(\frac{1-\cos(2x+\delta)}{1-\cos(2x-\delta)}\right).$$

Thus, $\partial_x \rho_\delta(x) \geq 0$ whenever

$$\cos(2x-\delta) \geq \cos(2x+\delta). \tag{A.9}$$

To check (A.9), note that

$$\cos(2x-\delta) - \cos(2x+\delta) = 2\sin(2x)\sin\delta.$$

Since $x \in [0,\frac{\pi}{2}]$, we have $2x \in [0,\pi]$ and hence $\sin(2x) \geq 0$, while $\delta \in (0,\pi)$ gives $\sin\delta > 0$. The difference is therefore nonnegative, so $\partial_x\rho_\delta(x) \geq 0$ for every $x \in [0,\frac{\pi}{2}]$ with $x \neq \frac{\delta}{2}$. Since ρ_δ is continuous, it is increasing on this interval, and by symmetry it attains its maximum at $x = \frac{\pi}{2}$. □

We prove Lemma 2.20 before Lemma 2.19, since the latter's proof relies on it.

Proof of Lemma 2.20: Let $I_{\delta,x}$ be the interval $[0,\pi] \cap (x-\delta, x+\delta)$, so that

$$\mathcal{I}(\delta) = \sup_{x\in[0,\pi]} \int_{I_{\delta,x}} K_n(x,y)\,dy.$$

Recall the expression for K_n obtained in Lemma 2.12,

$$K_n(x,y) = \frac{1}{2\pi}\log\left(\frac{1-\cos(x+y)}{1-\cos(x-y)}\right) + \frac{6}{\pi}\sum_{k\geq 1}\frac{1}{n^2k^3-4k}\sin(kx)\sin(ky).$$

We bound the logarithmic and Fourier parts of the kernel K_n separately.

From (2.14) we obtain

$$\log\left(\frac{1-\cos(x+y)}{1-\cos(x-y)}\right) = 2\sum_{k\geq 1}\frac{1}{k}\Big(\cos(k(x-y)) - \cos(k(x+y))\Big) = 4\sum_{k\geq 1}\frac{1}{k}\sin(kx)\sin(ky). \tag{A.10}$$

Accordingly, we define

$$F(x) := \int_{I_{\delta,x}} \sum_{k\geq 1} \frac{1}{k} \sin(kx)\sin(ky)\, dy.$$

We claim that F attains its maximum at $x = \frac{\pi}{2}$, which we prove in four steps.

First, F is symmetric with respect to $\frac{\pi}{2}$. Indeed, we have

$$\sin\big(k(\pi - x)\big)\sin\big(k(\pi - y)\big) = (-1)^{2k+2}\sin(kx)\sin(ky) = \sin(kx)\sin(ky).$$

Moreover, the change of variables $y \mapsto \pi - y$ maps $I_{\delta,\pi-x}$ onto $I_{\delta,x}$. Therefore, $F(\pi - x) = F(x)$, and it is enough to study F on $[0, \frac{\pi}{2}]$.

Second, F is increasing on $[\delta, \frac{\pi}{2}]$. For $x \in [\delta, \frac{\pi}{2}]$, we have $I_{\delta,x} = (x - \delta, x + \delta)$. Integrating F term by term,

$$\begin{aligned} F(x) &= \sum_{k\geq 1} \frac{1}{k^2} \sin(kx)[\cos(k(x-\delta)) - \cos(k(x+\delta))] \\ &= \sum_{k\geq 1} \frac{1}{k^2} \left[\frac{1}{2}\sin(k(2x-\delta)) - \frac{1}{2}\sin(k(2x+\delta)) + \sin(k\delta)\right] \\ &= \mathrm{Cl}_2(\delta) + \frac{1}{2}\mathrm{Cl}_2(2x-\delta) - \frac{1}{2}\mathrm{Cl}_2(2x+\delta) = \mathrm{Cl}_2(\delta) + \frac{1}{2}\rho_\delta(x), \end{aligned} \tag{A.11}$$

where ρ_δ and Cl_2 are the functions defined in Lemma A.1. By that lemma, ρ_δ is increasing on $[0, \frac{\pi}{2}]$, and hence so is F.

Third, F is increasing on $[0, \delta]$. For $x \in [0, \delta]$, we have $I_{\delta,x} = [0, x + \delta)$ and

$$\begin{aligned} F(x) &= \sum_{k\geq 1} \frac{1}{k^2} \sin(kx)\,[1 - \cos(k(x+\delta))] \\ &= \mathrm{Cl}_2(x) - \frac{1}{2}\mathrm{Cl}_2(2x+\delta) + \frac{1}{2}\mathrm{Cl}_2(\delta), \quad x \in [0,\delta]. \end{aligned} \tag{A.12}$$

Differentiating (A.12) with respect to x and using (A.8), we obtain

$$\partial_x F(x) = \frac{1}{2}\log\left(\frac{2 - 2\cos(2x+\delta)}{2 - 2\cos x}\right).$$

For $x \in (0, \delta]$, we have $0 < x \leq 2x + \delta \leq 3\delta \leq \pi$, and cos is decreasing on $[0, \pi]$, so the argument of the logarithm is at least 1 and $\partial_x F(x) \geq 0$. Here, we have used that $\delta < \frac{\pi}{3}$. Since F is continuous on $[0, \delta]$, it is increasing there.

Fourth, both expressions, (A.11) and (A.12), coincide at $x = \delta$. Hence, F is continuous on $[0, \frac{\pi}{2}]$ and increasing there. Together with the symmetry from the first step and using $\mathrm{Cl}_2(\pi + \delta) = -\mathrm{Cl}_2(\pi - \delta)$, we find that

$$\sup_{x\in[0,\pi]} F(x) = F\big(\tfrac{\pi}{2}\big) = \mathrm{Cl}_2(\delta) + \frac{1}{2}\rho_\delta\big(\tfrac{\pi}{2}\big) = \mathrm{Cl}_2(\delta) + \mathrm{Cl}_2(\pi - \delta). \tag{A.13}$$

Observe now that

$$\begin{aligned} \mathrm{Cl}_2(x) = \sum_{k\geq 1} \frac{1}{k^2}\sin(kx) &= \sum_{k\geq 1} \frac{1}{(2k-1)^2}\sin((2k-1)x) + \sum_{k\geq 1}\frac{1}{(2k)^2}\sin(2kx) \\ &= \sum_{k\geq 1} \frac{1}{(2k-1)^2}\sin((2k-1)x) + \frac{1}{4}\mathrm{Cl}_2(2x). \end{aligned} \tag{A.14}$$

Consequently,

$$\begin{aligned}\mathrm{Cl}_2(\delta)+\mathrm{Cl}_2(\pi-\delta) &= \sum_{k\geq 1}\frac{1}{k^2}\sin(k\delta)+\sum_{k\geq 1}\frac{1}{k^2}\sin\big(k(\pi-\delta)\big)\\ &= 2\sum_{k\geq 1}\frac{1}{(2k-1)^2}\sin\big((2k-1)\delta\big) = 2\mathrm{Cl}_2(\delta)-\frac{1}{2}\mathrm{Cl}_2(2\delta), \end{aligned} \tag{A.15}$$

where we have used (A.14) in the last equality. Therefore, from (A.8) and (A.15), we have that

$$\partial_\delta[\mathrm{Cl}_2(\delta)+\mathrm{Cl}_2(\pi-\delta)] = \partial_\delta\Big[2\mathrm{Cl}_2(\delta)-\frac{1}{2}\mathrm{Cl}_2(2\delta)\Big] = \frac{1}{2}\log\Big(\frac{2-2\cos(2\delta)}{(2-2\cos\delta)^2}\Big) = \log\cot(\delta/2).$$

Since $\cos(\delta/2)\leq 1$ and $\sin(\delta/2)\geq \delta/\pi$ for every $\delta\in(0,\pi)$, we have

$$\partial_\delta[\mathrm{Cl}_2(\delta)+\mathrm{Cl}_2(\pi-\delta)] \leq -\log\delta+\log\pi. \tag{A.16}$$

Hence, using (A.10) and combining estimates (A.13) and (A.16) with the identity $\mathrm{Cl}_2(0)+\mathrm{Cl}_2(\pi)=0$, we find that

$$\sup_{x\in[0,\pi]}\int_{I_{\delta,x}}\log\Big(\frac{1-\cos(x+y)}{1-\cos(x-y)}\Big)\,dy \leq 4\int_0^\delta \log\cot(t/2)\,dt \leq -4\delta\log\delta+4\delta(1+\log\pi). \tag{A.17}$$

Next, we estimate the Fourier series. For $n\geq 3$ and every $k\geq 1$, we have

$$n^2k^3-4k \geq 9k^3-4k = k(9k^2-4)\geq 5k>0,$$

so the series is bounded uniformly in x and y by

$$\left|\sum_{k\geq 1}\frac{\sin(kx)\sin(ky)}{n^2k^3-4k}\right| \leq \sum_{k\geq 1}\frac{1}{9k^3-4k}<\infty.$$

Since the length of $I_{\delta,x}$ is at most 2δ, this yields

$$\sup_{x\in[0,\pi]}\int_{I_{\delta,x}}\sum_{k\geq 1}\frac{\sin(kx)\sin(ky)}{n^2k^3-4k}\,dy \lesssim \delta. \tag{A.18}$$

Finally, from inequalities (A.17) and (A.18), we deduce that

$$\mathcal{I}(\delta)\lesssim \sup_{x\in[0,\pi]}\int_{I_{\delta,x}}\log\Big(\frac{1-\cos(x+y)}{1-\cos(x-y)}\Big)\,dy+\sup_{x\in[0,\pi]}\int_{I_{\delta,x}}\sum_{k\geq 1}\frac{\sin(kx)\sin(ky)}{n^2k^3-4k}\,dy\lesssim \delta(1-\log\delta).$$

□

Proof of Lemma 2.19: We are computing a limit as $\delta\to 0^+$, so we can assume $\delta<\frac{\pi}{3}$. Using the estimate in Lemma 2.20, we obtain

$$\lim_{\delta\to 0^+}\mathcal{I}(\delta)^\alpha\,(1+\cot(\delta/2))^{1-\alpha}\lesssim \lim_{\delta\to 0^+}\frac{(\delta(1-\log\delta))^\alpha}{(1+\cot(\delta/2))^{\alpha-1}}\lesssim \lim_{\delta\to 0^+}(\delta(1-\log\delta))^\alpha\,\delta^{\alpha-1}.$$

We conclude that, for $\delta>0$ small,

$$\mathcal{I}(\delta)^\alpha\,(1+\cot(\delta/2))^{1-\alpha}\lesssim \delta^{2\alpha-1}(-\log\delta)^\alpha = \Big(-\delta^{2-\alpha^{-1}}\log\delta\Big)^\alpha.$$

The right hand side tends to 0 when $\delta\to 0^+$, because

$$\lim_{\delta\to 0^+}\delta^b\log\delta=0 \qquad \text{for all } b>0,$$

and here $b=2-\alpha^{-1}$, which is positive if and only if $\alpha>\frac{1}{2}$. The left hand side is nonnegative, since $K_n\geq 0$ by Lemma 2.14 gives $\mathcal{I}\geq 0$, and $1+\cot(\delta/2)>0$ for δ small. The limit is therefore zero. □

III. Linear instability of a Burgers–Hilbert traveling wave

This chapter is based on the preprint [CGP26], written jointly with Á. Castro and J. Gómez-Serrano. The text has been adapted to the format and notation of this thesis.

III.1 Introduction

In [MW83], Marsden and Weinstein introduced the Burgers–Hilbert (BH) equation,

$$\begin{cases} \partial_t u + Hu + u\partial_x u = 0, \quad u : \mathbb{T} \times [0,\infty) \to \mathbb{R}, \\ u(x,0) = u_0(x), \end{cases} \tag{1.1}$$

as a quadratic approximation of the vortex patch problem for the 2D Euler equations.

A traveling wave $v : \mathbb{T} \to \mathbb{R}$, with speed $c \in \mathbb{R}$, is a solution u of (1.1) of the form

$$u(x,t) = v(x+ct),$$

and hence

$$c\partial_x v + Hv + v\partial_x v = 0. \tag{1.2}$$

We impose the additional condition

$$\int_{-\pi}^{\pi} v(x)\,dx = 0.$$

In this chapter, we focus on the stability of traveling waves of (1.1) which leads to studying the spectral properties of the operator $L_v f := c\partial_x f + Hf + \partial_x(vf)$, which will also be denoted $L_v := c\partial_x + H + \partial_x(v\,\cdot)$.

We briefly review some of the main results concerning (1.1). In [BH10], the validity of (1.1) for describing surface waves on a planar discontinuity in vorticity in two-dimensional inviscid flow was studied. Later, in [Hun+22], it was proved that the approximation remains valid on cubically nonlinear timescales.

Several works have addressed the question of whether the Hilbert transform can prevent the formation of singularities in (1.1). Numerical simulations were carried out in [BH10; Hun18; KS15], loss of $C^{1+\delta}$ regularity was proved in [CCG10], shock formation was established in [SW22], and the existence of asymptotically self-similar shocks was shown in [Yan21].

Bressan and Nguyen established the global existence of weak solutions in [BN14] for initial data $f_0 \in L^2(\mathbb{R})$, with $f(x,t) \in L^\infty(\mathbb{R}) \cap L^2(\mathbb{R})$ for all $t > 0$. In [BZ17], piecewise continuous solutions are constructed locally in time. Such solutions were shown to be unique in [KV20].

Branches of traveling waves v_n, bifurcating from zero, were constructed in [Hun18]. These solutions are C^∞, $\frac{2\pi}{n}$-periodic, and have speeds $c_n = \frac{1}{n}$. Later, in [CCZ23], it was shown that these solutions are analytic. By introducing the asymptotic expansions

$$v(x) = \varepsilon \cos x + \sum_{k=2}^{\infty} u_k(x)\varepsilon^k, \qquad c = 1 + \sum_{k=1}^{\infty} c_k \varepsilon^k, \tag{1.3}$$

it was shown in [CCZ23] that the branch of solutions could be extended beyond $\varepsilon = 0.22$ and that the series (1.3) cannot converge for $\varepsilon > \frac{2}{e}$. The existence of a maximal wave with speed $c \in [1.1, 1.121]$ and with an $x\log(|x|)$ singularity at the origin was proved in [DG23].

The stability of these traveling waves was also studied in [CCZ23]. In particular, the authors showed that near the beginning of the branches, namely, for small ε in (1.3), the eigenvalues of the operator $c\partial_x + H + \partial_x(v\,\cdot)$ are purely imaginary. Moreover, this phenomenon was used to enhance the lifespan of perturbations of a traveling wave. For lifespan enhancement of perturbations of the zero solution, see [HI12] and [Hun+15].

The main results of this chapter are Theorems 2.1 and 3.2. Taken together, they imply the following:

Theorem 1.1 *There exist a speed $\widetilde{c}$ and a real, smooth, even, and zero-mean function $\widetilde{v} \neq 0$, with minimal period $\frac{2\pi}{3}$, solving* (1.2)*, such that the linear operator $\widetilde{c}\partial_x + H + \partial_x(\widetilde{v}\,\cdot)$ has eigenvalues with both negative and positive real parts.*

Let us emphasize that the stability of traveling waves for related models was considered by Maspero and Radakovic in [MR25]. They work on $\mathbb{R}$ instead of $\mathbb{T}$ and consider the equation

$$\partial_t u + u\partial_x u + \mathcal{M}\partial_x u = 0, \tag{1.4}$$

where $\widehat{\mathcal{M}} = m(\xi)$, with $m(\xi) \in C^r(\mathbb{R})$, $r \geq 3$, real-valued, even, satisfying, for some $m \geq -1$, that $C_1\langle\xi\rangle^m \leq |m(\xi)| \leq C_2\langle\xi\rangle^m$, where $\xi \gg 1$ and $\inf_{n\in\mathbb{N}^+-\{1\}} |m(n) - m(1)| > c_0$.

The stability of small traveling waves of (1.4) is studied in [MR25] when subjected to perturbations with very small frequency (modulational instability). Roughly speaking, this corresponds to the analysis of the stability of $\frac{2\pi}{m}$-periodic traveling waves for very large m in our setting. Maspero and Radakovic completely characterize the spectrum near the origin of the complex plane of the operator $c\partial_x + \mathcal{M}\partial_x + \partial_x(v\,\cdot)$, and, in particular, find an unstable eigenvalue (see also [Joh13; HJ15a; HJ15b; HP16; PCN17]). In contrast, we focus on a large-amplitude traveling wave and small values of m, focusing only on the case $m = 3$.

III.1.1 Computational aspects of the proof

We emphasize that part of the proof of Theorem 1.1 is computer-assisted. There has been an emergence of proofs using those techniques in the last decade. The main idea is to substitute floating point numbers on a computer by rigorous bounds, which are then propagated through every operation that the computer performs taking into account any error made throughout the process.

More concretely, we use the computer as part of the proof in the following settings:

- Bounds for the approximate traveling wave and the approximate eigenfunction, including estimates for their residuals and norms. See Lemmas 2.3, 3.10, and 3.15.

- Non-degeneracy checks, including the non-vanishing of the Fourier coefficients of the traveling wave approximation in Lemma 4.3, validated enclosures of the roots of the associated polynomial $z^N P(z)$ in Lemma C.3, and the exclusion of certain integer obstructions in Lemmas 4.12 and 5.25.

- Bounds for certain explicit norms involving $\beta = (c + v^{\mathrm{ap}})^{-1}$. See Lemmas 4.18, 4.19, 4.20, 4.21, 5.27, and 5.28.
- Upper bounds for the sum of explicit quantities, obtained in Lemmas 4.25 and 5.31.
- Working with piecewise polynomial approximations of certain explicit functions, and deriving estimates for the approximation errors. These estimates are obtained in Lemmas D.1, D.7, D.8, and D.10.
- Singular value estimates for the finite-dimensional matrices $M^{\mathrm{tor}}_{\mathrm{exis}}$ and $M^{\mathrm{tor}}_{\mathrm{stab}}$, which are used throughout the argument. In particular, they enter the proofs of Lemmas 4.16, 5.16, 5.22, and 5.23.

In a broader context (computer-assisted proofs in PDE), the reduction of the proof of existence to a fixed-point argument has also been particularly successful in the context of *radii polynomials*, developed in [BL08; DLM07; GLM08] and later used in [Ber+21; CGL18].

There are several other works that incorporate computer-assisted proofs. We highlight the following representative examples: the Euler–Poisson system [Guo+22], extreme Stokes waves [Kob10], the De Gregorio model [CHH21], the SQG equation [CCG20a], and Boussinesq-type models, including applications to the 3D Euler equations [EP25; CH22; CH25b; CH25a]. Further applications include Whitham-type equations [Cad25], the Kuramoto–Sivashinsky equation [AK10; FL17; GL17; Fig+17; Zgl04; ZM01], compressible Euler and Navier–Stokes equations [BCG25], and the incompressible Navier–Stokes equations [AGK21; BP24].

The supplementary material, including the code, is available at https://github.com/MiguelMGPascualCaballo/bhtw . Larger data files are available at https://doi.org/10.5281/zenodo.19250315 .

III.1.2 The vortex patch problem

Let us briefly discuss the motion of vortex patches for the 2D Euler equations, which provides the main motivation for our study. These solutions correspond to vorticities of the form

$$\omega(\mathbf{x}, t) = \begin{cases} \varpi & \mathbf{x} \in \Omega(t) \\ 0 & \mathbf{x} \in \Omega^c(t) \end{cases},$$

where $\Omega(t)$ is a time-dependent domain transported by the flow. It was first proved by Chemin in [Che93; Che95] that if the boundary of $\Omega(0)$ is $C^{1+\delta}$, then there exists a global patch solution to the 2D Euler equations whose boundary remains of class $C^{1+\delta}$. A different proof of this result can be found in [BC93].

The existence of traveling waves for the BH equation is closely related to the existence of V-states for the vortex patch problem. A V-state is a simply connected patch of vorticity whose evolution consists of a rigid rotation with constant angular velocity. The properties of such solutions have been extensively studied in the mathematical literature. Their existence was first observed numerically in [DZ78] and later proved analytically by Burbea [Bur82] and Hmidi, Mateu, and Verdera in [HMV13]. In fact, there exist branches of m-fold symmetric solutions bifurcating from the circle at angular velocity $\Omega_m = \varpi \frac{m-1}{2m}$ for every $m \geq 2$. C^∞ regularity of the boundary was established in [HMV13], and analyticity in [CCG16]. There is a huge literature dealing with patches for 2D Euler, including many extensions of these results. For further references, see [Fra00; Góm+21; Hmi15; HMW20; Hoz+16a; Hoz+16b; HM16b; CLZ21; Cao+21; CJ22; EJ23; EJ20a; Gar20; Gar21; GH23; GHS20; HH21; HW22; HM17; Tur85; Par22; GPS25; EJ25; GHR25; GHM24; RS25]. More recently, KAM theory has been applied to construct quasi-periodic solutions, see [CF13; EPT23; BM21; BHM23; HR25; HHR24b; HHM25; HHR24a; GHH26]. Interval arithmetic has already been used for patches. In fact, the existence of non-convex V-states was proved in [CG25] (see also [DG23]).

A classical example of a V-state are Kirchhoff ellipses [Kir74], which correspond to the branch $m = 2$. The ellipses rotate with angular velocity

$$\Omega = \varpi \frac{\lambda}{2(1+\lambda)},$$

where λ denotes the aspect ratio.

Remarkably, thanks to the explicit form of these solutions, Guo, Hallstrom, and Spirn in [GHS04] proved that for some λ, the Kirchhoff ellipses become unstable. In particular, as part of the proof, they showed that the operator arising from the linearization of the boundary evolution equation for a patch around a rotating ellipse, that is, the analogue of the operator L_v, has an eigenvalue with negative real part.

However, the validity of this result has not been verified for higher symmetries. That is, it is expected that for $m \geq 3$, the corresponding branches of solutions also become unstable at some point, but no such result is currently available. For more information on the properties of Kirchhoff ellipses, see [CCG16; HM16b; HM16a; WXZ22] .

In this chapter, we focus on the stability of $\frac{2\pi}{3}$-periodic traveling waves for the BH equation, which correspond to threefold symmetric V-states for the 2D Euler equations.

III.1.3 Organization of the chapter

The chapter is organized as follows. In Section III.2, we introduce a traveling wave with speed given by (2.1) and postpone Proposition 2.2 for later proof. In Section III.3, we prove the existence of an eigenvalue with nonzero real part, while postponing Lemmas 3.12–3.14. In Section III.4, we establish Proposition 2.2, and in Section III.5 we prove Lemmas 3.12–3.14.

Appendix III.A contains classical estimates on the torus, Appendix III.B presents computations involving the Hilbert transform, and Appendix III.C describes an explicit polynomial arising in the analysis. Appendix III.D describes the approach to certain explicit functions, and Appendix III.E details the computer-assisted arguments and the supplementary data.

III.1.4 Functional setting and notation

We consider complex-valued functions. When needed, we will explicitly indicate that a function is real-valued.

In this chapter, we will use two conventions for the L^2 norm, defined by

$$\|f\|^2_{L^2(\mathbb{T})} := \sum_{k\in\mathbb{Z}} |\widehat{f_k}|^2, \qquad \|f\|^2_{L^2([-\pi,\pi])} := \int_{-\pi}^{\pi} |f(x)|^2\, dx,$$

where

$$\widehat{f_k} := \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-ikx}\, dx.$$

Throughout the chapter, we will work with functions $f \in C^1([-\pi,\pi])$ which are not necessarily periodic, and hence not in $C(\mathbb{T})$. In this case, we write $\|\partial_x f\|_{L^2([-\pi,\pi])}$ to emphasize that we consider the classical derivative of f on $(-\pi,\pi)$.

If f admits a weak derivative on the torus belonging to $L^2(\mathbb{T})$, we instead write $\|\partial_x f\|_{L^2(\mathbb{T})}$.

Moreover, the $\|\cdot\|_{L^2([-\pi,\pi])}$ and $\|\cdot\|_{L^2(\mathbb{T})}$ differ by a multiplicative constant. We have,

$$\|f\|^2_{L^2(\mathbb{T})} = \frac{1}{2\pi}\int_{-\pi}^{\pi} |f(x)|^2\, dx.$$

We denote by $H^s(\mathbb{T})$ the Sobolev space defined as the closure of $C^\infty(\mathbb{T})$ with respect to the norm

$$\|f\|^2_{H^s(\mathbb{T})} := \frac{1}{2\pi}\int_{-\pi}^{\pi} |f(x)|^2 + |\Lambda^s f(x)|^2\, dx = \sum_{k\in\mathbb{Z}} (1+|k|^{2s})|\widehat{f_k}|^2$$

In the next section, we work with the following Sobolev spaces

$$H^s_{\text{even}} = \{f \in H^s(\mathbb{T}) : \ f(x) = f(-x) \text{ a.e. in } \mathbb{T} \text{ and } \widehat{f}_0 = 0\},$$
$$H^s_{\text{odd}} = \{f \in H^s(\mathbb{T}) : \ f(x) = -f(-x) \text{ a.e. in } \mathbb{T}\}.$$

We emphasize that both H^s_{even} and H^s_{odd} consist of zero-mean functions.

We endow these spaces with the following norm

$$\|f\|^2_{H^s_{\text{odd (or even)}}} := \sum_{k\geq 1} k^{2s}|\widehat{f}_k|^2 = \frac{1}{4\pi}\int_{-\pi}^{\pi} |\Lambda^s f(x)|^2\, dx,$$

A function $f \in H^s_{\text{even}}$ can be represented by

$$f(x) = \sum_{k\geq 1} a_k \cos(kx),$$

and then

$$\|f\|^2_{H^s_{\text{even}}} = \frac{1}{4}\sum_{k\geq 1} k^{2s}|a_k|^2.$$

The operator $\partial_x^{-1} : H^k_{\text{odd (or even)}} \to H^{k+1}_{\text{even (or odd)}}$ is given by

$$\partial_x^{-1} f(x) = \int_0^x f(y)\, dy - \frac{1}{2\pi}\int_{-\pi}^{\pi}\int_0^x f(y)\, dy\, dx.$$

$\dot{\mathcal{H}}$ denotes the space of holomorphic functions on the unit disk $\mathbb{D}$ that vanish at zero. We note that if $F \in \dot{\mathcal{H}}$ then F admits the expansion

$$F(z) = \sum_{k\geq 1} F_k z^k \tag{1.5}$$

with $F_k \in \mathbb{C}$.

In addition, we introduce the Hardy spaces

$$\dot{\mathcal{H}}^s = \{F \in \dot{\mathcal{H}} : \ F|_{\mathbb{T}} \in H^s(\mathbb{T})\}, \tag{1.6}$$

where $F|_{\mathbb{T}} = F(e^{ix})$. These spaces are endowed with the norm

$$\|F\|^2_{\dot{\mathcal{H}}^s} := \sum_{k\geq 1} |F_k|^2 k^{2s} = \sum_{k\geq 1} |\widehat{f}_k|^2 k^{2s}.$$

We will use the following notation

$$f^+ := \tfrac{1}{2}(f + iHf), \qquad f^- := \tfrac{1}{2}(f^\sharp + iHf^\sharp),$$

where $f^\sharp(x) := f(-x)$.

Notice that for $f \in H^s(\mathbb{T})$, with $s > \frac{1}{2}$, the Fourier series of f^+ and f^- are

$$f^+(x) = \sum_{k\geq 1} \widehat{f}_k e^{ikx}, \qquad f^-(x) = \sum_{k\geq 1} \widehat{f}_{-k} e^{ikx}.$$

Furthermore, replacing e^{ix} by z, we also find holomorphic extensions $F^+, F^- \in \dot{\mathcal{H}}^s$ of f^+ and f^- to the complex unit disk,

$$F^+(z) = \sum_{k\geq 1} \widehat{f}_k z^k, \qquad F^-(z) = \sum_{k\geq 1} \widehat{f}_{-k} z^k. \tag{1.7}$$

Remark 1.2 *The definition in* (1.7) *will slightly change in Definition 5.2 from Section III.5.2.1.*

For every $F \in \dot{\mathcal{H}}^s$, there exists $f \in H^s_{\text{odd}}$ such that $f^+ = F|_{\mathbb{T}}$. This can be checked by taking $f(x) = F|_{\mathbb{T}}(x) - F|_{\mathbb{T}}(-x)$ and the following computation

$$Hf = HF|_{\mathbb{T}}(x) - HF|^\sharp_{\mathbb{T}}(x) = HF|_{\mathbb{T}}(x) + HF|_{\mathbb{T}}(-x) = -i(F|_{\mathbb{T}}(x) + F|_{\mathbb{T}}(-x)).$$

For $n \geq 2$ and $l = 1, ..., n-1$, we will also use the following Fourier projectors

$$\widehat{(\mathcal{P}_{l,n} f)}_k = \widehat{f}_{nk+l}.$$

We view vectors as *column vectors*. In addition, for each $v \in \mathbb{C}^d$, we write $v^* := \overline{v}^{\,t}$, so that v^* is a *row vector* whose entries are the complex conjugates of those of v. Thus, for $v_1, v_2 \in \mathbb{C}^d$, their scalar product can be written as

$$(v_1, v_2)_{\ell^2} = v_1^* v_2.$$

We denote by $\|v\|_{\ell^2}$ the standard $\ell^2(\mathbb{C}^d)$ norm of $v \in \mathbb{C}^d$.

Finally, given a matrix $A \in \mathbb{C}^{d\times d}$, we denote by

$$\sigma_1(A) \leq \sigma_2(A) \leq \cdots \leq \sigma_d(A)$$

its singular values, arranged in *increasing order*. This convention differs from the more common decreasing ordering.

III.2 Traveling wave existence

In this section, we prove the existence of a smooth traveling wave v for the Burgers–Hilbert equation (1.1), traveling at a fixed speed c. In particular, we will take

$$c := 1.109643839000099. \tag{2.1}$$

More precisely, we shall prove the following theorem.

Theorem 2.1 *There exists a smooth, even, real, and zero-mean traveling wave $v \neq 0$ of* (1.1) *with speed c. In addition, v is not $\frac{2\pi}{n}$-periodic for any $n \geq 2$. Since $\frac{1}{n}(v(nx), c)$ also solves* (1.2)*, we obtain traveling waves of minimal period $\frac{2\pi}{n}$, for every $n \in \mathbb{N}^+$.*

Section III.2 is devoted to the proof of Theorem 2.1. From now on, the speed c will remain fixed.

First of all, let us clarify that the existence of a branch of traveling wave $\frac{2\pi}{n}$-periodic solutions bifurcating from zero can be proved by using the classical Crandall–Rabinowitz theorem (see [CCZ23]). Moreover, it is shown in [CCZ23] that the branch exists on $[1, c^*)$, for some finite c^*.[1] However, this value is not sufficient to find instabilities and we need to go further up to c.

Our strategy for proving Theorem 2.1 consists in showing that there exists a unique traveling wave v close to an explicit approximation v^{ap} that we have constructed using numerical simulations. This is based on a fixed-point argument.

We write v as a perturbation of the explicit approximation v^{ap},

$$v = v^{\text{ap}} + \omega. \tag{2.2}$$

Here, v^{ap} is explicit and defined as

$$v^{\text{ap}}(x) := \sum_{j=1}^{N} v_j^{\text{ap}} \cos(jx). \tag{2.3}$$

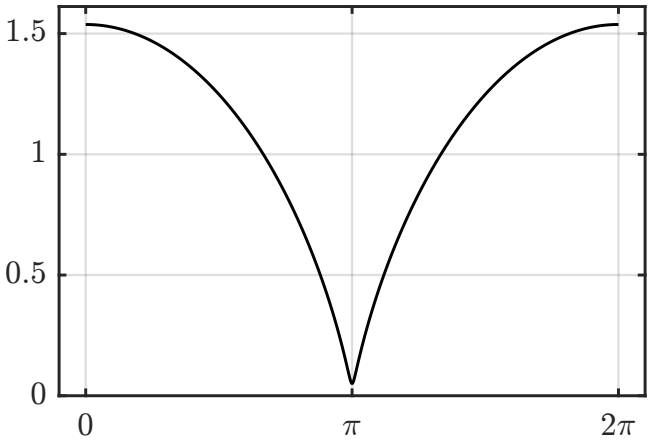


Figure III.1: Graph of $c + v^{\mathrm{ap}}$.

where the real coefficients v_j^{ap}, $1 \le j \le N$, with $N = 1000$, can be found in the supplementary material. The graph of $c + v^{\mathrm{ap}}$ is shown in Figure III.1.

Plugging (2.2) into (1.2), we obtain a new equation for ω

$$\begin{aligned} 0 &= c\partial_x v + Hv + v\partial_x v \\ &= (c\partial_x v^{\mathrm{ap}} + Hv^{\mathrm{ap}} + v^{\mathrm{ap}}\partial_x v^{\mathrm{ap}}) + (c\partial_x \omega + H\omega + v^{\mathrm{ap}}\partial_x \omega + \omega\partial_x v^{\mathrm{ap}}) + \omega\partial_x \omega \\ &= (c\partial_x v^{\mathrm{ap}} + Hv^{\mathrm{ap}} + v^{\mathrm{ap}}\partial_x v^{\mathrm{ap}}) + (c\partial_x \omega + H\omega + \partial_x(v^{\mathrm{ap}}\omega)) + \tfrac{1}{2}\partial_x(\omega^2) \end{aligned} \tag{2.4}$$

We denote by

$$\begin{aligned} \xi_{v^{\mathrm{ap}}} &:= c\partial_x v^{\mathrm{ap}} + Hv^{\mathrm{ap}} + v^{\mathrm{ap}}\partial_x v^{\mathrm{ap}}, \\ L_{v^{\mathrm{ap}}}\omega &:= c\partial_x \omega + H\omega + \partial_x(v^{\mathrm{ap}}\omega), \\ Q(\omega_1, \omega_2) &:= \tfrac{1}{2}\partial_x(\omega_1\omega_2). \end{aligned}$$

With this notation, equation (2.4) can be written as

$$\xi_{v^{\mathrm{ap}}} + L_{v^{\mathrm{ap}}}\omega + Q(\omega, \omega) = 0. \tag{2.5}$$

In order to prove the existence of $\omega \in H^1_{\mathrm{even}}$, one key step is to invert the linear operator $L_{v^{\mathrm{ap}}}$, so that we can write (2.5) in the form

$$\omega = -L_{v^{\mathrm{ap}}}^{-1}\big(\xi_{v^{\mathrm{ap}}} + Q(\omega, \omega)\big).$$

Nevertheless, this may not be possible. Observe that, if v is an even solution of equation (1.2), taking a derivative, we get

$$c\partial_x^2 v + H\partial_x v + \partial_x(v\partial_x v) = 0.$$

Then, $\partial_x v$ is an odd function in the kernel of the operator L_v given by

$$L_v f := c\partial_x f + Hf + \partial_x(vf). \tag{2.6}$$

This may create some difficulties when inverting $L_{v^{\mathrm{ap}}}$. To avoid them, we will work in spaces with defined parity. Furthermore, we also impose that ω has mean zero, as v^{ap}. This allows us to deduce that v is also even and zero-mean.

We will be able to prove the following proposition which is one of the main ingredients in our proof of the existence of traveling waves.

[1] Numerical simulations suggest that $c^* \simeq 1.01393404560815$.

Proposition 2.2 *Let v^{ap} be the function given in* (2.3) *and*

$$L_{v^{\mathrm{ap}}} : H^1_{\mathrm{even}} \to H^0_{\mathrm{odd}}$$

the operator defined in (2.6)*. Then, $L_{v^{\mathrm{ap}}}$ is invertible, and its inverse*

$$L^{-1}_{v^{\mathrm{ap}}} : H^0_{\mathrm{odd}} \to H^1_{\mathrm{even}}$$

satisfies

$$\|L^{-1}_{v^{\mathrm{ap}}}\|_{H^0_{\mathrm{odd}} \to H^1_{\mathrm{even}}} < 3.13328 \cdot 10^4.$$

Proof: Section III.4 is devoted to proving Proposition 2.2. □

We establish all the estimates required to apply the fixed-point argument in Proposition 2.5. We want to show that there exists ω satisfying equation (2.5). Using (4.2), the problem is equivalent to finding a fixed point for T_{exis}, where

$$T_{\mathrm{exis}}\omega := -L^{-1}_{v^{\mathrm{ap}}}(\xi_{v^{\mathrm{ap}}} + Q(\omega, \omega)). \tag{2.7}$$

Therefore, we aim to obtain bounds for $\xi_{v^{\mathrm{ap}}}$ and Q to be able to apply Proposition 2.5.

We first compute a bound for $\xi_{v^{\mathrm{ap}}}$.

Lemma 2.3 *Let $\xi_{v^{\mathrm{ap}}}$ be the odd function given by*

$$\xi_{v^{\mathrm{ap}}} := c\partial_x v^{\mathrm{ap}} + Hv^{\mathrm{ap}} + v^{\mathrm{ap}}\partial_x v^{\mathrm{ap}}.$$

Then, $\xi_{v^{\mathrm{ap}}}$ satisfies

$$\|\xi_{v^{\mathrm{ap}}}\|_{H^0_{\mathrm{odd}}} < 8.00338 \cdot 10^{-14}.$$

Proof: The proof is computer-assisted and the code can be found in the supplementary material. We refer to the Appendix III.E for details about the implementation. □

Lemma 2.4 *The following estimate holds*

$$\|Q\|_{H^1_{\mathrm{even}} \times H^1_{\mathrm{even}} \to H^0_{\mathrm{odd}}} \leq 2^{1/2}\zeta(2)^{1/2}.$$

Proof: We have

$$\|Q(\omega, \omega)\|_{H^0_{\mathrm{odd}}} = \frac{1}{2}\|\omega^2\|_{H^1_{\mathrm{even}}} \leq 2^{1/2}\zeta(2)^{1/2}\|\omega\|^2_{H^1_{\mathrm{even}}},$$

where we have used the estimate in Corollary A.2. □

Proposition 2.5 *The operator $T_{\mathrm{exis}} : H^1_{\mathrm{even}} \to H^1_{\mathrm{even}}$ is a contraction on*

$$X := \{g \in H^1_{\mathrm{even}} : \ \|g\|_{H^1_{\mathrm{even}}} \leq 2.50805 \cdot 10^{-9}\}.$$

Consequently, there exists a traveling wave v with speed c for the Burgers–Hilbert equation. Moreover, the difference between v and our approximation v^{ap}, denoted by $\omega := v - v^{\mathrm{ap}}$, satisfies

$$\|\omega\|_{H^1_{\mathrm{even}}} < 2.50805 \cdot 10^{-9}.$$

Proof: We first prove that $T_{\text{exis}}(X) \subset X$, and then we derive an upper bound smaller than 1 for the Lipschitz constant $\ell_{T_{\text{exis}}}$ of T_{exis} on X. These two facts imply that T_{exis} is a contraction on X. Since $X \neq \varnothing$ is a closed subset of the complete metric space H^1_{even}, it follows that T_{exis} admits a fixed point ω in X.

We now verify both assumptions. Observe that

$$\begin{aligned}\|T_{\text{exis}}\omega\|_{H^1_{\text{even}}} &= \|L^{-1}_{v^{\text{ap}}}(\xi_{v^{\text{ap}}} + Q(\omega,\omega))\|_{H^1_{\text{even}}} \leq \|L^{-1}_{v^{\text{ap}}}\|_{H^0_{\text{odd}}\to H^1_{\text{even}}}\|\xi_{v^{\text{ap}}} + Q(\omega,\omega)\|_{H^0_{\text{odd}}} \\ &\leq \|L^{-1}_{v^{\text{ap}}}\|_{H^0_{\text{odd}}\to H^1_{\text{even}}}\big(\|\xi_{v^{\text{ap}}}\|_{H^0_{\text{odd}}} + \|\omega\|^2_{H^1_{\text{even}}}\|Q\|_{H^1_{\text{even}}\times H^1_{\text{even}}\to H^0_{\text{odd}}}\big). \end{aligned} \tag{2.8}$$

We have shown in Lemma 2.3, Lemma 2.4, and Proposition 2.2 that

$$\|\xi_{v^{\text{ap}}}\|_{H^0_{\text{odd}}} \leq 8.00338 \cdot 10^{-14}, \tag{2.9a}$$

$$\|Q\|_{H^1_{\text{even}}\times H^1_{\text{even}}\to H^0_{\text{odd}}} \leq 2^{1/2}\zeta(2)^{1/2}, \tag{2.9b}$$

$$\|L^{-1}_{v^{\text{ap}}}\|_{H^0_{\text{odd}}\to H^1_{\text{even}}} < 3.13328 \cdot 10^{4}. \tag{2.9c}$$

Now, plugging (2.9) into (2.8), we obtain for every $\omega \in X$ that

$$\|T_{\text{exis}}\omega\|_{H^1_{\text{even}}} < 2.50805 \cdot 10^{-9}. \tag{2.10}$$

Consequently, we deduce that $T_{\text{exis}}X \subset X$.

We now derive a bound for the Lipschitz constant $\ell_{T_{\text{exis}}}$. Let $\omega_1, \omega_2 \in X$, then

$$\begin{aligned} T_{\text{exis}}\omega_1 - T_{\text{exis}}\omega_2 &= L^{-1}_{v^{\text{ap}}}[\xi_{v^{\text{ap}}} + Q(\omega_2,\omega_2)] - L^{-1}_{v^{\text{ap}}}[\xi_{v^{\text{ap}}} + Q(\omega_1,\omega_1)] \\ &= L^{-1}_{v^{\text{ap}}}[Q(\omega_2,\omega_2) - Q(\omega_2,\omega_1)] + L^{-1}_{v^{\text{ap}}}[Q(\omega_2,\omega_1) - Q(\omega_1,\omega_1)]. \end{aligned} \tag{2.11}$$

We denote by

$$I_1 := L^{-1}_{v^{\text{ap}}}[Q(\omega_2,\omega_2) - Q(\omega_2,\omega_1)] \qquad \text{and} \qquad I_2 := L^{-1}_{v^{\text{ap}}}[Q(\omega_2,\omega_1) - Q(\omega_1,\omega_1)].$$

Observe that

$$\left.\begin{aligned} \|I_1\|_{H^1_{\text{even}}} &\leq \|L^{-1}_{v^{\text{ap}}}\|_{H^0_{\text{odd}}\to H^1_{\text{even}}}\|Q\|_{H^1_{\text{even}}\times H^1_{\text{even}}\to H^0_{\text{odd}}}\|\omega_2\|_{H^1_{\text{even}}}\|\omega_2 - \omega_1\|_{H^1_{\text{even}}}, \\ \|I_2\|_{H^1_{\text{even}}} &\leq \|L^{-1}_{v^{\text{ap}}}\|_{H^0_{\text{odd}}\to H^1_{\text{even}}}\|Q\|_{H^1_{\text{even}}\times H^1_{\text{even}}\to H^0_{\text{odd}}}\|\omega_1\|_{H^1_{\text{even}}}\|\omega_2 - \omega_1\|_{H^1_{\text{even}}}. \end{aligned}\right\} \tag{2.12}$$

Plugging (2.9b) and (2.9c) into (2.12), we derive from (2.11) that

$$\|T_{\text{exis}}\omega_1 - T_{\text{exis}}\omega_2\|_{H^1_{\text{even}}} < 3 \cdot 10^{-4}\|\omega_2 - \omega_1\|_{H^1_{\text{even}}}.$$

Consequently, we found a bound for $\ell_{T_{\text{exis}}}$ smaller than 1,

$$\ell_{T_{\text{exis}}} < 3 \cdot 10^{-4} < 1.$$

□

We now conclude the proof of Theorem 2.1. First, observe that v is 2π-periodic, but not $\frac{2\pi}{n}$-periodic for any $n \geq 2$. To see this, if this were the case, then we would have

$$\frac{1}{\pi}\int_{-\pi}^{\pi} v(x)\cos x\, dx = 0.$$

However, we compute

$$\begin{aligned} \frac{1}{\pi}\int_{-\pi}^{\pi} v(x)\cos x\, dx &= \frac{1}{\pi}\int_{-\pi}^{\pi} v^{\text{ap}}(x)\cos x\, dx + \frac{1}{\pi}\int_{-\pi}^{\pi} \omega(x)\cos x\, dx \\ &\geq v_1^{\text{ap}} - \frac{1}{\sqrt{\pi}}\|\omega\|_{L^2([-\pi,\pi])} > 0.54 - 2 \cdot 2.50805 \cdot 10^{-9} > 0, \end{aligned}$$

where we have used that the first cosine coefficient of v^{ap} satisfies $v_1^{\mathrm{ap}} > 0.54$ and the Cauchy–Schwarz inequality. Hence the nonvanishing first Fourier mode of v implies that the minimal period is 2π. In addition, since $v_1^{\mathrm{ap}} > 0.54$, we also have

$$\|v\|_{H^1_{\mathrm{even}}} \geq \|v^{\mathrm{ap}}\|_{H^1_{\mathrm{even}}} - \|\omega\|_{H^1_{\mathrm{even}}} > \frac{1}{2} v_1^{\mathrm{ap}} - 2.50805 \cdot 10^{-9} > 0,$$

so that $v \neq 0$.

Second, in Proposition 2.5, we show the existence of a fixed point ω of T_{exis} in $X \subset H^1_{\mathrm{even}}$. Hence, $v = v^{\mathrm{ap}} + \omega \in H^1_{\mathrm{even}}$ solves (1.2).

However, this traveling wave is in fact smooth. This follows from Lemma 4.18, which provides an upper bound for $\beta := \frac{1}{c+v^{\mathrm{ap}}}$. Therefore,

$$c + v^{\mathrm{ap}} > \frac{1}{19.9056}.$$

In particular,

$$c + v \geq c + v^{\mathrm{ap}} - \|\omega\|_{L^\infty(\mathbb{T})} > \frac{1}{19.9056} - 2\zeta(2)^{1/2} 2.50805 \cdot 10^{-9} > 0,$$

where we have used Lemma A.1.

Since $v \in H^1_{\mathrm{even}}$ solves (1.2), we have

$$\partial_x v = -\frac{1}{c+v} H v.$$

A bootstrap argument then shows that all derivatives of v belong to L^2, and hence are continuous.

Finally, notice that the traveling wave v is real-valued, since T_{exis} is a real operator.

III.3 Traveling wave stability

In Section III.2, we proved that there exists a traveling wave v with speed c for the Burgers–Hilbert equation (1.1), where c is given by (2.1).

Here, we plug in the ansatz

$$u(x,t) = v(x+ct) + w(x+ct,t)$$

into (1.1). Therefore, we obtain the following equation for w

$$\partial_t w + w\partial_x w + \partial_x(vw) + Hw + c\partial_x w = 0$$

Integrating, we find

$$\partial_t f + \frac{1}{2}(\partial_x f)^2 + v\partial_x f + Hf + c\partial_x f = C(t).$$

where $\partial_x f = w$. If we want to consider f to have mean zero, we simply take

$$C(t) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \Big(\frac{1}{2} \big(\partial_x f(x)\big)^2 + v\partial_x f(x) \Big)\, dx.$$

At the linear level, we get that

$$\partial_t f + \mathcal{L}^{c,v}_{\mathrm{stab}} f = 0, \qquad \mathcal{L}^{c,v}_{\mathrm{stab}} f := (c+v)\partial_x f + Hf - \frac{1}{2\pi} \int_{-\pi}^{\pi} v(x)\partial_x f(x)\, dx. \tag{3.1}$$

Remark 3.1 *$\mathcal{L}^{\mathrm{ap}}_{\mathrm{exis}}$, defined in (4.1), and $\mathcal{L}^{c,v^{\mathrm{ap}}}_{\mathrm{stab}}$ have the same expression, but we restrict $\mathcal{L}^{\mathrm{ap}}_{\mathrm{exis}}$ to odd functions.*

From v we can construct new traveling waves v^n. Indeed, $v^n(x) := \frac{1}{n}v(nx)$ is a solution of (1.2) with speed $\frac{c}{n}$.

The main theorem of the chapter is as follows

Theorem 3.2 *The operator $\mathcal{L}_{\text{stab}}^{c/3,v^3}$ admits an eigenvalue $-\lambda$ with negative real part and a smooth associated eigenfunction $f_{-\lambda}$.*

Hence, $(-\lambda, \partial_x f_{-\lambda})$ is an eigenpair of L_v.

A numerical approximation of $f_{-\lambda}$ is shown in Figure III.2.

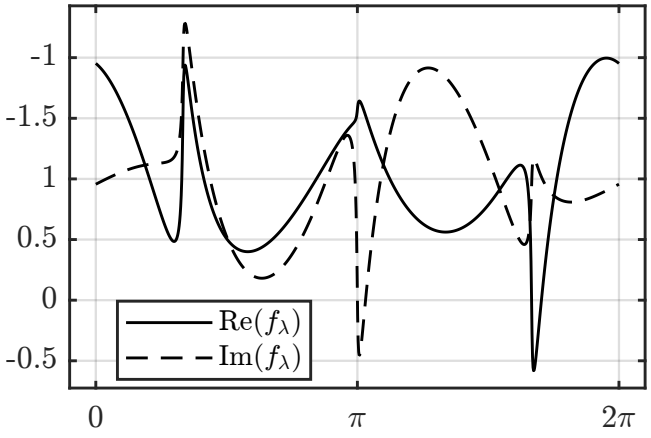


Figure III.2: Numerical approximation of $f_{-\lambda}$.

Remark 3.3 *If f is an eigenfunction of $\mathcal{L}_{\text{stab}}^{c/3,v^3}$ associated with the eigenvalue λ, then the function $x \mapsto f(-x)$ is also an eigenfunction of $\mathcal{L}_{\text{stab}}^{c/3,v^3}$ associated with the eigenvalue $-\lambda$.*

Remark 3.4 *Notice that $\mathcal{L}_{\text{stab}}^{c/3,v^3}$ is a real operator. Therefore, if f is an eigenfunction associated with the eigenvalue λ and $\text{Re}(\lambda) < 0$, then*

$$(t,x) \mapsto \text{Re}(e^{-\lambda t} f(x))$$

is an unstable solution of (3.1).

Remark 3.5 *Let us emphasize that we cannot guarantee that the solution $\left(\frac{c}{3}, v^3\right)$ belongs to the branch of $\frac{2\pi}{3}$-periodic functions bifurcating from 0. However, nonrigorous numerical evidence indicates that this is the case.*

The rest of Section III.3 is devoted to proving Theorem 3.2.

In the following lemma, we analyze the Fourier structure of $\mathcal{L}_{\text{stab}}^{c/n,v^n}$.

Lemma 3.6 *Let $n \in \mathbb{N}$ with $n \geq 2$ and let $f, g : \mathbb{T} \to \mathbb{C}$ be smooth functions with*

$$\widehat{f}_{nk} = \widehat{g}_{nk} = 0 \quad \text{for all } k \in \mathbb{Z}.$$

Let $V : \mathbb{T} \to \mathbb{C}$ be a smooth function and $V^n := \frac{1}{n}V(nx)$.
Then, f and g satisfy

$$g = V^n \partial_x f + Hf - \frac{1}{2\pi}\int_{-\pi}^{\pi} V^n(x)\partial_x f(x)dx \tag{3.2}$$

if and only if for every $1 \leq l \leq n-1$, $f^{l,n} := \mathcal{P}_{l,n} f$ and $g^{l,n} := \mathcal{P}_{l,n} g$ with $\theta := \frac{l}{n}$, satisfy

$$g^{l,n} = V\partial_x f^{l,n} + i\theta V f^{l,n} + Hf^{l,n} - i\widehat{f_0^{l,n}}. \tag{3.3}$$

Remark 3.7 *We exclude the case $l = 0$. Its behavior is the same as for $n = 1$, and we currently have no results concerning its stability. However, we conjecture that the eigenvalues of v with speed 1 remain stable up to the critical speed.*

The numerical spectra in Figures III.3 and III.4 suggest that considering such a high value of the speed c is necessary to detect the instability.

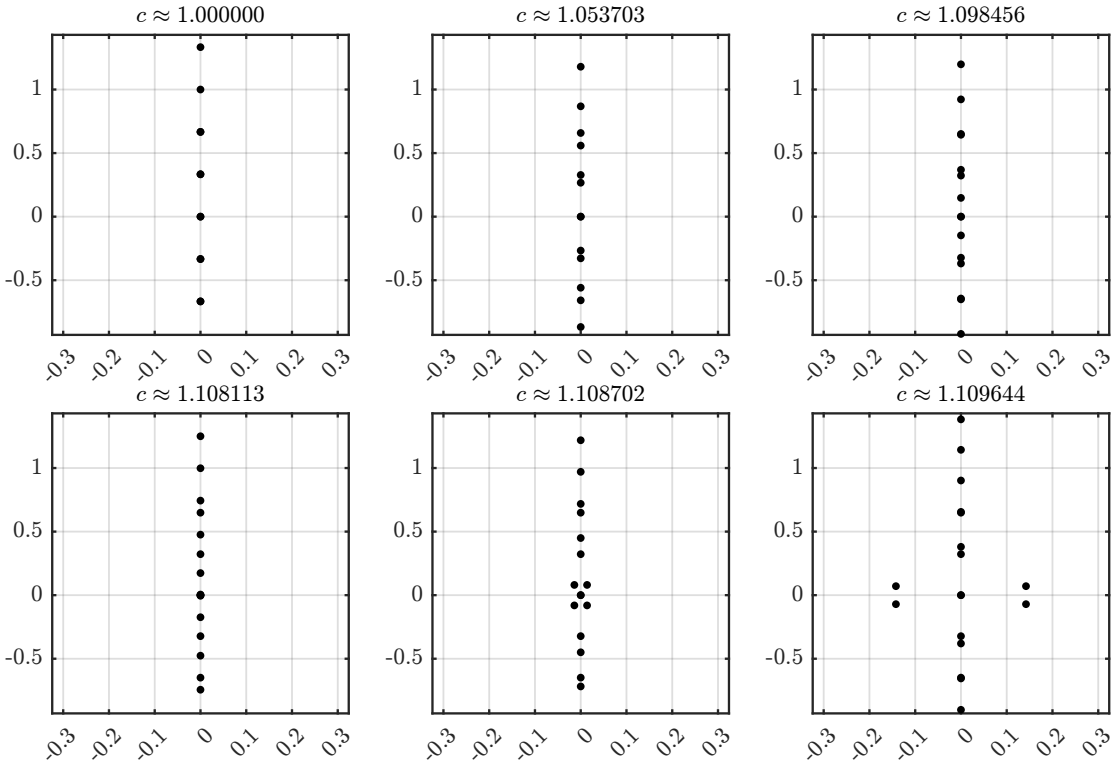


Figure III.3: Numeric approximation of the spectrum of $\mathcal{L}_{\text{stab}}^{c/3,v^3}$ for different values of c.

Proof: We project (3.2) onto e^{ikx} with $k \in \mathbb{Z}$ such that n does not divide k,

$$\begin{aligned}\widehat{g}_k &= \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-ikx}\left(V^n(x)\partial_x f(x) + Hf(x)\right)dx \\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-ikx}\frac{1}{n}\sum_{j\in\mathbb{Z}}\widehat{V}_j e^{injx}\sum_{l\in\mathbb{Z}} il\widehat{f}_l e^{ilx}\,dx - i\mathrm{sgn}(k)\widehat{f}_k \\ &= \frac{i}{n}\sum_{j\in\mathbb{Z}}\sum_{l\in\mathbb{Z}} l\widehat{V}_j\widehat{f}_l\delta_{k,nj+l} - i\mathrm{sgn}(k)\widehat{f}_k = \frac{i}{n}\sum_{j\in\mathbb{Z}}(k-nj)\widehat{V}_j\widehat{f}_{k-nj} - i\mathrm{sgn}(k)\widehat{f}_k. \end{aligned} \tag{3.4}$$

Fourier modes of $\mathcal{P}_{l,n}f$ are defined as

$$(\widehat{\mathcal{P}_{l,n}f})_k := \widehat{f}_{nk+l}.$$

Then, fixing $\theta := \frac{l}{n}$, denoting by $f^\theta := \mathcal{P}_{l,n}f$, and using identity (3.4), we obtain

$$\begin{aligned}\widehat{g_k^{l,n}} &= \frac{i}{n}\sum_{j\in\mathbb{Z}}(nk+l-nj)\widehat{V}_j\widehat{f}_{nk+l-nj} - i\mathrm{sgn}(nk+l)\widehat{f}_{nk+l} \\ &= i\sum_{j\in\mathbb{Z}}(j+\theta)\widehat{V}_{k-j}\widehat{f_j^{l,n}} - i(\mathrm{sgn}(k)+\delta_{k,0})\widehat{f_k^{l,n}}. \end{aligned} \tag{3.5}$$

Here, we used that $\mathrm{sgn}(nk+l) - \mathrm{sgn}(k) = \delta_{k,0}$ since $1 \le l \le n-1$.

Finally, we note that (3.5) is the projection of (3.3) onto e^{ikx}. □

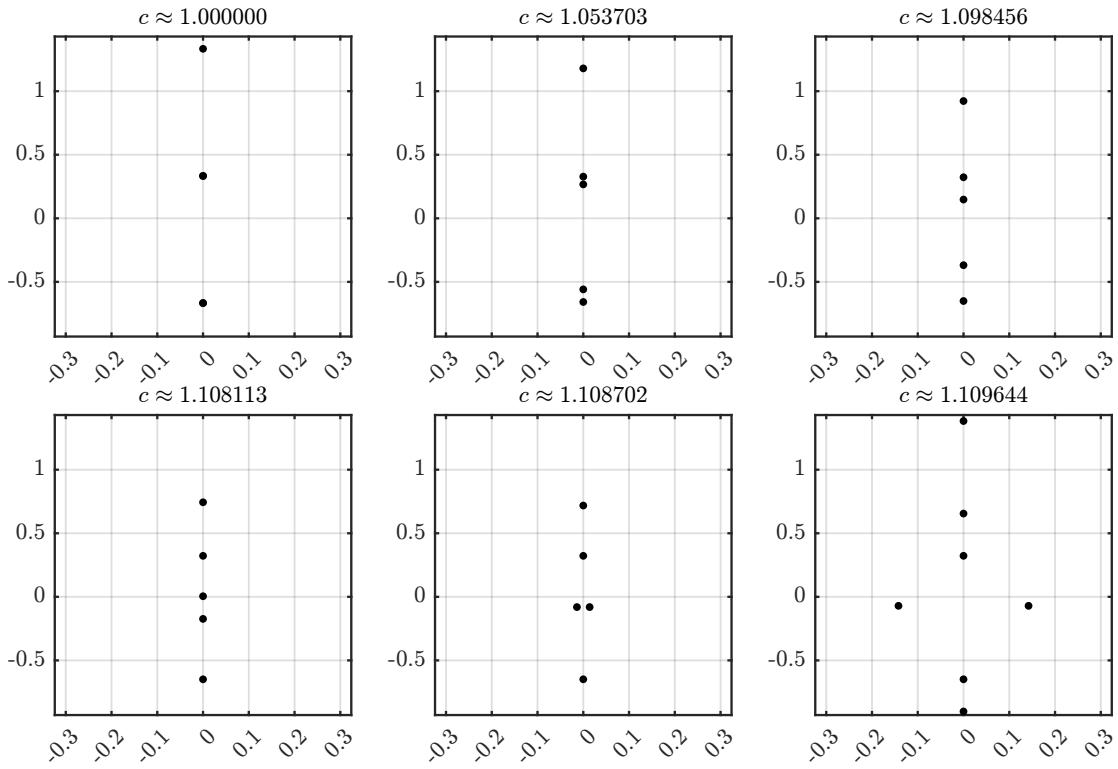


Figure III.4: Numeric approximation of the spectrum of $\mathcal{L}^{1/3}_{\text{stab}}$ for different values of c.

By Lemma 3.6 we can reduce our eigenvalue problem

$$\mathcal{L}^{c/n,v^n}_{\text{stab}} f = \lambda f$$

to

$$\mathcal{L}_\theta f = \lambda f \tag{3.6}$$

where

$$\mathcal{L}_\theta f := (c+v)\partial_x f + i\theta(c+v)f + Hf - i\widehat{f}_0. \tag{3.7}$$

In order to find eigenpairs (λ, f) for (3.6), we introduce a new operator, which approximates $\mathcal{L}_\theta$:

$$\mathcal{L}^{\text{ap}}_\theta f := (c+v^{\text{ap}})\partial_x f + i\theta(c+v^{\text{ap}})f + Hf - i\widehat{f}_0. \tag{3.8}$$

The reason for introducing this operator is that it contains v^{ap} instead of v, and v^{ap} is known explicitly. We can then construct a suitable approximation $(\lambda^{\text{ap}}, f^{\text{ap}})$ satisfying

$$\mathcal{L}^{\text{ap}}_\theta f^{\text{ap}} - \lambda^{\text{ap}} f^{\text{ap}} = \xi_{f^{\text{ap}}}. \tag{3.9}$$

Here λ^{ap} is given by

$$\lambda^{\text{ap}} = 0.142128757204011 - 0.0707987316896479i, \tag{3.10}$$

and the expression of f^{ap} can be found in the supplementary material. A numerical approximation of f^{ap} is shown in Figure III.5.

We now denote

$$\mathcal{D}_\theta := \mathcal{L}_\theta - \mathcal{L}^{\text{ap}}_\theta.$$

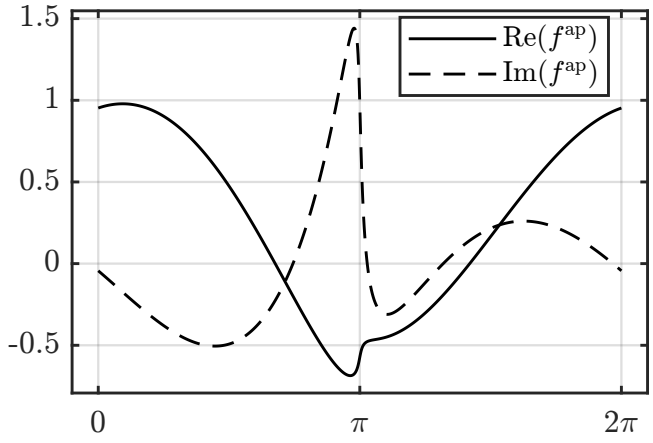


Figure III.5: Numerical approximation of f^{ap}.

Introducing the following ansatz into (3.6):

$$\lambda \mapsto \lambda^{\mathrm{ap}} + \eta, \qquad\qquad f \mapsto f^{\mathrm{ap}} + f,$$

we obtain

$$0 = (\mathcal{L}_\theta^{\mathrm{ap}} f^{\mathrm{ap}} - \lambda^{\mathrm{ap}} f^{\mathrm{ap}}) + (\mathcal{L}_\theta^{\mathrm{ap}} f - \lambda^{\mathrm{ap}} f) + \mathcal{D}_\theta(f^{\mathrm{ap}} + f) - \eta(f^{\mathrm{ap}} + f). \tag{3.11}$$

Hence, from (3.11) and (3.9), we finally obtain

$$(\mathcal{L}_\theta^{\mathrm{ap}} - \lambda^{\mathrm{ap}} I)f = (\eta I - \mathcal{D}_\theta)(f^{\mathrm{ap}} + f) - \xi_{f^{\mathrm{ap}}}. \tag{3.12}$$

III.3.1 Fixed point argument to prove linear instability

In this section, we derive the estimates required to apply the fixed-point argument in Proposition 3.11. Our goal is to show the existence of η and f satisfying the equation (3.12). This problem can be reformulated as finding a fixed point of the operator T^θ_{stab}, defined by

$$T^\theta_{\mathrm{stab}} f := (\mathcal{L}_\theta^{\mathrm{ap}} - \lambda^{\mathrm{ap}} I)^{-1}[(\eta(f) I - \mathcal{D}_\theta)(f^{\mathrm{ap}} + f) - \xi_{f^{\mathrm{ap}}}]. \tag{3.13}$$

where $\eta(f) : H^1(\mathbb{T}) \to \mathbb{C}$ will be defined later in Section III.5.3.

The operator $(\mathcal{L}_\theta^{\mathrm{ap}} - \lambda^{\mathrm{ap}} I)^{-1}$ is nearly non-invertible. However, with a particular choice of η in terms of f, we can overcome this problem. With this choice, T^θ_{stab}, with $\theta = \frac{1}{3}$, will become a contraction on an appropriate closed ball of $H^1(\mathbb{T})$.

Accordingly, we aim to obtain bounds for $\mathcal{D}_\theta$, f^{ap}, $\xi_{f^{\mathrm{ap}}}$, and $(\mathcal{L}_\theta^{\mathrm{ap}} - \lambda^{\mathrm{ap}} I)^{-1}$, and to verify that the latter operator is indeed well defined.

Lemma 3.8 *The following estimate holds*

$$\|\mathcal{D}_\theta\|_{H^1(\mathbb{T}) \to L^2(\mathbb{T})} \leq 2\zeta(2)^{1/2}(1+\theta^2)^{1/2}\|v - v^{\mathrm{ap}}\|_{H^1_{\mathrm{even}}}.$$

Proof: By definition, the operator $\mathcal{D}_\theta$ is given by

$$\begin{aligned}
\mathcal{D}_\theta f &:= \mathcal{L}_\theta f - \mathcal{L}_\theta^{\mathrm{ap}} f \\
&= \big[(c+v)\partial_x f + i\theta(c+v)f + Hf - i\widehat{f}_0\big] - \big[(c+v^{\mathrm{ap}})\partial_x f + i\theta(c+v^{\mathrm{ap}})f + Hf - i\widehat{f}_0\big] \\
&= (v - v^{\mathrm{ap}})(\partial_x f + i\theta f).
\end{aligned}$$

Taking norms,

$$\|\mathcal{D}_\theta f\|_{L^2(\mathbb{T})} \leq \|v - v^{\mathrm{ap}}\|_{L^\infty(\mathbb{T})}(\|\partial_x f\|_{L^2(\mathbb{T})} + \theta\|f\|_{L^2(\mathbb{T})}).$$

Note that we can apply Peter–Paul inequality in the optimal way

$$\begin{aligned}(\|\partial_x f\|_{L^2(\mathbb{T})} + \theta\|f\|_{L^2(\mathbb{T})})^2 =& \|\partial_x f\|^2_{L^2(\mathbb{T})} + \theta^2\|f\|^2_{L^2(\mathbb{T})} + 2\theta\|\partial_x f\|_{L^2(\mathbb{T})}\|f\|_{L^2(\mathbb{T})} \\ \leq& (1+\vartheta)\|\partial_x f\|^2_{L^2(\mathbb{T})} + (1+\vartheta^{-1})\theta^2\|f\|^2_{L^2(\mathbb{T})}.\end{aligned}$$

Taking $\vartheta = \theta^2$, we get

$$(\|\partial_x f\|_{L^2(\mathbb{T})} + \theta\|f\|_{L^2(\mathbb{T})})^2 \leq (1+\theta^2)(\|\partial_x f\|^2_{L^2(\mathbb{T})} + \|f\|^2_{L^2(\mathbb{T})}) = (1+\theta^2)\|f\|^2_{H^1(\mathbb{T})}.$$

Finally, by Lemma A.1

$$\|\mathcal{D}_\theta f\|_{L^2(\mathbb{T})} \leq 2\zeta(2)^{\frac{1}{2}}(1+\theta^2)^{\frac{1}{2}}\|v - v^{\mathrm{ap}}\|_{H^1_{\mathrm{even}}}\|f\|_{H^1(\mathbb{T})}.$$

□

Corollary 3.9 *For $\theta = \frac{1}{3}$, we have*

$$\|\mathcal{D}_\theta\|_{H^1(\mathbb{T})\to L^2(\mathbb{T})} \leq 6.78140 \cdot 10^{-9}.$$

Proof: We apply Lemma 3.8 and Proposition 2.5. □

Lemma 3.10 *The following estimates hold*

$$\|f^{\mathrm{ap}}\|^2_{L^2(\mathbb{T})} < 0.526963, \qquad \|f^{\mathrm{ap}}\|^2_{H^1(\mathbb{T})} < 4.54832, \qquad \|\xi_{f^{\mathrm{ap}}}\|^2_{L^2(\mathbb{T})} < 1.60881 \cdot 10^{-20}.$$

Proof: The proof is computer-assisted and the code can be found in the supplementary material. We refer to the Appendix III.E for details about the implementation. □

Proposition 3.11 *For $\theta = \frac{1}{3}$, the operator $T^\theta_{\mathrm{stab}} : H^1(\mathbb{T}) \to H^1(\mathbb{T})$ is a contraction on*

$$X := \{g \in H^1(\mathbb{T}) : \|g\|_{H^1(\mathbb{T})} \leq 10^{-3}\}.$$

Therefore, the operator $\mathcal{L}_\theta$ admits an eigenvalue λ with positive real part. In fact,

$$\mathrm{Re}(\lambda) > 0.142127.$$

The proof of Proposition 3.11 is based on the following lemmas.

Lemma 3.12 *$T^\theta_{stab} : H^1(\mathbb{T}) \to H^1(\mathbb{T})$ is a continuous operator that maps*

$$X = \{f \in H^1(\mathbb{T}) : \|f\|_{H^1(\mathbb{T})} \leq 10^{-3}\}$$

into itself.

Proof: The proof will be given in Section III.5. □

Lemma 3.13 *The following estimates hold*

$$\|T^\theta_{\mathrm{stab}} f_1 - T^\theta_{\mathrm{stab}} f_2\|_{H^1(\mathbb{T})} \leq 0.12\|f_1 - f_2\|_{H^1(\mathbb{T})},$$

for all $f_1, f_2 \in X$.

Proof: The proof will be given in Section III.5. □

Lemma 3.14 *For all $f \in X$, it holds that*

$$\mathrm{Re}(\lambda^{\mathrm{ap}}) - |\eta(f)| > 0.142127.$$

Proof: The proof will be given in Section III.5. □

Lemma 3.15 *It holds that*

$$\|f^{\mathrm{ap}}\|_{H^1(\mathbb{T})} > 10^{-3}.$$

Proof: The proof is computer-assisted and the code can be found in the supplementary material. We refer to the Appendix III.E for details about the implementation. □

Proof: By Lemmas 3.12and 3.13, we can apply the Banach fixed-point theorem. Hence, there exists $f_0 \in H^1(\mathbb{T})$, a fixed point for $T^{\theta}_{\mathrm{stab}}$, i.e.

$$(\mathcal{L}^{\mathrm{ap}}_{\theta} - \lambda^{\mathrm{ap}} I) f_0 = [(\eta(f_0) I - \mathcal{D}_{\theta})(f_0 + f^{\mathrm{ap}}) - \xi_{f^{\mathrm{ap}}}].$$

Consequently, $\lambda := \lambda^{\mathrm{ap}} + \eta(f_0)$ and $f^{\mathrm{ap}} + f_0$ solve the equation (3.6).

Moreover, Lemma 3.14 establishes a bound a bound for $\eta(f_0)$ which ensures that λ still has positive real part.

Finally, Lemma 3.15 shows that $f^{\mathrm{ap}} + f_0 \neq 0$. □

III.3.2 Conclusion of the proof of Theorem 3.2

Proposition 3.11 states that the operator $\mathcal{L}_{\theta}$ admits an eigenpair (λ, f) with

$$\mathrm{Re}(\lambda) > 0.142127.$$

Applying Lemma 3.6 with $n = 3$ and $l = 1$, hence $\theta = \frac{1}{3}$, we deduce that λ is an eigenvalue of $\mathcal{L}^{c/3, v^3}_{\mathrm{stab}}$. We construct the associated eigenfunction $\widetilde{f}$ as follows

$$\widetilde{f}(x) = e^{ix} f(3x).$$

Finally, substituting $x \mapsto -x$, we find that $-\lambda$ is an unstable eigenvalue for $\mathcal{L}^{c/3, v^3}_{\mathrm{stab}}$ with eigenfunction $f_{-\lambda}(x) = \widetilde{f}(-x)$.

Again, as at the end of Section III.2, a bootstrap argument shows that $f_{-\lambda}$ is, in fact, smooth.

III.4 Proof of Proposition 2.2

We define the operator $\mathcal{L}^{\mathrm{ap}}_{\mathrm{exis}} : H^2_{\mathrm{odd}} \to H^1_{\mathrm{even}}$ as

$$\mathcal{L}^{\mathrm{ap}}_{\mathrm{exis}} f(x) = Hf(x) + (c + v^{\mathrm{ap}}(x)) \partial_x f(x) - \frac{1}{2\pi} \int_{-\pi}^{\pi} v^{\mathrm{ap}}(x) \partial_x f(x)\, dx. \tag{4.1}$$

Then, for any smooth function $f : \mathbb{T} \to \mathbb{R}$, we find that

$$\partial_x \mathcal{L}^{\mathrm{ap}}_{\mathrm{exis}} f = L_{v^{\mathrm{ap}}} \partial_x f. \tag{4.2}$$

Therefore,

$$L_{v^{\mathrm{ap}}} = \partial_x \mathcal{L}^{\mathrm{ap}}_{\mathrm{exis}} \partial_x^{-1}. \tag{4.3}$$

Let us explain a possible strategy to study the equation $\mathcal{L}^{\mathrm{ap}}_{\mathrm{exis}} f = g$. If we apply the Hilbert transform to the equation and consider the function $f^+ := \frac{1}{2}(f + iHf)$, as we will see later, one can find a first-order ODE for f^+ which depends on N Fourier coefficients of f^+. This equation can be easily integrated. Computing the Fourier transform of the solution, one can arrive at a linear system that must be satisfied by these Fourier coefficients. One may think that the problem has been reduced to showing that this system is invertible. However, there are additional conditions to

check in order to guarantee that the function is periodic and all non-positive Fourier modes vanish. To overcome this problem, we will use a slightly different strategy that requires the introduction of a new operator.

The next lemma, Lemma 4.1, states that studying the invertibility of $\mathcal{L}^{\mathrm{ap}}_{\mathrm{exis}}$ is equivalent to solving a local problem in the unit complex disk.

We first define the operator $\mathcal{L}^{\mathrm{ap,\,hol}}_{\mathrm{exis}} : \dot{\mathcal{H}} \to \dot{\mathcal{H}}$ as

$$\mathcal{L}^{\mathrm{ap,\,hol}}_{\mathrm{exis}} F(z) := izP(z)\partial_z F(z) - iF(z) - \sum_{k=1}^{N} F_k \mathcal{G}_k(z), \tag{4.4}$$

where we are using (1.5), and $P(z)$ and $\mathcal{G}(F)(z)$ are given by

$$P(z) := \; c + \frac{1}{2}\sum_{j=1}^{N} v_j^{\mathrm{ap}}(z^j + z^{-j}), \tag{4.5}$$

$$\mathcal{G}_k(z) := \frac{i}{2} k v_k^{\mathrm{ap}} - \frac{i}{2}\sum_{m=k+1}^{N} k v_m^{\mathrm{ap}}(z^{m-k} - z^{-(m-k)}). \tag{4.6}$$

The operator $\mathcal{L}^{\mathrm{ap,\,hol}}_{\mathrm{exis}}$ is well defined. Indeed, for any $F \in \dot{\mathcal{H}}$, the expression in (4.4) defines a holomorphic function vanishing at the origin, since all terms involving non-positive powers of z cancel out.

Lemma 4.1 *The operator $\mathcal{L}^{\mathrm{ap,\,hol}}_{\mathrm{exis}} : \dot{\mathcal{H}}^2 \to \dot{\mathcal{H}}^1$ is invertible if and only if $\mathcal{L}^{\mathrm{ap}}_{\mathrm{exis}} : H^2_{\mathrm{odd}} \to H^1_{\mathrm{even}}$ is invertible.*

Furthermore, assuming both are invertible, the following identity holds:

$$\|(\mathcal{L}^{\mathrm{ap,\,hol}}_{\mathrm{exis}})^{-1}\|_{\dot{\mathcal{H}}^1 \to \dot{\mathcal{H}}^2} = \|(\mathcal{L}^{\mathrm{ap}}_{\mathrm{exis}})^{-1}\|_{H^1_{\mathrm{even}} \to H^2_{\mathrm{odd}}}.$$

Proof: First, we show that if $f \in H^2_{\mathrm{odd}}$ satisfies

$$Hf(x) + (c + v^{\mathrm{ap}}(x))\partial_x f(x) - \frac{1}{2\pi}\int_{-\pi}^{\pi} v^{\mathrm{ap}}(x)\partial_x f(x)\,dx = g, \tag{4.7}$$

then f^+ also satisfies

$$(c + v^{\mathrm{ap}}(x))\partial_x f^+(x) = if^+(x) + \mathcal{J}(x) + g^+(x) \tag{4.8}$$

with

$$\mathcal{J}(x) := \frac{i}{2}\sum_{j=1}^{N} j v_j^{\mathrm{ap}} \widehat{f^+}_j - \frac{i}{2}\sum_{j=1}^{N} v_j^{\mathrm{ap}} \sum_{m=1}^{j-1} m \widehat{f^+}_m (e^{i(j-m)x} - e^{-i(j-m)x}).$$

From (4.7), we obtain for $f^+ = \frac{1}{2}(f + iHf)$ that

$$-if^+(x) + (c + v^{\mathrm{ap}}(x))\partial_x f^+(x) + \frac{i}{2}[H, v^{\mathrm{ap}}]\partial_x f(x) - \frac{1}{4\pi}\int_{-\pi}^{\pi} v^{\mathrm{ap}}(x)\partial_x f(x)\,dx = g^+. \tag{4.9}$$

We now compute the integral term in (4.9):

$$\frac{1}{4\pi}\int_{-\pi}^{\pi} v^{\mathrm{ap}}(x)\partial_x f(x)\,dx = -\frac{1}{4\pi}\int_{-\pi}^{\pi} \partial_x v^{\mathrm{ap}}(x) f(x)\,dx = \frac{1}{4\pi}\int_{-\pi}^{\pi}\sum_{j=1}^{N} j v_j^{\mathrm{ap}} \sin(jx) f(x)\,dx$$

By construction, f is odd, so $\widehat{f}_k = -\widehat{f}_{-k}$ and we obtain

$$\frac{1}{4\pi}\int_{-\pi}^{\pi} v^{\mathrm{ap}}(x)\partial_x f(x)\,dx = \frac{i}{2}\sum_{j=1}^{N} j v_j^{\mathrm{ap}} \widehat{f^+}_j. \tag{4.10}$$

The computation of $[H, v^{\rm ap}]\partial_x f$ is given in Corollary B.2. Hence, as f is odd, computations yield

$$\begin{aligned}\frac{i}{2}[H, v^{\rm ap}]\partial_x f(x) &= \frac{i}{2}\sum_{j=1}^{N} v_j^{\rm ap}\Big(-\frac{j}{2}(\widehat{f}_j + \widehat{f}_{-j}) - \sum_{m=1}^{j-1} m(\widehat{f}_m e^{-i(j-m)x} + \widehat{f}_{-m}e^{i(j-m)x})\Big)\\ &= \frac{i}{2}\sum_{j=1}^{N} v_j^{\rm ap}\sum_{m=1}^{j-1} m\widehat{f}_m(e^{i(j-m)x} - e^{-i(j-m)x}). \end{aligned} \tag{4.11}$$

Expressions (4.11) and (4.10), together with (4.9), yield (4.8).

Conversely, if $F|_{\mathbb{T}}$, for $F \in \dot{\mathcal{H}}^2$, satisfies

$$(c + v^{\rm ap}(x))\partial_x F|_{\mathbb{T}}(x) = iF|_{\mathbb{T}}(x) + \mathcal{J}(x) + G|_{\mathbb{T}}(x), \tag{4.12}$$

with

$$\mathcal{J}(x) := \frac{i}{2}\sum_{j=1}^{N} j v_j^{\rm ap}(\widehat{F|_{\mathbb{T}}})_j - \frac{i}{2}\sum_{j=1}^{N} v_j^{\rm ap}\sum_{m=1}^{j-1} m(\widehat{F|_{\mathbb{T}}})_m(e^{i(j-m)x} - e^{-i(j-m)x}).$$

and $G \in \dot{\mathcal{H}}^1$, then the function $f(x) = F|_{\mathbb{T}}(x) - F|_{\mathbb{T}}(-x)$ satisfies (4.7) with $g(x) = G|_{\mathbb{T}}(x) + G|_{\mathbb{T}}(-x) \in H^1_{\rm even}$. To check this, we just note that

$$\begin{aligned}\partial_x f(x) &= \partial_x F|_{\mathbb{T}}(x) + \partial_x F|_{\mathbb{T}}(-x),\\ Hf(x) &= -i\big(F|_{\mathbb{T}}(x) + F|_{\mathbb{T}}(-x)\big),\\ \mathcal{J}(x) + \mathcal{J}(-x) &= \frac{1}{2\pi}\int_{-\pi}^{\pi} v^{\rm ap}(x)\partial_x f(x)\,dx,\end{aligned}$$

where we have used (4.10) and $\widehat{f^+}_n = \widehat{f}_n = (\widehat{F|_{\mathbb{T}}})_n$, for $n \geq 1$, in the identity for $\mathcal{J}$.

Let us now assume that $\mathcal{L}^{\rm ap,\,hol}_{\rm exis}$ is invertible and that $g \in H^1_{\rm even}$. We define $G \in \dot{\mathcal{H}}^1$ as the analytic extension of g^+ to the interior of the unit disk. Then, there exists a unique $F \in \dot{\mathcal{H}}^2$ such that $\mathcal{L}^{\rm ap,\,hol}_{\rm exis}F = G$.

The restriction $F|_{\mathbb{T}}$ will satisfy (4.12) and therefore $f = F|_{\mathbb{T}}(x) - F|_{\mathbb{T}}(-x) \in H^2_{\rm odd}$ solves

$$\mathcal{L}^{\rm ap}_{\rm exis} f = G|_{\mathbb{T}}(x) + G|_{\mathbb{T}}(-x) = g(x).$$

Furthermore, this solution must be unique. Otherwise, if there exists $f_1 \neq f$ solving $\mathcal{L}^{\rm ap}_{\rm exis}f_1 = g$, the holomorphic extension of f_1^+, F_1 satisfies $\mathcal{L}^{\rm ap,\,hol}_{\rm exis}F_1 = G$ and then $F_1 = F$, which yields $f_1 = f$.

Finally, let us assume that $\mathcal{L}^{\rm ap}_{\rm exis}$ is invertible and that $G \in \dot{\mathcal{H}}^1$. Take $g = G|_{\mathbb{T}}(x) + G|_{\mathbb{T}}(-x)$ and solve

$$\mathcal{L}^{\rm ap}_{\rm exis} f = g,$$

with $f \in H^2_{\rm odd}$. Then f^+ solves (4.8) and its holomorphic extension $F \in \dot{\mathcal{H}}^2$ solves $\mathcal{L}^{\rm ap,\,hol}_{\rm exis}F = g^+ = G$. This solution must also be unique since, if there exists another solution $\widetilde{F}$, then $tif := \widetilde{F}|_{\mathbb{T}}(x) - \widetilde{F}|_{\mathbb{T}}(-x)$ solves $\mathcal{L}^{\rm ap,\,hol}_{\rm exis}\widetilde{f} = g$ and therefore $\widetilde{f} = f$, which implies $\widetilde{F} = F$.

Moreover, we have

$$\|g\|^2_{H^1_{\rm even}} = \sum_{k\geq 1} k^2|\widehat{g}_k|^2 = \|G\|^2_{\dot{\mathcal{H}}^1} \qquad \text{and} \qquad \|f\|^2_{H^2_{\rm odd}} = \sum_{k\geq 1} k^4|\widehat{f}_k|^2 = \|F\|^2_{\dot{\mathcal{H}}^2}.$$

Hence, the norms of both operators coincide. □

We study $\mathcal{L}^{\rm ap,\,hol}_{\rm exis}$ instead of $\mathcal{L}^{\rm ap}_{\rm exis}$. In Section III.4.1, we analyze $\mathcal{L}^{\rm ap,\,hol}_{\rm exis}$. First, in Section III.4.1.1, we establish some theoretical results in Fuchsian theory. Then, in Section III.4.1.2, we prove that $\mathcal{L}^{\rm ap,\,hol}_{\rm exis}$ admits an inverse when considered between appropriate functional spaces. In Section III.4.1.3, we derive suitable bounds. Finally, we combine these results to prove Proposition 2.2 in Section III.4.2.

III.4.1 The linear operator $\mathcal{L}_{\text{exis}}^{\text{ap, hol}}$

In order to study the operator $\mathcal{L}_{\text{exis}}^{\text{ap, hol}}$, given in (4.4), our first idea is to momentarily disregard the Taylor coefficients F_j in expression (4.4) and replace them with general constants.

Lemma 4.2 *Let $a_k \in \mathbb{C}$ for $k = 1, \dots, N$, and let $F \in \dot{\mathcal{H}}$ and $G \in \dot{\mathcal{H}}$.*
Assume that F and G satisfy

$$izP(z)\partial_z F(z) = iF(z) + \frac{i}{2}\sum_{j=1}^{N} j v_j^{\text{ap}} a_j - \frac{i}{2}\sum_{j=2}^{N} v_j^{\text{ap}} \sum_{m=1}^{j-1} m(z^{j-m} - z^{-(j-m)})a_m + G(z). \tag{4.13}$$

Then, for every $k = 1, \dots, N$, we have

$$F_k := \frac{1}{k!}\frac{d^k F}{dz^k}\bigg|_{z=0} = a_k.$$

Proof: We argue by comparing Taylor coefficients at $z = 0$.
Multiplying (4.13) by z^N and using that $F(0) = G(0) = 0$, we obtain

$$iz^{N+1}P(z)\partial_z F(z) = \frac{i}{2}z^N \sum_{j=1}^{N} j v_j^{\text{ap}} a_j + \frac{i}{2}\sum_{j=2}^{N} v_j^{\text{ap}} \sum_{m=1}^{j-1} m z^{N-(j-m)} a_m + \mathcal{O}(z^{N+1}). \tag{4.14}$$

Expanding terms in (4.14) yields

$$\frac{i}{2}\sum_{j=1}^{N}\sum_{l=0}^{j} v_j^{\text{ap}} l F_l z^{N+l-j} = \frac{i}{2}z^N \sum_{j=1}^{N} j v_j^{\text{ap}} a_j + \frac{i}{2}\sum_{j=2}^{N} v_j^{\text{ap}} \sum_{m=1}^{j-1} m z^{N-(j-m)} a_m + \mathcal{O}(z^{N+1}). \tag{4.15}$$

Fix $k \in \{1, \dots, N-1\}$, apply $\frac{1}{k!}\frac{d^k}{dz^k}$ to (4.15), and evaluate at $z = 0$. We have

$$\frac{i}{2}\sum_{j=1}^{N}\sum_{l=0}^{j-1} l v_j^{\text{ap}} F_l \delta_{k,N+l-j} = \frac{i}{2}\sum_{j=2}^{N} v_j^{\text{ap}} \sum_{m=1}^{j-1} m a_m \delta_{k,N+m-j}$$

Hence,

$$0 = \sum_{j=N+1-k}^{N} (k+j-N) v_j^{\text{ap}} (F_{k+j-N} - a_{k+j-N}) = \sum_{j=1}^{k} j v_{j+N-k}^{\text{ap}} (F_j - a_j). \tag{4.16}$$

We now use the following lemma

Lemma 4.3 *It holds for every $j = 1, \dots, N$ that*

$$v_j^{\text{ap}} \neq 0.$$

Proof: The proof is computer-assisted and the code can be found in the supplementary material. We refer to the Appendix III.E for details about the implementation. □

Since $v_j^{\text{ap}} \neq 0$ for every $j = 1, \dots, N$ and (4.16) is a triangular system, we conclude that $F_k = a_k$ for $k = 1, \dots, N-1$.
Finally, we apply $\frac{1}{N!}\frac{d^N}{dz^N}$ to (4.15) and evaluate at $z = 0$:

$$\frac{i}{2}\sum_{j=1}^{N} j v_j^{\text{ap}} F_j = \frac{i}{2}\sum_{j=1}^{N} j v_j^{\text{ap}} a_j.$$

Since $F_j = a_j$ for $j = 1, \dots, N-1$, we conclude that

$$0 = N v_N^{\text{ap}} (F_N - a_N),$$

and thus $F_N = a_N$ because $v_N^{\text{ap}} \neq 0$. □

Our goal is to reduce the study of $\mathcal{L}_{\text{exis}}^{\text{ap, hol}}$ to a finite dimensional problem. The next section is devoted to developing some tools we need to prove that the operator $\mathcal{L}_{\text{exis}}^{\text{ap, hol}} : \dot{\mathcal{H}}^2 \to \dot{\mathcal{H}}^1$ is one-to-one.

III.4.1.1 Fuchsian theory

The tools in this subsection allow us to determine when the solutions of a singular first-order ODE are holomorphic in a simply connected domain $\Omega \subset \mathbb{C}$. We focus on the affine case. See [Inc56; WW96] for background on linear equations with regular singularities.

Let $\Omega \subset \mathbb{C}$ be open, connected, and simply connected, and let $A, B : \Omega \to \mathbb{C}$ be holomorphic (and continuous on $\overline{\Omega}$ when needed). Fix $z_0 \in \Omega$. We study conditions on A and B ensuring that a solution F of

$$(z - z_0)\partial_z F(z) + A(z)F(z) = B(z) \tag{4.17}$$

extends holomorphically across z_0.

For any $\omega_0 \in \Omega$, $\omega_0 \neq z_0$, equation (4.17) is regular at ω_0 and admits a one-parameter family of holomorphic solutions in a neighborhood of ω_0. Any singular behavior can only occur at z_0. By analytic continuation, solutions defined away from z_0 can be extended along paths in $\Omega - \{z_0\}$, which allows us to analyze their behavior near z_0.

We now return to the analytic development. We begin with the regular (non-singular) case, which also illustrates the analytic continuation mechanism.

Lemma 4.4 *Let $\Omega \subset \mathbb{C}$ be open and simply connected, and let A, B be holomorphic on Ω. Then for any $\omega_0 \in \Omega$ and any $c_0 \in \mathbb{C}$, there exists a unique holomorphic function $F : \Omega \to \mathbb{C}$ solving*

$$\begin{cases} \partial_z F + AF = B, \\ F(\omega_0) = c_0. \end{cases} \tag{4.18}$$

Proof: Define

$$F_h(z) := \exp\Big(- \int_{\omega_0}^{z} A(y)\, dy \Big),$$

where the integral is taken along any smooth path in Ω joining ω_0 to z. Since Ω is simply connected and A is holomorphic, the integral is path-independent and F_h is well defined and holomorphic on Ω, with $F_h(\omega_0) = 1$.

Note that $\partial_z\Big(\frac{1}{F_h}\Big) = A\frac{1}{F_h}$. Multiplying (4.18) by $\frac{1}{F_h}$ gives

$$\frac{B}{F_h} = \partial_z F \frac{1}{F_h} + F\partial_z\Big(\frac{1}{F_h}\Big) = \partial_z\Big(\frac{F}{F_h}\Big).$$

Integrating and using $F(\omega_0) = c_0$ yields

$$F(z) = c_0 F_h(z) + F_h(z) \int_{\omega_0}^{z} \frac{B(y)}{F_h(y)}\, dy.$$

This function is holomorphic on Ω and is the unique solution to (4.18). □

Although solutions to (4.17) need not be holomorphic at z_0, there are special cases (for instance, the trivial solution $F = 0$ when $B = 0$). Our goal is to give criteria that guarantee holomorphicity at the singular point.

Lemma 4.5 *Let $\Omega \subset \mathbb{C}$ be open and simply connected, and let $A : \Omega \to \mathbb{C}$ be holomorphic. Fix $z_0 \in \Omega$ and consider the homogeneous equation*

$$(z - z_0)\partial_z F_h(z) + A(z)F_h(z) = 0. \tag{4.19}$$

Then for any simply connected set $U \subset \Omega - \{z_0\}$, every solution on U is a complex multiple of

$$F_h(z) := (z - z_0)^{-A(z_0)}\psi(z), \tag{4.20}$$

where ψ admits a holomorphic extension to Ω and satisfies $\psi(z_0) = 1$, and where

$$(z - z_0)^{-A(z_0)} := \exp(-A(z_0)\log(z - z_0))$$

is defined using a branch of $\log(z - z_0)$ on U.

In particular, if $A(z_0) \notin \mathbb{Z}^- \cup \{0\}$, the only solution that extends holomorphically to z_0 is the trivial zero solution.

Proof: Write

$$A(z) = A(z_0) + (z - z_0)A^{(1)}(z),$$

where $A^{(1)}$ is holomorphic on Ω. Fix a simply connected $U \subset \Omega - \{z_0\}$ and a branch of $\log(z - z_0)$ on U. Solving by separation of variables gives

$$F_h(z) = \exp\Big(-\int_{\omega_0}^{z} \frac{A(y)}{y - z_0}\,dy\Big) = \exp\Big(-A(z_0)\int_{\omega_0}^{z} \frac{dy}{y - z_0}\Big)\exp\Big(-\int_{\omega_0}^{z} A^{(1)}(y)\,dy\Big),$$

where $\omega_0 \in U$ is fixed and all integrals are taken along paths in U. Using the chosen branch of $\log(z - z_0)$,

$$\int_{\omega_0}^{z} \frac{dy}{y - z_0} = \log(z - z_0) - \log(\omega_0 - z_0).$$

Hence

$$F_h(z) = C(z - z_0)^{-A(z_0)}\psi(z),$$

where $C \in \mathbb{C}$ is constant and

$$\psi(z) := \exp\Big(-\int_{z_0}^{z} A^{(1)}(y)\,dy\Big) \tag{4.21}$$

is holomorphic on Ω with $\psi(z_0) = 1$ (the integral is path-independent in Ω). The last assertion follows because $(z - z_0)^{-A(z_0)}$ is holomorphic at z_0 if and only if $-A(z_0) \in \mathbb{Z}^+ \cup \{0\}$. □

We now turn to the affine equation (4.17). When $A(z_0) \notin \mathbb{Z}^- \cup \{0\}$, one can construct a holomorphic solution near z_0 by a convergent power series, and then extend it holomorphically to all of Ω by analytic continuation.

Lemma 4.6 *Let $\Omega \subset \mathbb{C}$ be open and simply connected and let $A, B : \Omega \to \mathbb{C}$ be holomorphic. Fix $z_0 \in \Omega$ and assume $A(z_0) \notin \mathbb{Z}^- \cup \{0\}$. Then there exists a holomorphic function $F_p : \Omega \to \mathbb{C}$ solving* (4.17).

Proof: Let ψ be as in (4.21) and set

$$\psi_B(z) := \frac{B(z)}{\psi(z)}.$$

Since ψ never vanishes, ψ_B is holomorphic on Ω. Choose $\delta > 0$ such that $D_\delta(z_0) \subset \Omega$ and write the Taylor expansion

$$\psi_B(z) = \sum_{k\geq 0} \psi_{B,k}(z - z_0)^k$$

on $D_\delta(z_0)$.

Define, for $z \in D_\delta(z_0)$,

$$F_p(z) := \psi(z) \sum_{k\geq 0} \frac{\psi_{B,k}}{k + A(z_0)}(z - z_0)^k. \tag{4.22}$$

Since $A(z_0) \notin \mathbb{Z}^- \cup \{0\}$, all denominators are nonzero and the series has the same radius of convergence as ψ_B. In particular, F_p is holomorphic on $D_\delta(z_0)$.

We claim that F_p solves (4.17) on $D_\delta(z_0)$. Using

$$(z - z_0)\partial_z\psi(z) = -(A(z) - A(z_0))\psi(z),$$

we compute

$$\begin{aligned}
(z - z_0)\partial_z F_p(z) &= (z - z_0)\Big[\partial_z\psi(z) \sum_{k\geq 0} \frac{\psi_{B,k}}{k + A(z_0)}(z - z_0)^k + \psi(z) \sum_{k\geq 1} \frac{\psi_{B,k}}{k + A(z_0)} k(z - z_0)^{k-1}\Big] \\
&= -(A(z) - A(z_0))\psi(z) \sum_{k\geq 0} \frac{\psi_{B,k}}{k + A(z_0)}(z - z_0)^k + \psi(z) \sum_{k\geq 1} \frac{\psi_{B,k}}{k + A(z_0)} k(z - z_0)^k \\
&= -A(z)F_p(z) + \psi(z)\Big(\psi_{B,0} + \sum_{k\geq 1} \psi_{B,k}(z - z_0)^k\Big) \\
&= -A(z)F_p(z) + \psi(z)\psi_B(z) = -A(z)F_p(z) + B(z).
\end{aligned}$$

which is exactly (4.17).

We define the open and simply connected sets $U_1, U_2 \subset \Omega$ as follows. First, if $\Omega = \mathbb{C}$, we set

$$\begin{aligned}
U_1 &:= \{z_0 + re^{ix} : 0 < r,\, x \in (0, 2\pi)\}, \\
U_2 &:= \{z_0 + re^{ix} : 0 < r,\, x \in (-\pi, \pi)\}.
\end{aligned}$$

Otherwise, if $\Omega \neq \mathbb{C}$, we apply the Riemann mapping theorem (see [Ahl79, Chapter 6, §1]), so there exists a biholomorphic function $\varphi : \mathbb{D} \to \Omega$, with $\varphi(0) = z_0$. Then we define U_1 and U_2 by

$$\begin{aligned}
U_1 &:= \varphi(\{re^{ix} : 0 < r < 1,\, x \in (0, 2\pi)\}), \\
U_2 &:= \varphi(\{re^{ix} : 0 < r < 1,\, x \in (-\pi, \pi)\}).
\end{aligned}$$

In particular, $D_\delta(z_0) \cap U_1 \cap U_2 \neq \varnothing$, and $D_\delta(z_0) \cup (U_1 \cap U_2)$ is connected.

Consider any point $\omega_0 \in U_1 \cap U_2 \cap D_\delta(z_0) - \{z_0\}$ and set $c_0 := F_p(\omega_0)$. Since U_1 and U_2 are simply connected, and $\frac{A}{z-z_0}$ and $\frac{B}{z-z_0}$ are holomorphic in both domains, Lemma 4.4 yields unique holomorphic solutions

$$F_p^{(1)} : U_1 \to \mathbb{C}, \qquad F_p^{(2)} : U_2 \to \mathbb{C},$$

of (4.17) satisfying $F_p^{(1)}(\omega_0) = F_p^{(2)}(\omega_0) = c_0$. By uniqueness, $F_p^{(j)}$ coincides with F_p on $D_\delta(z_0) \cap U_j$. Consequently, $F_p^{(j)}$ admits a holomorphic extension to $D_\delta(z_0) \cup U_j$, which will be denoted again by $F_p^{(j)}$. These functions $F_p^{(1)}$ and $F_p^{(2)}$ coincide on $D_\delta(z_0) \cup (U_1 \cap U_2)$ since they already coincide on $D_\delta(z_0)$ and $D_\delta(z_0) \cup (U_1 \cap U_2)$ is connected.

Therefore, $F_p^{(1)}$ and $F_p^{(2)}$ glue together to define a single holomorphic function, extending F_p to $D_\delta(z_0) \cup U_1 \cup U_2 = \Omega$, which solves (4.17). □

Corollary 4.7 *Let Ω, A, B, z_0 be as in Lemma 4.6. Then for any simply connected $U \subset \Omega - \{z_0\}$, every holomorphic solution F of (4.17) on U can be written as*

$$F(z) = CF_h(z) + F_p(z),$$

where $C \in \mathbb{C}$, F_h is given by (4.20), and F_p is the holomorphic solution given in Lemma 4.6. In particular, there exists a unique holomorphic $F : \Omega \to \mathbb{C}$ solving (4.17), corresponding to $C = 0$.

Proof: If F solves (4.17) on U, then $F - F_p$ solves the homogeneous equation (4.19) on U, hence $F - F_p = CF_h$ by Lemma 4.5. If F extends holomorphically across z_0, then the homogeneous term must also extend holomorphically. Under $A(z_0) \notin \mathbb{Z}^- \cup \{0\}$, Lemma 4.5 implies $C = 0$. Uniqueness follows. □

Now assume $A(z_0) \notin \mathbb{Z}$. We give a useful criterion in terms of the restriction to a small circle around z_0.

Lemma 4.8 *Let $\Omega \subset \mathbb{C}$ be open and simply connected and let $\widetilde{A}, \widetilde{B} : \Omega \to \mathbb{C}$ be holomorphic. Fix $z_0 \in \Omega$ and assume $\widetilde{A}(z_0) \notin \mathbb{Z}$. Let $\delta > 0$ be such that $\overline{D}_\delta(z_0) \subset \Omega$, and define*

$$\gamma(x) = z_0 + \delta e^{i(x+x_0)}, \quad x \in [0, 2\pi],\ x_0 \in \mathbb{R}.$$

Consider the ODE on $[0, 2\pi]$:

$$\partial_x f(x) + i\widetilde{A}(\gamma(x))f(x) = i\widetilde{B}(\gamma(x)). \tag{4.23}$$

Let f be a solution of (4.23) and set $c_0 := f(0)$. Let $F_p : \Omega \to \mathbb{C}$ be the unique holomorphic solution of (4.17) given by Corollary 4.7, with $A = \widetilde{A}$ and $B = \widetilde{B}$. Then $f = F_p \circ \gamma$ if and only if $f(2\pi) = f(0)$.

Proof: Let $F_p : \Omega \to \mathbb{C}$ be the unique holomorphic function solving (4.17), see Corollary 4.7, and define $f := F_p \circ \gamma$. It follows that f satisfies (4.23) and

$$f(2\pi) = F_p(\gamma(2\pi)) = F_p(\gamma(0)) = f(0).$$

Now assume that f solves (4.23) and satisfies $f(2\pi) = f(0)$. We define the open and simply connected set $U \subset \Omega - \{z_0\}$ given by

$$U := \{z_0 + re^{i(x+x_0)} : r \in (0, \delta),\ x \in (0, 2\pi)\}.$$

Fix the branch of $\log(z - z_0)$ on U determined by

$$\log(re^{i(x+x_0)}) = \log(r) + i(x + x_0).$$

With this choice, $\log(z - z_0)$ is holomorphic on U and has well-defined boundary limits as $x \to 0^+$ and $x \to 2\pi^-$.

By Corollary 4.7 we find that every solution to (4.17) in U is written as

$$F = CF_h + F_p,$$

where $C \in \mathbb{C}$, $F_p : \Omega \to \mathbb{C}$ is holomorphic and $F_h : U \to \mathbb{C}$ is given in (4.20) with $A = \widetilde{A}$.

Fix any $x_1 \in (0, 2\pi)$ and set $\omega_1 := \gamma(x_1) = z_0 + \delta e^{i(x_1+x_0)} \neq \gamma(0)$. Choose C so that $F(\omega_1) = f(x_1)$, and hence $f = F \circ \gamma$. Our goal is to show that this choice forces $C = 0$, so that F is the unique holomorphic solution given by Corollary 4.7.

Since F_p is holomorphic on a neighborhood of $\overline{D}_\delta(z_0)$, the function $F_p \circ \gamma$ extends continuously to $[0, 2\pi]$ and satisfies $F_p(\gamma(2\pi)) = F_p(\gamma(0))$. Consequently

$$\begin{aligned} 0 = f(2\pi) - f(0) &= \lim_{x\to 2\pi^-} F\big(z_0 + \delta e^{i(x+x_0)}\big) - \lim_{x\to 0^+} F\big(z_0 + \delta e^{i(x+x_0)}\big) \\ &= C\Big(\lim_{x\to 2\pi^-} F_h(z_0 + \delta e^{i(x+x_0)}) - \lim_{x\to 0^+} F_h\big(z_0 + \delta e^{i(x+x_0)}\big)\Big). \end{aligned}$$

Therefore, it follows for every $z \in U$:

$$F_h(z) = (z - z_0)^{-\widetilde{A}(z_0)}\psi(z) = \exp\left(-\widetilde{A}(z_0)\int_{\omega_1}^{z} \frac{dy}{y - z_0}\right)\psi(z),$$

where the integral is defined along paths contained in U.

Indeed, working with γ yields

$$\begin{aligned} 0 &= C\Big(\lim_{x\to 2\pi^-} F_h(z_0 + \delta e^{i(x+x_0)}) - \lim_{x\to 0^+} F_h(z_0 + \delta e^{i(x+x_0)})\Big) \\ &= C\left[\lim_{x\to 2\pi^-} \exp\left(-i\widetilde{A}(z_0)\int_{x_1}^{x} ds\right) - \lim_{x\to 0^+} \exp\left(-i\widetilde{A}(z_0)\int_{x_1}^{x} ds\right)\right]\psi(z_0 + \delta e^{ix_0}) \\ &= C(e^{-i\widetilde{A}(z_0)2\pi} - 1)\psi(z_0 + \delta e^{ix_0}). \end{aligned}$$

By definition (see (4.21)), ψ is the exponential of a holomorphic function. Moreover, since $\widetilde{A}(z_0) \notin \mathbb{Z}$ by assumption, we have $e^{-i\widetilde{A}(z_0)2\pi} \neq 1$. Therefore, the only possibility is $C = 0$, and hence $F = F_p$ is holomorphic and $f = F_p \circ \gamma$. □

We introduce the following notation, which will be used in the next lemma. Let $z_0, z_1 \in \mathbb{C}$ and let r satisfy $0 < r < |z_0 - z_1|$. We define the *racket set* associated to z_0, z_1 and r by

$$R_{z_0,z_1}(r) := \big\{z \in \mathbb{C} : |z - z_1| = r\big\} \cup \big\{(1 - x)z_0 + x\big(z_1 + r\tfrac{z_0 - z_1}{|z_0 - z_1|}\big) : x \in [0, 1]\big\}. \tag{4.24}$$

Note that the point

$$z_1 + r\frac{z_0 - z_1}{|z_0 - z_1|}$$

is the unique intersection point between the circle of center z_1 and radius r and the line segment joining z_0 and z_1. For a geometric illustration of the set $R_{z_0,z_1}(r)$, see Figure III.6.

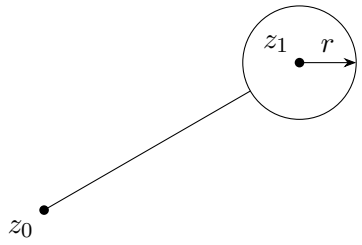


Figure III.6: The racket set $R_{z_0,z_1}(r)$.

The next lemma is technical and will be used to prove the main result of this section.

Lemma 4.9 *Let $\mathbb{D} \subset \mathbb{C}$ be the open unit disk and let $\{z_j\}_{j=1}^n \subset \mathbb{D}$ be n distinct points. Then there exists a point $\omega_0 \in \partial\mathbb{D}$ and a collection of open and convex sets $\{C_j\}_{j=1}^n$ covering $\mathbb{D}$ such that*

$$C_j \cap \{z_1, \ldots, z_n\} = \{z_j\}.$$

Moreover, ω_0 is an accumulation point for every C_j.
Additionally, there exists $r > 0$ such that the racket set $R_{\omega_0,z_j}(r)$ satisfies

$$R_{\omega_0,z_j}(r) \subset C_j \cup \{\omega_0\} \quad \textit{for all } j$$

Proof: We first choose a boundary point $\omega_0 \in \partial\mathbb{D}$ which is not collinear with any pair of distinct points z_j, z_k. For each pair $j \neq k$, the line through z_j and z_k meets $\partial\mathbb{D}$ in at most two points, so the set of such *bad* boundary points is finite. Fix ω_0 outside this set.

For each $j = 1, \ldots, n$, there exist unique numbers $t_j > 0$ and $\theta_j \in (0, \pi)$ such that

$$z_j = \omega_0(1 + it_j e^{i\theta_j}).$$

Each $z_j \in \mathbb{D}$ satisfies $\mathrm{Re}(\overline{\omega_0} z_j - 1) < 0$, so $\overline{\omega_0} z_j - 1$ has a unique polar form $\overline{\omega_0} z_j - 1 = it_j e^{i\theta_j}$ with $t_j > 0$ and $\theta_j \in (0, \pi)$.

By the choice of ω_0, the angles $\theta_1, \ldots, \theta_n$ are pairwise distinct. Relabel the points z_j so that

$$0 < \theta_1 < \theta_2 < \ldots < \theta_n < \pi, \tag{4.25}$$

and set $\theta_0 := 0$ and $\theta_{n+1} := \pi$.

For each $j = 1, \ldots, n$, define the set C_j as the intersection of an open cone with vertex at ω_0 and the $\mathbb{D}$:

$$C_j := \{\omega_0(1 + ite^{i\theta}) : t > 0,\ \theta \in (\theta_{j-1}, \theta_{j+1})\} \cap \mathbb{D}. \tag{4.26}$$

These sets $\{C_j\}$ cover $\mathbb{D}$. Each C_j is open and convex, as it is the intersection of open and convex sets, and contains z_j.

Finally, let $\delta_j > 0$ such that

$$D_{\delta_j}(z_j) \subset C_j.$$

We set $r := \min_j\{\delta_j\}$, so that

$$R_{\omega_0, z_j}(r) \subset C_j \cup \{\omega_0\} \qquad \text{for all } j.$$

For a geometric illustration of the set C_j and $R_{\omega_0,z_j}(r)$ for a fixed j, see Figure III.7.

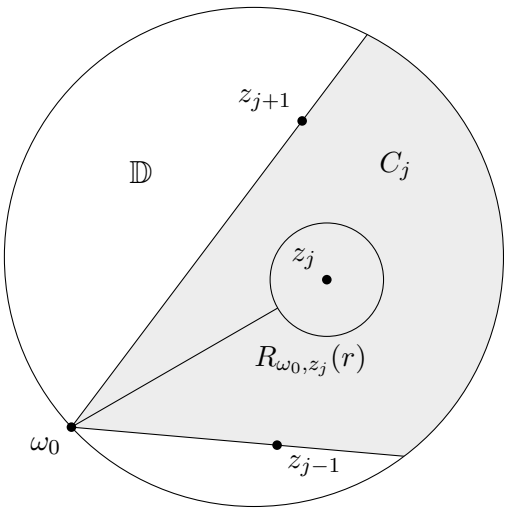


Figure III.7: $R_{\omega_0,z_j}(r)$ and C_j.

□

In order to prove the main lemma of this section, we introduce the following notation. Let $a, b, c, d \in \mathbb{R}$ with $a < b$ and $c < d$. Let $\gamma_1 : [a, b] \to \mathbb{C}$ and $\gamma_2 : [c, d] \to \mathbb{C}$ be continuous paths such that $\gamma_1(b) = \gamma_2(c)$. We denote by $\gamma_1 * \gamma_2 : [a, b + d - c] \to \mathbb{C}$ and by $\overline{\gamma}_1 : [a, b] \to \mathbb{C}$ the continuous paths defined by

$$\gamma_1 * \gamma_2(x) = \begin{cases} \gamma_1(x), & x \in [a, b], \\ \gamma_2(x - b + c), & x \in [b, b + d - c], \end{cases} \qquad \overline{\gamma}_1(x) = \gamma_1(b + a - x), \quad x \in [a, b]. \tag{4.27}$$

This parametrization avoids re-scaling. Moreover, the operation $*$ is associative.

We now state the main result of Section III.4.1.1.

Lemma 4.10 *Let $\mathbb{D}$ denote the open unit disk, and let $\widetilde{Z} := \{z_j\}_{j=1}^n \subset \mathbb{D}$ be a finite set of pairwise distinct points. Define*

$$Q(z) := \prod_{j=1}^{n}(z - z_j).$$

Let $A, B : \mathbb{D} \to \mathbb{C}$ be holomorphic functions extending continuously to $\overline{\mathbb{D}}$, and assume that

$$\frac{A(z_j)}{\partial_z Q(z_j)} \notin \mathbb{Z} \qquad \text{for all } j = 1, \dots, n. \tag{4.28}$$

Let ω_0, $\{C_j\}_{j=1}^n$, r, and $R_{\omega_0, z_j}(r)$ be as in Lemma 4.9. For each $j = 1, \dots, n$, define the paths given by $\gamma_j : [0,1] \to \mathbb{C}$ and $\alpha_j : [0, 2\pi] \to \mathbb{C}$ by

$$\gamma_j(x) = (1-x)\omega_0 + x\Big(z_j + r\frac{\omega_0 - z_j}{|\omega_0 - z_j|}\Big) \qquad \text{and} \qquad \alpha_j(x) = z_j + re^{ix}\frac{\omega_0 - z_j}{|\omega_0 - z_j|}.$$

Using the notation of (4.27)*, set*

$$\tau_j := \gamma_j * \alpha_j * \overline{\gamma}_j : [0, 2\pi + 2] \to \mathbb{C}. \tag{4.29}$$

For each j, consider the problem on $(0, 2\pi + 2) - \{1, 2\pi + 1\}$ given by

$$\frac{Q \circ \tau_j}{\partial_x \tau_j}\partial_x f_j + (A \circ \tau_j) f_j = B \circ \tau_j. \tag{4.30}$$

We also consider the equation on $\mathbb{D}$

$$Q\partial_z F + AF = B. \tag{4.31}$$

Let $c_0 \in \mathbb{C}$. For each $j = 1, \dots, n$ let $f_j : [0, 2\pi + 2] \to \mathbb{C}$ be the unique continuous function with $f_j(0) = c_0$, solving (4.30)*. Then there exists a holomorphic function $F : \mathbb{D} \to \mathbb{C}$, whose derivative extends continuously to $\overline{\mathbb{D}}$, solving* (4.31) *and satisfying $F(\omega_0) = c_0$ if and only if*

$$f_j(2\pi + 2) = c_0 \qquad \text{for all } j = 1, \dots, n.$$

Moreover, such a function F, if it exists, is unique.

Remark 4.11 *γ_j and α_j have been defined so that*

$$\gamma_j([0,1]) \cup \alpha_j([0, 2\pi]) = R_{\omega_0, z_j}(r).$$

Proof: Assume first that there exists a holomorphic function $F : \mathbb{D} \to \mathbb{C}$, with derivative continuous on $\overline{\mathbb{D}}$, solving (4.31) and satisfying $F(\omega_0) = c_0$. Fix $j \in \{1, \dots, n\}$ and define

$$g_j(x) := F(\tau_j(x)), \qquad x \in [0, 2\pi + 2].$$

Then g_j is continuous and, by change of variables along τ_j, it satisfies (4.30) on $(0, 2\pi+2) - \{1, 2\pi+1\}$. Since $g_j(0) = F(\omega_0) = c_0 = f_j(0)$, uniqueness of solutions to (4.30) yields

$$g_j = f_j \quad \text{on } [0, 2\pi + 2].$$

In particular,

$$f_j(2\pi+2) = g_j(2\pi+2) = F(\tau_j(2\pi+2)) = F(\omega_0) = c_0,$$

which proves the necessity.

Conversely, assume that for each $j = 1, \dots, n$, the continuous solution $f_j : [0, 2\pi+2] \to \mathbb{C}$ of (4.30) satisfies that $f_j(0) = f_j(2\pi+2) = c_0$. Fix j. We first claim that

$$f_j(1) = f_j(2\pi+1).$$

To see this, on $(0,1)$ the equation reads

$$\frac{Q\circ\gamma_j}{\partial_x\gamma_j}\partial_x f_j + (A\circ\gamma_j)f_j = B\circ\gamma_j,$$

while on $(2\pi+1, 2\pi+2)$ it reads

$$\frac{Q\circ\overline{\gamma}_j}{\partial_x\overline{\gamma}_j}\partial_x f_j + (A\circ\overline{\gamma}_j)f_j = B\circ\overline{\gamma}_j.$$

Because $\gamma_j([0,1]) \subset R_{\omega_0,z_j}(r) \subset C_j$, we know that Q never vanishes on this curve. Hence both equations admit unique solutions once an initial condition is prescribed.

Define $\phi_j^0 : (0,1) \to \mathbb{C}$ by $\phi_j^0(x) := f_j(2\pi+2-x)$. A direct change of variables shows that ϕ_j^0 satisfies the same ODE as f_j on $(0,1)$. Since $\phi_j^0(0) = f_j(2\pi+2) = c_0 = f_j(0)$, uniqueness yields $f_j = \phi_j^0$ on $(0,1)$, hence $f_j(1) = f_j(2\pi+1)$.

By Lemma 4.9, we know that the set C_j is convex, hence simply connected, and by the choice of r we have

$$C_j \cap \{z_1, \dots, z_n\} = \{z_j\},$$

Here, we assume the points z_j have been relabeled so (4.25) holds.

Define $\phi_j^1 : [0, 2\pi] \to \mathbb{C}$ by $\phi_j^1(x) := f_j(x+1)$ and observe that Q has exactly one zero in C_j, which is simple. Note that (4.28) is equivalent to

$$\lim_{z\to z_j}(z-z_j)\frac{A(z)}{Q(z)} = \frac{A(z_j)}{\partial_z Q(z_j)} \notin \mathbb{Z} \qquad \text{for all } j.$$

Then, applying Lemma 4.8 with $\Omega = C_j$ and $z_0 = z_j$, and

$$\widetilde{A}(z) = A(z)\prod_{\substack{l=1\\ l\neq j}}^{n}(z-z_l)^{-1}, \qquad \widetilde{B}(z) = B(z)\prod_{\substack{l=1\\ l\neq j}}^{n}(z-z_l)^{-1},$$

we obtain that the unique holomorphic function $F_j : C_j \to \mathbb{C}$ satisfies

$$(F_j\circ\alpha_j)(x) = \phi_j^1(x) = f_j(x+1), \qquad x \in [0, 2\pi],$$

where F_j is given by Corollary 4.7 and solves (4.31).

For every $\varepsilon > 0$, define

$$V_\varepsilon := D_\varepsilon(\omega_0) \cap \mathbb{D}.$$

Let $\varepsilon > 0$ be small enough so that $z_j \notin V_\varepsilon$ for all j. In particular, Q has no zeros on V_ε. As in the proof of Lemma 4.4, we solve (4.31) on V_ε by integration. For $z \in V_\varepsilon$, define

$$F_h^\varepsilon(z) := \exp\Big(-\int_{\omega_0}^{z}\frac{A(y)}{Q(y)}\,dy\Big),$$

and

$$F^\varepsilon(z) := c_0 F_h^\varepsilon(z) + F_h^\varepsilon(z) \int_{\omega_0}^z \frac{B(y)}{Q(y)F_h^\varepsilon(y)}\, dy,$$

where the integrals are taken along any smooth path in V_ε. Although $\omega_0 \notin V_\varepsilon$, these integrals are well defined because A and B are continuous at ω_0. Since V_ε is simply connected and A and B holomorphic, the integrals are path independent, and F^ε is holomorphic on V_ε and solves (4.31) there.

For each j, the functions F^ε and F_j coincide on $\gamma_j([0,1]) \cap V_\varepsilon$, since both solve the same initial value problem along γ_j with initial condition $F(\omega_0) = c_0$. By uniqueness,

$$F^\varepsilon = F_j \quad \text{on } C_j \cap V_\varepsilon.$$

Consequently, each function F_j extends holomorphically to the set $C_j \cup V_\varepsilon$. We continue to denote this extension by F_j. Now let $j \neq k$. The set $V_\varepsilon \cap (C_j \cup C_k)$ is connected, since every point of C_j admits a path contained in $C_j \cup V_\varepsilon$ joining it to V_ε. Since F_j and F_k coincide on V_ε, it follows by analytic continuation that they coincide on the whole set $V_\varepsilon \cap (C_j \cup C_k)$. In particular, F_k extends holomorphically to the domain of F_j, and vice versa.

Hence there exists a holomorphic function $F : \mathbb{D} \to \mathbb{C}$ such that

$$F|_{C_j} = F_j, \qquad F|_{V_\varepsilon} = F^\varepsilon.$$

Since A and B are continuous on $\overline{\mathbb{D}}$ and holomorphic on $\mathbb{D}$, we find that F admits a continuous extension to $\overline{\mathbb{D}}$. Finally, the equation (4.31) implies

$$\partial_z F(z) = \frac{B(z) - A(z)F(z)}{Q(z)}, \qquad z \in \mathbb{D}.$$

Because Q has no zeros on $\partial\mathbb{D}$, the right-hand side extends continuously to $\partial\mathbb{D}$, and therefore $\partial_z F$ extends continuously to $\overline{\mathbb{D}}$. Uniqueness follows from Corollary 4.7. □

III.4.1.2 Invertibility of $\mathcal{L}_{\text{exis}}^{\text{ap, hol}}$

In order to show that $\mathcal{L}_{\text{exis}}^{\text{ap, hol}} : \dot{\mathcal{H}}^2 \to \dot{\mathcal{H}}^1$ is invertible, we first need to understand the roots of the polynomial $z^N P(z)$.

Lemma 4.12 *Let P be the function defined in* (4.5)*, and denote by Z the set of roots of $z^N P(z)$ lying inside the unit disk $\mathbb{D}$. Then Z contains exactly N simple roots. Moreover, for every root $\alpha \in Z$, we have*

$$A(\alpha) \prod_{\substack{\alpha' \in Z \\ \alpha' \neq \alpha}} (\alpha - \alpha')^{-1} \notin \mathbb{Z}, \tag{4.32}$$

where the function A is defined by

$$A(z) := -2\frac{z^{N-1}}{v_N^{\text{ap}}} \prod_{\alpha \in Z} (z - \alpha^{-1})^{-1}.$$

Proof: The proof is computer-assisted and the code can be found in the supplementary material. We refer to the Appendix III.E for details about the implementation. □

Lemma 4.10 will be used to establish the invertibility of $\mathcal{L}_{\text{exis}}^{\text{ap, hol}}$. We introduce some notation that will be used to prove the next proposition.

Definition 4.13 *Let $Z = \{z_j\}_{j=1}^N$ be the set of roots of $z^N P(z)$ lying inside the open unit disk, and let ω_0, Q, and the paths $\{\tau_j\}$ be as in Lemma 4.10 with $\widetilde{Z} := Z$.*

For $G \in \mathcal{H} \cup \{\mathcal{G}_k : k = 1, \dots, N\}$ and $j = 1, \dots, N$, we define $J_{\tau_j} G$ as the unique continuous solution of (4.30) with initial condition $J_{\tau_j} G(0) = 0$ and coefficients

$$A(z) := -2\frac{z^{N-1}}{v_N^{\mathrm{ap}}} \prod_{\alpha \in Z} (z - \alpha^{-1})^{-1}, \qquad B(z) := -2i\frac{z^{N-1}}{v_N^{\mathrm{ap}}} G(z) \prod_{\alpha \in Z} (z - \alpha^{-1})^{-1}.$$

For each $j = 1, \dots, N$, we also define Π_{τ_j} as the unique continuous solution to (4.30) with initial condition $\Pi_{\tau_j}(0) = 1$ and

$$A(z) := -2\frac{z^{N-1}}{v_N^{\mathrm{ap}}} \prod_{\alpha \in Z} (z - \alpha^{-1})^{-1}, \qquad B(z) := 0.$$

Define $\gamma_{\omega_0}(x) := \omega_0 e^{ix}$ and

$$\beta_{\omega_0} := \frac{1}{P \circ \gamma_{\omega_0}}.$$

For $G \in C(\mathbb{T})$, we define $J_{\gamma_{\omega_0}} G$ as the unique continuous solution with $J_{\gamma_{\omega_0}} G(0) = 0$ of

$$\partial_x J_{\gamma_{\omega_0}} G = i\beta_{\omega_0} J_{\gamma_{\omega_0}} G + \beta_{\omega_0} (G \circ \gamma_{\omega_0}),$$

and $\Pi_{\gamma_{\omega_0}}$ as the unique continuous solution with $\Pi_{\gamma_{\omega_0}}(0) = 1$ of

$$\partial_x \Pi_{\gamma_{\omega_0}} = i\beta_{\omega_0} \Pi_{\gamma_{\omega_0}}.$$

At the end of the section, we prove that $\mathcal{L}_{\mathrm{exis}}^{\mathrm{ap,\,hol}} : \dot{\mathcal{H}}^2 \to \dot{\mathcal{H}}^1$ admits a bounded inverse. We will also need the following lemma, which provides a criterion for the invertibility of $\mathcal{L}_{\mathrm{exis}}^{\mathrm{ap,\,hol}}$ in a different class of function spaces and is based on Lemma 4.10.

Proposition 4.14 *Using the notation from Definition 4.13 and $\mathcal{G}_k$ from (4.6), define the $(N+1) \times (N+1)$ matrix $M_{\mathrm{exis}}^{\mathrm{hol}}$ by*

$$(M_{\mathrm{exis}}^{\mathrm{hol}})_{jk} = \begin{cases} J_{\tau_j}\mathcal{G}_k(2\pi + 2) - \underbrace{J_{\tau_j}\mathcal{G}_k(0)}_{0}, & 1 \le j \le N,\ 1 \le k \le N, \\ \Pi_{\tau_j}(2\pi + 2) - \underbrace{\Pi_{\tau_j}(0)}_{1}, & 1 \le j \le N,\ k = N+1, \\ \dfrac{1}{2\pi}\displaystyle\int_{-\pi}^{\pi} J_{\gamma_{\omega_0}}\mathcal{G}_k(x)\,dx, & j = N+1,\ 1 \le k \le N, \\ \dfrac{1}{2\pi}\displaystyle\int_{-\pi}^{\pi} \Pi_{\gamma_{\omega_0}}(x)\,dx, & j = N+1,\ k = N+1. \end{cases} \tag{4.33}$$

Then the matrix $M_{\mathrm{exis}}^{\mathrm{hol}}$ is nonsingular if and only if, for every holomorphic function $G : \mathbb{D} \to \mathbb{C}$ that is continuous on $\overline{\mathbb{D}}$ and satisfies $G(0) = 0$, there exists a unique holomorphic function $F : \mathbb{D} \to \mathbb{C}$ whose derivative $\partial_z F$ is continuous on $\overline{\mathbb{D}}$, satisfying $F(0) = 0$ and solving

$$\mathcal{L}_{\mathrm{exis}}^{\mathrm{ap,\,hol}} F = G. \tag{4.34}$$

Proof: Let $G : \mathbb{D} \to \mathbb{C}$ be holomorphic in $\mathbb{D}$ and continuous on $\overline{\mathbb{D}}$, with $G(0) = 0$. We study equation (4.34) and seek a holomorphic function $F : \mathbb{D} \to \mathbb{C}$ whose derivative $\partial_z F$ is continuous on $\overline{\mathbb{D}}$ and satisfies $F(0) = 0$.

Writing the equation explicitly, we have

$$izP(z)\partial_z F(z) = iF(z) + \sum_{k=1}^{N} F_k \mathcal{G}_k(z) + G(z),$$

where $F(z) = \sum_{k\geq 1} F_k z^k$.

Multiplying by $-iz^{N-1}$, we obtain

$$z^N P(z)\partial_z F(z) = z^{N-1}F(z) - iz^{N-1}\Big(\sum_{k=1}^{N} F_k \mathcal{G}_k(z) + G(z)\Big). \tag{4.35}$$

Since $z^{N-1}\sum_{k=1}^{N} F_k\mathcal{G}_k$ is holomorphic in $\mathbb{D}$, the singular points of (4.35) inside $\mathbb{D}$ are precisely the roots

$$Z = \{z_j\}_{j=1}^{N}$$

of $z^N P(z)$ lying in the open unit disk.

We introduce auxiliary unknowns $a_1, \ldots, a_N \in \mathbb{C}$ and consider the following affine problem

$$izP(z)\partial_z F(z) = iF(z) + \sum_{k=1}^{N} a_k \mathcal{G}_k(z) + G(z). \tag{4.36}$$

The equation (4.36) reduces to a linear first-order equation of the form (4.31) with

$$A(z) := -2\frac{z^{N-1}}{v_N^{\mathrm{ap}}}\prod_{\alpha\in Z}\frac{1}{z-\alpha^{-1}} \quad \text{and} \quad B(z) := -2i\frac{z^{N-1}}{v_N^{\mathrm{ap}}}\Big(\sum_{k=1}^{N} a_k\mathcal{G}_k(z) + G(z)\Big)\prod_{\alpha\in Z}\frac{1}{z-\alpha^{-1}}.$$

For each fixed $j = 1, \ldots, N$, solving (4.36) along τ_j yields a unique continuous solution ϕ_j. This solution can be expressed as

$$\phi_j = F(\omega_0)\Pi_{\tau_j} + \sum_{k=1}^{N} a_k J_{\tau_j}\mathcal{G}_k + J_{\tau_j}G. \tag{4.37}$$

Lemma 4.12 implies that condition (4.28) holds. Moreover, A is continuous on $\overline{\mathbb{D}}$, since it is a finite product of terms of the form

$$(z-\alpha^{-1})^{-1} \qquad \text{where } |\alpha| < 1.$$

Therefore, Lemma 4.10 applies, and the equation (4.36) admits a holomorphic solution F in $\mathbb{D}$, with $\partial_z F$ continuous on $\overline{\mathbb{D}}$, if and only if each function ϕ_j satisfies the condition

$$\phi_j(2\pi + 2) = \phi_j(0), \qquad j = 1, \ldots, N. \tag{4.38}$$

Using (4.37), these conditions yield N affine linear equations in the unknowns $a_1, \ldots, a_N, F(\omega_0)$.

Restricting (4.36) to γ_{ω_0} yields the equation

$$\partial_x\phi = i\beta_{\omega_0}\phi + \beta_{\omega_0}\Big[\sum_{k=1}^{N} a_k(\mathcal{G}_k\circ\gamma_{\omega_0}) + (G\circ\gamma_{\omega_0})\Big].$$

The solutions are of the form

$$\phi = \Pi_{\gamma_{\omega_0}}F(\omega_0) + \sum_{k=1}^{N} a_k J_{\gamma_{\omega_0}}\mathcal{G}_k + J_{\gamma_{\omega_0}}G.$$

Imposing the mean-zero condition

$$\frac{1}{2\pi}\int_0^{2\pi}\phi(x)\,dx = 0$$

yields one additional affine linear equation.

Collecting the N closing conditions (4.38) and the mean condition above, we obtain a linear system of $N+1$ equations in the $N+1$ unknowns $(a_1,\dots,a_N,F(\omega_0))$, whose coefficient matrix is exactly $M_{\text{exis}}^{\text{hol}}$.

If $M_{\text{exis}}^{\text{hol}}$ is nonsingular, the system admits a unique solution $(a_1,\dots,a_N,F(\omega_0))$. Consequently, there exists a unique holomorphic function $F:\mathbb{D}\to\mathbb{C}$ such that $\partial_z F\in C(\overline{\mathbb{D}})$,

$$F(0) = \frac{1}{2\pi}\int_0^{2\pi}F(e^{ix})\,dx = 0,$$

and F solves (4.36). By Lemma 4.2, the coefficients satisfy

$$a_k = F_k \qquad \text{for all } k,$$

and therefore F is the unique solution of (4.34).

Conversely, if for every admissible G there exists a unique solution F of (4.34), then the homogeneous system associated with $M_{\text{exis}}^{\text{hol}}$ admits only the trivial solution, and hence $M_{\text{exis}}^{\text{hol}}$ is nonsingular. □

We have observed numerically that for large N (our case applies), the singular values of $M_{\text{exis}}^{\text{hol}}$ cluster near zero.

This behavior is problematic for our analysis using interval arithmetic techniques, since it becomes difficult to certify that the smallest singular value does not vanish.

Moreover, the norm of $(M_{\text{exis}}^{\text{hol}})^{-1}$, which we need to know to establish Proposition 2.2, would be extremely large. To overcome these difficulties, we introduce an alternative $(N+1)\times(N+1)$ matrix, denoted by $M_{\text{exis}}^{\text{tor}}$, whose singular values accumulate near 1. We then show that

$$\det M_{\text{exis}}^{\text{tor}} \neq 0 \;\Rightarrow\; \det M_{\text{exis}}^{\text{hol}} \neq 0. \tag{4.39}$$

In addition, this new matrix allows us to compute an explicit bound for the inverse norm appearing in Proposition 2.2.

We introduce the following notation. Let $\beta:\mathbb{R}\to\mathbb{R}$ and $\Pi:\mathbb{R}\to\mathbb{C}$ be defined by

$$\beta(x) := \frac{1}{c+v^{\text{ap}}(x)} \qquad \text{and} \qquad \Pi(x) := \exp\!\left(i\int_0^x \beta(y)\,dy\right). \tag{4.40}$$

Both functions β and Π are continuous on $\mathbb{R}$, since $c+v^{\text{ap}}(x)$ has no zeros on the real line, since $c+v^{\text{ap}}(x) = P(e^{ix})$ and the polynomial $z^N P(z)$ does not vanish on the unit circle.

Let J denote the operator defined by

$$Jg(x) := \Pi(x)\int_0^x \beta(y)\frac{g(y)}{\Pi(y)}\,dy. \tag{4.41}$$

We now introduce the new criterion.

Lemma 4.15 *Let β, Π, and J be defined in* (4.40) *and* (4.41).

For every $k=1,\dots,N$, define

$$g_k(x) := \frac{i}{2}kv_k^{\text{ap}} + \sum_{m=1}^{N-k} kv_{k+m}^{\text{ap}}\sin(mx). \tag{4.42}$$

For each $k = 1, \dots, N+1$, set

$$h_k := \begin{cases} Jg_k, & 1 \le k \le N, \\ \Pi, & k = N+1. \end{cases} \tag{4.43}$$

Finally, define the square matrix $M^{\mathrm{tor}}_{\mathrm{exis}}$ of order $N+1$ by

$$(M^{\mathrm{tor}}_{\mathrm{exis}})_{jk} = (\widehat{h_k})_{j-1} - \delta_{k,j-1}, \qquad 1 \le j,k \le N+1. \tag{4.44}$$

Then the implication (4.39) *holds.*

Proof: Assume that $\det M^{\mathrm{hol}}_{\mathrm{exis}} = 0$ and fix $G = 0$. Then, by Proposition 4.14, there exists a nontrivial subspace V of holomorphic functions on $\mathbb{D}$ with continuous derivative on $\overline{\mathbb{D}}$, such that for every $F \in V$ we have that F (with $G = 0$) satisfies (4.34).

Fix $0 \neq F \in V$ and set the function $f : \mathbb{R} \to \mathbb{C}$ by

$$f(x) := F(e^{ix}).$$

A change of variables shows that f satisfies

$$(c + v^{\mathrm{ap}}(x))\partial_x f(x) = if(x) + \sum_{k=1}^{N} \widehat{f}_k g_k(x).$$

Therefore, using Duhamel's formula, we find that

$$f(x) = \Pi(x)f(0) + \Pi(x)\int_0^x \frac{\beta(y)}{\Pi(y)} \sum_{k=1}^{N} \widehat{f}_k g_k(y)\, dy = \Pi(x)f(0) + \sum_{k=1}^{N} \widehat{f}_k Jg_k(x). \tag{4.45}$$

We project (4.45) onto the modes e^{ijx}, $j = 0, \dots, N$,

$$\widehat{f}_j = f(0)(\widehat{h_{N+1}})_j + \sum_{k=1}^{N} \widehat{f}_k (\widehat{h_k})_j.$$

Rearranging, we obtain for every $j = 0, \dots, N$ that

$$0 = f(0)(\widehat{h_{N+1}})_j + \sum_{k=1}^{N} \widehat{f}_k ((\widehat{h_k})_j - \delta_{k,j}). \tag{4.46}$$

Since $\widehat{f}_0 = 0$, taking $j = 0$ in (4.46) yields

$$0 = f(0)(\widehat{h_{N+1}})_0 + \sum_{k=1}^{N} \widehat{f}_k (\widehat{h_k})_0. \tag{4.47}$$

Equations (4.46) and (4.47) are equivalent to

$$M^{\mathrm{tor}}_{\mathrm{exis}} \begin{pmatrix} \widehat{f}_1 \\ \vdots \\ \widehat{f}_N \\ f(0) \end{pmatrix} = 0.$$

Finally, note that $f(0) = 0$ and $\widehat{f}_k = 0$ for every $k = 1, \dots, N$ is not possible by equation (4.45). Hence, we have $\det M^{\mathrm{tor}}_{\mathrm{exis}} = 0$. □

Lemma 4.16 *The smallest singular value σ_1 of $M_{\text{exis}}^{\text{tor}}$ satisfies*

$$\sigma_1^2 > 6.82641 \cdot 10^{-4}.$$

Proof: The proof is computer-assisted and the code can be found in the supplementary material. We refer to the Appendix III.E for details about the implementation. □

Corollary 4.17 *For every holomorphic function $G : \mathbb{D} \to \mathbb{C}$ that is continuous on $\overline{\mathbb{D}}$ and satisfies $G(0) = 0$, there exists a unique holomorphic function $F : \mathbb{D} \to \mathbb{C}$ with $F(0) = 0$ and $\partial_z F$ continuous on $\overline{\mathbb{D}}$, such that* (4.34) *holds.*

Proof: By definition, the singular values of $M_{\text{exis}}^{\text{tor}}$ are the square roots of the eigenvalues of the Hermitian matrix

$$M_{\text{exis}}^{\text{tor}\,*} M_{\text{exis}}^{\text{tor}}.$$

Since $\sigma_1 > 0$, all singular values are positive and therefore

$$|\det M_{\text{exis}}^{\text{tor}}|^2 = \prod_{k=1}^{N+1} \sigma_k^2 > 0.$$

In particular, $\det M_{\text{exis}}^{\text{tor}} \neq 0$.

Lemma 4.15 then implies $\det M_{\text{exis}}^{\text{hol}} \neq 0$, and Proposition 4.14 applies. □

III.4.1.3 Estimates of the explicit functions and operators

This section is devoted to proving explicit bounds for some functions that will allow us to obtain an estimate for the norm of $(\mathcal{L}_{\text{exis}}^{\text{ap, hol}})^{-1}$ from $\dot{\mathcal{H}}^1$ to $\dot{\mathcal{H}}^2$. We recall that the spaces $\dot{\mathcal{H}}^1$ and $\dot{\mathcal{H}}^2$ are defined in (1.6).

We collect such bounds in Lemmas 4.18–4.25.

Lemma 4.18 *The function $\beta : \mathbb{R} \to \mathbb{R}$ given in* (4.40) *satisfies*

$$0 < \beta(x) < 19.9056 \quad \textit{for all } x \in [-\pi, \pi].$$

Proof: The proof is computer-assisted and the code can be found in the supplementary material. We refer to the Appendix III.E for details about the implementation. □

Lemma 4.19 *Let κ_1 be defined by*

$$\kappa_1(x) := \int_0^x \beta(y)^2 \, dy.$$

Then

$$\|\kappa_1\|_{L^1([0,\pi])} < 4.03105.$$

Proof: The proof is computer-assisted and the code can be found in the supplementary material. We refer to the Appendix III.E for details about the implementation. □

Lemma 4.20 *The following estimate holds:*

$$\|\partial_x \beta + i\beta^2\|_{L^2([-\pi,\pi])}^2 < 12107.9.$$

Proof: The proof is computer-assisted and the code can be found in the supplementary material. We refer to the Appendix III.E for details about the implementation. □

Lemma 4.21 *Let κ_1 as in Lemma 4.19 and κ_2 be defined by*

$$\kappa_2(x) := [(\partial_x \beta(x))^2 + \beta(x)^4]\kappa_1(x).$$

Then

$$\|\kappa_2\|_{L^1([0,\pi])} < 77441.4.$$

Proof: The proof is computer-assisted and the code can be found in the supplementary material. We refer to the Appendix III.E for details about the implementation. □

Lemma 4.22 *Let $g \in L^2([-\pi,\pi])$. Then*

$$\|Jg\|^2_{L^2([-\pi,\pi])} \leq 4.03105\, \|g\|^2_{L^2([-\pi,\pi])}.$$

Proof: It suffices to prove the estimate for $g \in C(\mathbb{T})$, since $C(\mathbb{T})$ is dense in $L^2([-\pi,\pi])$.

We note that $|\Pi(x)| = 1$ for all $x \in \mathbb{R}$. Hence,

$$\begin{aligned}\|Jg\|^2_{L^2([-\pi,\pi])} &= \int_{-\pi}^{\pi} \Big|\Pi(x) \int_0^x \beta(y) \frac{g(y)}{\Pi(y)}\, dy\Big|^2 dx = \int_{-\pi}^{\pi} \Big| \int_0^x \beta(y) \frac{g(y)}{\Pi(y)}\, dy\Big|^2 dx \\ &\leq \int_0^{\pi} \|\beta g\|^2_{L^1([0,x])}\, dx + \int_{-\pi}^{0} \|\beta g\|^2_{L^1([x,0])}\, dx. \end{aligned} \tag{4.48}$$

We focus on the first integral. The computations for the second one are similar since β is an even function. Applying the Cauchy–Schwarz inequality yields

$$\int_0^{\pi} \|\beta g\|^2_{L^1([0,x])}\, dx \leq \int_0^{\pi} \|\beta\|^2_{L^2([0,x])} \|g\|^2_{L^2([0,x])}\, dx. \tag{4.49}$$

Let κ_1 be the function in Lemma 4.19, we obtain

$$\begin{aligned}\int_0^{\pi} \|\beta\|^2_{L^2([0,x])} \|g\|^2_{L^2([0,x])}\, dx &= \int_0^{\pi} \kappa_1(x) \int_0^x |g(y)|^2\, dy\, dx \\ &= \int_0^{\pi} |g(y)|^2 \int_y^{\pi} \kappa_1(x)\, dx\, dy \leq \|\kappa_1\|_{L^1([0,\pi])} \|g\|^2_{L^2([0,\pi])} \end{aligned} \tag{4.50}$$

Combining (4.49) and (4.50), we obtain

$$\int_0^{\pi} \|\beta g\|^2_{L^1([0,x])}\, dx \leq \|\kappa_1\|_{L^1([0,\pi])} \|g\|^2_{L^2([0,\pi])}.$$

Therefore, we get

$$\begin{aligned}\|Jg\|^2_{L^2([-\pi,\pi])} &\leq \|\kappa_1\|_{L^1([0,\pi])} \big(\|g\|^2_{L^2([0,\pi])} + \|g\|^2_{L^2([-\pi,0])}\big) \\ &\leq \|\kappa_1\|_{L^1([0,\pi])} \|g\|^2_{L^2([-\pi,\pi])}. \end{aligned} \tag{4.51}$$

Finally, we substitute the explicit estimate in Lemma 4.19 into (4.51) to obtain the bound of the statement. □

Lemma 4.23 *Let $G \in \dot{\mathcal{H}}^1$ and let $g := G|_{\mathbb{T}}$. Then*

$$\|\partial_x^2 Jg\|_{L^2([-\pi,\pi])} \leq 354.490 \|\partial_x g\|_{L^2([-\pi,\pi])}.$$

Remark 4.24 *Observe that Jg is not necessarily continuous on $\mathbb{T}$, even if it is smooth on $[-\pi,\pi]$.*

Proof: First, assume that $g = G|_{\mathbb{T}} \in C^1(\mathbb{T})$, so that $\partial_x^2 Jg$ is well defined pointwise. We prove the estimate in this case. Since smooth functions are dense in the space of traces of $\dot{\mathcal{H}}^1$ and the norms involved depend continuously on g, it follows that the estimate extends to all $G \in \dot{\mathcal{H}}^1$.

Fix $G \in \dot{\mathcal{H}}^1$ with $g := G|_{\mathbb{T}} \in C^1(\mathbb{T})$. Differentiating Jg twice yields

$$\partial_x^2 Jg = \beta \partial_x g + (\partial_x \beta + i\beta^2) g + (i\partial_x \beta - \beta^2) Jg,$$

because Jg satisfies $\partial_x Jg = \beta g + i\beta Jg$.

We compute

$$\|\partial_x^2 Jg\|_{L^2([-\pi,\pi])} \leq I_1 + I_2 + I_3, \tag{4.52}$$

where

$$I_1 := \|\beta \partial_x g\|_{L^2([-\pi,\pi])}, \qquad I_2 := \|(\partial_x \beta + i\beta^2) g\|_{L^2([-\pi,\pi])}, \qquad I_3 := \|(i\partial_x \beta - \beta^2) Jg\|_{L^2([-\pi,\pi])}.$$

We first bound I_1,

$$I_1 = \|\beta \partial_x g\|_{L^2([-\pi,\pi])} \leq C_1 \|\partial_x g\|_{L^2([-\pi,\pi])} \tag{4.53}$$

Here, $C_1 := \|\beta\|_{L^\infty(\mathbb{T})}$ is bounded by Lemma 4.18.

Let $C_2 \in \mathbb{R}$ be defined as

$$C_2 := \Big(\frac{\pi}{12}\Big)^{1/2} \|\partial_x \beta + i\beta^2\|_{L^2([-\pi,\pi])},$$

The constant C_2 is bounded by Lemma 4.20.

Then, we bound I_2:

$$I_2 = \|(\partial_x \beta + i\beta^2) g\|_{L^2([-\pi,\pi])} \leq \|\partial_x \beta + i\beta^2\|_{L^2([-\pi,\pi])} \|g\|_{L^\infty(\mathbb{T})} \leq C_2 \|\partial_x g\|_{L^2([-\pi,\pi])}. \tag{4.54}$$

Here, we have used Lemma A.3 in the last step.

Finally, we bound I_3. As in the proof of Lemma 4.22, we use that $|\Pi(x)| = 1$ for all $x \in \mathbb{R}$. Thus

$$I_3^2 \leq \underbrace{\int_0^\pi \big|i\partial_x \beta(x) - \beta(x)^2\big|^2 \|\beta g\|_{L^1([0,x])}^2 \, dx}_{I_{3+}^2} + \underbrace{\int_{-\pi}^0 \big|i\partial_x \beta(x) - \beta(x)^2\big|^2 \|\beta g\|_{L^1([x,0])}^2 \, dx}_{I_{3-}^2}.$$

We study I_{3+}^2. The computations for I_{3-}^2 are similar since β is even. Let κ_1 and κ_2 be as in Lemma 4.21, applying the same strategy used in the proof of Lemma 4.22, we obtain

$$\begin{aligned} I_{3+}^2 &= \int_0^\pi \big[(\partial_x \beta(x))^2 + \beta(x)^4\big] \|\beta g\|_{L^1([0,x])}^2 \, dx \leq \int_0^\pi \underbrace{\big[(\partial_x \beta(x))^2 + \beta(x)^4\big] \kappa_1(x)}_{\kappa_2(x)} \|g\|_{L^2([0,x])}^2 \, dx \\ &\leq \int_0^\pi |g(y)|^2 \int_y^\pi \kappa_2(x) \, dx \, dy \leq \|\kappa_2\|_{L^1([0,\pi])} \|g\|_{L^2([0,\pi])}^2. \end{aligned} \tag{4.55}$$

Define C_3 by

$$C_3 := \|\kappa_2\|_{L^1([0,\pi])}^{1/2}.$$

The constant C_3 is bounded by Lemma 4.21.

Therefore, we find that

$$\begin{aligned} I_3^2 &\leq \|\kappa_2\|_{L^1([0,\pi])} \big(\|g\|_{L^2([0,\pi])}^2 + \|g\|_{L^2([-\pi,0])}^2\big) \\ &\leq \|\kappa_2\|_{L^1([0,\pi])} \|g\|_{L^2([-\pi,\pi])}^2 \leq C_3^2 \|\partial_x g\|_{L^2([-\pi,\pi])}^2. \end{aligned} \tag{4.56}$$

In the last step, we used Parseval's identity in the following way:

$$\frac{1}{2\pi}\|g\|^2_{L^2([-\pi,\pi])} = \sum_{k\geq 1}|\widehat{g}_k|^2 \leq \sum_{k\geq 1} k^2|\widehat{g}_k|^2 = \frac{1}{2\pi}\|\partial_x g\|^2_{L^2([-\pi,\pi])},$$

where we used that $G \in \dot{\mathcal{H}}$, so that $\widehat{g}_k = 0$ for all $k \leq 0$.

Therefore, substituting (4.53), (4.54), and (4.56) into (4.52), we obtain

$$\|\partial_x^2 Jg\|_{L^2([-\pi,\pi])} \leq (C_1 + C_2 + C_3)\|\partial_x g\|_{L^2([-\pi,\pi])}.$$

□

The next lemma is the last one needed before proving Proposition 2.2 and is also computer-assisted.

Lemma 4.25 *For each $k = 1, \dots, N+1$, let h_k be as in (4.43). Then the following estimate holds:*

$$\sum_{k=1}^{N+1} \|\partial_x^2 h_k\|^2_{L^2([-\pi,\pi])} \leq 1.02110 \cdot 10^6.$$

Proof: The proof is computer-assisted and the code can be found in the supplementary material. We refer to the Appendix III.E for details about the implementation. □

III.4.2 Conclusion

We are now in a position to prove Proposition 2.2.

By Lemma 4.1, the operator $\mathcal{L}^{\mathrm{ap}}_{\mathrm{exis}} : H^2_{\mathrm{odd}} \to H^1_{\mathrm{even}}$ is invertible if and only if its holomorphic counterpart $\mathcal{L}^{\mathrm{ap,\,hol}}_{\mathrm{exis}} : \dot{\mathcal{H}}^2 \to \dot{\mathcal{H}}^1$ is invertible. In addition, the norms of the inverses coincide.

Let $G \in \dot{\mathcal{H}}^1$ and set $g := G|_{\mathbb{T}} \in H^1(\mathbb{T})$. In particular, g is continuous. Hence, by Corollary 4.17, we know that there exists a unique holomorphic function $F : \mathbb{D} \to \mathbb{C}$ with $F(0) = 0$ and $\partial_z F \in C(\overline{\mathbb{D}})$ that satisfies (4.34). Set $f := F|_{\mathbb{T}} \in C^1(\mathbb{T})$.

Making the change of variables $z = e^{ix}$ in (4.34), we find that f and g satisfy

$$(c + v^{\mathrm{ap}}(x))\partial_x f(x) = if(x) + \sum_{k=1}^{N} \widehat{f}_k g_k(x) + g(x),$$

where g_k is given in (4.42).

By Duhamel's formula, we have

$$f(x) = f(0)\Pi(x) + \sum_{k=1}^{N} \widehat{f}_k Jg_k(x) + Jg(x),$$

where the coefficients $\widehat{f}_k$, $f(0)$ and the matrix $M^{\mathrm{tor}}_{\mathrm{exis}}$ from (4.44) satisfy the relation

$$0 = M^{\mathrm{tor}}_{\mathrm{exis}} \begin{pmatrix} \widehat{f}_1 \\ \vdots \\ \widehat{f}_N \\ f(0) \end{pmatrix} + \begin{pmatrix} (\widehat{Jg})_0 \\ \vdots \\ (\widehat{Jg})_N \end{pmatrix}.$$

Hence, we obtain

$$f = \Gamma g + Jg, \tag{4.57}$$

where

$$\Gamma g(x) := -\begin{pmatrix} h_1(x) \\ \vdots \\ h_{N+1}(x) \end{pmatrix}^t (M_{\text{exis}}^{\text{tor}})^{-1} \begin{pmatrix} (\widehat{Jg})_0 \\ \vdots \\ (\widehat{Jg})_N \end{pmatrix}, \tag{4.58}$$

$h_k = Jg_k$ for $k = 1, \dots, N$, and $h_{N+1} = \Pi$.

Let

$$M_{\text{exis}}^{\text{tor}} = U\Sigma V^*$$

be a singular value decomposition, where $\Sigma = \operatorname{diag}(\sigma_1, \dots, \sigma_{N+1})$ is ordered increasingly.

Because V and U are unitary, for all $a, b \in \mathbb{C}^{N+1}$, we have

$$\begin{aligned}\Big| \sum_{k,l,m=1}^{N+1} a_k V_{k,l} \frac{1}{\sigma_l} \overline{U}_{m,l} b_m \Big|^2 &\le \|a\|_{\ell^2}^2 \sum_{k=1}^{N+1} \Big| \sum_{l,m=1}^{N+1} V_{k,l} \frac{1}{\sigma_l} \overline{U}_{m,l} b_m \Big|^2 = \|a\|_{\ell^2}^2 \sum_{l=1}^{N+1} \frac{1}{\sigma_l^2} \Big| \sum_{m=1}^{N+1} \overline{U}_{m,l} b_m \Big|^2 \\ &\le \frac{1}{\sigma_1^2} \|a\|_{\ell^2}^2 \sum_{l=1}^{N+1} \Big| \sum_{m=1}^{N+1} \overline{U}_{m,l} b_m \Big|^2 = \frac{1}{\sigma_1^2} \|a\|_{\ell^2}^2 \|b\|_{\ell^2}^2.\end{aligned}$$

In particular, taking $b_m = (\widehat{Jg})_{m-1},\ m = 1, \dots, N+1$ and $a_k = \partial_x^2 h_k(x),\ k = 1, \dots, N+1$ for a fixed $x \in [0, 2\pi]$, we have

$$\begin{aligned}\big|\partial_x^2 \Gamma(x)\big|^2 &= \Big| \sum_{k,l,m=1}^{N+1} \partial_x^2 h_k(x) V_{k,l} \frac{1}{\sigma_l} \overline{U}_{m,l} (\widehat{Jg})_{m-1} \Big|^2 \\ &\le \frac{1}{\sigma_1^2} \Big(\sum_{k=1}^{N+1} |\partial_x^2 h_k(x)|^2 \Big) \Big(\sum_{m=0}^{N} |(\widehat{Jg})_m|^2 \Big) \le \frac{1}{\sigma_1^2} \Big(\sum_{k=1}^{N+1} |\partial_x^2 h_k(x)|^2 \Big) \|Jg\|_{L^2(\mathbb{T})}^2,\end{aligned}$$

where we used Parseval's identity.

Taking the $L^2([-\pi, \pi])$ norm in (4.58) yields

$$\|\partial_x^2 \Gamma\|_{L^2([-\pi,\pi])}^2 \le \frac{1}{\sigma_1^2} \|Jg\|_{L^2(\mathbb{T})}^2 \sum_{k=1}^{N+1} \|\partial_x^2 h_k\|_{L^2([-\pi,\pi])}^2. \tag{4.59}$$

Let $C_1 \in \mathbb{R}$ be defined by

$$C_1^2 := \frac{1}{\sigma_1^2} \|J\|_{L^2(\mathbb{T}) \to L^2(\mathbb{T})}^2 \sum_{k=1}^{N+1} \|\partial_x^2 h_k\|_{L^2([-\pi,\pi])}^2,$$

Lemmas 4.16, 4.22, and 4.25 provide an explicit upper bound for C_1.

By Lemma 4.23, there exists an explicit constant $C_2 > 0$ such that

$$\|\partial_x^2 Jg\|_{L^2([-\pi,\pi])} \le C_2 \|\partial_x g\|_{L^2([-\pi,\pi])}. \tag{4.60}$$

We differentiate (4.57) twice and apply the estimates (4.59) and (4.60):

$$\|\partial_x^2 f\|_{L^2([-\pi,\pi])} \le \|\partial_x^2 \Gamma\|_{L^2([-\pi,\pi])} + \|\partial_x^2 Jg\|_{L^2([-\pi,\pi])} \le C_1 \|g\|_{L^2(\mathbb{T})} + C_2 \|\partial_x g\|_{L^2([-\pi,\pi])}.$$

Since $G \in \dot{\mathcal{H}}^1$, the trace $g = G|_{\mathbb{T}}$ has only positive Fourier modes. Hence,

$$\|g\|_{L^2(\mathbb{T})}^2 = \sum_{k \ge 1} |\widehat{g}_k|^2 \le \sum_{k \ge 1} k^2 |\widehat{g}_k|^2 = \frac{1}{2\pi} \|\partial_x g\|_{L^2([-\pi,\pi])}^2.$$

Define $C_3 := \frac{C_1}{\sqrt{2\pi}} + C_2$, so we get that

$$\|\partial_x^2 f\|_{L^2([-\pi,\pi])} \leq C_3 \|\partial_x g\|_{L^2([-\pi,\pi])}.$$

Consequently, since $f \in C^1(\mathbb{T})$ and $\partial_x^2 f \in L^2([-\pi,\pi])$, it follows that $f \in H^2(\mathbb{T})$. Additionally, since F is holomorphic in $\mathbb{D}$ and $F(0) = 0$, the function f has only positive Fourier modes. We then find that

$$\|F\|_{\dot{\mathcal{H}}^2}^2 = \frac{1}{2\pi}\|\partial_x^2 f\|_{L^2([-\pi,\pi])}^2 \leq \frac{C_3^2}{2\pi}\|\partial_x g\|_{L^2([-\pi,\pi])}^2 = C_3^2 \|G\|_{\dot{\mathcal{H}}^1}^2.$$

Therefore, by Lemma 4.1 and identity (4.3), we have that

$$\|L_{v^{\mathrm{ap}}}^{-1}\|_{H^0_{\mathrm{odd}} \to H^1_{\mathrm{even}}} = \|(\mathcal{L}^{\mathrm{ap}}_{\mathrm{exis}})^{-1}\|_{H^1_{\mathrm{even}} \to H^2_{\mathrm{odd}}} = \|(\mathcal{L}^{\mathrm{ap,\,hol}}_{\mathrm{exis}})^{-1}\|_{\dot{\mathcal{H}}^1 \to \dot{\mathcal{H}}^2} \leq C_3.$$

Finally, we substitute all the bounds and find that

$$\|L_{v^{\mathrm{ap}}}^{-1}\|_{H^0_{\mathrm{odd}} \to H^1_{\mathrm{even}}} \leq C_3 < 3.13328 \cdot 10^4.$$

III.5 Proof of Proposition 3.11: Lemmas 3.12–3.14

In this section we shall prove Lemmas 3.12, 3.13 and 3.14 which yield Proposition 3.11. Recall that Proposition 3.11 is the main ingredient which remains to show that the solution $(\frac{c}{3}, v^3)$ is linearly unstable.

III.5.1 The operator T^θ_{stab}

This section is devoted to the study of the operator T^θ_{stab}, given by

$$T^\theta_{\mathrm{stab}} f := \left(\mathcal{L}^{\mathrm{ap}}_\theta - \lambda^{\mathrm{ap}} I\right)^{-1}\left[(\eta(f)I - \mathcal{D}_\theta)(f^{\mathrm{ap}} + f) - \xi_{f^{\mathrm{ap}}}\right]. \tag{5.1}$$

Before proceeding with the analysis of T^θ_{stab}, it is necessary to justify that the definition (5.1) is indeed meaningful. More precisely, the expression involves the inverse operator

$$\left(\mathcal{L}^{\mathrm{ap}}_\theta - \lambda^{\mathrm{ap}} I\right)^{-1},$$

and *a priori* it is not clear that the operator $\mathcal{L}^{\mathrm{ap}}_\theta - \lambda^{\mathrm{ap}} I$ is invertible.

In addition, in order to define $\eta(f)$, we will need to further understand the structure of $\mathcal{L}^{\mathrm{ap}}_\theta - \lambda^{\mathrm{ap}} I$. Indeed, since λ^{ap} is an approximation of the eigenvalue λ of $\mathcal{L}_\theta$ (whose existence we aim to prove), and $\mathcal{L}^{\mathrm{ap}}_\theta$ is close to $\mathcal{L}_\theta$, the operator $\mathcal{L}^{\mathrm{ap}}_\theta - \lambda^{\mathrm{ap}} I$ is expected to be close to being non-invertible.

Nevertheless, by choosing $\eta(f)$ appropriately, we will remove the nearly singular direction, yielding uniform bounds for the inverse.

III.5.2 The linear operator $\mathcal{L}^{\mathrm{ap}}_\theta$

Our first goal in this section is to reduce the study of the operator $\mathcal{L}^{\mathrm{ap}}_\theta - \lambda^{\mathrm{ap}} I$, acting on periodic function spaces, to the study of an operator acting on Hardy spaces. Here, $\mathcal{L}^{\mathrm{ap}}_\theta$ is defined in (3.8).

To this end, we define the operator $\mathcal{L}^{\mathrm{ap,\,hol}}_{\theta,\lambda} : \mathcal{H} \oplus \mathcal{H} \to \mathcal{H} \oplus \mathcal{H}$, depending on two parameters $\theta \in (0,1)$ and $\lambda \in \mathbb{C}$, by

$$\mathcal{L}^{\mathrm{ap,\,hol}}_{\theta,\lambda}(F^+, F^-) := (\mathcal{L}^{\mathrm{ap,\,hol}}_{\theta,\lambda}(F^+, F^-)^1, \mathcal{L}^{\mathrm{ap,\,hol}}_{\theta,\lambda}(F^+, F^-)^2).$$

Set

$$D_z := z\partial_z, \qquad \theta^+ := \theta, \qquad \theta^- := 1 - \theta.$$

Then the components of $\mathcal{L}^{\mathrm{ap,\,hol}}_{\theta,\lambda}(F^+,F^-)$ are given by

$$\begin{cases} \mathcal{L}^{\mathrm{ap,\,hol}}_{\theta,\lambda}(F^+,F^-)^1 := \quad iPD_zF^+ - (i+\lambda)F^+ + i\theta^+PF^+ - \mathcal{G}^{\theta^+}(F^+,F^-)\ , & \text{(5.2a)} \\ \mathcal{L}^{\mathrm{ap,\,hol}}_{\theta,\lambda}(F^+,F^-)^2 := -(iPD_zF^- - (i-\lambda)F^- + i\theta^-PF^- - \mathcal{G}^{\theta^-}(F^-,F^+)). & \text{(5.2b)} \end{cases}$$

Recall that P is defined in (4.5). The meromorphic functions $\mathcal{G}^{\theta^\pm}(A,B)$ are defined for $A,B\in\mathcal{H}$ by

$$\mathcal{G}^{\theta}(A,B) := \sum_{k=1}^{N}\left(A_{k-1}\mathcal{G}^-_{k,1-\theta} + B_{k-1}\mathcal{G}^+_{k,\theta}\right) \quad \text{for } \theta\in(0,1), \tag{5.3}$$

where

$$\mathcal{G}^-_{k,\theta}(z) := (k-\theta)\frac{i}{2}\sum_{m=0}^{N-k} v^{\mathrm{ap}}_{m+k}z^{-(1+m)} \qquad \text{and} \qquad \mathcal{G}^+_{k,\theta}(z) := (k-\theta)\frac{i}{2}\sum_{m=0}^{N-k} v^{\mathrm{ap}}_{m+k}z^{m}. \tag{5.4}$$

The operator $\mathcal{L}^{\mathrm{ap,\,hol}}_{\theta,\lambda}(F^+,F^-)$ is well defined. Indeed, for any $(F^+,F^-)\in\mathcal{H}\oplus\mathcal{H}$, the expressions in (5.2) define holomorphic functions, since all terms involving negative powers of z cancel out.

Remark 5.1 *With this notation, we have*

$$\mathcal{G}^{\theta^+}(F^+,F^-)(z) := \frac{i}{2}\sum_{j=1}^{N} v^{\mathrm{ap}}_j \sum_{m=1}^{j}\left[(m-\theta^-)F^+_{m-1}z^{-(1+j-m)} + (m-\theta^+)F^-_{m-1}z^{j-m}\right],$$

$$\mathcal{G}^{\theta^-}(F^-,F^+)(z) := \frac{i}{2}\sum_{j=1}^{N} v^{\mathrm{ap}}_j \sum_{m=1}^{j}\left[(m-\theta^+)F^-_{m-1}z^{-(1+j-m)} + (m-\theta^-)F^+_{m-1}z^{j-m}\right].$$

Definition 5.2 *Let $f\in H^1(\mathbb{T})$. We define $F^+,F^-\in\mathcal{H}^1$ by*

$$F^+(z) := \sum_{k\geq 0}\widehat{f}_k z^k \qquad \text{and} \qquad F^-(z) := \sum_{k\geq 0}\widehat{f}_{-k-1}z^k.$$

Remark 5.3 *Unlike* (1.7)*, here we do not assume $\widehat{f}_0 = 0$. Thus, F^+ contains the Fourier modes $k\geq 0$, while F^- contains the modes $k\leq -1$.*

Lemma 5.4 *Let $\theta\in(0,1)$ and $\lambda\in\mathbb{C}$. Let $f\in H^2(\mathbb{T})$ and $g\in H^1(\mathbb{T})$, and define $F^\pm\in\mathcal{H}^2$ and $G^\pm\in\mathcal{H}^1$ as in Definition 5.2. Assume that f and g satisfy*

$$\mathcal{L}^{\mathrm{ap}}_{\theta} f - \lambda f = g, \tag{5.5}$$

Then $F^\pm$ and $G^\pm$ satisfy (5.2)*.*

Proof: Assume that f and g satisfy (5.5), that is

$$(c+v^{\mathrm{ap}})(\partial_x f + i\theta f) + Hf - i\widehat{f}_0 - \lambda f = g. \tag{5.6}$$

Define

$$f^+(x) := \sum_{k\geq 0}\widehat{f}_k e^{ikx} \qquad \text{and} \qquad f^-(x) := \sum_{k\geq 0}\widehat{f}_{-k-1}e^{ikx}.$$

Then, setting $f^\sharp(x) := e^{-ix}f(-x)$, we obtain

$$f^+(x) = \frac{1}{2}(f(x)+iHf(x)) + \frac{1}{2}\widehat{f}_0, \qquad f^-(x) = \frac{1}{2}(f^\sharp(x)+iHf^\sharp(x)) + \frac{1}{2}\widehat{f}_{-1}.$$

We first prove that f^+, f^-, g^+, and g^- satisfy (5.2) restricted to $\mathbb{T}$, namely

$$\begin{cases} (c+v^{\mathrm{ap}})(\partial_x f^+ + i\theta^+ f^+) = (i+\lambda)f^+ + g^+ + \mathcal{J}^{\theta^+}(f^+, f^-), & \text{(5.7a)} \\ (c+v^{\mathrm{ap}})(\partial_x f^- + i\theta^- f^-) = (i-\lambda)f^- - g^- + \mathcal{J}^{\theta^-}(f^-, f^+), & \text{(5.7b)} \end{cases}$$

where

$$\mathcal{J}^{\theta}(f^+, f^-)(x) := \frac{i}{2}\sum_{j=1}^{N} v_j^{\mathrm{ap}} \sum_{m=1}^{j} (m+\theta-1)\widehat{f^+}_{m-1} e^{-i(1+j-m)x} + (m-\theta)\widehat{f^-}_{m-1} e^{i(j-m)x}. \quad (5.8)$$

Taking the mean of (5.6) yields

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} \big(c+v^{\mathrm{ap}}(x)\big)\big(\partial_x f(x) + i\theta f(x)\big)dx - i\widehat{f}_0 - \lambda\widehat{f}_0 = \widehat{g}_0.$$

Hence,

$$\begin{aligned} i\widehat{f}_0 + \lambda\widehat{f}_0 + \widehat{g}_0 &= \frac{1}{2\pi}\int_{-\pi}^{\pi} \big(c+v^{\mathrm{ap}}(x)\big)\big(\partial_x f(x) + i\theta f(x)\big)dx \\ &= -\frac{1}{2\pi}\int_{-\pi}^{\pi} \partial_x v^{\mathrm{ap}}(x) f(x)\,dx + i\theta\frac{1}{2\pi}\int_{-\pi}^{\pi} \big(c+v^{\mathrm{ap}}(x)\big) f(x)\,dx \\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi} \sum_{j=1}^{N} j v_j^{\mathrm{ap}} \sin(jx) f(x)\,dx + i\theta\frac{1}{2\pi}\int_{-\pi}^{\pi} \Big(c + \sum_{j=1}^{N} v_j^{\mathrm{ap}}\cos(jx)\Big) f(x)\,dx \\ &= \frac{i}{2}\sum_{j=1}^{N} j v_j^{\mathrm{ap}}(\widehat{f}_j - \widehat{f}_{-j}) + i\theta\Big(c\widehat{f}_0 + \frac{1}{2}\sum_{j=1}^{N} v_j^{\mathrm{ap}}(\widehat{f}_j + \widehat{f}_{-j})\Big) \\ &= i\theta c\widehat{f}_0 + \frac{i}{2}\sum_{j=1}^{N} v_j^{\mathrm{ap}}((j+\theta)\widehat{f}_j - (j-\theta)\widehat{f}_{-j}). \end{aligned} \quad (5.9)$$

In addition, applying $\frac{1}{2}(I+iH)$ to (5.6) gives

$$\begin{aligned} \frac{1}{2}(I+iH)\big[(c+v^{\mathrm{ap}})(\partial_x f + i\theta f)\big] &= \frac{1}{2}(I+iH)\big[g - Hf + i\widehat{f}_0 + \lambda f\big] \\ &= (g^+ - \frac{1}{2}\widehat{g}_0) + i(f^+ - \widehat{f}_0) + \frac{i}{2}\widehat{f}_0 + \lambda(f^+ - \frac{1}{2}\widehat{f}_0) \\ &= g^+ + (i+\lambda)f^+ - \frac{1}{2}(i\widehat{f}_0 + \lambda\widehat{f}_0 + \widehat{g}_0), \end{aligned} \quad (5.10)$$

where we have used that

$$\begin{aligned} -\frac{1}{2}(I+iH)[Hf](x) &= -\frac{1}{2}\sum_{k\in\mathbb{Z}} (1+\mathrm{sgn}(k))(-i\mathrm{sgn}(k))\widehat{f}_k e^{ikx} \\ &= \frac{i}{2}\sum_{k\in\mathbb{Z}} (\mathrm{sgn}(k) + \mathrm{sgn}(k)^2)\widehat{f}_k e^{ikx} = i(f^+(x) - \widehat{f}_0). \end{aligned}$$

Define

$$\Xi^+ := (c+v^{\mathrm{ap}})(\partial_x f^+ + i\theta f^+) - (i+\lambda)f^+ - g^+.$$

Using the definition of f^+, we write

$$\begin{aligned} \Xi^+ &= \frac{1}{2}(c+v^{\mathrm{ap}})(I+iH)(\partial_x f + i\theta f) + \frac{i\theta}{2}(c+v^{\mathrm{ap}})\widehat{f}_0 - (i+\lambda)f^+ - g^+ \\ &= \frac{1}{2}(I+iH)\big[(c+v^{\mathrm{ap}})(\partial_x f + i\theta f)\big] - \frac{i}{2}[H, c+v^{\mathrm{ap}}](\partial_x f + i\theta f) \\ &\quad + \frac{i\theta}{2}(c+v^{\mathrm{ap}})\widehat{f}_0 - (i+\lambda)f^+ - g^+. \end{aligned} \quad (5.11)$$

Now, plugging (5.10) into (5.11), we obtain

$$\begin{aligned}\Xi^+ &= -\frac{1}{2}(i\widehat{f}_0 + \lambda\widehat{f}_0 + \widehat{g}_0) - \frac{i}{2}[H, c + v^{\mathrm{ap}}](\partial_x f + i\theta f) + \frac{i\theta}{2}(c + v^{\mathrm{ap}})\widehat{f}_0 \\ &= -\frac{i}{4}\sum_{j=1}^{N} v_j^{\mathrm{ap}}((j+\theta)\widehat{f}_j - (j-\theta)\widehat{f}_{-j}) - \frac{i}{2}[H, c + v^{\mathrm{ap}}](\partial_x f + i\theta f) + \frac{i\theta}{2}v^{\mathrm{ap}}\widehat{f}_0, \end{aligned} \tag{5.12}$$

where we used (5.9) in the last step.

The expression for the commutator is given in Corollary B.3. Substituting it into (5.12) yields

$$\begin{aligned}\Xi^+(x) &= -\frac{i}{4}\sum_{j=1}^{N} v_j^{\mathrm{ap}}((j+\theta)\widehat{f}_j - (j-\theta)\widehat{f}_{-j}) + \frac{i\theta}{2}v^{\mathrm{ap}}(x)\widehat{f}_0 \\ &\quad + \frac{i\theta}{4}\widehat{f}_0\sum_{j=1}^{N} v_j^{\mathrm{ap}}(e^{-ijx} - e^{ijx}) + \frac{i}{4}\sum_{j=1}^{N} v_j^{\mathrm{ap}}\Big((j+\theta)\widehat{f}_j + (j-\theta)\widehat{f}_{-j}\Big) \\ &\quad + \frac{i}{2}\sum_{j=1}^{N} v_j^{\mathrm{ap}}\sum_{m=1}^{j-1}\left((m+\theta)\widehat{f}_m e^{-i(j-m)x} + (m-\theta)\widehat{f}_{-m}e^{i(j-m)x}\right) \\ &= \frac{i}{2}\sum_{j=1}^{N} v_j^{\mathrm{ap}}(j-\theta)\widehat{f}_{-j} + \widehat{f}_0\frac{i\theta}{2}\sum_{j=1}^{N} v_j^{\mathrm{ap}}e^{-ijx} \\ &\quad + \frac{i}{2}\sum_{j=1}^{N} v_j^{\mathrm{ap}}\sum_{m=1}^{j-1}\left((m+\theta)\widehat{f}_m e^{-i(j-m)x} + (m-\theta)\widehat{f}_{-m}e^{i(j-m)x}\right) \\ &= \frac{i}{2}\sum_{j=1}^{N} v_j^{\mathrm{ap}}\sum_{m=1}^{j}\left((m-1+\theta)\widehat{f}_{m-1}e^{-i(1+j-m)x} + (m-\theta)\widehat{f}_{-m}e^{i(j-m)x}\right) = \mathcal{J}^{\theta^+}(f^+, f^-). \end{aligned}$$

We now derive the equation satisfied by f^-. Changing variables $x \mapsto -x$ in (5.6) and multiplying by e^{-ix}, we obtain

$$e^{-ix}\big(c + v^{\mathrm{ap}}(x)\big)\big(\partial_x f(-x) + i\theta f(-x)\big) + e^{-ix}Hf(-x) - i\widehat{f}_0 e^{-ix} - \lambda f^\sharp(x) = g^\sharp(x). \tag{5.13}$$

where $f^\sharp(x) := e^{-ix}f(-x)$ and $g^\sharp(x) := e^{-ix}g(-x)$. Using that

$$\begin{aligned} e^{-ix}\big(\partial_x f(-x) + i\theta f(-x)\big) &= -\partial_x f^\sharp(x) + i(\theta-1)f^\sharp(x), \\ e^{-ix}Hf(-x) &= -Hf^\sharp(x) + i\widehat{f}_{-1} + i\widehat{f}_0 e^{-ix}, \end{aligned} \tag{5.14}$$

and multiplying (5.13) by -1, we obtain

$$\big(c + v^{\mathrm{ap}}(x)\big)\big(\partial_x f^\sharp + i\theta^- f^\sharp\big) + Hf^\sharp - i\widehat{f}_{-1} + \lambda f^\sharp = -g^\sharp. \tag{5.15}$$

Since $\widehat{f^\sharp}_0 = \widehat{f}_{-1}$, we may repeat the previous argument to conclude that f^- satisfies (5.7b).

Finally, the functions f^+ and f^-, and their derivatives extend holomorphically to $\mathbb{D}$, and these extensions coincide with F^+ and F^-, respectively. The same holds for g, hence extending (5.7) from $\mathbb{T}$ to $\mathbb{D}$ yields (5.2). □

Looking at the Fourier modes of f in Definition 5.2, we see that the mapping $f \in H^1(\mathbb{T}) \to (F^+, F^-) \in \mathcal{H}^1 \oplus \mathcal{H}^1$ is a bijection. In fact, we can recover f via

$$f(x) := F^+|_{\mathbb{T}}(x) + e^{-ix}F^-|_{\mathbb{T}}(-x) = F^+(e^{ix}) + e^{-ix}F^-(e^{-ix}). \tag{5.16}$$

Here, the Sobolev exponent is chosen as 1 so that f is continuous.

In the next lemma, we prove that the converse of Lemma 5.4 also holds.

Lemma 5.5 *Let $\theta \in (0,1)$ and $\lambda \in \mathbb{C}$. Let $F^\pm \in \mathcal{H}^2$ and $G^\pm \in \mathcal{H}^1$, and define $f \in H^2(\mathbb{T})$ and $g \in H^1(\mathbb{T})$ via (5.16).*

If $(F^\pm, G^\pm)$ solves (5.2), then (f,g) satisfies (5.5).

Proof: Since $F^\pm$ and $G^\pm$ satisfy (5.2), we also know that $F^\pm|_{\mathbb{T}}$ and $G^\pm|_{\mathbb{T}}$ satisfy (5.7). Therefore, we shall deduce (5.6) from (5.7).

First, define

$$F|_{\mathbb{T}}^{-\sharp}(x) := e^{-ix}F^-|_{\mathbb{T}}(-x) \quad \text{and} \quad G|_{\mathbb{T}}^{-\sharp}(x) := e^{-ix}G^-|_{\mathbb{T}}(-x).$$

It follows from the second equation in (5.7b) and (5.14) that

$$(c+v^{\mathrm{ap}})(\partial_x F|_{\mathbb{T}}^{-\sharp} + i\theta^+ F|_{\mathbb{T}}^{-\sharp}) = (\lambda - i)F|_{\mathbb{T}}^{-\sharp} + G|_{\mathbb{T}}^{-\sharp} - \mathcal{J}^{\theta^-}(F^-|_{\mathbb{T}}, F^+|_{\mathbb{T}})^\sharp.$$

We now add the first equation of (5.7a), since $f = F|_{\mathbb{T}}^+ + F|_{\mathbb{T}}^{-\sharp}$, we obtain

$$(c+v^{\mathrm{ap}})(\partial_x f + i\theta^+ f) - \lambda f - g = i(F|_{\mathbb{T}}^+ - F|_{\mathbb{T}}^{-\sharp}) + \mathcal{J}^{\theta^+}(F|_{\mathbb{T}}^+, F|_{\mathbb{T}}^-) - \mathcal{J}^{\theta^-}(F^-|_{\mathbb{T}}, F^+|_{\mathbb{T}})^\sharp \quad (5.17)$$

Note that

$$\mathcal{J}^{\theta^+}(F|_{\mathbb{T}}^+, F|_{\mathbb{T}}^-)(x) = e^{-ix}\mathcal{J}^{\theta^-}(F^-|_{\mathbb{T}}, F^+|_{\mathbb{T}})(-x) = \mathcal{J}^{\theta^-}(F^-|_{\mathbb{T}}, F^+|_{\mathbb{T}})^\sharp(x).$$
$$Hf(x) = \sum_{k\in\mathbb{Z}} -i\mathrm{sgn}(k)\widehat{f}_k = \sum_{k\geq 1} -i\widehat{f}_k + \sum_{k\leq -1} i\widehat{f}_k = -iF|_{\mathbb{T}}^+ + i\widehat{f}_0 + iF|_{\mathbb{T}}^{-\sharp}.$$

Therefore, (5.17) is equivalent to (5.5). □

As we did in Section III.4.1, we temporarily ignore the Taylor coefficients F_k^+ and F_k^- appearing in (5.3) and replace them with arbitrary constants.

Lemma 5.6 *Let $\theta \in (0,1)$ and $\lambda \in \mathbb{C}$. Let $a_k^+, a_k^- \in \mathbb{C}$ for $k = 0, \dots, N-1$. Let $F^+, F^-, G^+, G^- \in \mathcal{H}$.*

Assume that the following identities hold

$$iPD_zF^+ = (i+\lambda)F^+ - i\theta^+ PF^+ + G^+ + \sum_{k=1}^N \left(a_{k-1}^+ \mathcal{G}_{k,\theta^-}^- + a_{k-1}^- \mathcal{G}_{k,\theta^+}^+\right)$$
$$iPD_zF^- = (i-\lambda)F^- - i\theta^- PF^- - G^- + \sum_{k=1}^N \left(a_{k-1}^- \mathcal{G}_{k,\theta^+}^- + a_{k-1}^+ \mathcal{G}_{k,\theta^-}^+\right),$$

where $D_z := z\partial_z$.

Then, for every $k = 0, \dots, N-1$, we have

$$F_k^+ := \frac{1}{k!}\frac{d^k}{dz^k}\bigg|_{z=0} F^+ = a_k^+, \qquad F_k^- := \frac{1}{k!}\frac{d^k}{dz^k}\bigg|_{z=0} F^- = a_k^-.$$

Proof: We prove the statement for F^+, since the argument for F^- is identical.

Recalling the definition of $\mathcal{G}_{k,\theta}^\pm$, the first equation is

$$izP(z)\partial_z F^+(z) = (i+\lambda)F^+(z) - i\theta^+ P(z)F^+(z) + G^+(z)$$
$$+ \frac{i}{2}\sum_{j=1}^N v_j^{\mathrm{ap}} \sum_{m=1}^j (m-\theta^-)z^{-(1+j-m)}a_{m-1}^+ + (m-\theta^+)z^{j-m}a_{m-1}^-.$$

Multiply by z^N:

$$iz^{N+1}P(z)\partial_z F^+(z) = (i+\lambda)z^N F^+(z) - i\theta^+ z^N P(z)F^+(z) + z^N G^+(z) + \frac{i}{2}\sum_{j=1}^{N} v_j^{\mathrm{ap}} \sum_{m=1}^{j} (m-\theta^-) z^{N-(1+j-m)} a_{m-1}^+ + \mathcal{O}(z^N). \tag{5.18}$$

We now compare coefficients of z^k for $k = 0, \dots, N-1$. Since $z^N G^+(z) = \mathcal{O}(z^N)$, and $z^N F^+(z) = \mathcal{O}(z^N)$, none of these terms contribute to the coefficient of z^k. From, (5.18) implies that

$$\frac{i}{2}\sum_{j=1}^{N}\sum_{l=0}^{j-1} v_j^{\mathrm{ap}}(l+\theta^+)F_l^+ z^{N+l-j} = \frac{i}{2}\sum_{j=1}^{N} v_j^{\mathrm{ap}} \sum_{m=1}^{j} (m-\theta^-) z^{N-(1+j-m)} a_{m-1}^+ + \mathcal{O}(z^N).$$

Consequently, for each $k = 0, \dots, N-1$ we apply $\frac{1}{k!}\frac{d^k}{dz^k}$ to both sides yielding

$$\frac{i}{2}\sum_{j=1}^{N}\sum_{l=0}^{j-1} v_j^{\mathrm{ap}}(l+\theta^+)F_l^+ \delta_{k,N+l-j} = \frac{i}{2}\sum_{j=1}^{N} v_j^{\mathrm{ap}} \sum_{l=0}^{j-1} (l+1-\theta^-) a_l^+ \delta_{k,N+l-j}.$$

Since $\theta^+ = 1-\theta^-$, it follows that

$$0 = \sum_{j=N-k}^{N} v_j^{\mathrm{ap}}(k+j-N+\theta^+)(F_{k+j-N}^+ - a_{k+j-N}^+) = \sum_{j=0}^{k} v_{j+N-k}^{\mathrm{ap}}(j+\theta^+)(F_j^+ - a_j^+).$$

Since $v_j^{\mathrm{ap}} \neq 0$ for all $j = 1, \dots, N$ (Lemma 4.3) and $\theta^+ \in (0,1)$, the above system is triangular and nonsingular, hence $F_k^+ = a_k^+$ for all $k = 0, \dots, N-1$. □

III.5.2.1 Invertibility of $\mathcal{L}_{\theta,\lambda}^{\mathrm{ap,\,hol}}$

In order to study the operator $\mathcal{L}_{\theta,\lambda}^{\mathrm{ap,\,hol}}$, we first need a result similar to Lemma 4.12, but adapted to this new operator.

Condition 5.7 *We say that $\theta \in (0,1)$ and $\lambda \in \mathbb{C}$ satisfy Condition 5.7 if for every $\alpha \in Z$, we have*

$$\lim_{z\to\alpha} i\frac{z-\alpha}{z}\left(\frac{i\pm\lambda}{P(z)} - i\theta^{\pm}\right) \notin \mathbb{Z}.$$

We recall that Z denotes the set of the roots of $z^N P(z)$ lying inside the unit disk $\mathbb{D}$.

We introduce some notation, analogous to Definition 4.13.

Definition 5.8 *Let $\theta \in (0,1)$ and $\lambda \in \mathbb{C}$. Let*

$$Z_0 := \{0\} \cup Z,$$

where $Z = \{z_j\}$ denotes the set of roots of $z^N P(z)$ lying inside the open unit disk, and let ω_0, Q, and the paths $\{\tau_j\}$ be as in Lemma 4.10 with $\widetilde{Z} = Z_0$.

Given $G \in \mathcal{H} \cup \{\mathcal{G}_{k,\theta}^{\pm} : \theta \in (0,1),\ k = 1, \dots, N\}$, define

$$B_G(z) := -2i\frac{z^N}{v_N^{\mathrm{ap}}} G(z) \prod_{\alpha\in Z} (z-\alpha^{-1})^{-1}.$$

We also define

$$A_{\lambda,\theta}(z) := 2i\frac{z^N}{v_N^{\mathrm{ap}}}[i+\lambda-i\theta P(z)]\prod_{\alpha\in Z}(z-\alpha^{-1})^{-1},$$

For each $j=1,\dots,N+1$, we define $J^+_{\tau_j}G$ as the unique continuous solution to (4.30) *along τ_j with initial condition $J^+_{\tau_j}G(0)=0$ and coefficients $A=A_{\lambda,\theta^+}$ and $B=B_G$.*

Analogously, for each $j=1,\dots,N+1$, we define $J^-_{\tau_j}G$ as the unique continuous solution of (4.30) *along τ_j with initial condition $J^-_{\tau_j}G(0)=0$ and coefficients $A=A_{-\lambda,\theta^-}$ and $B=B_G$.*

Moreover, for each $j=1,\dots,N+1$, we define $\Pi^+_{\tau_j}$ as the unique continuous solution of (4.30) *along τ_j with initial condition $\Pi^+_{\tau_j}(0)=1$ and coefficients $A=A_{\lambda,\theta^+}$ and $B=0$.*

Finally, for each $j=1,\dots,N+1$, we define $\Pi^-_{\tau_j}$ as the unique continuous solution of (4.30) *along τ_j with initial condition $\Pi^-_{\tau_j}(0)=1$ and coefficients $A=A_{-\lambda,\theta^-}$ and $B=0$.*

The next step is to establish an invertibility criterion for $\mathcal{L}^{\mathrm{ap,\,hol}}_{\theta,\lambda}$. The following result follows the same strategy used to prove Proposition 4.14.

Proposition 5.9 *Let $\theta\in(0,1)$ and $\lambda\in\mathbb{C}$ and assume Condition 5.7 holds, and set $b:=2\pi+2$. Using the notation from Definition 5.8 and $\mathcal{G}^{\pm}_{k,\theta}$ from* (5.4)*, define the $(2N+2)\times(2N+2)$ matrix $M^{\mathrm{hol}}_{\mathrm{stab}}(\lambda)$ by*

$$\big(M^{\mathrm{hol}}_{\mathrm{stab}}(\lambda)\big)_{jk}=\begin{cases} J^+_{\tau_j}\mathcal{G}^-_{k,\theta^-}(b), & 1\le j\le N+1,\quad 1\le k\le N\\ J^+_{\tau_j}\mathcal{G}^+_{k-N,\theta^+}(b), & 1\le j\le N+1,\quad N+1\le k\le 2N\\ \Pi^+_{\tau_j}(b)-1, & 1\le j\le N+1,\quad k=2N+1\\ J^-_{\tau_{j-N-1}}\mathcal{G}^+_{k,\theta^-}(b), & N+2\le j\le 2N+2,\quad 1\le k\le N\\ J^-_{\tau_{j-N-1}}\mathcal{G}^-_{k-N,\theta^+}(b), & N+2\le j\le 2N+2,\quad N+1\le k\le 2N\\ \Pi^-_{\tau_{j-N-1}}(b)-1, & N+2\le j\le 2N+2,\quad k=2N+2\\ 0, & \text{in other case}\end{cases}. \tag{5.19}$$

Then the matrix $M^{\mathrm{hol}}_{\mathrm{stab}}(\lambda)$ is nonsingular if and only if, for every holomorphic pair $(G^+,G^-)\in\mathcal{H}\oplus\mathcal{H}$ of functions, continuous on $\overline{\mathbb{D}}$, there exists a unique pair $(F^+,F^-)\in\mathcal{H}\oplus\mathcal{H}$ of functions, with $\partial_z F^{\pm}$ continuous on $\overline{\mathbb{D}}$, such that

$$\mathcal{L}^{\mathrm{ap,\,hol}}_{\theta,\lambda}(F^+,F^-)=(G^+,G^-). \tag{5.20}$$

Moreover, assume that the kernel of $M^{\mathrm{hol}}_{\mathrm{stab}}$ has dimension d. Then there exist d linearly independent pairs $\{(F^+_k,F^-_k)\}_{k=1}^d$ of holomorphic functions, continuous on $\overline{\mathbb{D}}$, such that

$$\mathcal{L}^{\mathrm{ap,\,hol}}_{\theta,\lambda}(F^+,F^-)=0. \tag{5.21}$$

Example 5.10 *With $N=2$, the matrix $M^{\mathrm{hol}}_{\mathrm{stab}}$ would be of order 6, given by*

$$M^{\mathrm{hol}}_{\mathrm{stab}}=\begin{pmatrix} J^+_{\tau_1}\mathcal{G}^-_{1,\theta^-}(b) & J^+_{\tau_1}\mathcal{G}^-_{2,\theta^-}(b) & J^+_{\tau_1}\mathcal{G}^+_{1,\theta^+}(b) & J^+_{\tau_1}\mathcal{G}^+_{2,\theta^+}(b) & \Pi^+_{\tau_1}(b)-1 & 0\\ J^+_{\tau_2}\mathcal{G}^-_{1,\theta^-}(b) & J^+_{\tau_2}\mathcal{G}^-_{2,\theta^-}(b) & J^+_{\tau_2}\mathcal{G}^+_{1,\theta^+}(b) & J^+_{\tau_2}\mathcal{G}^+_{2,\theta^+}(b) & \Pi^+_{\tau_2}(b)-1 & 0\\ J^+_{\tau_3}\mathcal{G}^-_{1,\theta^-}(b) & J^+_{\tau_3}\mathcal{G}^-_{2,\theta^-}(b) & J^+_{\tau_3}\mathcal{G}^+_{1,\theta^+}(b) & J^+_{\tau_3}\mathcal{G}^+_{2,\theta^+}(b) & \Pi^+_{\tau_3}(b)-1 & 0\\ J^-_{\tau_1}\mathcal{G}^+_{1,\theta^-}(b) & J^-_{\tau_1}\mathcal{G}^+_{2,\theta^-}(b) & J^-_{\tau_1}\mathcal{G}^-_{1,\theta^+}(b) & J^-_{\tau_1}\mathcal{G}^-_{2,\theta^+}(b) & 0 & \Pi^-_{\tau_1}(b)-1\\ J^-_{\tau_2}\mathcal{G}^+_{1,\theta^-}(b) & J^-_{\tau_2}\mathcal{G}^+_{2,\theta^-}(b) & J^-_{\tau_2}\mathcal{G}^-_{1,\theta^+}(b) & J^-_{\tau_2}\mathcal{G}^-_{2,\theta^+}(b) & 0 & \Pi^-_{\tau_2}(b)-1\\ J^-_{\tau_3}\mathcal{G}^+_{1,\theta^-}(b) & J^-_{\tau_3}\mathcal{G}^+_{2,\theta^-}(b) & J^-_{\tau_3}\mathcal{G}^-_{1,\theta^+}(b) & J^-_{\tau_3}\mathcal{G}^-_{2,\theta^+}(b) & 0 & \Pi^-_{\tau_3}(b)-1 \end{pmatrix}. \tag{5.22}$$

Proof: Let $G^+, G^- : \mathbb{D} \to \mathbb{C}$ be holomorphic functions, continuous on $\overline{\mathbb{D}}$. We study equation (5.20), and look for holomorphic functions $F^+, F^- : \mathbb{D} \to \mathbb{C}$ whose derivative $\partial_z F$ is continuous on $\overline{\mathbb{D}}$.

Multiplying (5.20) by z^N, and introducing auxiliary unknowns $a_1^\pm, \dots, a_N^\pm \in \mathbb{C}$, we obtain

$$\begin{cases} iz^{N+1}P(z)\partial_z F^+(z) = z^N\big(i+\lambda-i\theta^+P(z)\big)F^+(z) + z^N G^+(z) \\ \qquad\qquad + z^N \sum_{k=1}^N \big(a_k^+ \mathcal{G}^-_{k,\theta^-}(z) + a_k^- \mathcal{G}^+_{k,\theta^+}(z)\big), & (5.23a) \\ iz^{N+1}P(z)\partial_z F^-(z) = z^N\big(i-\lambda-i\theta^-P(z)\big)F^-(z) - z^N G^-(z) \\ \qquad\qquad + z^N \sum_{k=1}^N \big(a_k^- \mathcal{G}^-_{k,\theta^+}(z) + a_k^+ \mathcal{G}^+_{k,\theta^-}(z)\big). & (5.23b) \end{cases}$$

Note that the singular points of (5.23a) and (5.23b) correspond to Z_0.

The equation (5.23a) reduces to a linear first order equation of the form (4.31), where $A = A_{\lambda,\theta^+}$ and

$$B(z) := B_{G^+}(z) + \sum_{k=1}^N a_k^+ B_{\mathcal{G}^-_{k,\theta^-}}(z) + a_k^- B_{\mathcal{G}^+_{k,\theta^+}}(z).$$

In addition, (5.23b) reduces to another linear first order equation of the form (4.31), where $A = A_{-\lambda,\theta^-}$ and

$$B(z) := -B_{G^-}(z) + \sum_{k=1}^N a_k^+ B_{\mathcal{G}^+_{k,\theta^-}}(z) + a_k^- B_{\mathcal{G}^-_{k,\theta^+}}(z).$$

Since Condition 5.7 holds, it follows that (4.28) also holds. Hence, we may apply Lemma 4.10 to both equations.

For each fixed $j = 1, \dots, N+1$, solving (5.23) along τ_j yields unique continuous solutions $\phi_j^+, \phi_j^- : [0, 2\pi] \to \mathbb{C}$. These solutions can be expressed as

$$\phi_j^+(x) = F^+(\omega_0)\Pi^+_{\tau_j}(x) + J^+_{\tau_j}G^+(x) + \sum_{k=1}^N a_k^+ J^+_{\tau_j}\mathcal{G}^-_{k,\theta^-}(x) + a_k^- J^+_{\tau_j}\mathcal{G}^+_{k,\theta^+}(x),$$
$$\phi_j^-(x) = F^-(\omega_0)\Pi^-_{\tau_j}(x) - J^-_{\tau_j}G^-(x) + \sum_{k=1}^N a_k^+ J^-_{\tau_j}\mathcal{G}^+_{k,\theta^-}(x) + a_k^- J^-_{\tau_j}\mathcal{G}^-_{k,\theta^+}(x).$$

Lemma 4.10 states that F^+ and F^- are holomorphic functions on $\mathbb{D}$, continuous on $\overline{\mathbb{D}}$, and satisfying (5.23), if and only if each function $\phi_j^\pm$ satisfies the condition

$$\phi_j^\pm(b) = \phi_j^\pm(0), \qquad j = 1, \dots, N. \tag{5.24}$$

Therefore, F^+ and F^- are holomorphic if and only if

$$0 = F^+(\omega_0)\big(\Pi^+_{\tau_j}(b) - 1\big) + J^+_{\tau_j}G^+(b) + \sum_{k=1}^N a_k^+ J^+_{\tau_j}\mathcal{G}^-_{k,\theta^-}(b) + a_k^- J^+_{\tau_j}\mathcal{G}^+_{k,\theta^+}(b),$$
$$0 = F^-(\omega_0)\big(\Pi^-_{\tau_j}(b) - 1\big) - J^-_{\tau_j}G^-(b) + \sum_{k=1}^N a_k^+ J^-_{\tau_j}\mathcal{G}^+_{k,\theta^-}(b) + a_k^- J^-_{\tau_j}\mathcal{G}^-_{k,\theta^+}(b).$$

Writing these conditions gives $2N+2$ affine linear equations in the unknowns $a_1^{\pm},\dots,a^{\pm},F^{\pm}(\omega_0)$:

$$M_{\mathrm{stab}}^{\mathrm{hol}}\begin{pmatrix} a_1^+ \\ \vdots \\ a_N^+ \\ a_1^- \\ \vdots \\ a_N^- \\ F^+(\omega_0) \\ F^-(\omega_0)\end{pmatrix}+\begin{pmatrix} J_{\tau_1}^+G^+(b) \\ \vdots \\ J_{\tau_{N+1}}^+G^+(b) \\ -J_{\tau_1}^-G^-(b) \\ \vdots \\ -J_{\tau_{N+1}}^-G^-(b)\end{pmatrix}=0.$$

If $M_{\mathrm{stab}}^{\mathrm{hol}}$ is nonsingular, then the system admits a unique solution. Consequently, there exists a unique pair of holomorphic functions $F^+,F^-:\mathbb{D}\to\mathbb{C}$ such that $\partial_z F^+,\partial_z F^-\in C(\overline{\mathbb{D}})$, and satisfy (5.23). By Lemma 5.6, the coefficients satisfy

$$a_k^{\pm}=F_{k-1}^{\pm}\qquad\text{for all } k,$$

and therefore (F^+,F^-) is the unique solution of (5.20).

Conversely, if for every admissible pair $(G^+,G^-)\in\mathcal{H}\oplus\mathcal{H}$ there exists a unique $(F^+,F^-)\in\mathcal{H}\oplus\mathcal{H}$ of (5.20), by setting $(G^+,G^-)=0$, we see that then the homogeneous system associated with $M_{\mathrm{stab}}^{\mathrm{hol}}$ admits only the trivial solution, and hence $M_{\mathrm{stab}}^{\mathrm{hol}}$ is nonsingular.

Finally, assume that there exist d independent terms in the kernel of $M_{\mathrm{stab}}^{\mathrm{hol}}$: $\{v_k\}_{k=1}^d$. For each $k=1,\dots,d$, there exists a pair of functions $(F_k^+,F_k^-)\in\mathcal{H}\oplus\mathcal{H}$, whose derivative extends continuously to $\mathbb{D}$, and satisfying (5.21).

Suppose that these pairs are not independent, so there exists $c\in\mathbb{C}^{2N+2}$, with $c\neq 0$ such that

$$\sum_{k=1}^{d}c_k(F_k^+,F_k^-)=0.$$

Computing the difference between the evaluations at $b=2\pi+2$ and 0, we see that

$$\sum_{k=1}^{d}c_kv_k=0,$$

so we have a contradiction. □

Define

$$\Pi^+(x):=e^{(i+\lambda)\int_0^x\beta(y)\,dy-i\theta^+x}\qquad\text{and}\qquad\Pi^-(x):=e^{(i-\lambda)\int_0^x\beta(y)\,dy-i\theta^-x},\tag{5.25}$$

recall that β is defined by (4.40). Moreover, define the formal operators

$$J^+g(x):=\Pi^+(x)\int_0^x\frac{\beta(y)}{\Pi^+(y)}g(y)\,dy\qquad\text{and}\qquad J^-g(x):=\Pi^-(x)\int_0^x\frac{\beta(y)}{\Pi^-(y)}g(y)\,dy.\tag{5.26}$$

Remark 5.11 *The functions Π^+ and Π^- depend on λ and θ, and the same holds for the operators J^+ and J^-.*

Lemma 5.12 *Let $\Pi^{\pm}$ and $J^{\pm}$ be defined in* (5.25) *and* (5.26)*, respectively.*

For every $k=1,\dots,N$, define

$$g_{k,\theta}^-(x):=(k-\theta)\frac{i}{2}\sum_{m=0}^{N-k}v_{m+k}^{\mathrm{ap}}e^{-i(1+m)x}\qquad\text{and}\qquad g_{k,\theta}^+(x):=(k-\theta)\frac{i}{2}\sum_{m=0}^{N-k}v_{m+k}^{\mathrm{ap}}e^{imx}.\tag{5.27}$$

For each $k = 1, \dots, N$, set

$$\left.\begin{aligned} h_k^{-J^+} &:= J^+ g^-_{k,\theta^-}, & h_k^{+J^+} &:= J^+ g^+_{k,\theta^+}, \\ h_k^{-J^-} &:= J^- g^-_{k,\theta^+}, & h_k^{+J^-} &:= J^- g^+_{k,\theta^-}. \end{aligned}\right\} \tag{5.28}$$

Finally, define the square matrix $M^{\mathrm{tor}}_{\mathrm{stab}}(\lambda)$ of order $2N+2$ by

$$\big(M^{\mathrm{tor}}_{\mathrm{stab}}(\lambda)\big)_{jk} = \begin{cases} \widehat{(h_k^{-J^+})}_{j-2} - \delta_{j-1,k}, & 1 \le j \le N+1, \quad 1 \le k \le N \\ \widehat{(h_{k-N}^{+J^+})}_{j-2}, & 1 \le j \le N+1, \quad N+1 \le k \le 2N \\ \widehat{(\Pi^+)}_{j-2}, & 1 \le j \le N+1, \quad k = 2N+1 \\ \widehat{(h_k^{+J^-})}_{j-N-3}, & N+2 \le j \le 2N+2, \quad 1 \le k \le N \\ \widehat{(h_{k-N}^{-J^-})}_{j-N-3} - \delta_{j-2,k}, & N+2 \le j \le 2N+2, \quad N+1 \le k \le 2N \\ \widehat{(\Pi^-)}_{j-N-3}, & N+2 \le j \le 2N+2, \quad k = 2N+2 \\ 0, & \textit{in other case} \end{cases} . \tag{5.29}$$

Then

$$\dim\big(\ker\big(M^{\mathrm{hol}}_{\mathrm{stab}}(\lambda)\big)\big) \le \dim\big(\ker\big(M^{\mathrm{tor}}_{\mathrm{stab}}(\lambda)\big)\big).$$

Example 5.13 *With $N = 2$ the matrix $M^{\mathrm{tor}}_{\mathrm{stab}}$ would be of order 6. In this case,*

$$M^{\mathrm{tor}}_{\mathrm{stab}} = \begin{pmatrix} \widehat{(h_1^{-J^+})}_{-1} & \widehat{(h_2^{-J^+})}_{-1} & \widehat{(h_1^{+J^+})}_{-1} & \widehat{(h_2^{+J^+})}_{-1} & \widehat{(\Pi^+)}_{-1} & 0 \\ \widehat{(h_1^{-J^+})}_{0} - 1 & \widehat{(h_2^{-J^+})}_{0} & \widehat{(h_1^{+J^+})}_{0} & \widehat{(h_2^{+J^+})}_{0} & \widehat{(\Pi^+)}_{0} & 0 \\ \widehat{(h_1^{-J^+})}_{1} & \widehat{(h_2^{-J^+})}_{1} - 1 & \widehat{(h_1^{+J^+})}_{1} & \widehat{(h_2^{+J^+})}_{1} & \widehat{(\Pi^+)}_{1} & 0 \\ \widehat{(h_1^{+J^-})}_{-1} & \widehat{(h_2^{+J^-})}_{-1} & \widehat{(h_1^{-J^-})}_{-1} & \widehat{(h_2^{-J^-})}_{-1} & 0 & \widehat{(\Pi^-)}_{-1} \\ \widehat{(h_1^{+J^-})}_{0} & \widehat{(h_2^{+J^-})}_{0} & \widehat{(h_1^{-J^-})}_{0} - 1 & \widehat{(h_2^{-J^-})}_{0} & 0 & \widehat{(\Pi^-)}_{0} \\ \widehat{(h_1^{+J^-})}_{1} & \widehat{(h_2^{+J^-})}_{1} & \widehat{(h_1^{-J^-})}_{1} & \widehat{(h_2^{-J^-})}_{1} - 1 & 0 & \widehat{(\Pi^-)}_{1} \end{pmatrix}. \tag{5.30}$$

Proof: The argument is essentially the same as in the proof of Lemma 4.15. Let

$$d := \dim\big(\ker(M^{\mathrm{hol}}_{\mathrm{stab}}(\lambda))\big).$$

Fix a basis of $\ker(M^{\mathrm{hol}}_{\mathrm{stab}}(\lambda))$. By Proposition 5.9, there exist d linearly independent pairs (F^+, F^-) such that $\mathcal{L}^{\mathrm{ap,\,hol}}_{\theta,\lambda}(F^+, F^-) = 0$.

Fix such a pair (F^+, F^-). Then (5.7) is satisfied by $F^+|_{\mathbb{T}}$, $F^-|_{\mathbb{T}}$ with $g^+ = g^- = 0$. Applying Duhamel's formula to both equations yields

$$F^+|_{\mathbb{T}}(x) = \widehat{F^+|_{\mathbb{T}}}(0)\Pi^+(x) + \sum_{k=1}^{N} (\widehat{F^+|_{\mathbb{T}}})_{k-1} h_k^{-J^+}(x) + \sum_{k=1}^{N} (\widehat{F^-|_{\mathbb{T}}})_{k-1} h_k^{+J^+}(x), \tag{5.31a}$$

$$F^-|_{\mathbb{T}}(x) = \widehat{F^-|_{\mathbb{T}}}(0)\Pi^-(x) + \sum_{k=1}^{N} (\widehat{F^+|_{\mathbb{T}}})_{k-1} h_k^{+J^-}(x) + \sum_{k=1}^{N} (\widehat{F^-|_{\mathbb{T}}})_{k-1} h_k^{-J^-}(x). \tag{5.31b}$$

Projecting onto e^{ijx} for $j = -1, \dots, N-1$ and collecting the unknown Fourier coefficients, we obtain a linear system of the form

$$M^{\mathrm{tor}}_{\mathrm{stab}} \mathbf{c}(F^+, F^-) = 0,$$

where

$$\mathbf{c}(F^+,F^-) := \begin{pmatrix} \widetilde{\mathbf{c}}(F^+) \\ \widetilde{\mathbf{c}}(F^-) \\ F^+|_{\mathbb{T}}(0) \\ F^-|_{\mathbb{T}}(0) \end{pmatrix} \qquad \text{and} \qquad \widetilde{\mathbf{c}}(F) := \begin{pmatrix} (\widehat{F|_{\mathbb{T}}})_0 \\ \vdots \\ (\widehat{F|_{\mathbb{T}}})_{N-1}. \end{pmatrix}. \tag{5.32}$$

Since the d pairs (F^+,F^-) are linearly independent, (5.31) imply that the vectors $\mathbf{c}(F^+,F^-)$ are linearly independent as well. Hence $\dim\ker(M^{\mathrm{tor}}_{\mathrm{stab}}) \geq d$. □

Corollary 5.14 *Let $\theta \in (0,1)$ and $\lambda \in \mathbb{C}$. Assume Condition 5.7 holds and*

$$\det\left(M^{\mathrm{tor}}_{\mathrm{stab}}(\lambda)\right) \neq 0.$$

Then, for every holomorphic pair of functions $(G^+,G^-) \in \mathcal{H}\oplus\mathcal{H}$, that are continuous on $\overline{\mathbb{D}}$, there exists a unique pair of holomorphic functions $(F^+,F^-) \in \mathcal{H}\oplus\mathcal{H}$ with $\partial_z F^\pm$ continuous on $\overline{\mathbb{D}}$, such that (5.20) *holds.*

Proof: Since the dimension of $\ker\left(M^{\mathrm{tor}}_{\mathrm{stab}}(\lambda)\right)$ is 0, the same holds for $\ker\left(M^{\mathrm{hol}}_{\mathrm{stab}}(\lambda)\right)$ and Proposition 5.9 applies. □

III.5.2.2 Restricted invertibility of $\mathcal{L}^{\mathrm{ap,\,hol}}_{\theta,\lambda^{\mathrm{ap}}}$

The main idea of this section is to construct a suitable *pseudoinverse* that allows us to invert $\mathcal{L}^{\mathrm{ap,\,hol}}_{\theta,\lambda^{\mathrm{ap}}}$ controlling the norm.

Lemma 5.15 *The dependence of* $\det M^{\mathrm{tor}}_{\mathrm{stab}}$ *on λ is analytic.*

Proof: We show that each entry of $M^{\mathrm{tor}}_{\mathrm{stab}}$, defined in (5.29), is analytic in λ. Let $g:\mathbb{T}\to\mathbb{C}$ be continuous, let $C_0 \in \mathbb{C}$, and consider

$$f(x) := \Pi^+(x)C_0 + J^+g(x),$$

where Π^+ and J^+ are defined in (5.25) and (5.26), respectively.

The function Π^+ is analytic in λ and never vanishes, since it is an exponential function. As for J^+g, observe that it is the product of Π^+ with an antiderivative. Define

$$\Phi(x,\lambda) := \int_0^x \frac{\beta(y)}{\Pi^+(y)} g(y)\,dy.$$

Then, for every $x \in [a,b]$, the map $\Phi(x,\cdot)$ is holomorphic on $\mathbb{C}$. Indeed, for any triangular contour $\partial\Delta \subset \mathbb{C}$, continuity and boundedness on $[0,x]\times\partial\Delta$ allow us to interchange the order of integration, yielding

$$\int_{\partial\Delta} \Phi(x,\lambda)\,d\lambda = \int_0^x \int_{\partial\Delta} \frac{\beta(y)}{\Pi^+(y)} g(y)\,d\lambda\,dy = 0.$$

Hence, by Morera's theorem, $\Phi(x,\cdot)$ is holomorphic.

Consequently, the function f depends analytically on λ. Applying Morera's theorem once more, we obtain

$$\int_{\partial\Delta} \widehat{f}_k\,d\lambda = \frac{1}{2\pi}\int_{\partial\Delta}\int_{-\pi}^{\pi} e^{-ikx} f(x)\,dx\,d\lambda = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-ikx}\int_{\partial\Delta} f(x)\,d\lambda\,dx = 0.$$

Therefore, $\widehat{f}_k$ is holomorphic. □

Lemma 5.16 *Let $\theta = \frac{1}{3}$. Then there exists $\lambda \in \mathbb{C}$ such that*

$$\det\left(M_{\text{stab}}^{\text{tor}}(\lambda)\right) \neq 0.$$

Proof: The proof is computer-assisted and the code can be found in the supplementary material. We refer to the Appendix III.E for details about the implementation. □

Corollary 5.17 *The function $\det M_{\text{stab}}^{\text{tor}} : \mathbb{C} \to \mathbb{C}$ has only isolated zeros for $\theta = \frac{1}{3}$.*

Proof: By Lemma 5.15, the function $\det M_{\text{stab}}^{\text{tor}}$ is holomorphic. If it had a non-isolated zero, then by the Identity Theorem for holomorphic functions it would vanish identically on $\mathbb{C}$. However, there exists $\lambda \in \mathbb{C}$ such that $\det\left(M_{\text{stab}}^{\text{tor}}(\lambda)\right) \neq 0$, a contradiction. □

Definition 5.18 *Let $\theta \in (0,1)$ and $\lambda \in \mathbb{C}$. Let σ_1 be the smallest singular value of $M_{\text{stab}}^{\text{tor}}(\lambda)$ and assume that σ_1 has multiplicity one. Let $u_1, v_1 \in \mathbb{C}^{2N+2}$ be left and right singular vectors corresponding to σ_1, normalized so that $\|u_1\| = \|v_1\| = 1$. Then*

$$M_{\text{stab}}^{\text{tor}} v_1 = \sigma_1 u_1, \qquad M_{\text{stab}}^{\text{tor}\ *} u_1 = \sigma_1 v_1.$$

Let $\mathcal{U}_1 := \operatorname{span}(u_1)$, $\mathcal{V}_1 := \operatorname{span}(v_1)$ and let $P_{\mathcal{U}_1} : \mathbb{C}^{2N+2} \to \mathcal{U}_1$, $P_{\mathcal{V}_1} : \mathbb{C}^{2N+2} \to \mathcal{V}_1$ be the orthogonal projections. In particular, it holds that

$$P_{\mathcal{U}_1} = u_1 u_1^*, \qquad P_{\mathcal{V}_1} = v_1 v_1^*.$$

For any pair $(G^+, G^-) \in \mathcal{H}^0 \oplus \mathcal{H}^0$, define the vector $\mathbf{b}(G^+, G^-) \in \mathbb{C}^{2N+2}$ by

$$\mathbf{b}(G^+, G^-) := \begin{pmatrix} \mathbf{b}^+(G^+) \\ -\mathbf{b}^-(G^-) \end{pmatrix} \qquad \text{where} \quad \mathbf{b}^\pm(G^\pm) := \begin{pmatrix} (\widehat{J^\pm G^\pm|_{\mathbb{T}}})_{-1} \\ \vdots \\ (\widehat{J^\pm G^\pm|_{\mathbb{T}}})_{N-1} \end{pmatrix}. \tag{5.33}$$

Let $(G_{u_1}^+, G_{u_1}^-) \in \mathcal{H} \oplus \mathcal{H}$ be such that

$$u_1^* \mathbf{b}(G_{u_1}^+, G_{u_1}^-) \neq 0. \tag{5.34}$$

We define the operator

$$\mathcal{P}_\lambda[G_{u_1}^+, G_{u_1}^-] : \mathcal{H} \oplus \mathcal{H} \longrightarrow \mathcal{H} \oplus \mathcal{H}$$

by

$$\mathcal{P}_\lambda[G_{u_1}^+, G_{u_1}^-](G^+, G^-) := (G^+, G^-) - \frac{u_1^* \mathbf{b}(G^+, G^-)}{u_1^* \mathbf{b}(G_{u_1}^+, G_{u_1}^-)} (G_{u_1}^+, G_{u_1}^-).$$

Remark 5.19 *We write $\mathcal{P}_\lambda[G_{u_1}^+, G_{u_1}^-]$ to emphasize the dependence on λ. In fact, both u_1 and $\mathbf{b}$ also depend on λ.*

Remark 5.20 *Let $\lambda \in \mathbb{C}$ and let $\sigma_1(\lambda)$ denote the smallest singular value of $M_{\text{stab}}^{\text{tor}}(\lambda)$, and assume that $\sigma_1(\lambda)$ has multiplicity 1. The quantity $\sigma_1(\lambda)$ may be very small. In fact, we do not prove that it is nonzero for $\lambda = \lambda^{\text{ap}}$.*

Lemma 5.21 *Let $\lambda_0 \in \mathbb{C}$ and let $\overline{\sigma}_1, \underline{\sigma}_2 > 0$, with $\overline{\sigma}_1 < \underline{\sigma}_2$. Let V^{λ_0} be a neighborhood of λ_0 and let $M : V^{\lambda_0} \to \mathbb{C}^{r \times r}$ be a continuous map.*

Let

$$\sigma_1(M(\lambda)) \leq \cdots \leq \sigma_r(M(\lambda))$$

denote the singular values of $M(\lambda)$, counted with multiplicity. Assume that

$$\sigma_1(M(\lambda)) \le \overline{\sigma}_1, \qquad \sigma_2(M(\lambda)) \ge \underline{\sigma}_2$$

for all $\lambda \in V^{\lambda_0}$.

Then the maps

$$\lambda \mapsto P_{\mathcal{U}_1(\lambda)}, \qquad \lambda \mapsto P_{\mathcal{V}_1(\lambda)}$$

are continuous on V^{λ_0}.

Moreover, the map defined by

$$M^{\dagger}_{\mathrm{reg}}(\lambda) := M(\lambda)^*\big(M(\lambda)M(\lambda)^* + P_{\mathcal{U}_1(\lambda)}\big)^{-1}(I - P_{\mathcal{U}_1(\lambda)}) \tag{5.35}$$

is well defined and continuous on V^{λ_0} and satisfies, for all $\lambda \in V^{\lambda_0}$,

$$P_{\mathcal{V}_1} M^{\dagger}_{\mathrm{reg}} = 0 \qquad \text{and} \qquad M M^{\dagger}_{\mathrm{reg}} = I - P_{\mathcal{U}_1}. \tag{5.36}$$

Furthermore, the following estimate holds

$$\|M^{\dagger}_{\mathrm{reg}}(\lambda)u\|_{\ell^2} \le \frac{1}{\underline{\sigma}_2}\|u\|_{\ell^2} \quad \text{for all } \lambda \in V^{\lambda_0}. \tag{5.37}$$

Proof: The singular values of M may be computed as the nonnegative square roots of the eigenvalues of M^*M. By Theorem 5.5 from [Kat80, Chapter 2, §5.1], we know that the rank-one spectral projectors $\lambda \mapsto P_{\mathcal{U}_1(\lambda)}$ and $\lambda \mapsto P_{\mathcal{V}_1(\lambda)}$ are continuous on V^{λ_0}.

In addition, note that the matrix given by

$$MM^* + P_{\mathcal{U}_1}$$

is also continuous on V^{λ_0}.

For each fixed λ, take $M(\lambda) = U\Sigma V^*$. Then $P_{\mathcal{U}_1} = u_1 u_1^*$ and $P_{\mathcal{V}_1} = v_1 v_1^*$ for some unit vectors $u_1, v_1 \in \mathbb{C}^r$.

Therefore,

$$\det\left(MM^* + P_{\mathcal{U}_1}\right) = \det\left(\Sigma\Sigma^* + E_{11}\right) \ge \prod_{k=2}^{r} \sigma_k^2 \ge \underline{\sigma}_2^{2(r-1)} > 0,$$

where E_{11} denotes the square matrix of order r consisting of all zeros except for the $(1,1)$ entry, where it has a 1. Hence, $MM^* + P_{\mathcal{U}_1}$ is invertible for all $\lambda \in D_\delta(z_0)$, and its inverse depends continuously on λ.

Consequently, the function $M^{\dagger}_{\mathrm{reg}}(\lambda)$ from (5.35) is well defined and continuous on V^{λ_0}.

We now prove the identities in (5.36). The key observation is that, after passing to singular-vector coordinates, all the relevant middle factors are diagonal. Fix $\lambda \in V^{\lambda_0}$. Then

$$P_{\mathcal{V}_1} M^{\dagger}_{\mathrm{reg}} = P_{\mathcal{V}_1} M^*\big(MM^* + P_{\mathcal{U}_1}\big)^{-1}(I - P_{\mathcal{U}_1}) = V E_{11}\Sigma^*\big(\Sigma\Sigma^* + E_{11}\big)^{-1}(I - E_{11})U^* = 0.$$

Moreover,

$$\begin{aligned} M M^{\dagger}_{\mathrm{reg}} = MM^*\big(MM^* + P_{\mathcal{U}_1}\big)^{-1}(I - P_{\mathcal{U}_1}) &= U\Sigma\Sigma^*\big(\Sigma\Sigma^* + E_{11}\big)^{-1}(I - E_{11})U^* \\ &= U(I - E_{11})U^* = I - P_{\mathcal{U}_1}. \end{aligned}$$

Finally, (5.37) follows by applying the same conjugation argument for fixed λ

$$\begin{aligned} \|M^{\dagger}_{\mathrm{reg}}u\|^2_{\ell^2} &= \|M^*\big(MM^* + P_{\mathcal{U}_1}\big)^{-1}(I - P_{\mathcal{U}_1})u\|^2 = \|V\Sigma^*\big(\Sigma\Sigma^* + E_{11}\big)^{-1}(I - E_{11})U^*u\|^2 \\ &= \|\Sigma^*\big(\Sigma\Sigma^* + E_{11}\big)^{-1}(I - E_{11})U^*u\|^2 \le \sum_{k=2}^{r} \frac{1}{\sigma_k^2}|(U^*u)_k|^2 \le \frac{1}{\sigma_2^2}\|U^*u\|^2_{\ell^2} \le \frac{1}{\underline{\sigma}_2^2}\|u\|^2_{\ell^2}. \end{aligned}$$

□

Lemma 5.22 *Let $\theta = \frac{1}{3}$ and let $\sigma_1 \leq \cdots \leq \sigma_{2N+2}$ denote the singular values of $M^{\mathrm{tor}}_{\mathrm{stab}}(\lambda^{\mathrm{ap}})$, counted with multiplicity. Then it holds that*

$$\sigma_1^2 < \overline{\sigma}_1, \qquad \sigma_2^2 > \underline{\sigma}_2,$$

where

$$\overline{\sigma}_1 := 10^{-7}, \qquad \underline{\sigma}_2 := 1.31011 \cdot 10^{-3}.$$

Proof: The proof is computer-assisted and the code can be found in the supplementary material. We refer to the Appendix III.E for details about the implementation. □

Recall the approximate eigenvector f^{ap} of $\mathcal{L}_\theta^{\mathrm{ap}}$, see (3.9). We choose $(G^+_{u_1}, G^-_{u_1})$ in Definition 5.18 to be (F^+, F^-), where (F^+, F^-) are the functions introduced in Definition 5.2 with $f = f^{\mathrm{ap}}$.

The following lemma shows that this choice satisfies (5.34) for every λ sufficiently close to λ^{ap}. Moreover, we prove that the operator $\mathcal{P}_\lambda$ is continuous with respect to λ.

Lemma 5.23 *Let $\theta = \frac{1}{3}$ and let $(F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-})$ be the pair from Definition 5.2 corresponding to $f = f^{\mathrm{ap}}$, defined in* (3.9)*. It holds that*

$$\left|u_1(\lambda^{\mathrm{ap}})^* \mathbf{b}(F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-})\right| > 4.54415 \cdot 10^{-2},$$

b *is computed at $\lambda = \lambda^{\mathrm{ap}}$.*

Proof: The proof is computer-assisted and the code can be found in the supplementary material. We refer to the Appendix III.E for details about the implementation. □

Corollary 5.24 *Let $\theta = \frac{1}{3}$, let $(F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-})$ as in Lemma 5.23 and let $\mathcal{P}_\lambda[F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-}]$ be the operator from Definition 5.18.*

Then there is a choice for u_1 in terms of λ that makes the operator $\mathcal{P}_\lambda[F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-}] : \mathcal{H} \oplus \mathcal{H} \to \mathcal{H} \oplus \mathcal{H}$ continuous in a neighborhood $V^{\lambda^{\mathrm{ap}}}$ of λ^{ap}. In particular, it holds that

$$u_1(\lambda)^* \mathbf{b}\big(\mathcal{P}_\lambda[F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-}](G^+, G^-)\big) = 0 \tag{5.38}$$

for all $(G^+, G^-) \in \mathcal{H} \oplus \mathcal{H}$ and $\lambda \in V^{\lambda^{\mathrm{ap}}}$.

Proof: We know from Lemma 5.21 that the projection map $P_{\mathcal{U}_1}$ is continuous at λ^{ap}. Notice that the singular vector $u_1(\lambda)$, even if normalized, is only unique up to multiplication by a complex scalar of unit modulus. Let us kill this invariance. Fix a singular vector $u_1(\lambda^{\mathrm{ap}})$ and take a normalized $u_1(\lambda)$. We define $u_1^\sharp(\lambda)$ as

$$u_1^\sharp(\lambda) := \frac{\widetilde{u}_1(\lambda)}{\|\widetilde{u}_1(\lambda)\|_{\ell^2}},$$

where

$$\widetilde{u}_1(\lambda) := P_{\mathcal{U}_1(\lambda)} u_1(\lambda^{\mathrm{ap}}).$$

For the sake of notational simplicity we will denote $u_1(\lambda) := u_1^\sharp(\lambda)$.

By construction, the only possible obstruction to continuity is that $\widetilde{u}_1$ vanishes. However, we have that

$$\|\widetilde{u}_1(\lambda)\|_{\ell^2} \geq \|\widetilde{u}_1(\lambda^{\mathrm{ap}})\|_{\ell^2} - \|\widetilde{u}_1(\lambda) - \widetilde{u}_1(\lambda^{\mathrm{ap}})\|_{\ell^2} \geq 1 - \|P_{\mathcal{U}_1(\lambda)} - P_{\mathcal{U}_1(\lambda^{\mathrm{ap}})}\|.$$

Since the projection is continuous in λ, we can find a suitable neighborhood $V^{\lambda^{\mathrm{ap}}}$ of λ^{ap} such that

$$\|P_{\mathcal{U}_1(\lambda)} - P_{\mathcal{U}_1(\lambda^{\mathrm{ap}})}\| < 1 \quad \text{for all } \lambda \in V^{\lambda^{\mathrm{ap}}}.$$

Consequently, u_1 is continuous on $V^{\lambda^{\mathrm{ap}}}$. Note that the dependence of $\mathbf{b}(G^+, G^-)$ on λ is also continuous so we can shrink the neighborhood to ensure that

$$\left|u_1(\lambda)^* \mathbf{b}(F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-})\right| \neq 0 \quad \text{for all } \lambda \in V^{\lambda^{\mathrm{ap}}}.$$

In conclusion, the operator $\mathcal{P}_\lambda[F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-}]$ is continuous on $V^{\lambda^{\mathrm{ap}}}$.
Finally, a straightforward computation shows (5.38). □

Lemma 5.25 *Condition 5.7 holds for $\theta = \frac{1}{3}$ and $\lambda = \lambda^{\mathrm{ap}}$.*

Proof: The proof is computer-assisted and the code can be found in the supplementary material. We refer to the Appendix III.E for details about the implementation. □

Lemma 5.26 *Let $\theta = \frac{1}{3}$ and let $(F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-})$ as in Lemma 5.23, and $\mathcal{P}_\lambda[F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-}]$ be the operator defined in Definition 5.18.*

For every pair $(G^+, G^-) \in \mathcal{H} \oplus \mathcal{H}$ of functions, with $G^\pm$ continuous on $\overline{\mathbb{D}}$, there exists a unique pair $(F^+, F^-) \in \mathcal{H} \oplus \mathcal{H}$ whose derivatives are continuous on $\overline{\mathbb{D}}$ such that

$$\mathcal{L}^{\mathrm{ap,\,hol}}_{\theta,\lambda^{\mathrm{ap}}}(F^+, F^-) = \mathcal{P}_{\lambda^{\mathrm{ap}}}[F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-}](G^+, G^-), \tag{5.39}$$

and

$$M^{\mathrm{tor}}_{\mathrm{stab}}(\lambda^{\mathrm{ap}})\mathbf{c}(F^+, F^-) = -\mathbf{b}\mathcal{P}_{\lambda^{\mathrm{ap}}}[F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-}](G^+, G^-), \qquad P_{\mathcal{V}_1(\lambda^{\mathrm{ap}})}\mathbf{c}(F^+, F^-) = 0. \tag{5.40}$$

Here **c** *and* **b** *are defined at* (5.32) *and* (5.33) *respectively.*

Proof: By Corollary 5.17, we know that there exists a neighborhood $V^{\lambda^{\mathrm{ap}}}$ of λ^{ap} such that $\det M^{\mathrm{tor}}_{\mathrm{stab}}$ never vanishes on $V^{\lambda^{\mathrm{ap}}} - \{\lambda^{\mathrm{ap}}\}$. Moreover, from Corollary 5.24 and Lemma 5.25, we can choose this neighborhood small enough so that $\mathcal{P}_\lambda[F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-}]$ is continuous and Condition 5.7 is satisfied for $\lambda \in V^{\lambda^{\mathrm{ap}}}$.

Fix $\lambda \in V^{\lambda^{\mathrm{ap}}} - \{\lambda^{\mathrm{ap}}\}$ and apply Corollary 5.14 to $\mathcal{P}_\lambda[F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-}](G^+, G^-)$ to obtain $(F_\lambda^+, F_\lambda^-) \in \mathcal{H} \oplus \mathcal{H}$, with derivatives continuous on $\overline{\mathbb{D}}$, such that (5.39) holds.

We have that

$$\begin{cases} F_\lambda^+|_{\mathbb{T}}(x) = F_\lambda^+|_{\mathbb{T}}(0)\Pi_\lambda^+(x) + J_\lambda^+ \widetilde{G}_\lambda^+|_{\mathbb{T}}(x) \\ \qquad\qquad + \displaystyle\sum_{k=1}^{N} (\widehat{F_\lambda^+|_{\mathbb{T}}})_{k-1} h_k^{-J_\lambda^+}(x) + \sum_{k=1}^{N} (\widehat{F_\lambda^-|_{\mathbb{T}}})_{k-1} h_k^{+J_\lambda^+}(x), & \text{(5.41a)} \\ F_\lambda^-|_{\mathbb{T}}(x) = F_\lambda^-|_{\mathbb{T}}(0)\Pi_\lambda^-(x) - J_\lambda^- \widetilde{G}_\lambda^-|_{\mathbb{T}}(x) \\ \qquad\qquad + \displaystyle\sum_{k=1}^{N} (\widehat{F_\lambda^+|_{\mathbb{T}}})_{k-1} h_k^{+J_\lambda^-}(x) + \sum_{k=1}^{N} (\widehat{F_\lambda^-|_{\mathbb{T}}})_{k-1} h_k^{-J_\lambda^-}(x), & \text{(5.41b)} \end{cases}$$

we are emphasizing the quantities that depend on λ and $(\widetilde{G}_\lambda^+, \widetilde{G}_\lambda^-)$ is given by

$$(\widetilde{G}_\lambda^+, \widetilde{G}_\lambda^-) := \mathcal{P}_\lambda[F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-}](G^+, G^-).$$

Projecting both equations onto e^{ikx}, $k = -1, \ldots, N-1$, yields

$$M^{\mathrm{tor}}_{\mathrm{stab}}(\lambda)\mathbf{c}(F_\lambda^+, F_\lambda^-) + \mathbf{b}_\lambda(\widetilde{G}_\lambda^+, \widetilde{G}_\lambda^-) = 0. \tag{5.42}$$

Moreover, by (5.38) we know that

$$P_{\mathcal{U}_1(\lambda)}\mathbf{b}_\lambda(\widetilde{G}_\lambda^+, \widetilde{G}_\lambda^-) = 0.$$

Since $\det\left(M^{\mathrm{tor}}_{\mathrm{stab}}(\lambda)\right) \neq 0$, we know that $\mathbf{c}(F_\lambda^+, F_\lambda^-)$ is the only vector satisfying (5.42).

By Lemma 5.21, with $M = M^{\mathrm{tor}}_{\mathrm{stab}}$, we know that

$$\mathbf{c}(\lambda) := -M^{\dagger}_{\mathrm{reg}}(\lambda)\mathbf{b}_\lambda(\widetilde{G}^+_\lambda, \widetilde{G}^-_\lambda)$$

is continuous in λ. Moreover,

$$P_{\mathcal{V}_1}\mathbf{c}(\lambda) = 0 \qquad \text{and} \qquad M^{\mathrm{tor}}_{\mathrm{stab}}\mathbf{c}(\lambda) = -\mathbf{b}_\lambda(\widetilde{G}^+_\lambda, \widetilde{G}^-_\lambda).$$

Consequently, we have that

$$\mathbf{c}(F^+_\lambda, F^-_\lambda) = \mathbf{c}(\lambda).$$

Therefore, $(F^+_\lambda|_{\mathbb{T}}, F^-_\lambda|_{\mathbb{T}})$ is continuous in λ for all $\lambda \in V^{\lambda^{\mathrm{ap}}} - \{\lambda^{\mathrm{ap}}\}$, since (5.41) allows us to express these functions as a sum of continuous terms. By the dominated convergence theorem, for every $k < 0$ we obtain:

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} F^{\pm}_{\lambda^{\mathrm{ap}}}|_{\mathbb{T}}(x)e^{-ikx}\,dx = \frac{1}{2\pi}\int_{-\pi}^{\pi} \lim_{\lambda\to\lambda^{\mathrm{ap}}} F^{\pm}_{\lambda}|_{\mathbb{T}}(x)e^{-ikx}\,dx = \lim_{\lambda\to\lambda^{\mathrm{ap}}} \frac{1}{2\pi}\int_{-\pi}^{\pi} F^{\pm}_{\lambda}|_{\mathbb{T}}(x)e^{-ikx}\,dx = 0.$$

Hence, the functions $F^+_\lambda|_{\mathbb{T}}$ and $F^-_\lambda|_{\mathbb{T}}$ admit a holomorphic extension to $\mathbb{D}$ even for $\lambda = \lambda^{\mathrm{ap}}$.

Finally, regarding uniqueness, assume that $(F^+_{\lambda^{\mathrm{ap}}}, F^-_{\lambda^{\mathrm{ap}}}) \in \mathcal{H} \oplus \mathcal{H}$ satisfies

$$\mathcal{L}^{\mathrm{ap,\,hol}}_{\theta,\lambda^{\mathrm{ap}}}(F^+_{\lambda^{\mathrm{ap}}}, F^-_{\lambda^{\mathrm{ap}}}) = 0, \qquad \big(\mathbf{c}(F^+_{\lambda^{\mathrm{ap}}}, F^-_{\lambda^{\mathrm{ap}}}), v_1(\lambda^{\mathrm{ap}})\big) = 0.$$

Projecting the corresponding identities as above yields

$$M^{\mathrm{tor}}_{\mathrm{stab}}(\lambda^{\mathrm{ap}})\mathbf{c}(F^+_{\lambda^{\mathrm{ap}}}, F^-_{\lambda^{\mathrm{ap}}}) = 0.$$

By Lemma 5.22, we know that

$$\sigma_2(M^{\mathrm{tor}}_{\mathrm{stab}}(\lambda^{\mathrm{ap}})) \geq \underline{\sigma}_2 > 0,$$

which implies that $\dim\big(\ker(M^{\mathrm{tor}}_{\mathrm{stab}}(\lambda^{\mathrm{ap}}))\big) \leq 1$. Hence

$$\mathbf{c}(F^+_{\lambda^{\mathrm{ap}}}, F^-_{\lambda^{\mathrm{ap}}}) = \alpha v_1(\lambda^{\mathrm{ap}})$$

for some $\alpha \in \mathbb{C}$. Taking the inner product with $v_1(\lambda^{\mathrm{ap}})$ and using the constraint

$$P_{\mathcal{V}_1(\lambda^{\mathrm{ap}})}\mathbf{c}(F^+_{\lambda^{\mathrm{ap}}}, F^-_{\lambda^{\mathrm{ap}}}) = 0$$

yields $\alpha = 0$, so $\mathbf{c}(F^+_{\lambda^{\mathrm{ap}}}, F^-_{\lambda^{\mathrm{ap}}}) = 0$. □

III.5.2.3 Estimates of the explicit functions and operators

The next step is to find an explicit estimate for the restricted inverse. To this end, we establish several bounds in Lemmas 5.27–5.30.

Lemma 5.27 *Let κ^+_3 and κ^-_3 be the functions given by*

$$\kappa^{\pm}_3(x) := e^{\pm 2\mathrm{Re}(\lambda^{\mathrm{ap}})\int_0^x \beta(y)\,dy}\int_0^x \beta(y)^2 e^{\mp 2\mathrm{Re}(\lambda^{\mathrm{ap}})\int_0^y \beta(s)\,ds}\,dy.$$

Recall (5.25)*, so that*

$$|\Pi^{\pm}(x)| = e^{\pm \mathrm{Re}(\lambda^{\mathrm{ap}})\int_0^x \beta(y)\,dy}.$$

Then

$$\|\kappa^{\pm}_3\|_{L^1([0,\pi])} < 14.0518.$$

Proof: The proof is computer-assisted and the code can be found in the supplementary material. We refer to the Appendix III.E for details about the implementation. □

Lemma 5.28 *Let $\theta = \frac{1}{3}$. Given a function $\kappa : [0,\pi] \to \mathbb{R}$, define the functions $\kappa^+_{4,\kappa}$ and $\kappa^-_{4,\kappa}$ as*

$$\kappa^\pm_{4,\kappa}(x) := \left|(i \pm \lambda^{\mathrm{ap}})\beta(x) \mp i\theta\right|^2 \kappa(x).$$

Let κ^+_3 and κ^-_3 as in Lemma 5.27. Then the following estimates hold

$$\|\kappa^+_{4,\kappa^\pm_3}\|_{L^1([0,\pi])} < 524.203, \qquad \|\kappa^-_{4,\kappa^\pm_3}\|_{L^1([0,\pi])} < 782.958.$$

Proof: The proof is computer-assisted and the code can be found in the supplementary material. We refer to the Appendix III.E for details about the implementation. □

Lemma 5.29 *Let $g \in L^2([-\pi,\pi])$, $\lambda = \lambda^{\mathrm{ap}}$, and $\theta = \frac{1}{3}$. Then*

$$\|J^\pm g\|^2_{L^2([-\pi,\pi])} < 14.0518\, \|g\|^2_{L^2([-\pi,\pi])}.$$

Recall that $J^\pm$ is defined in (5.26).

Proof: We first note that

$$|\Pi^\pm(x)| = \exp\left(\pm \mathrm{Re}(\lambda^{\mathrm{ap}}) \int_0^x \beta(y)\, dy\right)$$

It follows that, on $[0,\pi]$,

$$\begin{aligned}\|J^+ g\|^2_{L^2([0,\pi])} &= \int_0^\pi |\Pi^+(x)|^2 \left|\int_0^x \frac{\beta(y)}{\Pi^+(y)} g(y)\, dy\right|^2 dx \\ &\le \int_0^\pi |\Pi^+(x)|^2 \int_0^x \frac{\beta(y)^2}{|\Pi^+(y)|^2}\, dy \int_0^x |g(y)|^2\, dy\, dx = \int_0^\pi \kappa^+_3(x) \|g\|^2_{L^2([0,x])}\, dx. \quad (5.43)\end{aligned}$$

where κ^+_3 is defined in Lemma 5.27.

Hence, applying Fubini's theorem to (5.43), we obtain

$$\|J^+ g\|^2_{L^2([0,\pi])} \le \int_0^\pi |g(y)|^2 \int_y^\pi \kappa^+_3(x)\, dx\, dy \le \|\kappa^+_3\|_{L^1([0,\pi])} \|g\|^2_{L^2([0,\pi])}.$$

An analogous argument for $x \in [-\pi, 0]$ yields

$$\|J^+ g\|^2_{L^2([-\pi,0])} \le \|\kappa^-_3\|_{L^1([0,\pi])} \|g\|^2_{L^2([-\pi,0])}.$$

Consequently,

$$\begin{aligned}\|J^+ g\|^2_{L^2([-\pi,\pi])} &\le \|\kappa^+_3\|_{L^1([0,\pi])} \|g\|^2_{L^2([0,\pi])} + \|\kappa^-_3\|_{L^1([0,\pi])} \|g\|^2_{L^2([-\pi,0])} \\ &\le \max\left\{\|\kappa^+_3\|_{L^1([0,\pi])}, \|\kappa^-_3\|_{L^1([0,\pi])}\right\} \|g\|^2_{L^2([-\pi,\pi])}.\end{aligned}$$

Finally, for J^- the roles of $\|\kappa^+_3\|_{L^1([0,\pi])}$ and $\|\kappa^-_3\|_{L^1([0,\pi])}$ are exchanged. Hence, we obtain the same estimate. □

Lemma 5.30 *Let $g \in L^2([-\pi,\pi])$, $\lambda = \lambda^{\mathrm{ap}}$, and $\theta = \frac{1}{3}$. Then*

$$\begin{aligned}\|\partial_x J^+ g\|_{L^2([-\pi,\pi])} &< 42.8011\, \|g\|_{L^2([-\pi,\pi])}, \\ \|\partial_x J^- g + i J^- g\|_{L^2([-\pi,\pi])} &< 47.8870\, \|g\|_{L^2([-\pi,\pi])}.\end{aligned}$$

Proof: First assume that $g \in C^0(\mathbb{T})$, so that $\partial_x J^{\pm} g$ is well defined pointwise. Since continuous functions are dense in $L^2([-\pi,\pi])$, the estimate extends to all $g \in L^2([-\pi,\pi])$.

Differentiating $J^{\pm} g$ yields

$$\partial_x J^{\pm} g = \big((i \pm \lambda^{\mathrm{ap}})\beta - i\theta^{\pm}\big) J^{\pm} g + \beta g.$$

Define

$$I_{1^+} := \|((i+\lambda^{\mathrm{ap}})\beta - i\theta^+)J^+ g\|_{L^2([-\pi,\pi])}, \qquad I_2 := \|\beta g\|_{L^2([-\pi,\pi])},$$
$$I_{1^-} := \|((i-\lambda^{\mathrm{ap}})\beta + i\theta^+)J^- g\|_{L^2([-\pi,\pi])}.$$

Thus, we estimate

$$\|\partial_x J^+ g\|_{L^2([-\pi,\pi])} \le I_{1^+} + I_2,$$
$$\|\partial_x J^- g + i J^- g\|_{L^2([-\pi,\pi])} \le I_{1^-} + I_2.$$

It is straightforward that

$$I_2 \le \|\beta\|_{L^\infty([-\pi,\pi])} \|g\|_{L^2([-\pi,\pi])}. \tag{5.44}$$

Lemma 4.18 provides an estimate for $\|\beta\|_{L^\infty([-\pi,\pi])}$.

To estimate I_{1^+}, recall (5.43),

$$\begin{aligned}
\|((i+\lambda^{\mathrm{ap}})\beta - i\theta^+)J^+ g\|^2_{L^2([0,\pi])} &= \int_0^\pi \left|((i+\lambda^{\mathrm{ap}})\beta(x) - i\theta^+)J^+ g(x)\right|^2 dx \\
&\le \int_0^\pi \underbrace{\left|(i+\lambda^{\mathrm{ap}})\beta(x) - i\theta^+\right|^2 \kappa_3^+(x)}_{\kappa^+_{4,\kappa_3^+}(x)} \|g\|^2_{L^2([0,x])}\, dx \\
&\le \int_0^\pi |g(y)|^2 \int_y^\pi \kappa^+_{4,\kappa_3^+}(x)\, dx\, dy \\
&\le \|g\|^2_{L^2([0,\pi])} \|\kappa^+_{4,\kappa_3^+}\|_{L^1([0,\pi])}.
\end{aligned} \tag{5.45}$$

By Lemma 5.28, we have an explicit bound for (5.45).

Now, for $x \in [-\pi, 0]$,

$$\|((i+\lambda^{\mathrm{ap}})\beta - i\theta^+)J^+ g\|^2_{L^2([-\pi,0])} \le \|g\|^2_{L^2([-\pi,0])} \|\kappa^+_{4,\kappa_3^-}\|_{L^1([0,\pi])}. \tag{5.46}$$

Therefore, we get from (5.45) and (5.46) that

$$I^2_{1^+} \le \max\left\{\|\kappa^+_{4,\kappa_3^+}\|_{L^1([0,\pi])}, \|\kappa^+_{4,\kappa_3^-}\|_{L^1([0,\pi])}\right\} \|g\|^2_{L^2([-\pi,\pi])}.$$

Combining these estimates yields the first bound in the statement. The second one follows in the same way

$$I^2_{1^-} \le \max\left\{\|\kappa^-_{4,\kappa_3^-}\|_{L^1([0,\pi])}, \|\kappa^-_{4,\kappa_3^+}\|_{L^1([0,\pi])}\right\} \|g\|^2_{L^2([-\pi,\pi])}.$$

□

Lemma 5.31 *The following estimates hold for $\lambda = \lambda^{\mathrm{ap}}$ and $\theta = \frac{1}{3}$,*

$$\|\Pi^+\|^2_{L^2([-\pi,\pi])} + \sum_{k=1}^N \|J^+ g^-_{k,\theta^-}\|^2_{L^2([-\pi,\pi])} + \|J^+ g^+_{k,\theta^+}\|^2_{L^2([-\pi,\pi])} \le 9.61596,$$

$$\|\Pi^-\|^2_{L^2([-\pi,\pi])} + \sum_{k=1}^N \|J^- g^-_{k,\theta^+}\|^2_{L^2([-\pi,\pi])} + \|J^- g^+_{k,\theta^-}\|^2_{L^2([-\pi,\pi])} \le 9.59380,$$

$$\|\partial_x\Pi^+\|^2_{L^2([-\pi,\pi])} + \sum_{k=1}^N \|\partial_x J^+ g^-_{k,\theta^-}\|^2_{L^2([-\pi,\pi])} + \|\partial_x J^+ g^+_{k,\theta^+}\|^2_{L^2([-\pi,\pi])} \le 247.212,$$

$$\|\widetilde{\partial_x}\Pi^-\|^2_{L^2([-\pi,\pi])} + \sum_{k=1}^N \|\widetilde{\partial_x} J^- g^-_{k,\theta^+}\|^2_{L^2([-\pi,\pi])} + \|\widetilde{\partial_x} J^- g^+_{k,\theta^-}\|^2_{L^2([-\pi,\pi])} \le 283.513,$$

where $\widetilde{\partial_x} f := \partial_x f + if$.

Proof: The proof is computer-assisted and the code can be found in the supplementary material. We refer to the Appendix III.E for details about the implementation. □

We introduce a new operator, which will be useful to specify the space in which we are working. For every $s \in \mathbb{R}$, let $i_H : H^s(\mathbb{T}) \to \mathcal{H}^s \oplus \mathcal{H}^s$ be the operator given by

$$i_H(g) := \Big(\sum_{k\geq 0} \widehat{g}_k z^k, \sum_{k\geq 0} \widehat{g}_{-k-1} z^k \Big). \tag{5.47}$$

Define $\mathcal{W}_{\lambda^{\mathrm{ap}}} \subset \mathcal{H}^0 \oplus \mathcal{H}^0$ and $W_{\lambda^{\mathrm{ap}}} \in L^2(\mathbb{T})$ by

$$\mathcal{W}_{\lambda^{\mathrm{ap}}} := \mathcal{P}_{\lambda^{\mathrm{ap}}}[F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-}](\mathcal{H}^0 \oplus \mathcal{H}^0). \tag{5.48}$$

$$W_{\lambda^{\mathrm{ap}}} := i_H^{-1} \mathcal{W}_{\lambda^{\mathrm{ap}}}. \tag{5.49}$$

Proposition 5.32 *Let $g \in W_{\lambda^{\mathrm{ap}}}$. Then there exists a unique $f \in H^1(\mathbb{T})$, with*

$$\mathcal{P}_{\mathcal{V}_1(\lambda^{\mathrm{ap}})} \mathbf{c}(i_H f) = 0,$$

such that

$$\mathcal{L}_\theta^{\mathrm{ap}} f - \lambda^{\mathrm{ap}} f = g.$$

In addition, the following estimate holds

$$\|f\|_{H^1(\mathbb{T})} \leq 1014.41 \, \|g\|_{H^0(\mathbb{T})}.$$

Proof: Since $H^1(\mathbb{T})$ is dense in $H^0(\mathbb{T})$, we will assume that $g \in H^1(\mathbb{T})$ and extend the result by density to all $H^0(\mathbb{T})$. Hence, we have $(G^+, G^-) := i_H(g) \in W_{\lambda^{\mathrm{ap}}} \cap (\mathcal{H}^1 \oplus \mathcal{H}^1)$. Then, $G^\pm$ is continuous on $\overline{\mathbb{D}}$ and we can apply Lemma 5.26:

There exists a unique pair $(F^+, F^-) \in \mathcal{H} \oplus \mathcal{H}$ of holomorphic functions, with derivatives continuous on $\overline{\mathbb{D}}$, such that

$$\mathcal{L}^{\mathrm{ap,\,hol}}_{\theta,\lambda^{\mathrm{ap}}}(F^+, F^-) = \mathcal{P}_{\lambda^{\mathrm{ap}}}[F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-}](G^+, G^-) = (G^+, G^-).$$

Now we seek for an explicit estimate. Set $(F^+, F^-) := i_H(f)$ and observe that

$$\|f\|^2_{H^1(\mathbb{T})} = \sum_{k\in\mathbb{Z}} (1+k^2)|\widehat{f}_k|^2 = \sum_{k\geq 0} (1+k^2)|F_k^+|^2 + \sum_{k\geq 0} (1+(k+1)^2)|F_k^-|^2.$$

Note that

$$\partial_x(e^{ix} F^-|_{\mathbb{T}}(x)) = \sum_{k\geq 0} F_k^- i(k+1) e^{i(k+1)x}.$$

Hence, by Parseval's identity,

$$\sum_{k\geq 0} (k+1)^2 |F_k^-|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left|\partial_x(e^{ix} F^-(x))\right|^2 dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left|\partial_x F^-(x) + iF^-(x)\right|^2 dx.$$

Thus, we obtain

$$\|f\|^2_{H^1(\mathbb{T})} = \|F^+|_{\mathbb{T}}\|^2_{L^2(\mathbb{T})} + \|\partial_x F^+|_{\mathbb{T}}\|^2_{L^2(\mathbb{T})} + \|F^-|_{\mathbb{T}}\|^2_{L^2(\mathbb{T})} + \|\partial_x F^-|_{\mathbb{T}} + iF^-|_{\mathbb{T}}\|^2_{L^2(\mathbb{T})} \tag{5.50}$$

We only explain how to work with F^+ using (5.41a), since F^- follows in the same way using (5.41b). Expression (5.41a) reads as follows:

$$F^+|_{\mathbb{T}}(x) = F^+|_{\mathbb{T}}(0)\Pi^+(x) + J^+G^+|_{\mathbb{T}}(x) + \sum_{k=1}^{N}(\widehat{F^+|_{\mathbb{T}}})_{k-1}h_k^{-J^+}(x) + \sum_{k=1}^{N}(\widehat{F^-|_{\mathbb{T}}})_{k-1}h_k^{+J^+}(x).$$

Define

$$\mathbf{h}^+ := \begin{pmatrix} h_1^{-J^+} & \cdots & h_N^{-J^+} & h_1^{+J^+} & \cdots & h_N^{+J^+} & \Pi^+ & 0 \end{pmatrix}$$

Hence, we obtain

$$F^+|_{\mathbb{T}}(x) = \mathbf{h}^+(x)\mathbf{c}(F^+,F^-) + J^+G^+|_{\mathbb{T}}(x),$$

where we know that

$$\mathbf{c}(F^+,F^-) = -M_{\mathrm{reg}}^{\dagger}\mathbf{b}(G^+,G^-).$$

Recall that $\mathbf{c}(F^+,F^-)$ is defined in (5.32), $\mathbf{b}$ is defined in (5.33), and $M_{\mathrm{reg}}^{\dagger}$ is the matrix given in Lemma 5.21 with $M = M_{\mathrm{stab}}^{\mathrm{tor}}$.

Using the same strategy from (4.59), we have that

$$\|\mathbf{h}^+M_{\mathrm{reg}}^{\dagger}\mathbf{b}(G^+,G^-)\|_{L^2([-\pi,\pi])} \le \frac{1}{\sigma_2(\lambda^{\mathrm{ap}})}C_{0^+}\|\mathbf{b}(G^+,G^-)\|_{\ell^2}. \tag{5.51}$$

where

$$C_{0^+}^2 := \|\Pi^+\|_{L^2([-\pi,\pi])}^2 + \sum_{k=1}^{N}\|h_k^{-J^+}\|_{L^2([-\pi,\pi])}^2 + \|h_k^{+J^+}\|_{L^2([-\pi,\pi])}^2.$$

Additionally, note that

$$\begin{aligned}\|\mathbf{b}(G^+,G^-)\|_{\ell^2}^2 &\le \|J^+G^+\|_{L^2(\mathbb{T})}^2 + \|J^-G^-\|_{L^2(\mathbb{T})}^2 \\ &\le \max\{\|J^+\|_{L^2(\mathbb{T})\to L^2(\mathbb{T})}, \|J^-\|_{L^2(\mathbb{T})\to L^2(\mathbb{T})}\}^2\big(\underbrace{\|G^+\|_{L^2(\mathbb{T})}^2 + \|G^-\|_{L^2(\mathbb{T})}^2}_{\|g\|_{L^2(\mathbb{T})}^2}\big).\end{aligned}$$

Moreover, by Lemma 5.29, we have the same explicit bound for the $L^2(\mathbb{T})$ estimates of J^+ and J^-. Let C_{J_0} be such that

$$\|\mathbf{b}(G^+,G^-)\|_{\ell^2} \le C_{J_0}\|g\|_{L^2(\mathbb{T})}. \tag{5.52}$$

Consequently, we have from (5.51)–(5.52) that

$$\begin{aligned}\|F^+|_{\mathbb{T}}\|_{L^2(\mathbb{T})} &\le \frac{1}{\sqrt{2\pi}}\Big(\frac{C_{0^+}}{\sigma_2(\lambda^{\mathrm{ap}})}\|\mathbf{b}(G^+,G^-)\|_{\ell^2} + \|J^+G^+|_{\mathbb{T}}\|_{L^2([-\pi,\pi])}\Big) \\ &\le \frac{1}{\sqrt{2\pi}}\frac{C_{0^+}C_{J_0}}{\sigma_2(\lambda^{\mathrm{ap}})}\|g\|_{L^2(\mathbb{T})} + C_{J_0}\|G^+|_{\mathbb{T}}\|_{L^2(\mathbb{T})}.\end{aligned} \tag{5.53}$$

Let C_{0^-}, C_{1^+}, and C_{1^-} be defined by

$$C_{0^-}^2 := \|\Pi^-\|_{L^2([-\pi,\pi])}^2 + \sum_{k=1}^{N}\Big(\|h_k^{+J^-}\|_{L^2([-\pi,\pi])}^2 + \|h_k^{-J^-}\|_{L^2([-\pi,\pi])}^2\Big),$$

$$C_{1^+}^2 := \|\partial_x\Pi^+\|_{L^2([-\pi,\pi])}^2 + \sum_{k=1}^{N}\Big(\|\partial_x h_k^{-J^+}\|_{L^2([-\pi,\pi])}^2 + \|\partial_x h_k^{+J^+}\|_{L^2([-\pi,\pi])}^2\Big),$$

$$C_{1^-}^2 := \|\widetilde{\partial_x}\Pi^-\|_{L^2([-\pi,\pi])}^2 + \sum_{k=1}^{N}\Big(\|\widetilde{\partial_x} h_k^{+J^-}\|_{L^2([-\pi,\pi])}^2 + \|\widetilde{\partial_x} h_k^{-J^-}\|_{L^2([-\pi,\pi])}^2\Big),$$

where $\widetilde{\partial_x} f := \partial_x f + if$. Also, define $C_{J_{1^+}}$ and $C_{J_{1^-}}$ by

$$C_{J_{1^+}} := \sup_{\|g\|_{L^2([-\pi,\pi])}=1} \|\partial_x J^+ g\|_{L^2([-\pi,\pi])} \qquad \text{and} \qquad C_{J_{1^-}} := \sup_{\|g\|_{L^2([-\pi,\pi])}=1} \|\widetilde{\partial_x} J^- g\|_{L^2([-\pi,\pi])}.$$

A similar computation to (5.53) shows

$$\begin{aligned}
\|F^-|_{\mathbb{T}}\|_{L^2(\mathbb{T})} &\le \frac{1}{\sqrt{2\pi}} \frac{C_{0^-} C_{J_0}}{\sigma_2(\lambda^{\mathrm{ap}})} \|g\|_{L^2(\mathbb{T})} + C_{J_0} \|G^-|_{\mathbb{T}}\|_{L^2(\mathbb{T})}, \\
\|\partial_x F^+|_{\mathbb{T}}\|_{L^2(\mathbb{T})} &\le \frac{1}{\sqrt{2\pi}} \frac{C_{1^+} C_{J_0}}{\sigma_2(\lambda^{\mathrm{ap}})} \|g\|_{L^2(\mathbb{T})} + C_{J_{1^+}} \|G^+|_{\mathbb{T}}\|_{L^2([-\pi,\pi])}, \\
\|\widetilde{\partial_x} F^-|_{\mathbb{T}}\|_{L^2(\mathbb{T})} &\le \frac{1}{\sqrt{2\pi}} \frac{C_{1^-} C_{J_0}}{\sigma_2(\lambda^{\mathrm{ap}})} \|g\|_{L^2(\mathbb{T})} + C_{J_{1^-}} \|G^-|_{\mathbb{T}}\|_{L^2([-\pi,\pi])}.
\end{aligned}$$

Combining these estimates into (5.50) yields

$$\begin{aligned}
\|f\|_{H^1(\mathbb{T})}^2 \le a\|g\|_{L^2(\mathbb{T})}^2 + b^+ \|G^+|_{\mathbb{T}}\|_{L^2(\mathbb{T})}^2 + b^- \|G^-|_{\mathbb{T}}\|_{L^2(\mathbb{T})}^2 \\
+ 2c^+ \|g\|_{L^2(\mathbb{T})} \|G^+|_{\mathbb{T}}\|_{L^2(\mathbb{T})} + 2c^- \|g\|_{L^2(\mathbb{T})} \|G^-|_{\mathbb{T}}\|_{L^2(\mathbb{T})}.
\end{aligned}$$

where a, b^+, b^-, c^+, and c^- are given by

$$\begin{aligned}
a &:= \frac{1}{2\pi} \frac{C_{J_0}^2}{\sigma_2(\lambda^{\mathrm{ap}})^2} (C_{0^+}^2 + C_{0^-}^2 + C_{1^+}^2 + C_{1^-}^2), \\
b^\pm &:= C_{J_0}^2 + C_{J_{1^\pm}}^2, \qquad c^\pm := \frac{1}{\sqrt{2\pi}} \frac{C_{J_0}}{\sigma_2(\lambda^{\mathrm{ap}})} (C_{0^\pm} C_{J_0} + C_{1^\pm} C_{J_{1^\pm}}).
\end{aligned}$$

Finally, define the quantities ϑ_1, ϑ_2 by

$$\vartheta_+ := 0.637881, \qquad \vartheta_- := 0.770135.$$

Consequently, we have that

$$\begin{aligned}
\|f\|_{H^1(\mathbb{T})}^2 &\le (a + c^+\vartheta_+ + c^-\vartheta_-)\|g\|_{L^2(\mathbb{T})}^2 + (b^+ + \vartheta_+^{-1} c^+)\|G^+|_{\mathbb{T}}\|_{L^2(\mathbb{T})}^2 + (b^- + \vartheta_-^{-1} c^-)\|G^-|_{\mathbb{T}}\|_{L^2(\mathbb{T})}^2 \\
&\le \big(a + c^+\vartheta_+ + c^-\vartheta_- + \max\{b^+ + \vartheta_+^{-1} c^+, b^- + \vartheta_-^{-1} c^-\}\big)\|g\|_{L^2(\mathbb{T})}^2.
\end{aligned}$$

Here, every quantity has an explicit upper bound. We substitute them and check the inequality in the statement. □

III.5.3 The choice for η

In this section we fix $\theta = \frac{1}{3}$ and $\lambda = \lambda^{\mathrm{ap}}$. We describe the construction of the function η from (3.13) and compute some estimates required to close our argument.

Definition 5.33 *Let $f \in H^1(\mathbb{T})$ and $i_H : H^1(\mathbb{T}) \to \mathcal{H}^1 \oplus \mathcal{H}^1$ as in* (5.47)*. Define the operator P_{ub} as the composition given by*

$$P_{ub} := u_1^*(\lambda^{\mathrm{ap}}) \circ \mathbf{b} \circ i_H. \tag{5.54}$$

Then we define $\eta(f)$ by

$$\eta(f) := \frac{P_{ub}(\mathcal{D}_\theta(f + f^{\mathrm{ap}}) + \xi_{f^{\mathrm{ap}}})}{P_{ub}(f + f^{\mathrm{ap}})}.$$

Remark 5.34 *With this choice of $\eta(f)$, we have that*

$$P_{ub}\big[(\eta(f)I - \mathcal{D}_\theta)(f + f^{\mathrm{ap}}) - \xi_{f^{\mathrm{ap}}}\big] = 0.$$

Moreover,

$$(\eta(f)I - \mathcal{D}_\theta)(f + f^{\mathrm{ap}}) - \xi_{f^{\mathrm{ap}}} \in W_{\lambda^{\mathrm{ap}}}.$$

Recall the definition of $W_{\lambda^{\mathrm{ap}}}$ in (5.49).

Lemma 5.35 *Recall Lemma 5.23 and let $x_{\max} > 0$ such that*

$$x_{\max} < \frac{|P_{ub}(f^{\mathrm{ap}})|}{C_{J_0}},$$

where $C_{J_0} \geq \max\{\|J^+\|_{L^2(\mathbb{T})\to L^2(\mathbb{T})}, \|J^-\|_{L^2(\mathbb{T})\to L^2(\mathbb{T})}\}$. Let $B_{x_{\max}}$ be defined as

$$B_{x_{\max}} := \{f \in H^1(\mathbb{T}) : \|f\|_{H^1(\mathbb{T})} < x_{\max}\}.$$

Then, the operator $\eta|_{B_{x_{\max}}}$ is well defined. Moreover, for every $f \in B_{x_{\max}}$, it holds that

$$|\eta(f)| \leq q_\eta(\|f\|_{H^1(\mathbb{T})}),$$

where $q_\eta : [0, x_{\max}) \to \mathbb{R}$ is given by

$$q_\eta(x) := \frac{(x + \|f^{\mathrm{ap}}\|_{H^1(\mathbb{T})})\|\mathcal{D}_\theta\|_{H^1(\mathbb{T})\to L^2(\mathbb{T})} + \|\xi_{f^{\mathrm{ap}}}\|_{L^2(\mathbb{T})}}{x_{\max} - x}. \tag{5.55}$$

Remark 5.36 *We can compute an upper bound for $x_{\max}$ using Lemma 5.29 and Lemma 5.23. Moreover, in* (5.55) *we also have explicit bounds for every term by Corollary 3.9 and Lemma 3.10.*

Proof: We start by finding an upper bound for the numerator appearing in η. For each function $f \in H^1(\mathbb{T})$, denote by $(F^+, F^-) = i_H(f)$, it follows that

$$\begin{aligned}|P_{ub}(f)|^2 &\leq \|\mathbf{b}(F^+, F^-)\|_{\ell^2}^2 \leq \|J^+\|_{L^2(\mathbb{T})\to L^2(\mathbb{T})}^2\|F^+\|_{L^2(\mathbb{T})}^2 + \|J^-\|_{L^2(\mathbb{T})\to L^2(\mathbb{T})}^2\|F^-\|_{L^2(\mathbb{T})}^2 \\ &\leq \max\{\|J^+\|_{L^2(\mathbb{T})\to L^2(\mathbb{T})}, \|J^-\|_{L^2(\mathbb{T})\to L^2(\mathbb{T})}\}^2\|f\|_{L^2(\mathbb{T})}^2.\end{aligned}$$

Here, we know that $\|J^-\|_{L^2(\mathbb{T})\to L^2(\mathbb{T})}$ is finite since Lemma 5.29 provides an explicit estimate.

Then there exists $C_{J_0} > 0$ such that

$$|P_{ub}(f)| \leq C_{J_0}\|f\|_{L^2(\mathbb{T})}. \tag{5.56}$$

Hence, the numerator is estimated as follows

$$\begin{aligned}\big|P_{ub}(\mathcal{D}_\theta(f + f^{\mathrm{ap}}) + \xi_{f^{\mathrm{ap}}})\big| &\leq C_{J_0}\|\mathcal{D}_\theta(f + f^{\mathrm{ap}}) + \xi_{f^{\mathrm{ap}}}\|_{L^2(\mathbb{T})} \\ &\leq C_{J_0}\big((\|f\|_{H^1(\mathbb{T})} + \|f^{\mathrm{ap}}\|_{H^1(\mathbb{T})})\|\mathcal{D}_\theta\|_{H^1(\mathbb{T})\to L^2(\mathbb{T})} + \|\xi_{f^{\mathrm{ap}}}\|_{L^2(\mathbb{T})}\big). \quad (5.57)\end{aligned}$$

Now, we get a lower bound for the denominator. Assume $\|f\|_{H^1(\mathbb{T})} < x_{\max}$. Then

$$\begin{aligned}|P_{ub}(f + f^{\mathrm{ap}})| &\geq |P_{ub}(f^{\mathrm{ap}})| - |P_{ub}(f)| \geq |P_{ub}(f^{\mathrm{ap}})| - C_{J_0}\|f\|_{L^2(\mathbb{T})} \\ &\geq |P_{ub}(f^{\mathrm{ap}})| - C_{J_0}\|f\|_{H^1(\mathbb{T})} > C_{J_0}(x_{\max} - \|f\|_{H^1(\mathbb{T})}). \quad (5.58)\end{aligned}$$

Finally, (5.55) follows from (5.57) and (5.58). □

We will also need an estimate for the derivative of η.

Lemma 5.37 *Let $x_{\max}$ as in Lemma 5.35. Then, for every $f \in H^1(\mathbb{T})$ with $\|f\|_{H^1(\mathbb{T})} < x_{\max}$, the linear operator $d_f\eta$ satisfies that*

$$|d_f\eta(h)| \leq q_{d\eta}(\|f\|_{H^1(\mathbb{T})})\|h\|_{H^1(\mathbb{T})}, \quad \text{for all } h \in H^1(\mathbb{T}),$$

where the function $q_{d\eta} : [0, x_{\max}] \to \mathbb{R}^+$ is given by

$$q_{d\eta}(x) := \frac{(2x + \|f^{\mathrm{ap}}\|_{L^2(\mathbb{T})} + \|f^{\mathrm{ap}}\|_{H^1(\mathbb{T})})\|\mathcal{D}_\theta\|_{H^1(\mathbb{T})\to L^2(\mathbb{T})} + \|\xi_{f^{\mathrm{ap}}}\|_{L^2(\mathbb{T})}}{(x_{\max} - x)^2}.$$

Proof: The computation is straightforward

$$\begin{aligned} d_f\eta(h) &= \frac{P_{ub}(\mathcal{D}_\theta(h))}{P_{ub}(f + f^{\mathrm{ap}})} - \frac{P_{ub}(\mathcal{D}_\theta(f + f^{\mathrm{ap}}) + \xi_{f^{\mathrm{ap}}})}{\big[P_{ub}(f + f^{\mathrm{ap}})\big]^2} P_{ub}(h) \\ &= \frac{\big[P_{ub}(\mathcal{D}_\theta(h))\big]\big[P_{ub}(f + f^{\mathrm{ap}})\big] - \big[P_{ub}(\mathcal{D}_\theta(f + f^{\mathrm{ap}}) + \xi_{f^{\mathrm{ap}}})\big]\big[P_{ub}(h)\big]}{\big[P_{ub}(f + f^{\mathrm{ap}})\big]^2}. \end{aligned}$$

Hence, taking the absolute value, and using (5.56) and (5.58) yields

$$\begin{aligned} |d_f\eta(h)| &\leq \frac{\|\mathcal{D}_\theta(h)\|_{L^2(\mathbb{T})}\|f + f^{\mathrm{ap}}\|_{L^2(\mathbb{T})} + \|\mathcal{D}_\theta(f + f^{\mathrm{ap}}) + \xi_{f^{\mathrm{ap}}}\|_{L^2(\mathbb{T})}\|h\|_{L^2(\mathbb{T})}}{\big[x_{\max} - \|f\|_{H^1(\mathbb{T})}\big]^2} \\ &\leq \frac{(2\|f\|_{H^1(\mathbb{T})} + \|f^{\mathrm{ap}}\|_{L^2(\mathbb{T})} + \|f^{\mathrm{ap}}\|_{H^1(\mathbb{T})})\|\mathcal{D}_\theta\|_{H^1(\mathbb{T})\to L^2(\mathbb{T})} + \|\xi_{f^{\mathrm{ap}}}\|_{L^2(\mathbb{T})}}{\big[x_{\max} - \|f\|_{H^1(\mathbb{T})}\big]^2}\|h\|_{H^1(\mathbb{T})}. \end{aligned}$$

□

III.5.4 Proofs of Lemmas 3.12–3.14

Fix $f \in H^1(\mathbb{T})$, with $\|f\|_{H^1(\mathbb{T})} \leq 10^{-3}$. In particular, we have that

$$\|f\|_{H^1(\mathbb{T})} < x_{\max}.$$

Moreover, the function q_η defined in (5.55) is increasing on $[0, x_{\max})$. Hence, we obtain

$$|\eta(f)| \leq q_\eta(\|f\|_{H^1(\mathbb{T})}) \leq q_\eta(10^{-3}).$$

This estimate allows us to prove Lemma 3.14 since we have explicit estimates for every quantity appearing in (5.55). Therefore, we check that

$$\mathrm{Re}(\lambda^{\mathrm{ap}}) - q_\eta(10^{-3}) > 0.142127.$$

We now turn to Lemma 3.12. Recall (5.1), i.e.,

$$T^\theta_{\mathrm{stab}} f := (\mathcal{L}^{\mathrm{ap}}_\theta - \lambda^{\mathrm{ap}} I)^{-1}[(\eta(f)I - \mathcal{D}_\theta)(f^{\mathrm{ap}} + f) - \xi_{f^{\mathrm{ap}}}]. \tag{5.59}$$

Taking norms in (5.59) and using the estimate in Proposition 5.32, we obtain

$$\begin{aligned} \|T^\theta_{\mathrm{stab}}(f)\|_{H^1(\mathbb{T})} &\leq 1014.41\|(\eta(f)I - \mathcal{D}_\theta)(f^{\mathrm{ap}} + f) - \xi_{f^{\mathrm{ap}}}\|_{L^2(\mathbb{T})} \\ &\leq 1014.41\Big(q_\eta(\|f\|_{H^1(\mathbb{T})})\|f^{\mathrm{ap}} + f\|_{L^2(\mathbb{T})} \\ &\qquad + \|\mathcal{D}_\theta\|_{H^1(\mathbb{T})\to L^2(\mathbb{T})}\|f^{\mathrm{ap}} + f\|_{H^1(\mathbb{T})} + \|\xi_{f^{\mathrm{ap}}}\|_{L^2(\mathbb{T})}\Big). \end{aligned} \tag{5.60}$$

Here, q_η is the increasing function defined in Lemma 5.35, which we have applied in the last step. Hence, inserting the estimates in Corollary 3.9 and Lemma 3.10 into (5.60) yields for all $f \in X$:

$$0 \leq \|T^\theta_{\mathrm{stab}}(f)\|_{H^1(\mathbb{T})} \leq 10^{-3} \;\Rightarrow\; T^\theta_{\mathrm{stab}}(X) \subset X.$$

Finally, we prove Lemma 3.13. We begin by deriving an estimate for $d_f T^\theta_{\rm stab}$, with $f \in X$. Differentiating (5.59), we obtain

$$d_f T^\theta_{\rm stab}(h) = (\mathcal{L}^{\rm ap}_\theta - \lambda^{\rm ap} I)^{-1}[(f + f^{\rm ap}) d_f\eta(h) + \eta(f)h - \mathcal{D}_\theta h]. \tag{5.61}$$

To apply Proposition 5.32, it remains to check that the term inside the brackets belongs to $W_{\lambda^{\rm ap}}$. Observe that the map

$$(\eta(f)I - \mathcal{D}_\theta)(f^{\rm ap} + f) - \xi_{f^{\rm ap}} : H^1(\mathbb{T}) \to W_{\lambda^{\rm ap}},$$

takes values in $W_{\lambda^{\rm ap}}$. Therefore, its Fréchet derivative at f also takes values in $W_{\lambda^{\rm ap}}$. Consequently,

$$\begin{aligned}\|d_f T^\theta_{\rm stab}(h)\|_{H^1(\mathbb{T})} &\le 1014.41 \big\|[(f + f^{\rm ap}) d_f\eta(h) + \eta(f)h - \mathcal{D}_\theta h]\big\|_{L^2(\mathbb{T})} \\ &\le 1014.41 \Big((\|f\|_{H^1(\mathbb{T})} + \|f^{\rm ap}\|_{L^2(\mathbb{T})}) q_{d\eta}(\|f\|_{H^1(\mathbb{T})}) \\ &\qquad + q_\eta(\|f\|_{H^1(\mathbb{T})}) + \|\mathcal{D}_\theta\|_{H^1(\mathbb{T})\to L^2(\mathbb{T})} \Big) \|h\|_{H^1(\mathbb{T})}. \end{aligned} \tag{5.62}$$

Here, $q_{d\eta}$ is the increasing function from Lemma 5.37. We have applied this lemma and Lemma 5.35 again. Inserting the estimates in Corollary 3.9 and Lemma 3.10 into (5.62) yields

$$\|d_f T^\theta_{\rm stab}(h)\|_{H^1(\mathbb{T})} \le 0.12 \|h\|_{H^1(\mathbb{T})}.$$

III.A Classical inequalities

In this section, we establish some classical a priori estimates.

Lemma A.1 *Let $f \in H^1_{\text{odd or (even)}}$. Then the function f is continuous on $\mathbb{T}$ and*

$$\|f\|^2_{L^\infty(\mathbb{T})} \le 4\zeta(2) \|f\|^2_{H^1_{\text{odd or (even)}}}.$$

Proof: Continuity of f follows from Morrey's inequality $\mathcal{C}^{s-\frac12} \supset H^s(\mathbb{T})$, $s > \frac12$.

Applying the Cauchy–Schwarz inequality yields

$$\|f\|^2_{L^\infty(\mathbb{T})} \le \Big(\sum_{k\neq 0} |\widehat{f}_k| \Big)^2 \le \Big(\sum_{k\neq 0} \frac{1}{k^2} \Big) \Big(\sum_{k\in\mathbb{Z}} k^2 |\widehat{f}_k|^2 \Big) = 2\zeta(2) \|\partial_x f\|^2_{L^2(\mathbb{T})} = 4\zeta(2) \|f\|^2_{H^1_{\text{odd or (even)}}}.$$

□

Corollary A.2 *Let $f, g \in H^1_{\rm even}$. Then $fg \in H^1_{\rm even}$ and*

$$\|fg\|^2_{H^1_{\rm even}} \le 8\zeta(2) \|f\|^2_{H^1_{\rm even}} \|g\|^2_{H^1_{\rm even}}.$$

Proof: We use

$$\begin{aligned}\|fg\|^2_{H^1_{\rm even}} \le 2\|\partial_x fg\|^2_{H^0_{\rm odd}} + 2\|f\partial_x g\|^2_{H^0_{\rm odd}} &\le 2\|g\|^2_{L^\infty(\mathbb{T})} \|f\|^2_{H^1_{\rm even}} + 2\|f\|^2_{L^\infty(\mathbb{T})} \|g\|^2_{H^1_{\rm even}} \\ &\le 8\zeta(2) \|f\|^2_{H^1_{\rm even}} \|g\|^2_{H^1_{\rm even}}.\end{aligned}$$

The last step is a consequence of Lemma A.1. □

Lemma A.3 *Let $F \in \dot{\mathcal{H}}^1$. Then $f := F|_{\mathbb{T}}$ is continuous on $\mathbb{T}$ and*

$$\|f\|^2_{L^\infty(\mathbb{T})} \le \frac{\pi}{12} \|\partial_x f\|^2_{L^2([-\pi,\pi])}.$$

Proof: We proceed as in the proof of Lemma A.1. Since $f = F|_{\mathbb{T}}$ has only positive Fourier modes, we have

$$\|f\|^2_{L^\infty(\mathbb{T})} \le \Big(\sum_{k>0} |\widehat{f}_k|\Big)^2 \le \Big(\sum_{k>0} \frac{1}{k^2}\Big)\Big(\sum_{k>} k^2|\widehat{f}_k|^2\Big) = \zeta(2)\|\partial_x f\|^2_{L^2(\mathbb{T})} = \frac{\pi}{12}\|\partial_x f\|^2_{L^2([-\pi,\pi])}.$$

Here we used the Basel identity $\zeta(2) = \frac{\pi^2}{6}$ and the relation

$$\|f\|^2_{L^2(\mathbb{T})} = \frac{1}{2\pi}\|f\|^2_{L^2([-\pi,\pi])} \qquad \text{for all } f \in L^2(\mathbb{T}).$$

□

III.B Commutator of Hilbert transform and cosine

In this section, we do some of the computations required to prove Lemmas 4.1 and 5.4.

Lemma B.1 *Let $j \in \mathbb{N}$ and $f \in H^s(\mathbb{T})$, with $s > \frac{1}{2}$. Then*

$$[H, \cos(j\,\cdot)]f(x) = \frac{i}{2}\widehat{f}_0(e^{-ijx} - e^{ijx}) + \frac{i}{2}(\widehat{f}_j - \widehat{f}_{-j}) + i\sum_{m=1}^{j-1}(\widehat{f}_m e^{-i(j-m)x} - \widehat{f}_{-m}e^{i(j-m)x}).$$

Proof: Expand f in Fourier series:

$$f(x) = \sum_{m\in\mathbb{Z}} \widehat{f}_m e^{imx}.$$

A direct computation yields

$$\begin{aligned}
[H, \cos(j\,\cdot)]f(x) &= \frac{1}{2}[H, e^{ij\cdot} + e^{-ij\cdot}]f(x) = \frac{1}{2}\sum_{m\in\mathbb{Z}} \widehat{f}_m [H, e^{ij\cdot} + e^{-ij\cdot}]e^{imx} \\
&= -\frac{i}{2}\sum_{m\in\mathbb{Z}} \widehat{f}_m \Big[(\operatorname{sgn}(j+m) - \operatorname{sgn}(m))e^{i(j+m)x} \\
&\qquad\qquad + (\operatorname{sgn}(m-j) - \operatorname{sgn}(m))e^{-i(j-m)x}\Big] \\
&= -\frac{i}{2}\sum_{m=-j}^{0} \widehat{f}_m(\operatorname{sgn}(j+m) - \operatorname{sgn}(m))e^{i(j+m)x} \\
&\quad + \frac{i}{2}\sum_{m=0}^{j} \widehat{f}_m(\operatorname{sgn}(j-m) + \operatorname{sgn}(m))e^{-i(j-m)x} \\
&= \frac{i}{2}\sum_{m=0}^{j} (\operatorname{sgn}(j-m) + \operatorname{sgn}(m))(\widehat{f}_m e^{-i(j-m)x} - \widehat{f}_{-m}e^{i(j-m)x}) \\
&= \frac{i}{2}(e^{-ijx} - e^{ijx})\widehat{f}_0 + \frac{i}{2}(\widehat{f}_j - \widehat{f}_{-j}) + i\sum_{m=1}^{j-1}(\widehat{f}_m e^{-i(j-m)x} - \widehat{f}_{-m}e^{i(j-m)x}).
\end{aligned}$$

□

Corollary B.2 *Let f be a function in $H^s(\mathbb{T})$, $s > \frac{3}{2}$. We have that*

$$[H, v^{\mathrm{ap}}]\partial_x f(x) = \sum_{j=1}^{N} v_j^{\mathrm{ap}}\Big(-\frac{j}{2}(\widehat{f}_j + \widehat{f}_{-j}) - \sum_{m=1}^{j-1} m(\widehat{f}_m e^{-i(j-m)x} + \widehat{f}_{-m}e^{i(j-m)x})\Big).$$

Proof: $\partial_x f$ is zero-mean. Then, substituting $\partial_x f$ into Lemma B.1 we obtain

$$[H, \cos(j\,\cdot)]\partial_x f(x) = -\frac{j}{2}(\widehat{f}_j + \widehat{f}_{-j}) - \sum_{m=1}^{j-1} m(\widehat{f}_m e^{-i(j-m)x} + \widehat{f}_{-m} e^{i(j-m)x}).$$

Therefore, we get that

$$\begin{aligned}[H, v^{\mathrm{ap}}]\partial_x f(x) &= \sum_{j=1}^{N} v_j^{\mathrm{ap}} [H, \cos(j\,\cdot)]\partial_x f(x) \\ &= \sum_{j=1}^{N} v_j^{\mathrm{ap}} \Big(-\frac{j}{2}(\widehat{f}_j + \widehat{f}_{-j}) - \sum_{m=1}^{j-1} m(\widehat{f}_m e^{-i(j-m)x} + \widehat{f}_{-m} e^{i(j-m)x}) \Big).\end{aligned}$$

□

Corollary B.3 *Let f be a function in $H^s(\mathbb{T})$, $s > \frac{3}{2}$. We have that*

$$\begin{aligned}[H, v^{\mathrm{ap}}](\partial_x f + i\theta f)(x) = &-\frac{\theta}{2}\widehat{f}_0 \sum_{j=1}^{N} v_j^{\mathrm{ap}}(e^{-ijx} - e^{ijx}) - \frac{1}{2}\sum_{j=1}^{N} v_j^{\mathrm{ap}} \Big((j+\theta)\widehat{f}_j + (j-\theta)\widehat{f}_{-j} \Big) \\ &- \sum_{j=1}^{N} v_j^{\mathrm{ap}} \sum_{m=1}^{j-1} ((m+\theta)\widehat{f}_m e^{-i(j-m)x} + (m-\theta)\widehat{f}_{-m} e^{i(j-m)x}).\end{aligned}$$

Proof: A direct substitution yields the desired formula:

$$\begin{aligned}[H, v^{\mathrm{ap}}](\partial_x f + i\theta f)(x) = &\sum_{j=1}^{N} v_j^{\mathrm{ap}} \Big(-\frac{j}{2}(\widehat{f}_j + \widehat{f}_{-j}) - \sum_{m=1}^{j-1} m(\widehat{f}_m e^{-i(j-m)x} + \widehat{f}_{-m} e^{i(j-m)x}) \Big) \\ &+ i\theta \sum_{j=1}^{N} v_j^{\mathrm{ap}} \Big(\frac{i}{2}\widehat{f}_0(e^{-ijx} - e^{ijx}) + \frac{i}{2}(\widehat{f}_j - \widehat{f}_{-j}) \\ &\qquad + i\sum_{m=1}^{j-1} (\widehat{f}_m e^{-i(j-m)x} - \widehat{f}_{-m} e^{i(j-m)x}) \Big) \\ = &-\frac{\theta}{2}\widehat{f}_0 \sum_{j=1}^{N} v_j^{\mathrm{ap}}(e^{-ijx} - e^{ijx}) - \frac{1}{2}\sum_{j=1}^{N} v_j^{\mathrm{ap}}((j+\theta)\widehat{f}_j + (j-\theta)\widehat{f}_{-j}) \\ &- \sum_{j=1}^{N} v_j^{\mathrm{ap}} \sum_{m=1}^{j-1} ((m+\theta)\widehat{f}_m e^{-i(j-m)x} + (m-\theta)\widehat{f}_{-m} e^{i(j-m)x}).\end{aligned}$$

□

III.C Palindromic polynomial

In this section, we explain how to work rigorously with the roots of the polynomial $z^N P(z)$, where P is defined at (4.5). Furthermore, we also prove Lemma 4.12 and Lemma 5.25.

Remark C.1 *The polynomial $z^N P(z)$ is related to c and v^{ap} by the following identity on $\mathbb{T}$*

$$P(e^{ix}) = c + v^{\mathrm{ap}}(x).$$

A first observation about $z^N P(z)$ is that it is palindromic, since

$$z^N P(z) = cz^N + \frac{1}{2}\sum_{j=1}^{N} v_j^{\mathrm{ap}}(z^{N+j} + z^{N-j}).$$

Such polynomials have an interesting property, as the next lemma states.

Lemma C.2 *Let $d \in \mathbb{N}$ and $Q(z) = \sum_{k=0}^{d} Q_k z^k$ be a polynomial with complex coefficients. Assume Q is palindromic, i.e.*

$$Q_k = Q_{d-k} \quad \text{for every } k = 0, \dots, d.$$

Then, for every root $\alpha \in \mathbb{C}$ of Q, the inverse α^{-1} is also a root of Q.

Proof: Since the degree of Q is d, we know that $0 \neq Q_d = Q_0$. Consequently, 0 is never a root of Q.

We now compute $z^d Q(z^{-1})$ for $z \neq 0$:

$$z^d Q(z^{-1}) = z^d \sum_{k=0}^{d} Q_k z^{-k} = \sum_{k=0}^{d} Q_k z^{d-k} = \sum_{k=0}^{d} Q_{d-k} z^k = \sum_{k=0}^{d} Q_k z^k = Q(z).$$

Thus, if α is a root of Q, we have $\alpha^d Q(\alpha^{-1}) = 0$, so α^{-1} is again a root of Q. □

In the supplementary material we attach a file with approximations of N roots of $z^N P(z)$ inside the unit disk. The next lemma shows how accurate these approximations are.

Lemma C.3 *For every approximated root $\widetilde{\alpha}$ of $z^N P(z)$ lying inside the unit complex disk, and whose explicit expression is given in the supplementary material, there exists a unique root $\alpha \in D_r(\widetilde{\alpha})$, where $\overline{D}_r(\widetilde{\alpha})$ denotes the closed disk with center $\widetilde{\alpha}$ and radius $r = 10^{-41}$.*

Furthermore, given any other approximated root $\widetilde{\alpha}'$, the disks $D_r(\widetilde{\alpha})$ and $D_r(\widetilde{\alpha}')$ are disjoint.

In addition, we also know that no disk $\overline{D}_r(\widetilde{\alpha})$ intersects the unit circle.

Proof: Although this proof is computer-assisted, we include it here rather than in Appendix III.E. We denote by $Q(z) := z^N P(z)$ and

$$Q(z) =: \sum_{j=0}^{2N} Q_j z^j.$$

Let $\widetilde{\alpha} \in \mathbb{C}$ be one of the numerical approximations provided in the supplementary material. The goal is to prove that there exists a unique root α of Q in the closed disk $\overline{D}_r(\widetilde{\alpha})$, with $r = 10^{-41}$.

This verification is performed by the routine `enclose_roots()` in `lemmas.py`. The method proceeds by looking for a correction $\delta \in \mathbb{C}$ such that

$$Q(\widetilde{\alpha} + \delta) = 0.$$

Expanding Q in Taylor series at $\widetilde{\alpha}$ yields

$$0 = Q(\widetilde{\alpha}) + \delta \partial_z Q(\widetilde{\alpha}) + R(\widetilde{\alpha}, \delta),$$

where the remainder is

$$R(z, \delta) := \sum_{k=2}^{2N} \frac{\delta^k}{k!} \partial_z^k Q(z).$$

The code defines the operator $T_{\widetilde{\alpha}} : \overline{D}_r(0) \to \mathbb{C}$, given by

$$T_{\widetilde{\alpha}}(\delta) := -\frac{1}{\partial_z Q(\widetilde{\alpha})} (Q(\widetilde{\alpha}) + R(\widetilde{\alpha}, \delta)),$$

so that fixed points of $T_{\widetilde{\alpha}}$, when added to $\widetilde{\alpha}$, correspond exactly to roots of Q near $\widetilde{\alpha}$.

To verify that $T_{\widetilde{\alpha}}$ is a contraction, the code first computes coefficient-based upper bounds for the derivatives $\partial_z^k Q$ on the unit disk. These bounds are implemented using the explicit formula

$$|\partial_z^k Q(z)| \le \sum_{j=0}^{2N} \frac{j!}{(j-k)!}|Q_j|, \qquad |z| \le 1,$$

which depends only on the coefficients of Q.

Using these bounds, the routine constructs explicit enclosures for both the remainder $R(\widetilde{\alpha}, \delta)$ and its δ–derivative, valid for all $|\delta| \le r$. The two contraction conditions

$$T_{\widetilde{\alpha}}(\overline{D}_r(0)) \subset \overline{D}_r(0), \qquad |\partial_z T_{\widetilde{\alpha}}(\overline{D}_r(0))| < 1$$

are then verified numerically using ball arithmetic, ensuring full rigor.

Once these conditions are certified, the contraction mapping theorem guarantees the existence of a unique fixed point $\delta \in \overline{D}_r(0)$, and hence a unique root $\alpha = \widetilde{\alpha} + \delta$ of Q inside $\overline{D}_r(\widetilde{\alpha})$.

Finally, the same routine checks that the disks corresponding to distinct approximations $\widetilde{\alpha}$ are pairwise disjoint and strictly contained in the open unit disk. □

Corollary C.4 *The polynomial $z^N P(z)$ has exactly N roots inside the unit complex open disk $\mathbb{D}$. Recall this set is denoted by Z.*

Proof: We know that there are at least N simple roots inside the unit complex disk. Then, applying Lemma C.2 we find other N different roots with modulus greater than one. Since the degree of $z^N P(z)$ is $2N$, we have found every root and there are exactly N contained in $\mathbb{D}$. □

III.D Treatment of the solutions of the differential equations

Several parts of the proof require working with the solutions of linear first-order ODEs on $[-\pi, \pi]$. In particular, we refer to the 'h functions' from (4.43), (5.25), and (5.28). Even though these solutions are explicit, they involve heavy computational operations, which make it difficult to work directly with them.

To overcome this problem, we provide high-accuracy approximations, which are more manageable, of the solutions in the supplementary data. The error between the exact and the approximate solution will be measured by evaluating explicit residuals and applying Duhamel-type representations.

III.D.1 Existence argument

Recall the definition of h_k with $k = 1, \dots, N+1$ in (4.43). These functions arise in the construction of the approximate inverse operator used in the existence argument, and their accurate numerical evaluation is a key ingredient in our proof.

In this section, we explain how to construct continuous approximations, h_k^{ap}, of h_k and provide rigorous bounds for the errors

$$\tau_k := h_k - h_k^{\mathrm{ap}}.$$

More precisely, we prove the following lemma.

Lemma D.1 *For each $k = 1, \dots, N+1$, the approximation error satisfies*

$$\|\tau_k\|^2_{L^2([-\pi,\pi])} < \rho_k.$$

Here, the quantities ρ_k are explicit and can be found in the file `exis_resis.txt` *from the supplementary material.*

In order to construct the approximations h_k^{ap}, we partition the domain $[-\pi,\pi]$ into subintervals of varying length, and denote this partition by $\mathcal{J}$. Let $J=[x_0-r,x_0+r]\in\mathcal{J}$ be one of these intervals. For every $k=1,\dots,N+1$, we provide in the supplementary material `exis_data.txt` the coefficients $h_{N+1,l}^{\mathrm{ap},J}$, for $l=1,\dots,5$. We define with them the function $h_k^{\mathrm{ap},J}$, given by

$$h_k^{\mathrm{ap},J}(x):=\delta_{k,N+1}+\sum_{l=1}^{4}h_{k,l}^{\mathrm{ap},J}(x-x_0)^l.$$

Observe that, by definition, each h_k solves

$$\left.\begin{aligned}(c+v^{\mathrm{ap}}(x))\partial_x h_k(x)&=ih_k(x)+g_k(x),\\ h_k(0)&=\delta_{k,N+1},\end{aligned}\right\}\tag{D.1}$$

where g_k is given by (4.42) for $k=1,\dots,N$ and $g_{N+1}=0$. The function h_{N+1} corresponds to a homogeneous equation, while the remaining solutions include an affine term.

The data defining the functions $h_k^{\mathrm{ap,J}}$ have been constructed to approximate the functions $h_k^J:J\to\mathbb{C}$, solving

$$\left.\begin{aligned}(c+v^{\mathrm{ap}}(x))\partial_x h_k^J(x)&=ih_k^J(x)+g_k(x),\\ h_k^J(x_0)&=\delta_{k,N+1},\end{aligned}\right\}\tag{D.2}$$

The coefficients $h_{k,l}^{\mathrm{ap},J}$ have been chosen so that they minimize the function

$$(h_{k,1}^{\mathrm{ap},J},h_{k,2}^{\mathrm{ap},J},h_{k,3}^{\mathrm{ap},J},h_{k,4}^{\mathrm{ap},J})\mapsto\|(c+v^{\mathrm{ap}})\partial_x h_k^{\mathrm{ap},J}-ih_k^{\mathrm{ap},J}-g_k\|^2_{L^2(J)}.$$

With these pieces, we are able to construct continuous approximations of h_k, namely h_k^{ap}, for every $k=1,\dots,N+1$.

The construction of h_k from h_k^{ap} is done using the following idea. Let $J_1,J_2\in\mathcal{J}$ be two consecutive intervals with a common endpoint $x_*:=J_1\cap J_2$. On each interval J, we have:

- A local approximation $h_{N+1}^{\mathrm{ap},J}$ on J of the function h_{N+1}^J solving (D.2);
- For each $k=1,\dots,N$, a local approximation $h_k^{\mathrm{ap},J}$ of the function h_k^J solving (D.2).

Since the equation is linear, any other smooth solution $\widetilde{h}_k^J$ on J to (D.2) is written as

$$\widetilde{h}_k^J=Ch_{N+1}^J+h_k^J,$$

for a suitable constant $C\in\mathbb{C}$. This allows us to choose C on each subinterval so that h_k^{ap} is continuous and satisfies the same initial condition as h_k, that is $h_k(0)=\delta_{k,N+1}$.

Fix $k=1,\dots,N+1$ and assume that we have already constructed on J_1 the approximation $h_k^{\mathrm{ap}}|_{J_1}$, of the function h_k solving (D.1). To extend it to J_2 by continuity, we choose C so that the value at the interface x_* coincides:

$$h_k^{\mathrm{ap}}|_{J_1}(x_*)=Ch_{N+1}^{\mathrm{ap},J_2}(x_*)+h_k^{\mathrm{ap},J_2}(x_*).$$

Equivalently,

$$C=\frac{h_k^{\mathrm{ap}}|_{J_1}(x_*)-h_k^{\mathrm{ap},J_2}(x_*)}{h_{N+1}^{\mathrm{ap},J_2}(x_*)}.$$

Hence, we can continuously extend $h_k^{\mathrm{ap}}|_{J_1}$ to $h_k^{\mathrm{ap}}|_{J_1\cup J_2}$

$$h_k^{\mathrm{ap}}|_{J_2}(x):=Ch_{N+1}^{\mathrm{ap},J_2}(x)+h_k^{\mathrm{ap},J_2}(x).\tag{D.3}$$

We apply this procedure iteratively from $x = 0$, imposing

$$h_k^{\mathrm{ap}}(0) = \delta_{k,N+1} \tag{D.4}$$

thereby obtaining a continuous approximation on the whole interval $[-\pi, \pi]$.

The routine `_ode_values()` from `load_data.py` splits the domain into positive and negative parts using `_split_blocks_by_domain()`, and constructs the solutions on each domain using `_cons_sol_0_x`.

Working with the approximations. Fix one of the intervals $J = [x_0 - r, x_0 + r] \in \mathcal{J}$ and $k = 1, \dots, N+1$. On J, the approximating function $h^{\mathrm{ap}} := h_k^{\mathrm{ap}}$ is a polynomial of degree $N_r = 5$ of the form

$$h^{\mathrm{ap}}(x) = \sum_{l=0}^{N_r} \alpha_l^J (x - x_0)^l, \tag{D.5}$$

where the coefficients α_l^J have been determined from the stored data `exis_data.txt` and the iterative process from (D.3) and (D.4).

Definition D.2 *Let $J = [x_0 - r, x_0 + r]$. Fix $l \geq 0$ and $j \in \mathbb{Z}$, define:*

$$C_J^{l,j} := \int_J (x - x_0)^l \cos(jx)\, dx, \qquad S_J^{l,j} := \int_J (x - x_0)^l \sin(jx)\, dx, \qquad \overline{E}_J^{l,j} := \int_J (x - x_0)^l e^{-ijx}\, dx.$$

With the notation introduced in Definition D.2, j-th Fourier coefficient of $h^{\mathrm{ap}}(x)$ is computed as follows:

$$(\widehat{h^{\mathrm{ap}}})_j = \frac{1}{2\pi} \sum_{J \in \mathcal{J}} \sum_{l=0}^{N_r} \alpha_l^J \overline{E}_J^{l,j}.$$

Note D.3 *The integrals from Definition D.2 are thought to be precomputed. For example, the routine* `build_CS_for_jj()` *from* `auxiliar_funcs.py` *does this job. In particular constructs $C_J^{l,j}$ and $S_J^{l,j}$ with l from 0 up to a specified order, and j fixed. In addition, the function* `precompute_expo_vals()`*, which builds, for l from 0 up to a specified order, and $j =$* `n_min`*,...,*`n_min` + `total`*, the quantities*

$$\overline{E}_J^{l,j} = C_J^{l,j} - iS_J^{l,j}.$$

Proof of Lemma D.1:

Although this proof contains some computer-assisted parts, we present it here rather than in Appendix III.E.

The goal of this section is to provide a rigorous L^2 bound for the approximation error

$$\tau_k := h_k - h_k^{\mathrm{ap}}, \qquad k = 1, \dots, N+1.$$

More precisely, we prove that for every $k = 1, \dots, N+1$,

$$\|\tau_k\|^2_{L^2([-\pi,\pi])} < \rho_k,$$

where the corresponding quantities ρ_k are listed in the file `exis_resis.txt`.

Notice that the function τ_k satisfies a new ODE on each J,

$$\begin{aligned}
(c + v^{\mathrm{ap}}(x))\partial_x \tau_k(x) &= (c + v^{\mathrm{ap}}(x))\partial_x h_k(x) - (c + v^{\mathrm{ap}}(x))\partial_x h_k^{\mathrm{ap}}(x) \\
&= i h_k(x) + g_k(x) - (c + v^{\mathrm{ap}}(x))\partial_x h_k^{\mathrm{ap}}(x) \\
&= i(\tau_k + h_k^{\mathrm{ap}}(x)) + g_k(x) - (c + v^{\mathrm{ap}}(x))\partial_x h_k^{\mathrm{ap}}(x) \\
&= i\tau_k(x) - \xi_k(x),
\end{aligned} \tag{D.6}$$

where the function ξ_k is explicit and defined by

$$\xi_k := (c + v^{\mathrm{ap}})\partial_x h_k^{\mathrm{ap}} - i h_k^{\mathrm{ap}} - g_k. \tag{D.7}$$

Since h_k is smooth and h_k^{ap} is piecewise smooth, the derivative of τ_k will not be continuous on every point. However, we can obtain τ_k from $\xi_k(x)$ using Duhamel's formula,

$$\begin{aligned}\tau_k(x) &= \underbrace{\tau_k(0)}_{0} - \Pi(x)\int_0^x \frac{\beta(y)}{\Pi(y)}\xi_k(y)\,dy \\ &= -J\xi_k(x),\end{aligned} \tag{D.8}$$

where J is defined in (4.41).

Remark D.4 *Continuity of τ_k is needed to apply Duhamel's formula. In our construction, h_k^{ap} is continuous on $[-\pi,\pi]$, and therefore $\tau_k = h_k - h_k^{\mathrm{ap}}$ is continuous as well.*

Therefore, we apply Lemma 4.22 to obtain a $L^2([-\pi,\pi])$ bound of τ_k in terms of ξ_k.
The whole routine is carried out by the function `ode_exis_L2()` from `lemmas.py`. □

Decomposition of $\|\xi_k\|^2_{L^2([-\pi,\pi])}$. This section is devoted to explain the formulas we use to estimate $\|\xi_k\|_{L^2([-\pi,\pi])}$ in the proof of Lemma D.1.

We decompose the $L^2([-\pi,\pi])$-norm into a sum of local L^2 contributions over the subintervals $J \in \mathcal{J}$. Accordingly, for each fixed interval J we expand $\|\xi_k\|_{L^2(J)}$ as follows:

$$\|\xi_k\|^2_{L^2(J)} = \sum_{l=1}^{5} I_l^{\mathrm{exis}}(h_k^{\mathrm{ap}}; J) + \|g_k\|^2_{L^2(J)}$$

where

$$I_1^{\mathrm{exis}}(h; J) := \|h\|^2_{L^2(J)}, \tag{D.9a}$$

$$I_2^{\mathrm{exis}}(h; J) := \|(c + v^{\mathrm{ap}})\partial_x h\|^2_{L^2(J)}, \tag{D.9b}$$

$$I_3^{\mathrm{exis}}(h; J) := 2\mathrm{Re}\big((c + v^{\mathrm{ap}})\partial_x h, -ih\big)_{L^2(J)}, \tag{D.9c}$$

$$I_{4,k}^{\mathrm{exis}}(h; J) := 2\mathrm{Re}\big(ih, g_k\big)_{L^2(J)} \tag{D.9d}$$

$$I_{5,k}^{\mathrm{exis}}(h; J) := 2\mathrm{Re}\big((c + v^{\mathrm{ap}})\partial_x h, -g_k\big)_{L^2(J)}. \tag{D.9e}$$

Recall that g_k is given by

$$g_k(x) := \frac{i}{2} k v_k^{\mathrm{ap}} + k\sum_{m=1}^{N-k} v_{k+m}^{\mathrm{ap}} \sin(mx).$$

Notice that we can compute $\|g_k\|_{L^2([-\pi,\pi])}$ using Parseval's identity,

$$\frac{1}{2\pi}\|g_k\|^2_{L^2([-\pi,\pi])} = \sum_{j\in\mathbb{Z}} |(\widehat{g_k})_j|^2 = \frac{k^2}{4}(v_k^{\mathrm{ap}})^2 + \frac{k^2}{2}\sum_{m=1}^{N-k}(v_{k+m}^{\mathrm{ap}})^2, \tag{D.10}$$

We compute $\|g_k\|^2_{L^2([-\pi,\pi])}$ for all $k = 1, \dots, N+1$, where $g_{N+1} = 0$, in `gk_exis_norm_sq()`.

In order to obtain formulas for (D.9), we will write h on J as

$$h(x) := \sum_{l=0}^{N_r} \alpha_l (x - x_0)^l. \tag{D.11}$$

The quantity $I_1^{\text{exis}}(h;J)$ is computed as follows:

$$I_1^{\text{exis}}(h;J) = \|h\|^2_{L^2(J)} = \sum_{\substack{l_1=0\\ l_2=0}}^{N_r} \alpha_{l_1}\overline{\alpha_{l_2}} I\!I_1^{\text{exis}}(l_1,l_2;J),$$

where $I\!I_1^{\text{exis}}$ is given by

$$I\!I_1^{\text{exis}}(l_1,l_2;J) := C_J^{l_1+l_2,0}, \tag{D.12}$$

and $C_J^{l,j}$ is defined in Definition D.2:

$$C_J^{l_1+l_2,0} = \int_J (x-x_0)^{l_1+l_2}\,dx = \frac{r^{l_1+l_2+1}}{l_1+l_2+1}(1+(-1)^{l_1+l_2}).$$

$I_2^{\text{exis}}(h;J)$ and $I_3^{\text{exis}}(h;J)$ are computed in a similar way. Define $v_0^{\text{ap}} := c$ and

$$D[v,v](d) := \sum_{j_1-j_2=d} v_{j_1}\overline{v_{j_2}}, \qquad S[v,v](s) := \sum_{j_1+j_2=s} v_{j_1}\overline{v_{j_2}}, \tag{D.13}$$

$$\begin{aligned} I\!I_2^{\text{exis}}(l_1,l_2;J) &:= \frac{l_1l_2}{2}\sum_{\substack{j_1=0\\ j_2=0}}^{N} v_{j_1}^{\text{ap}} v_{j_2}^{\text{ap}} (C_J^{l_1+l_2-2,j_1-j_2} + C_J^{l_1+l_2-2,j_1+j_2}) \\ &= \frac{l_1l_2}{2}\Big(\sum_{d=-N}^{N} D[v^{\text{ap}},v^{\text{ap}}](d) C_J^{l_1+l_2-2,d} + \sum_{s=0}^{2N} S[v^{\text{ap}},v^{\text{ap}}](s) C_J^{l_1+l_2-2,s}\Big). \end{aligned} \tag{D.14}$$

Remark D.5 *We only have defined $C_J^{l,j}$ for $l \geq 0$. In* (D.14)*, $l_1+l_2-2<0$ implies $l_1l_2=0$, hence $I\!I_2^{\text{exis}}(l_1,l_2;J)=0$. For simplicity, we adopt the convention of allowing $C_J^{l,j}$ with negative l whenever it is multiplied by a zero coefficient throughout the rest of the section.*

Then, for $I_2^{\text{exis}}(h;J)$ we obtain

$$\begin{aligned} I_2^{\text{exis}}(h;J) &= \|(c+v^{\text{ap}})\partial_x h\|^2_{L^2(J)} = \sum_{\substack{l_1=0\\ l_2=0}}^{N_r}\sum_{\substack{j_1=0\\ j_2=0}}^{N} \int_J l_1l_2 v_{j_1}^{\text{ap}} v_{j_2}^{\text{ap}} \alpha_{l_1}\overline{\alpha_{l_2}} \cos(j_1x)\cos(j_2x)(x-x_0)^{l_1+l_2-2}\,dx \\ &= \sum_{\substack{l_1=0\\ l_2=0}}^{N_r} \alpha_{l_1}\overline{\alpha_{l_2}} \frac{l_1l_2}{2}\sum_{\substack{j_1=0\\ j_2=0}}^{N} v_{j_1}^{\text{ap}} v_{j_2}^{\text{ap}} (C_J^{l_1+l_2-2,j_1-j_2} + C_J^{l_1+l_2-2,j_1+j_2}) = \sum_{\substack{l_1=0\\ l_2=0}}^{N_r} \alpha_{l_1}\overline{\alpha_{l_2}} I\!I_2^{\text{exis}}(l_1,l_2;J). \end{aligned}$$

For $I_3^{\text{exis}}(h;J)$ we have that

$$\begin{aligned} I_3^{\text{exis}}(h;J) &= 2\text{Re}\big((c+v^{\text{ap}})\partial_x h, -ih\big)_{L^2(J)} = 2\text{Re}\int_J i(c+v^{\text{ap}}(x))\partial_x h(x)\overline{h(x)}\,dx \\ &= 2\text{Re}\sum_{\substack{l_1=0\\ l_2=0}}^{N_r}\sum_{j=0}^{N} il_1 v_j^{\text{ap}} \alpha_{l_1}\overline{\alpha_{l_2}} C_J^{l_1+l_2-1,j} = \sum_{\substack{l_1=0\\ l_2=0}}^{N_r} \text{Re}\big(\alpha_{l_1}\overline{\alpha_{l_2}} I\!I_3^{\text{exis}}(l_1,l_2;J)\big). \end{aligned}$$

where $I\!I_3^{\text{exis}}(l_1,l_2;J)$ is defined as

$$I\!I_3^{\text{exis}}(l_1,l_2;J) := 2il_1\sum_{j=0}^{N} v_j^{\text{ap}} C_J^{l_1+l_2-1,j}, \tag{D.15}$$

Finally, we proceed to compute $I^{\text{exis}}_{4,k}(h;J)$ and $I^{\text{exis}}_{5,k}(h;J)$. Here, we will also need $S^{l,j}_J$ from Definition D.2. Then,

$$I^{\text{exis}}_{4,k}(h;J) = 2\text{Re}\int_J (ih(x)\overline{g_k(x)})\,dx = 2k\sum_{l=0}^{N_r}\text{Re}\int_J i\alpha_l\Big(-\frac{i}{2}v^{\text{ap}}_k + \sum_{m=1}^{N-k} v^{\text{ap}}_{m+k}\sin(mx)\Big)(x-x_0)^l\,dx$$
$$= \sum_{l=0}^{N_r}\text{Re}\big(\alpha_l \mathit{\Pi}^{\text{exis}}_4(l,k;J)\big),$$

where

$$\mathit{\Pi}^{\text{exis}}_4(l,k;J) := 2k\Big(\frac{1}{2}v^{\text{ap}}_k C^{l,0}_J + i\sum_{m=1}^{N-k} v^{\text{ap}}_{m+k}S^{l,m}_J\Big). \tag{D.16}$$

Finally, we have for $I^{\text{exis}}_{5,k}(h;J)$ the computations below

$$I^{\text{exis}}_{5,k}(h;J) = -2\text{Re}\int_J (c+v^{\text{ap}}(x))\partial_x h(x)\overline{g_k(x)}\,dx = \sum_{l=0}^{N_r}\text{Re}(\alpha_l \mathit{\Pi}^{\text{exis}}_{51,k}(l;J)),$$

where

$$\mathit{\Pi}^{\text{exis}}_{51,k}(l;J) := \sum_{j=0}^{N} lkv^{\text{ap}}_j\Big(iC^{l-1,j}_J v^{\text{ap}}_k - \sum_{m=1}^{N-k} v^{\text{ap}}_{m+k}\big(S^{l-1,m+j}_J + S^{l-1,m-j}_J\big)\Big)$$
$$= ilkv^{\text{ap}}_k\sum_{j=0}^{N} v^{\text{ap}}_j C^{l-1,j}_J - lk\sum_{m=1}^{N-k} v^{\text{ap}}_{m+k}\mathit{III}^{\text{exis}}_{5,m}(l;J), \tag{D.17a}$$

$$\mathit{III}^{\text{exis}}_{5,m}(l;J) := \sum_{j=0}^{N} v^{\text{ap}}_j\big(S^{l-1,m+j}_J + S^{l-1,m-j}_J\big). \tag{D.17b}$$

Note D.6 *In order to compute* (D.9) *for every $k = 1,\dots,N+1$, we precompute some data. In particular, there is a different routine for each* (D.12)–(D.17) *and most of them are precomputed in* `pre_resis_exis()` *for each different interval J. We then use these precomputed data to compute the sum of the terms of* (D.9) *for a fixed interval J, this part is carried out by the routine* `I_sols_resi_exis_from_precomp()`.

III.D.2 Stability argument

This section is analogous to Appendix III.D.1, but now working with (5.25) and (5.28) instead of (4.43). Indeed, in (5.25) and (5.28) we construct $4N+2$ functions depending on the parameters λ and θ. We introduce the following notation, which fixes an ordering of these functions:

$$h_k = \begin{cases} h^{+J^+}_k, & 1\le k\le N \\ h^{-J^+}_{k-N}, & N+1\le k\le 2N \\ h^{+J^-}_{k-2N}, & 2N+1\le k\le 3N \\ h^{-J^-}_{k-3N}, & 3N+1\le k\le 4N \\ \Pi^+, & k = 4N+1, \\ \Pi^-, & k = 4N+2. \end{cases} \tag{D.18}$$

Since we have already described how we construct the continuous approximations in Appendix III.D.1, our goal in this section is to prove an estimate for the difference between h_k and the functions approximating it: h^{ap}_k. The data used to construct the h^{ap}_k with $\lambda = \lambda^{\text{ap}}$ is found

in `sing_p_data.txt` and `sing_m_data.txt`, while the data used to construct the approximations with $\lambda = 1.035i$ is stored in `regu_p_data.txt` and `regu_m_data.txt`.

These functions are solutions of the following ODEs

$$(c + v^{\mathrm{ap}}(x))\partial_x h_k^{\pm J^+}(x) = (i+\lambda)h_k^{\pm J^+}(x) - i\theta^+(c + v^{\mathrm{ap}}(x))h_k^{\pm J^+}(x) + g_{k,\theta^\pm}^{\pm},$$
$$(c + v^{\mathrm{ap}}(x))\partial_x h_k^{\pm J^-}(x) = (i-\lambda)h_k^{\pm J^-}(x) - i\theta^-(c + v^{\mathrm{ap}}(x))h_k^{\pm J^+}(x) + g_{k,\theta^\mp}^{\pm},$$

where $g_{k,\theta}^+$ and $g_{k,\theta}^-$ are given in (5.27) for $k = 1, \dots, N$ and $g_{N+1,\theta}^+ = g_{N+1,\theta}^- = 0$.

The homogeneous equation for h_k with $k = 1, \dots, 2N$ is the equation that $h_{4N+1} = \Pi^+$ satisfies, and the homogeneous equation for h_k with $k = 2N+1, \dots, 4N$, is the equation satisfied by $h_{4N+2} = \Pi^-$. Hence, we use the routine `_ode_values()` to construct the continuous approximations by joining the pieces as we did in (D.3) and (D.4). We then arrange them as in (D.18) using `_stab_fixed()`.

The two lemmas we show are the following:

Lemma D.7 *Let $\lambda = 1.035i$ and $\theta = \frac{1}{3}$, let h_k be defined as in* (D.18)*, and let h_k^{ap} be constructed from* `regu_p_data.txt` *and* `regu_m_data.txt`*. Then it holds that*

$$\|h_k - h_k^{\mathrm{ap}}\|^2_{L^2([-\pi,\pi])} < \rho_k,$$

where the quantities ρ_k are explicit and can be found in the file `regu_resis.txt` *from the supplementary material.*

Lemma D.8 *Let $\lambda = \lambda^{\mathrm{ap}}$ and $\theta = \frac{1}{3}$, let h_k be defined as in* (D.18)*, and let h_k^{ap} be constructed from* `sing_p_data.txt` *and* `sing_m_data.txt`*. Then it holds that*

$$\|h_k - h_k^{\mathrm{ap}}\|^2_{L^2([-\pi,\pi])} < \rho_k,$$

where the quantities ρ_k are explicit and can be found in the file `sing_resis.txt` *from the supplementary material.*

Proof of Lemma D.7 and Lemma D.8: The argument is the same as in the proof of Lemma D.1. We estimate the $L^2([-\pi,\pi])$-norm of the residuals and use L^2 estimates for $J^\pm$.

Observe that $|\Pi(x)| = 1$ for $x \in \mathbb{R}$ and $\lambda = 1.035i$. Hence, we use Lemma 4.22 to estimate $J^\pm$, instead of Lemma 5.29, which is used for $\lambda = \lambda^{\mathrm{ap}}$.

The computations are carried out by `ode_stab_resis_sing()` and `ode_stab_resis_regu()`. The former calls the latter with the required parameters. □

We introduce additional notation that will be useful for computing the residual terms. Recall the definition (5.28). Namely,

$$h_k^{-J^+} := J^+ g_{k,\theta^-}^-, \qquad h_k^{+J^+} := J^+ g_{k,\theta^+}^+,$$
$$h_k^{-J^-} := J^- g_{k,\theta^+}^-, \qquad h_k^{+J^-} := J^- g_{k,\theta^-}^+,$$

where

$$\theta^- := 1 - \theta, \qquad \theta^+ := \theta,$$

and, for $k = 1, \dots, N$ and $\theta \in \mathbb{R}$,

$$g_{k,\theta}^-(x) := (k-\theta)\frac{i}{2}\sum_{m=0}^{N-k} v_{m+k}^{\mathrm{ap}} e^{-i(m+1)x}, \qquad g_{k,\theta}^+(x) := (k-\theta)\frac{i}{2}\sum_{m=0}^{N-k} v_{m+k}^{\mathrm{ap}} e^{imx}.$$

For $s \in \{\pm 1\}$,

$$J^{(s)} := \begin{cases} J^+, & s = 1, \\ J^-, & s = -1, \end{cases} \qquad g_{k,\theta}^{(s)} := \begin{cases} g_{k,\theta}^+, & s = 1, \\ g_{k,\theta}^-, & s = -1, \end{cases} \qquad \theta^{(s)} := \begin{cases} \theta^+, & s = 1, \\ \theta^-, & s = -1. \end{cases}$$

With this notation, for $s_J, s_g \in \{\pm 1\}$, define

$$h_k^{(s_J,s_g)} := h_k^{(s_g)J^{(s_J)}} = J^{(s_J)} g_{k,\theta^{(s_J s_g)}}^{(s_g)}.$$

Define $\xi_k^{(s_J,s_g)}$ by

$$\begin{aligned} \xi_k^{(s_J,s_g)}(x) &:= (c + v^{\mathrm{ap}}(x))\partial_x h_k^{(s_J,s_g)}(x) \\ &\quad - \Big((i + s_J\lambda^{\mathrm{ap}}) - i\theta^{(s_J)}(c + v^{\mathrm{ap}}(x))\Big) h_k^{(s_J,s_g)}(x) - g_{k,\theta^{(s_J s_g)}}^{(s_g)}(x). \end{aligned} \tag{D.19}$$

The study of Π^+ and Π^- corresponds to the particular case of (D.19) where the affine term vanishes, i.e. with $g_{k,\theta^{(s_J s_g)}}^{(s_g)} = 0$.

Decomposition of $\|\xi_k^{(s_J,s_g)}\|_{L^2([-\pi,\pi])}^2$. To get a lighter notation, we set $v_0^{\mathrm{ap}} := c$ and

$$W_j^{(s_J)} := \begin{cases} i + s_J\lambda^{\mathrm{ap}} - i\theta^{(s_J)}c, & j = 0, \\ -i\theta^{(s_J)} v_j^{\mathrm{ap}}, & j = 1, \dots, N. \end{cases} \qquad W^{(s_J)}(x) := \sum_{j=0}^{N} W_j^{(s_J)} \cos(jx). \tag{D.20}$$

If we write (D.19) with this notation, we obtain

$$[\xi_k^{(s_J,s_g)} h](x) = (c + v^{\mathrm{ap}}(x))\partial_x h(x) - W^{(s_J)} h(x) - g_{k,\theta^{(s_J s_g)}}^{(s_g)}(x). \tag{D.21}$$

Hence, taking the $L^2(J)$-norm yields

$$\begin{aligned} \|\xi_k^{(s_J,s_g)}\|_{L^2(J)}^2 &= I_{1,s_J}^{\mathrm{stab}}(h_k^{(s_J,s_g)}; J) + I_2^{\mathrm{stab}}(h_k^{(s_J,s_g)}; J) + I_{3,s_J}^{\mathrm{stab}}(h_k^{(s_J,s_g)}; J) \\ &\quad + \sum_{l=4}^{5} I_{l,s_J,s_g,k}^{\mathrm{stab}}(h_k^{(s_J,s_g)}; J) + \|g_{k,\theta^{(s_J s_g)}}^{(s_g)}\|_{L^2(J)}^2, \end{aligned}$$

where

$$I_{1,s_J}^{\mathrm{stab}}(h; J) := \|W^{(s_J)} h\|_{L^2(J)}^2, \tag{D.22a}$$

$$I_2^{\mathrm{stab}}(h; J) := \|(c + v^{\mathrm{ap}})\partial_x h\|_{L^2(J)}^2 = I_2^{\mathrm{exis}}(h; J), \tag{D.22b}$$

$$I_{3,s_J}^{\mathrm{stab}}(h; J) := 2\mathrm{Re}\big(- W^{(s_J)} h, (c + v^{\mathrm{ap}})\partial_x h\big)_{L^2(J)}, \tag{D.22c}$$

$$I_{4,s_J,s_g,k}^{\mathrm{stab}}(h; J) := 2\mathrm{Re}\big(W^{(s_J)} h, g_{k,\theta^{(s_J s_g)}}^{(s_g)}\big)_{L^2(J)}, \tag{D.22d}$$

$$I_{5,s_J,s_g,k}^{\mathrm{stab}}(h; J) := 2\mathrm{Re}\big((c + v^{\mathrm{ap}})\partial_x h, -g_{k,\theta^{(s_J s_g)}}^{(s_g)}\big)_{L^2(J)}. \tag{D.22e}$$

As we did in (D.10), we can compute directly $\|g_{k,\theta^{(s_J s_g)}}^{(s_g)}\|_{L^2([-\pi,\pi])}^2$ using Parseval's identity. To this end, note that

$$g_{k,\theta^{(s_J s_g)}}^{(s_g)} = (k - \theta^{(s_J s_g)}) \frac{i}{2} \sum_{m=0}^{N-k} v_{m+k}^{\mathrm{ap}} e^{i\nu_{m,s_g} x}$$

where, for convenience, we have set

$$\nu_{m,s_g} := \frac{(2m+1)s_g - 1}{2}.$$

Therefore, for every $k = 1, \dots, N$ and $s_J, s_g \in \{-1, +1\}$, we obtain

$$\frac{1}{2\pi}\|g^{(s_g)}_{k,\theta^{(s_J s_g)}}\|^2_{L^2([-\pi,\pi])} = \frac{1}{4}\big(k - \theta^{(s_J s_g)}\big)^2 \sum_{m=0}^{N-k} |v^{\mathrm{ap}}_{m+k}|^2. \tag{D.23}$$

The other terms are computed using the local expression (D.11) of h on J. Observe that $I_2^{\mathrm{stab}}(h;J) = I_2^{\mathrm{exis}}(h;J)$ so we have that

$$I_2^{\mathrm{stab}}(h;J) = \sum_{\substack{l_1=0\\ l_2=0}}^{N_r} \alpha_{l_1}\overline{\alpha_{l_2}} I\!I_2^{\mathrm{exis}}(l_1,l_2;J).$$

For $I_1^{\mathrm{stab}}(h;J)$, we obtain

$$I_{1,s_J}^{\mathrm{stab}}(h;J) = \sum_{\substack{l_1=0\\ l_2=0}}^{N_r} \alpha_{l_1}\overline{\alpha_{l_2}} I\!I_{1,s_J}^{\mathrm{stab}}(l_1,l_2;J),$$

where

$$\begin{aligned} I\!I_{1,s_J}^{\mathrm{stab}}(l_1,l_2;J) &= \int_J \sum_{\substack{j_1=0\\ j_2=0}}^{N} W_{j_1}^{(s_J)}\overline{W_{j_2}^{(s_J)}} \cos(j_1 x)\cos(j_2 x)(x-x_0)^{l_1+l_2}\,dx \\ &= \frac{1}{2}\sum_{\substack{j_1=0\\ j_2=0}}^{N} W_{j_1}^{(s_J)}\overline{W_{j_2}^{(s_J)}}\big(C_J^{l_1+l_2,j_1-j_2} + C_J^{l_1+l_2,j_1+j_2}\big) \\ &= \frac{1}{2}\Big(\sum_{d=-N}^{N} D[W^{(s_J)},W^{(s_J)}](d) C_J^{l_1+l_2,d} + \sum_{s=0}^{2N} S[W^{(s_J)},W^{(s_J)}](s) C_J^{l_1+l_2,s}\Big) \end{aligned} \tag{D.24}$$

Recall that $D[W^{(s_J)},W^{(s_J)}]$ and $S[W^{(s_J)},W^{(s_J)}]$ are defined using (D.13).

For $I_{3,s_J}^{\mathrm{stab}}(h;J)$, we find that

$$I_{3,s_J}^{\mathrm{stab}}(h;J) = 2\mathrm{Re}\big(-W^{(s_J)}h, (c+v^{\mathrm{ap}})\partial_x h\big)_{L^2(J)} = \sum_{\substack{l_1=0\\ l_2=0}}^{N_r} \mathrm{Re}\big(\alpha_{l_1}\overline{\alpha_{l_2}} I\!I_{3,s_J}^{\mathrm{stab}}(l_1,l_2;J)\big),$$

where

$$\begin{aligned} I\!I_{3,s_J}^{\mathrm{stab}}(l_1,l_2;J) &:= -2l_2 \int_J \sum_{\substack{j_1=0\\ j_2=0}}^{N} W_{j_1}^{(s_J)} v^{\mathrm{ap}}_{j_2} \cos(j_1 x)\cos(j_2 x)(x-x_0)^{l_1+l_2-1}\,dx \\ &= -l_2 \sum_{\substack{j_1=0\\ j_2=0}}^{N} W_{j_1}^{(s_J)} v^{\mathrm{ap}}_{j_2}\big(C_J^{l_1+l_2-1,j_1-j_2} + C_J^{l_1+l_2-1,j_1+j_2}\big) \\ &= -l_2\Big(\sum_{d=-N}^{N} D[W^{(s_J)},v^{\mathrm{ap}}](d) C_J^{l_1+l_2-1,d} + \sum_{s=0}^{2N} S[W^{(s_J)},v^{\mathrm{ap}}](s) C_J^{l_1+l_2-1,s}\Big). \end{aligned} \tag{D.25}$$

For $I_{4,s_J,s_g,k}^{\mathrm{stab}}$, it yields

$$I_{4,s_J,s_g,k}^{\mathrm{stab}}(h;J) := 2\mathrm{Re}\big(W^{(s_J)}h, g^{(s_g)}_{k,\theta^{(s_J s_g)}}\big)_{L^2(J)} = \sum_{l=0}^{N_r} \mathrm{Re}\big(\alpha_l I\!I_{4,s_J,s_g,k}^{\mathrm{stab}}(l;J)\big)$$

where

$$\begin{aligned} \mathit{II}^{\mathrm{stab}}_{4,s_J,s_g,k}(l;J) &:= -i(k-\theta^{(s_Js_g)})\int_J \sum_{j=0}^{N}\sum_{m=0}^{N-k} W_j^{(s_J)} v^{\mathrm{ap}}_{m+k}\cos(jx)e^{-i\nu_{m,s_g}x}(x-x_0)^l\,dx \\ &= -i\frac{k-\theta^{(s_Js_g)}}{2}\sum_{m=0}^{N-k} v^{\mathrm{ap}}_{m+k}\mathit{III}^{\mathrm{stab}}_{4,m,s_J,s_g}(l;J), \end{aligned} \tag{D.26a}$$

$$\mathit{III}^{\mathrm{stab}}_{4,m,s_J,s_g}(l;J) := \sum_{j=0}^{N} W_j^{(s_J)}\big(\overline{E}_J^{\,l,\nu_{m,s_g}-j} + \overline{E}_J^{\,l,\nu_{m,s_g}+j}\big). \tag{D.26b}$$

Finally, we compute $I^{\mathrm{stab}}_{5,s_J,s_g,k}$, getting

$$I^{\mathrm{stab}}_{5,s_J,s_g,k}(h;J) := 2\mathrm{Re}\big((c+v^{\mathrm{ap}})\partial_x h, -g^{(s_g)}_{k,\theta^{(s_Js_g)}}\big)_{L^2(J)} = \sum_{l=0}^{N_r}\mathrm{Re}\big(\alpha_l \mathit{II}^{\mathrm{stab}}_{5,s_J,s_g,k}(l;J)\big),$$

where

$$\begin{aligned} \mathit{II}^{\mathrm{stab}}_{5,s_J,s_g,k}(l;J) &:= il(k-\theta^{(s_Js_g)})\int_J \sum_{j=0}^{N} v_j^{\mathrm{ap}}\sum_{m=0}^{N-k} v^{\mathrm{ap}}_{m+k}\cos(jx)e^{-i\nu_{m,s_g}x}(x-x_0)^{l-1}\,dx \\ &= il\frac{k-\theta^{(s_Js_g)}}{2}\sum_{m=0}^{N-k} v^{\mathrm{ap}}_{m+k}\mathit{III}^{\mathrm{stab}}_{5,m,s_g}(l;J). \end{aligned} \tag{D.27a}$$

$$\mathit{III}^{\mathrm{stab}}_{5,m,s_g}(l;J) := \sum_{j=0}^{N} v_j^{\mathrm{ap}}\big(\overline{E}_J^{\,l-1,\nu_{m,s_g}-j} + \overline{E}_J^{\,l-1,\nu_{m,s_g}+j}\big). \tag{D.27b}$$

Note D.9 *In order to compute* (D.22) *for every $k = 1,\dots,N+1$, we precompute some data. In particular, there is a different routine for each* (D.24)–(D.27) *and most of them are precomputed in* `pre_resis_stab()` *for each different interval J. We then use these precomputed data to compute the sum of the terms of* (D.22) *for a fixed interval J, this part is carried out by the routine* `I_sols_resi_stab_from_precomp()`.

III.D.2.1 The functions $J^+F_v^{\mathrm{ap}+}|_{\mathbb{T}}$ and $J^-F_v^{\mathrm{ap}-}|_{\mathbb{T}}$

The functions $F_v^{\mathrm{ap}+}$ and $F_v^{\mathrm{ap}-}$ are given in Definition 5.2, with $f = f^{\mathrm{ap}}$ (see (3.9)). From the stored data in `Jfv_p_data.txt` and `Jfv_m_data.txt`, we construct continuous approximations of $J^+F_v^{\mathrm{ap}+}|_{\mathbb{T}}$ and $J^-F_v^{\mathrm{ap}-}|_{\mathbb{T}}$, respectively. These approximations, denoted by h_+^{ap} and h_-^{ap}, are obtained following the construction introduced in (D.3) and (D.4).

The goal of this section is proving the following lemma.

Lemma D.10 *Let $\lambda = \lambda^{\mathrm{ap}}$ and $\theta = \frac{1}{3}$. Let $F_v^{\mathrm{ap}+}$ and $F_v^{\mathrm{ap}-}$ given in Definition 5.2 with $f = f^{\mathrm{ap}}$, where f^{ap} is provided in the supplementary material.*

Then

$$\|J^+F_v^{\mathrm{ap}+}|_{\mathbb{T}} - h_+^{\mathrm{ap}}\|^2_{L^2([-\pi,\pi])} < \rho_+ \qquad \text{and} \qquad \|J^-F_v^{\mathrm{ap}-}|_{\mathbb{T}} - h_-^{\mathrm{ap}}\|^2_{L^2([-\pi,\pi])} < \rho_-.$$

The quantities ρ_+ and ρ_- are explicit and are given in the file `Jfv_resis.txt` *in the supplementary material.*

Proof: The proof is the same as for Lemma D.8, but the residual is computed with a different formula. □

From f^{ap}, given by

$$f^{\mathrm{ap}}(x) = \sum_{m=-N}^{N-1} \widehat{f_m^{\mathrm{ap}}} e^{imx},$$

define the functions

$$f^{\mathrm{ap}+}(x) := \sum_{m=0}^{N-1} \widehat{f_m^{\mathrm{ap}}} e^{imx}, \qquad f^{\mathrm{ap}-}(x) := \sum_{m=0}^{N-1} \widehat{f_{-m-1}^{\mathrm{ap}}} e^{imx}.$$

Hence, the residuals ξ^+ and ξ^- are given by

$$\xi^{\pm}(x) := \big(c + v^{\mathrm{ap}}(x)\big)\partial_x h_{\pm}^{\mathrm{ap}}(x) - W^{(\pm 1)} h_{\pm}^{\mathrm{ap}}(x) - \sum_{m=0}^{N-1} \widehat{f_m^{\mathrm{ap}\pm}} e^{imx}, \tag{D.28a}$$

where $W^{(1)}$ and $W^{(-1)}$ are given in (D.20).

Hence, taking the $L^2(J)$-norm yields

$$\|\xi^{\pm}\|_{L^2(J)}^2 = I_{1,\pm 1}^{\mathrm{stab}}(h_{\pm}^{\mathrm{ap}}; J) + I_2^{\mathrm{stab}}(h_{\pm}^{\mathrm{ap}}; J) + I_{3,\pm 1}^{\mathrm{stab}}(h_{\pm}^{\mathrm{ap}}; J) + \sum_{l=4}^{5} I_{l,\pm 1, f^{\mathrm{ap}}}^{\mathrm{stab}}(h_{\pm}^{\mathrm{ap}}; J) + \|f^{\mathrm{ap}\pm}\|_{L^2(J)}^2,$$

The terms $I_{1,\pm 1}^{\mathrm{stab}}, I_2^{\mathrm{stab}}$, and $I_{3,\pm 1}^{\mathrm{stab}}$ have been also studied, see (D.22a)–(D.22c). In addition, as we did in (D.10) and (D.23), we can compute directly the sum over J of $\|f^{\mathrm{ap}\pm}\|_{L^2(J)}^2$ using Parseval's identity. In this case, we obtain

$$\frac{1}{2\pi}\|f^{\mathrm{ap}+}\|_{L^2([-\pi,\pi])}^2 = \sum_{m=0}^{N-1} |\widehat{f_m^{\mathrm{ap}}}|^2, \qquad \frac{1}{2\pi}\|f^{\mathrm{ap}-}\|_{L^2([-\pi,\pi])}^2 = \sum_{m=-N}^{-1} |\widehat{f_m^{\mathrm{ap}}}|^2. \tag{D.29}$$

Finally, we provide the computations of $I_{l,+1,f^{\mathrm{ap}}}^{\mathrm{stab}}$ and $I_{l,-1,f^{\mathrm{ap}}}^{\mathrm{stab}}$. Using the notation from (D.11), we find that

$$\begin{aligned}
I_{4,s_J,f^{\mathrm{ap}}}^{\mathrm{stab}}(h; J) &:= 2\mathrm{Re}\big(W^{(s_J)}h, f^{\mathrm{ap}(s_J)}\big) = \sum_{l=0}^{N_r} \mathrm{Re}\big(\alpha_l I\!I_{4,s_J,f^{\mathrm{ap}}}^{\mathrm{stab}}(l; J)\big), \\
I_{5,s_J,f^{\mathrm{ap}}}^{\mathrm{stab}}(h; J) &:= 2\mathrm{Re}\big((c + v^{\mathrm{ap}})\partial_x h, -f^{\mathrm{ap}(s_J)}\big) = \sum_{l=0}^{N_r} \mathrm{Re}\big(\alpha_l I\!I_{5,s_J,f^{\mathrm{ap}}}^{\mathrm{stab}}(l; J)\big),
\end{aligned}$$

where

$$\begin{aligned}
I\!I_{4,s_J,f^{\mathrm{ap}}}^{\mathrm{stab}}(l; J) &:= \int_J \sum_{j=0}^{N} \sum_{m=0}^{N-1} W_j^{(s_J)} \overline{\widehat{f_m^{\mathrm{ap}(s_J)}}} \cos(jx) e^{-imx} (x - x_0)^l \, dx \\
&= \sum_{j=0}^{N} \sum_{m=0}^{N-1} W_j^{(s_J)} \overline{\widehat{f_m^{\mathrm{ap}(s_J)}}} \big(\overline{E}_J^{l,m-j} + \overline{E}_J^{l,m+j}\big), \qquad \text{(D.30)} \\
I\!I_{5,s_J,f^{\mathrm{ap}}}^{\mathrm{stab}}(l; J) &:= -\int_J \sum_{j=0}^{N} \sum_{m=0}^{N-1} l v_j^{\mathrm{ap}} \overline{\widehat{f_m^{\mathrm{ap}(s_J)}}} \cos(jx) e^{-imx} (x - x_0)^{l-1} \, dx \\
&= -\sum_{j=0}^{N} \sum_{m=0}^{N-1} l v_j^{\mathrm{ap}} \overline{\widehat{f_m^{\mathrm{ap}(s_J)}}} \big(\overline{E}_J^{l-1,m-j} + \overline{E}_J^{l-1,m+j}\big). \qquad \text{(D.31)}
\end{aligned}$$

III.E Computer-assisted proofs

This appendix presents the computer-assisted results used in the chapter. We describe the computational procedures, algorithms, and verification strategies used in the proofs, so that the computer-assisted component is transparent and reproducible.

All computer-assisted steps are carried out using rigorous numerical methods, based on interval arithmetic, ensuring that every computed interval encloses the exact real value. The programs used for these computations were written in Python using the SageMath mathematical software system, which provides the interval arithmetic routines used throughout the verification. In particular, numerical approximations are never used heuristically: each algorithm is designed to produce mathematically certified bounds, which are then combined with analytic arguments to establish the stated results.

The appendix is organized as follows. We first describe the organization of the data files used by the programs, indicating the role of each file and the information it contains. Next, for every lemma that requires computer-assisted verification, we describe the corresponding verification procedure and explain how the relevant bounds are obtained.

The material in this appendix is not required for understanding the conceptual arguments of the chapter, but it is essential for verifying the correctness of the computer-assisted claims. Readers interested in reproducibility or independent verification are encouraged to consult this section.

All source code and most data files used in the computations are available at https://github.com/MiguelMGPascualCaballo/bhtw . Larger data files are provided separately on Zenodo at https://doi.org/10.5281/zenodo.19250315 . To reproduce the results, the Zenodo files should be placed in the directory `supplementary_data/`.

III.E.1 Supplementary data

All auxiliary numerical data used in the computer-assisted proofs are stored in plain text files stored in the directory `supplementary_data/`. These files are loaded at runtime by the routines defined in `load_data.py`.

All numerical quantities are interpreted using interval arithmetic. Real numbers are converted to `RealBallField` instances (`RBF`) and complex numbers to `ComplexBallField` instances (`CBF`), both with 200 bits of precision. Consequently, every value used in the computation represents an enclosure of the corresponding exact quantity. Numerical values may appear either in standard decimal notation or in base-10 scientific notation (for instance `1.23e-6` meaning $1.23 \cdot 10^{-6}$). In all cases they are parsed as elements of the real ball field (`RBF`).

Tabulated real data. Some files consist of rows of real numbers separated by whitespace. These are read by the routine `_load_tab_data()`, which converts each entry to an element of `RBF`. In the files listed below each line contains a single real number.

- `tw_data.txt` contains the speed c of the traveling wave and the cosine Fourier coefficients. The first line stores c, while the $(j+1)$-th line stores v_j^{ap} for $j = 1, \dots, N$. Recall that $N = 1000$.
- `exis_resis.txt` contains the $N+1$ quantities ρ_k appearing in Lemma D.1.
- `regu_resis.txt` contains the $4N+2$ quantities ρ_k appearing in Lemma D.7.
- `sing_resis.txt` contains the $4N+2$ quantities ρ_k appearing in Lemma D.8.
- `Jfv_resis.txt` contains the two quantities ρ_+ and ρ_- appearing in Lemma D.10.
- `gk_Linf_bounds.txt` contains L^∞ bounds for the real part of the g_k functions given in (4.42). In particular, we use the data from this file in the proof of Lemma 4.25.

Tabulated complex data. Several files contain complex numbers written in the form `re,im`, representing the real and imaginary parts of a complex number. These files are read by the routine `_load_tab_data_complex()`, which parses each token and constructs the corresponding element of `CBF`.

- `eigen_data.txt` contains 2 lines, each consisting of a complex number. The first line contains λ^{ap}, while the second one contains $1.035i$.
- `polynomial_zeros.txt` contains N lines, each containing a complex number. Each number is approximating a root of $z^N P(z)$ inside the unit complex disk. Recall that P is defined in (4.5).
- `sing_u1.txt` contains $2N + 2$ lines, each one with a complex number. These numbers represent the coefficients of an explicit approximation of the left singular vector associated with the smallest singular value of $M^{\mathrm{tor}}_{\mathrm{stab}}(\lambda^{\mathrm{ap}})$. In particular the loader routine `eigen_u1()` also normalizes the approximation so that it has norm 1.
- `exis_V.txt` contains an approximation of the V matrix in the singular value decomposition of $M^{\mathrm{tor}}_{\mathrm{exis}}$.
- `regu_V.txt` contains an approximation of the V matrix in the singular value decomposition of $M^{\mathrm{tor}}_{\mathrm{stab}}(1.035i)$.
- `sing_V.txt` contains an approximation of the V matrix in the singular value decomposition of $M^{\mathrm{tor}}_{\mathrm{stab}}(\lambda^{\mathrm{ap}})$. We also remark that we construct f^{ap} from the last column of V.

Construction of f^{ap}. Let $V \in \mathbb{C}^{(2N+2)\times(2N+2)}$ be the complex matrix constructed from `sing_V.txt`. We define the Fourier coefficients of f^{ap} as

$$\widehat{f^{\mathrm{ap}}_k} = \begin{cases} V_{N-k,2N+2} & -N \le k \le -1, \\ V_{k+1,2N+2} & 0 \le k \le N-1. \end{cases}$$

Then f^{ap} is defined by

$$f^{\mathrm{ap}} = \sum_{k=-N}^{N-1} \widehat{f^{\mathrm{ap}}_k} e^{ikx}.$$

ODE approximation data. The polynomial approximations used in the rigorous integration of the ODE are stored in files organized in blocks. Each block corresponds to a dyadic subinterval of the $[-\pi, \pi]$ and has the form

```
J j k
VALUES
...
END
```

where `j` and `k` determine the interval $J_{j,k}$ defined by

$$J_{j,k} := \frac{2\pi}{2^j}[k-1, k] - \pi.$$

The keyword `VALUES` is followed by a sequence of lines, each line containing $N_r = 5$ complex coefficients. The first data line corresponds to the coefficients of the homogeneous solution, while the rest of the lines correspond to the affine ones. The block is terminated by the keyword `END`.

In this section we only explain how the local polynomial data on an interval J is stored. The procedure used to reconstruct the approximations from the stored data was described in Appendix III.D.

- `exis_data.txt` contains the necessary data to construct the approximations h_k^{ap}, $k = 1, \dots, N+1$ described in Section III.D.1.

  Let $\mathcal{J}$ denote the collection of all subintervals used in the construction, namely the set of intervals $J_{j,k}$. Let $J \in \mathcal{J}$ be one of these intervals, and let $x_0 \in \mathbb{R}$ and $r > 0$ be such that

  $$J = [x_0 - r, x_0 + r]. \tag{E.1}$$

  On J, we store after `VALUES` and before `END` $N+1$ lines, each one corresponding to one of the functions $h_k^{\mathrm{ap},J}$. More precisely, the first line stores the non-constant coefficients of the homogeneous approximation $h_{N+1}^{\mathrm{ap},J}$, while the subsequent N lines store the non-constant coefficients of the affine approximations $h_k^{\mathrm{ap},J}$ for $k = 1, \dots, N$. In each line we can find 5 pairs of coefficients separated by spaces. These pairs represent the coefficients $h_{k,l}^{\mathrm{ap},J}$, $l = 1, \dots, 5$ and have the format `re,im`.

- `sing_p_data.txt` and `sing_m_data.txt` contain the data required to construct the approximations h_k^{ap}, $k = 1, \dots, 4N+2$, described in Section III.D.2, with $\lambda = \lambda^{\mathrm{ap}}$ and $\theta = \frac{1}{3}$.

  The structure is the same as in `exis_data.txt`, so we only explain how the ordering of the lines corresponds to (D.18).

  The first line of `sing_p_data.txt` corresponds to h_{4N+1}, while the remaining lines in this file correspond to the functions h_k, $k = 1, \dots, 2N$.

  Similarly, the first line of `sing_m_data.txt` corresponds to h_{4N+2}, while the remaining lines correspond to the functions h_k, $k = 2N+1, \dots, 4N$.

- `regu_p_data.txt` and `regu_m_data.txt`: As for `sing_p_data.txt` and `sing_m_data.txt`, we use the same framework, with $\lambda = 1.035i$ instead of λ^{ap}.

- `Jfv_p_data.txt` and `Jfv_m_data.txt`: These files contain approximations of $J^+ F_v^{\mathrm{ap},+}\big|_{\mathbb{T}}$ and $J^- F_v^{\mathrm{ap},-}\big|_{\mathbb{T}}$, respectively. The framework is the same as described above, but in this case each block consists of a single line.

Bounds. Finally, the file `bounds.txt` contains the remaining numerical bounds used throughout the chapter. These are read by the routine `load_bounds()`, which converts the numerical values into `RBF` elements and stores them in a dictionary indexed by their labels.

The file is organized as sequence of entries, one per line. Each line contains several fields separated by the symbol `|`. Lines that are empty or start with the character `#` are ignored. From each entry the program extracts three pieces of information: a label identifying the bound, the numerical value of the bound, and a tag indicating the side of the inequality. Thus a typical entry has the form

```
label|...|value|...|side|...|description
```

where `label` is the identifier used in the code, `value` is the numerical bound written in decimal notation, and `side` indicates whether the bound corresponds to the lower or upper side of the corresponding inequality. In addition, `description` is a brief description of the bound to improve readability.

III.E.2 Objects, methods and routines

We briefly describe the main classes and routines used in the implementation.

Classes. The following classes are defined in `classes.py`.

- `FourierRealSeries`. This class represents real Fourier series and implements basic operations with them, such as evaluation, differentiation, and algebraic manipulations of the coefficients.
- `Functions_1D`. A class representing one-dimensional real-valued functions by storing callable black-box evaluators for the function and a finite number of its derivatives. It also supports basic algebraic manipulations, with the corresponding derivatives computed automatically. In particular, products are handled through Leibniz rule, and inverses through a Faà di Bruno formula. We also include a constructor that creates a `Functions_1D` instance from a `FourierRealSeries`.

Methods. The following classes are defined in `methods.py`.

- `integral_1D` constructs a callable instance for one-dimensional integration, which encloses integrals of `Functions_1D` objects over an interval I.

  We only include the method `gauss_lege_2dots()`, which implements a Gauss–Legendre quadrature. In particular, it uses the enclosure

  $$\int_I f(x)\,dx \in r\big(f(x_0 + r\omega) + f(x_0 - r\omega)\big) + \frac{r^5}{135}\partial_x^4 f(I), \tag{E.2}$$

  where $I := [x_0 - r, x_0 + r]$ and $\omega := \frac{\sqrt{3}}{3}$.

- `image_1D` constructs a callable object that encloses the images of functions represented as `Functions_1D` objects. We only include the method `taylor_method()`, which, for a fixed order k, implements a Taylor enclosure. In particular, if $k = 2$, we use

  $$f(I) \subset f(x_0) + [-r, r]\partial_x f(x_0) + \frac{1}{2}[0, r^2]\partial_x^2 f(I). \tag{E.3}$$

  where $I := [x_0 - r, x_0 + r]$.

Routines. We describe a selection of the routines from `auxiliar_funcs.py`. We do not cover every function, but focus on those that are more elaborated.

- `max_prev_esti()`. Let $[a, b] \subset \mathbb{R}$ be a compact interval, $B \in \mathbb{R}$, and let $f : [a, b] \to \mathbb{R}$. This routine verifies whether

  $$f(x) \leq B, \quad \text{for all } x \in [a, b]. \tag{E.4}$$

  The function f is represented as a `Functions_1D` instance, the domain as a list containing the endpoints of the interval, and B as a `RealBallField` instance.

  In order to prove (E.4), assume we have a routine, which we call `method()`, such that given a compact interval I and a function $f : I \to \mathbb{R}$, it returns an interval J satisfying

  $$J \supset f(I).$$

  Smaller intervals I are expected to produce tighter enclosures J.

  The algorithm starts with the list `intervals` containing only $[a, b]$. While `intervals` is nonempty, an interval I is removed from the list and $f(I)$ is enclosed using `method()`. If $f(I) > B$, the routine stops and returns `False`. If $J \ni B$, the interval I is bisected and both subintervals are appended to the list. Otherwise the interval is accepted.

  If the list becomes empty, the inequality holds on the whole domain $[a, b]$ and the routine returns `True`. Two additional stopping criteria control the total number of iterations and the number of intervals stored in the list.

- `integ_adaptive_1D()`. Let $[a,b] \subset \mathbb{R}$ be a compact interval and $f : [a,b] \to \mathbb{R}$. This routine produces an enclosure of the integral

$$\int_{[a,b]} f(x)\,dx.$$

The function f is represented as a `Functions_1D` instance and the domain as a list containing the endpoints of the interval.

Assume we have a routine `method()` which, given a compact interval I and a function $f : I \to \mathbb{R}$, returns an interval J such that

$$J \supset \int_I f(x)\,dx.$$

As before, smaller intervals are expected to yield tighter enclosures.

The adaptive subdivision strategy is the same as in `max_prev_esti()`. Starting from the list `intervals` containing $[a,b]$, intervals are removed, evaluated using `method()`, and either accepted or bisected according to prescribed tolerances.[2]

Accepted intervals contribute their enclosure to the total integral. When the list becomes empty, the whole domain has been processed and the routine returns the final enclosure. As before, two stopping conditions limit the total number of iterations and the number of stored intervals.

- `triangular_integral()`. Let $f, g : [0,\pi] \to \mathbb{R}$, and assume that $f(x) \geq 0$ for all $x \in [0,\pi]$. This routine produces an enclosure of the triangular integral

$$\int_0^\pi \int_0^x f(x)g(y)\,dy\,dx.$$

The input `integrand` represents the function f, while `mult_term` represents the function g. Both are given as `Functions_1D` instances and are assumed to provide derivatives up to order four.

The routine is based on the identity

$$\int_0^\pi \int_0^x f(x)g(y)\,dy\,dx = \int_0^\pi g(x)\iota(x)\,dx,$$

where

$$\iota(x) = \int_x^\pi f(t)\,dt.$$

In particular,

$$\iota(\pi) = 0, \qquad \partial_x \iota(x) = -f(x).$$

The algorithm starts from the interval $[0,\pi]$ and maintains a stack of subintervals, processed from right to left. For each interval $I := [x_0 - r, x_0 + r]$, the values of ι at the relevant quadrature nodes are enclosed by integrating f over suitable subintervals. We use the Gauss–Legendre enclosure formula (E.2) to enclose

$$\int_I \iota(y)g(y)\,dy.$$

[2]Rigor does not come from the tolerances. They are only used as a criterion to accept or reject an interval.

Hence, we need to evaluate ι at $x_0 - r\omega$ and $x_0 + r\omega$ with $\omega = \frac{\sqrt{3}}{3}$ and enclose $\partial_x^4(\iota g)(I)$. To this end, we need to know how to enclose the function ι up to four derivatives since

$$\partial_x^4(\iota g) = \partial_x^4 \iota g + 4\partial_x^3 \iota \partial_x g + 6\partial_x^2 \iota \partial_x^2 g + 4\partial_x \iota \partial_x^3 g + \iota \partial_x^4 g.$$

Since ι is an antiderivative of $-f$, its derivatives can be enclosed directly from those of f. Because f is nonnegative, ι is decreasing, and therefore

$$\iota(I) = [\iota(x_0 + r), \iota(x_0 - r)].$$

Each evaluation of ι is obtained by a cumulative integral, computed using the enclosure (E.2).

If the resulting local error enclosure satisfies a prescribed tolerance, the interval is accepted and its contribution is added to the total integral. Otherwise, the interval is bisected and both subintervals are returned to the stack, with the right one processed first so that the reconstruction of ι remains consistent.

When all intervals have been accepted, the routine returns the final enclosure of the triangular integral. As before, the total number of iterations is bounded by a safety criterion.

- `gersh_smallest_eig()` implements Gershgorin's theorem to produce a lower bound for the real parts of the eigenvalues of a given matrix.
- `gersh_gap_one_two()` implements Gershgorin's theorem to compute an upper bound for the real part of the eigenvalue associated with the Gershgorin disk corresponding to the last row. It also returns a lower bound for the real parts of the remaining eigenvalues.

III.E.3 Details of the proofs

The routine `substituting_estimates()` can be found in the file `lemmas.py`. Although it does not prove any lemma, it collects all substitutions used throughout the chapter. The function returns `True` if no errors are detected.

Proof of Lemma 2.3: The coefficients of v^{ap} are provided in the supplementary material. The function $\xi_{v^{\mathrm{ap}}}$ is defined as the residual obtained by substituting v^{ap} into the Burgers–Hilbert equation, namely

$$\xi_{v^{\mathrm{ap}}}(x) = c\partial_x v^{\mathrm{ap}}(x) + Hv^{\mathrm{ap}}(x) + v^{\mathrm{ap}}(x)\partial_x v^{\mathrm{ap}}(x).$$

We represent v^{ap} as a `FourierRealSeries` object, which stores Fourier coefficients and implements algebraic operations directly at the coefficient level, including differentiation `dx()`, multiplication `__mul__()`, and the Hilbert transform `hx()`.

An instance representing $\xi_{v^{\mathrm{ap}}}$ is constructed by the routine `equ_tw_eval_symbolic()` from the speed c and the Fourier coefficients of v^{ap}. Its H^0_{odd}-norm is then obtained from these Fourier coefficients with `norm_homogeneous()`. Since this routine returns the full $H^0(\mathbb{T})$ norm, the result is finally divided by $\sqrt{2}$.

Because all operations are performed on interval-valued coefficients and the norm is evaluated algebraically, the resulting enclosure of $\|\xi_{v^{\mathrm{ap}}}\|_{H^0_{\mathrm{odd}}}$ is fully rigorous. All these steps are carried out by the routine `norm_xi_twap_sq()` in `lemmas.py`.

The verified enclosure is strictly smaller than the constant stated in the lemma, which completes the proof. □

Proof of Lemmas 3.10 and 3.15: The proof of this lemmas is done in `norm_xi_fvap_sq()` from `lemmas.py`.

The computation of the residual is carried out coefficient by coefficient, following the same strategy used in the proof of Lemma 2.3. In that lemma we used the class `FourierRealSeries` in order to represent the action of the operator on the Fourier coefficients.

In the present case we proceed in the same spirit, but without introducing an auxiliary class. Instead, the computation is implemented directly by manipulating the Fourier coefficients of the approximate eigenvector.

Let f^{ap} denote the approximate eigenvector with Fourier coefficients stored in the vector `fvap`. The $L^2(\mathbb{T})$ norm is computed directly as

$$\|f^{\mathrm{ap}}\|^2_{L^2(\mathbb{T})} = \sum_{k=-N}^{N-1} |\widehat{f^{\mathrm{ap}}_k}|^2,$$

which in the code corresponds to the function `norm_vector_sq()`, from `auxiliar_funcs.py`, applied to `fvap`.

Similarly, the $H^1(\mathbb{T})$ norm is evaluated using the Fourier representation

$$\|f^{\mathrm{ap}}\|^2_{H^1(\mathbb{T})} = \sum_{k=-N}^{N-1} (1+k^2)|\widehat{f^{\mathrm{ap}}_k}|^2.$$

This is implemented by multiplying each Fourier coefficient by the weight $1+k^2$ and summing the resulting contributions.

Finally, the residual is computed from its definition, that is

$$\xi_{f^{\mathrm{ap}}} = \mathcal{L}^{\mathrm{ap}}_\theta f^{\mathrm{ap}} - \lambda^{\mathrm{ap}} f^{\mathrm{ap}}.$$

The operator $\mathcal{L}^{\mathrm{ap}}_\theta$ is applied directly at the level of Fourier coefficients: the derivative, the Hilbert transform, and the convolution with the Fourier coefficients of the wave profile are computed mode by mode. The resulting coefficients are split into those that remain inside the truncated Fourier window and those that fall outside it. The $L^2(\mathbb{T})$ norm of the residual is then obtained as the sum of the squared norms of these two contributions. □

Proof of Lemma 4.3: The Fourier coefficients of v^{ap} are read from the file `tw_data.txt` and represented using interval arithmetic. After reading the file, we also verify that none of these intervals contains 0. The routine checking this is `coeffs_are_not_zero()` from `lemmas.py`. □

Proof of Lemma 4.12: Let $\alpha \in Z$. Since α is a simple root, we may write

$$\begin{aligned} A(\alpha) \prod_{\substack{\alpha' \in Z \\ \alpha' \neq \alpha}} (\alpha - \alpha')^{-1} &= -2\frac{z^{N-1}}{v^{\mathrm{ap}}_N} \prod_{\alpha \in Z} (z - \alpha^{-1})^{-1} \prod_{\substack{\alpha' \in Z \\ \alpha' \neq \alpha}} (\alpha - \alpha')^{-1} \\ &= \lim_{z \to \alpha} -(z-\alpha)\frac{z^{N-1}}{z^N P(z)} = -\frac{\alpha^{N-1}}{\partial_z(z^N P(z))(\alpha)}. \end{aligned}$$

This quantity is evaluated using the interval enclosure of α provided by Lemma C.3 together with interval arithmetic for the evaluation of $\partial_z(z^N P(z))$. The corresponding verification is implemented in the routine `proots_exis()` in `lemmas.py`, which checks that the resulting complex interval cannot contain any integer.

The computation certifies that the above expression is never an integer for any $\alpha \in Z$. □

Proof of Lemma 4.16: Recall from (4.44) that $M^{\mathrm{tor}}_{\mathrm{exis}}$ is defined in terms of finitely many Fourier coefficients of the functions h_k introduced in (4.43). For each $k = 1, \ldots, N+1$, we have constructed an explicit approximation h^{ap}_k. The corresponding numerical data are stored in the file

`exis_data.txt` and described in Appendix III.D.1, where we explain how we compute the Fourier modes of h_k^{ap} and provide explicit $L^2([-\pi,\pi])$ bounds for τ_k, defined by

$$\tau_k := h_k - h_k^{\mathrm{ap}}, \qquad k = 1, \dots, N+1.$$

Lemma D.1 provides rigorous bounds for $\|\tau_k\|^2_{L^2([-\pi,\pi])}$ in terms of the stored quantities ρ_k, listed in `exis_resis.txt`.

Using the h_k^{ap} functions, we construct the matrix

$$M_{\mathrm{exis}}^{\mathrm{thin}} \in \mathbb{C}^{(N+1)\times(N+1)},$$

with entries

$$(M_{\mathrm{exis}}^{\mathrm{thin}})_{jk} = (\widehat{h_k^{\mathrm{ap}}})_{j-1} - \delta_{k,j-1}, \qquad 1 \le j,k \le N+1. \tag{E.5}$$

By construction,

$$M_{\mathrm{exis}}^{\mathrm{tor}} = M_{\mathrm{exis}}^{\mathrm{thin}} + R,$$

where the error matrix R is given by

$$R_{jk} = (\widehat{\tau_k})_{j-1}, \qquad 1 \le j,k \le N+1.$$

A direct expansion shows that

$$M_{\mathrm{exis}}^{\mathrm{tor}\,*} M_{\mathrm{exis}}^{\mathrm{tor}} - M_{\mathrm{exis}}^{\mathrm{thin}*} M_{\mathrm{exis}}^{\mathrm{thin}} = R^* M_{\mathrm{exis}}^{\mathrm{thin}} + M_{\mathrm{exis}}^{\mathrm{thin}*} R + R^* R \tag{E.6}$$

For each $k = 1, \dots, N+1$, we set

$$r_k := \Big(\frac{\rho_k}{2\pi}\Big)^{1/2}, \qquad c_k := \Big(\sum_{j=1}^{N+1} |(M_{\mathrm{exis}}^{\mathrm{thin}})_{jk}|^2\Big)^{1/2}.$$

By Parseval's identity and Lemma D.1,

$$\sum_{j=1}^{N+1} |R_{jk}|^2 \le \frac{1}{2\pi}\|\tau_k\|^2_{L^2([-\pi,\pi])} \le \frac{1}{2\pi}\rho_k = r_k^2.$$

Applying Cauchy–Schwarz, we obtain for every j,k

$$|(R^* M_{\mathrm{exis}}^{\mathrm{thin}})_{jk}| \le r_j c_k, \qquad |(M_{\mathrm{exis}}^{\mathrm{thin}*} R)_{jk}| \le c_j r_k, \qquad |(R^* R)_{jk}| \le r_j r_k.$$

Hence, combining these estimates into (E.6) yields

$$\big|(M_{\mathrm{exis}}^{\mathrm{tor}\,*} M_{\mathrm{exis}}^{\mathrm{tor}} - M_{\mathrm{exis}}^{\mathrm{thin}*} M_{\mathrm{exis}}^{\mathrm{thin}})_{jk}\big| \le r_j c_k + c_j r_k + r_j r_k. \tag{E.7}$$

We define the matrix $B \in \mathbb{R}^{(N+1)\times(N+1)}$ by

$$B_{jk} := r_j c_k + r_k c_j + r_j r_k,$$

and its Frobenius norm

$$\|B\|_F := \Big(\sum_{j,k=1}^{N+1} B_{jk}^2\Big)^{1/2}.$$

Since $\|E\|_2 \le \|E\|_F$ for any matrix E, we conclude from (E.7) that

$$\|M_{\mathrm{exis}}^{\mathrm{tor}\,*} M_{\mathrm{exis}}^{\mathrm{tor}} - M_{\mathrm{exis}}^{\mathrm{thin}*} M_{\mathrm{exis}}^{\mathrm{thin}}\|_2 \le \|B\|_F. \tag{E.8}$$

By Weyl's theorem on eigenvalue perturbations for Hermitian matrices, see for example [Bha97, Corollary III.2.6], we deduce that

$$\begin{aligned}\lambda_{\min}({M_{\text{exis}}^{\text{tor}}}^* M_{\text{exis}}^{\text{tor}}) &\geq \lambda_{\min}({M_{\text{exis}}^{\text{thin}}}^* M_{\text{exis}}^{\text{thin}}) - \|{M_{\text{exis}}^{\text{tor}}}^* M_{\text{exis}}^{\text{tor}} - {M_{\text{exis}}^{\text{thin}}}^* M_{\text{exis}}^{\text{thin}}\|_2 \\ &\geq \lambda_{\min}({M_{\text{exis}}^{\text{thin}}}^* M_{\text{exis}}^{\text{thin}}) - \|B\|_F.\end{aligned}$$

Hence, a lower bound on $\lambda_{\min}({M_{\text{exis}}^{\text{thin}}}^* M_{\text{exis}}^{\text{thin}})$ and B_F immediately yields a rigorous lower bound for

$$\sigma_1^2 = \lambda_{\min}({M_{\text{exis}}^{\text{tor}}}^* M_{\text{exis}}^{\text{tor}}).$$

To obtain a bound for $\lambda_{\min}({M_{\text{exis}}^{\text{thin}}}^* M_{\text{exis}}^{\text{thin}})$, we apply Gershgorin's theorem [Ger31] to a nearly diagonalized representation of ${M_{\text{exis}}^{\text{thin}}}^* M_{\text{exis}}^{\text{thin}}$. To this end, we use the matrix

$$V_{\text{exis}}^{\text{ap}} \in \mathbb{C}^{(N+1)\times(N+1)},$$

provided in the supplementary material, and define

$$\widetilde{S} := {V_{\text{exis}}^{\text{ap}}}^* {M_{\text{exis}}^{\text{thin}}}^* M_{\text{exis}}^{\text{thin}} V_{\text{exis}}^{\text{ap}}.$$

The matrix $\widetilde{S}$ is nearly diagonal and Hermitian. However, its spectrum is not the same as ${M_{\text{exis}}^{\text{thin}}}^* M_{\text{exis}}^{\text{thin}}$ because $V_{\text{exis}}^{\text{ap}}$ is not unitary. However, it is close to a unitary one V. This is proved in Lemma 2.4 from [GO21] for orthogonal matrices, the argument for the unitary case follows the same argument with very small changes. Moreover, immediately after the lemma, the authors provide a method to rigorously construct an enclosure of

$$S := V^* {M_{\text{exis}}^{\text{thin}}}^* M_{\text{exis}}^{\text{thin}} V.$$

from the computed enclosure of $\widetilde{S}$. Applying Gershgorin's theorem to S provides a lower bound for its smallest eigenvalue.

The computer-assisted part of the argument is done in `svd_exis()` from `lemmas.py` and proceeds as follows. The matrix $M_{\text{exis}}^{\text{thin}}$ is constructed directly from the data stored in `exis_data.txt`, this is done in `construct_exis_mat()`. Using the certified $L^2([-\pi,\pi])$-bounds ρ_k, stored in `exis_resis.txt`, the routine `mat_exis_add_radii()` constructs `to_rest`, an upper bound for $\|B\|_F$,

To obtain a lower bound on $\lambda_{\min}({M_{\text{exis}}^{\text{thin}}}^* M_{\text{exis}}^{\text{thin}})$ we first compute $\widetilde{S}$. To this end, we use $V_{\text{exis}}^{\text{ap}}$, stored in `exis_V.txt`. Then the function `S_from_ti_S_for_Gersh()` constructs S, and finally the routine `gersh_smallest_eig()` computes `min_val`, a lower bound for the eigenvalues of S, using Gershgorin's theorem.

Finally, we compute the difference between `min_val` and `to_rest`, yielding

$$\sigma_1^2 > 6.82641 \cdot 10^{-4}.$$

□

Proof of Lemma 4.18: The function β is represented as the `Functions_1D` object with derivatives up to order 2, constructed from the cosine Fourier representation of v^{ap} and the parameter c, by `beta_func_constructor()`.

The maximum bound $B := 19.9056$ of β is then verified by the adaptive routine `max_prev_esti`, we also show that β is positive applying `max_prev_esti()` to $-\beta$ with $B = 0$. The method we choose is based on a second order Taylor, see (E.3).

The whole computation is carried out in `beta_max()` from `lemmas.py`. □

Proof of Lemma 4.19: Observe that

$$\|\kappa_1\|_{L^1([0,\pi])} = \int_0^\pi \int_0^x \beta(y)^2 \, dy \, dx = \int_0^\pi \int_y^\pi \beta(y)^2 \, dx \, dy = \int_0^\pi \beta(y)^2 (\pi - y) \, dy.$$

The function `aux_kappa1_func_constructor()` builds $(\pi - x)\beta(x)^2$ as a `Functions_1D` with derivatives up to order 4. In order to construct β^2, the routine `aux_kappa1_func_constructor()` uses `beta_sq_func_constructor()`, where we use a Fourier representation of $(c + v^{\mathrm{ap}})^2$.

We then compute an enclosure of the integral

$$\int_0^\pi (\pi - x)\beta(x)^2 \, dx$$

using `integ_adaptive_1D()` and `gauss_lege_2dots()`.

All these steps are implemented in the routine `kappa1()` from `lemmas.py`. Finally, the resulting enclosure is compared with the bound in the statement. □

Proof of Lemma 4.20: Since β is $\mathbb{R}$-valued, we have that

$$\left|\partial_x \beta(x) + i\beta(x)^2\right|^2 = \left(\partial_x \beta(x)\right)^2 + \beta(x)^4,$$

and therefore,

$$\|\partial_x \beta + i\beta^2\|^2_{L^2([-\pi,\pi])} = \int_{-\pi}^\pi \left(\partial_x \beta(x)\right)^2 + \beta(x)^4 \, dx.$$

The computer-assisted verification proceeds as follows. We construct a `Functions_1D` object, with derivatives up to order 4, for the nonnegative integrand

$$x \longmapsto \left(\partial_x \beta(x)\right)^2 + \beta(x)^4$$

using `betamod_func_sq_constructor()`. Internally, this constructor builds the equivalent expression

$$\left(\partial_x \beta(x)\right)^2 + \beta(x)^4 = \frac{1 + (\partial_x v^{\mathrm{ap}}(x))^2}{(c + v^{\mathrm{ap}}(x))^4},$$

and creates `FourierRealSeries` instances for the numerator and the denominator.

We then compute an enclosure of the integral

$$\int_0^\pi \left(\partial_x \beta(x)\right)^2 + \beta(x)^4 \, dx$$

using `integ_adaptive_1D()` and `gauss_lege_2dots()`.

Finally, since the integrand is even, the integral over $[-\pi, \pi]$ equals twice the integral over $[0, \pi]$, and hence

$$\|\partial_x \beta + i\beta^2\|^2_{L^2(\mathbb{T})} = 2 \int_0^\pi \left(\partial_x \beta(x)\right)^2 + \beta(x)^4 \, dx.$$

All these steps are implemented in the routine `betamod_L2_sq()` from `lemmas.py`. Finally, the resulting enclosure is compared with the bound in the statement. □

Proof of Lemma 4.21: Our computer-assisted proof evaluates the integral

$$\|\kappa_2\|_{L^1([0,\pi])} = \int_0^\pi \int_0^x \left[(\partial_x \beta(x))^2 + \beta(x)^4\right],$$

using `triangular_integral()`.

We then compare the resulting enclosure of the integral with the bound in the statement. The whole routine is called `kappa2()` and is found in `lemmas.py`. □

Proof of Lemma 4.25: For every $k = 1, \dots, N+1$, the function h_k solves

$$\partial_x h_k(x) = \beta(x)\big(i h_k(x) + g_k(x)\big),$$

where g_k is given by (4.42) for $k = 1, \dots, N$ and $g_{N+1} = 0$. Hence,

$$\partial_x^2 h_k = \beta \partial_x g_k + (\partial_x \beta + i\beta^2) g_k + (i\partial_x \beta - \beta^2) h_k.$$

First, we have for $h_{N+1} = \Pi$ that

$$\|\partial_x^2 \Pi\|_{L^2([-\pi,\pi])} = \|(i\partial_x \beta - \beta^2)\Pi\|_{L^2([-\pi,\pi])} = \|i\partial_x \beta - \beta^2\|_{L^2([-\pi,\pi])},$$

Lemma 4.20 provides an explicit upper bound for $\|i\partial_x \beta - \beta^2\|^2_{L^2([-\pi,\pi])}$.

We now estimate the remaining terms. We have for every $k = 1, \dots, N$ that

$$\|\partial_x^2 h_k\|_{L^2([-\pi,\pi])} \le I_1 + I_2 + I_3,$$

where

$$\begin{aligned} I_1 &:= \|\beta\|_{L^\infty([-\pi,\pi])} \|\partial_x g_k\|_{L^2([-\pi,\pi])}, \\ I_2 &:= \|\partial_x \beta + i\beta^2\|_{L^2([-\pi,\pi])} \|g_k\|_{L^\infty([-\pi,\pi])}, \\ I_3 &:= \|\kappa_2\|^{1/2}_{L^1([0,\pi])} \|g_k\|_{L^2([-\pi,\pi])}. \end{aligned}$$

The quantities $\|\beta\|_{L^\infty([-\pi,\pi])}$, $\|\partial_x \beta + i\beta^2\|_{L^2([-\pi,\pi])}$ and $\|\kappa_2\|_{L^1([0,\pi])}$ are bounded using Lemmas 4.18, 4.20, and 4.21, respectively.

Next, we compute explicit expressions for the norms of g_k. A direct computation yields

$$\|\partial_x g_k\|^2_{L^2([-\pi,\pi])} = k^2 \pi \sum_{m=1}^{N-k} m^2 (v^{\mathrm{ap}}_{k+m})^2 \qquad \text{and} \qquad \|g_k\|^2_{L^2([-\pi,\pi])} = k^2 \pi \Big(\frac{1}{2} (v^{\mathrm{ap}}_k)^2 + \sum_{m=1}^{N-k} (v^{\mathrm{ap}}_{k+m})^2 \Big).$$

To bound $\|g_k\|_{L^\infty([-\pi,\pi])}$, we use the estimates stored in `gk_Linf_bounds.txt`. More precisely, we verify using `max_prev_esti()` that the values C_k, $k = 1, \dots, N$, satisfy

$$\|\mathrm{Re}(g_k)\|_{L^\infty([0,\pi])} < C_k.$$

Hence, we have

$$\|g_k\|^2_{L^\infty([-\pi,\pi])} \le \|\mathrm{Im}(g_k)\|^2_{L^\infty([-\pi,\pi])} + \|\mathrm{Re}(g_k)\|^2_{L^\infty([-\pi,\pi])} \le \Big(\frac{k}{2} v^{\mathrm{ap}}_k \Big)^2 + C_k^2.$$

Combining the above estimates yields the claimed bound. The full verification is carried out in the routine `hk_norm_exis_H2sq()` in `lemmas.py`.

□

Proof of Lemma 5.16: We fix $\lambda = 1.035i$ and show that

$$\det\big(M^{\mathrm{tor}}_{\mathrm{stab}}(\lambda)^* M^{\mathrm{tor}}_{\mathrm{stab}}(\lambda)\big) > 0.$$

We proceed as in the proof of Lemma 4.16. The entries of $M^{\mathrm{tor}}_{\mathrm{stab}}(\lambda)$ are the Fourier coefficients of certain functions, namely h_k from (D.18). With this notation, the matrix $M^{\mathrm{tor}}_{\mathrm{stab}}(\lambda)$ from (5.29) can be written as

$$M^{\mathrm{tor}}_{\mathrm{stab}}(\lambda)_{jk} = \begin{cases} (\widehat{h_{k+N}})_{j-2} - \delta_{j-1,k}, & 1 \le j \le N+1,\ 1 \le k \le N, \\ (\widehat{h_k})_{j-2}, & 1 \le j \le N+1,\ N+1 \le k \le 2N, \\ (\widehat{h_{4N+1}})_{j-2}, & 1 \le j \le N+1,\ k = 2N+1, \\ (\widehat{h_{k+2N}})_{j-N-3}, & N+1 \le j \le 2N+2,\ 1 \le k \le N, \\ (\widehat{h_{k+3N}})_{j-N-3} - \delta_{j-2,k}, & N+1 \le j \le 2N+2,\ N+1 \le k \le 2N, \\ (\widehat{h_{4N+2}})_{j-N-3}, & N+1 \le j \le 2N+2,\ k = 2N+2, \end{cases} \tag{E.9}$$

Hence, using the notation from Lemma D.7, we have

$$\|h_k\|_{L^2([-\pi,\pi])} \leq \|h_k^{\mathrm{ap}}\|_{L^2([-\pi,\pi])} + \rho_k^{1/2}.$$

Consequently, following the same strategy as in the proof of Lemma 4.16, we obtain a rigorous lower bound on the smallest singular value of $M_{\mathrm{stab}}^{\mathrm{tor}}(\lambda)$, which is strictly positive. The only routines that differ are `construct_stab_mat()` and `mat_stab_add_radii()`.

The routine `construct_stab_mat()` constructs $M_{\mathrm{stab}}^{\mathrm{thin}}$ by substituting h_k by h_k^{ap} in (E.9). We define

$$R := M_{\mathrm{stab}}^{\mathrm{tor}} - M_{\mathrm{stab}}^{\mathrm{thin}},$$

so the routine `mat_stab_add_radii()` uses the fact that

$$2\pi \sum_{j=1}^{2N+2} |R_{jk}|^2 \leq \begin{cases} \rho_{k+N} + \rho_{k+2N}, & 1 \leq k \leq N, \\ \rho_k + \rho_{k+3N}, & N+1 \leq k \leq 2N, \\ \rho_{4N+1}, & k = 2N+1, \\ \rho_{4N+2}, & k = 2N+2, \end{cases} \tag{E.10}$$

to bound the following quantity

$$\|M_{\mathrm{stab}}^{\mathrm{thin}*} R + R M_{\mathrm{stab}}^{\mathrm{thin}*} + R^* R\|_F.$$

The full routine is implemented in `svd_regu()` from `lemmas.py`. □

Proof of Lemma 5.22: As in the proof of Lemma 5.16, we adopt the notation from (D.18). However, here we work with $\lambda = \lambda^{\mathrm{ap}}$ instead of $\lambda = 1.035i$, and we use Lemma D.8 rather than Lemma D.7. These are not the only differences from the argument used in the proof of Lemma 5.16. In the present proof, we seek rigorous bounds for the two smallest singular values of a matrix. Consequently, we run the routine `gersh_gap_one_two()` instead of `gersh_smallest_eig()`. The full routine is `svd_sing()` from `lemmas.py`. □

Proof of Lemma 5.23: We load an explicit approximation of u_1 stored in `sing_u1.txt`, after a renormalization, we denote it by u_1^{ap}, i.e. $\|u_1^{\mathrm{ap}}\|_{\ell^2} = 1$. Let $\omega \in \mathbb{C}$ with $|\omega| = 1$ and define $\delta := u_1 - \omega^{-1} u_1^{\mathrm{ap}}$, so we get

$$\begin{aligned} |u_1(\lambda)^* \mathbf{b}(F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-})| &= |(\mathbf{b}(F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-}), u_1(\lambda))_{\ell^2}| \\ &\geq |(\mathbf{b}(F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-}), \omega^{-1} u_1^{\mathrm{ap}})_{\ell^2}| - |(\mathbf{b}(F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-}), \delta)_{\ell^2}| \\ &\geq |(\mathbf{b}(F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-}), u_1^{\mathrm{ap}})_{\ell^2}| - \|\mathbf{b}(F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-})\|_{\ell^2} \|\delta\|_{\ell^2}, \end{aligned}$$

where we have used Cauchy–Schwarz inequality in the last step. We can use Lemma 3.10 and Lemma 5.29 to obtain an upper estimate on $\|\mathbf{b}(F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-})\|_{\ell^2}$. Using these lemmas, we obtain

$$\begin{aligned} \|\mathbf{b}(F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-})\|_{\ell^2}^2 &\leq \|J^+ F_v^{\mathrm{ap}+}\|_{L^2(\mathbb{T})}^2 + \|J^- F_v^{\mathrm{ap}-}\|_{L^2(\mathbb{T})}^2 \\ &\leq \max\left\{\|J^+\|_{L^2(\mathbb{T})\to L^2(\mathbb{T})}, \|J^-\|_{L^2(\mathbb{T})\to L^2(\mathbb{T})}\right\}^2 \|f^{\mathrm{ap}}\|_{L^2(\mathbb{T})}^2. \end{aligned}$$

We now bound $\|\delta\|_{\ell^2}$. Set

$$\omega := \frac{(u_1^{\mathrm{ap}}, u_1)_{\ell^2}}{|(u_1^{\mathrm{ap}}, u_1)_{\ell^2}|}. \tag{E.11}$$

We claim that the denominator $|(u_1^{\mathrm{ap}}, u_1)_{\ell^2}|$ does not vanish, we will show this after (E.14).

For this choice of ω, we find that

$$\begin{aligned}\|\delta\|_{\ell^2}^2 = \|u_1 - \omega^{-1}u_1^{\mathrm{ap}}\|_{\ell^2}^2 &= \|u_1\|_{\ell^2}^2 + \|u_1^{\mathrm{ap}}\|_{\ell^2}^2 - 2\mathrm{Re}(\omega^{-1}u_1^{\mathrm{ap}}, u_1)_{\ell^2} \\ &= 2 - 2|(u_1^{\mathrm{ap}}, u_1)_{\ell^2}| \leq 2(1 - |(u_1^{\mathrm{ap}}, u_1)_{\ell^2}|^2),\end{aligned} \tag{E.12}$$

where we have used that $|(u_1^{\mathrm{ap}}, u_1)_{\ell^2}| \leq 1$ in the last step.

Define $\xi_{u_1^{\mathrm{ap}}} \in \mathbb{C}^{2N+2}$ by

$$\begin{aligned}\xi_{u_1^{\mathrm{ap}}} &:= M_{\mathrm{stab}}^{\mathrm{tor}} M_{\mathrm{stab}}^{\mathrm{tor}\ *} u_1^{\mathrm{ap}} - \sigma_1^2 u_1^{\mathrm{ap}} = \sum_{k=1}^{2N+2} (M_{\mathrm{stab}}^{\mathrm{tor}} M_{\mathrm{stab}}^{\mathrm{tor}\ *} - \sigma_1^2 I)(u_1^{\mathrm{ap}}, u_k)_{\ell^2} u_k \\ &= \sum_{k=1}^{2N+2} (\sigma_k^2 - \sigma_1^2)(u_1^{\mathrm{ap}}, u_k)_{\ell^2} u_k = \sum_{k=2}^{2N+2} (\sigma_k^2 - \sigma_1^2)(u_1^{\mathrm{ap}}, u_k)_{\ell^2} u_k.\end{aligned} \tag{E.13}$$

Since

$$(\sigma_2^2 - \sigma_1^2)^2 \sum_{k=2}^{2N+2} |(u_1^{\mathrm{ap}}, u_k)_{\ell^2}|^2 = (\sigma_2^2 - \sigma_1^2)^2 (1 - |(u_1^{\mathrm{ap}}, u_1)_{\ell^2}|^2),$$

taking the norm in (E.13) yields

$$\|\xi_{u_1^{\mathrm{ap}}}\|_{\ell^2}^2 \geq (\sigma_2^2 - \sigma_1^2)^2 (1 - |(u_1^{\mathrm{ap}}, u_1)_{\ell^2}|^2). \tag{E.14}$$

The denominator $|(u_1^{\mathrm{ap}}, u_1)_{\ell^2}|$ in (E.11) does not vanish. If it did we would have that

$$\|\xi_{u_1^{\mathrm{ap}}}\|_{\ell^2} \geq \sigma_2^2 - \sigma_1^2,$$

so we check that this does not happen.

From (E.12) and (E.14) we get the upper estimate

$$\|\delta\|_{\ell^2} \leq \frac{\sqrt{2}}{\sigma_2^2 - \sigma_1^2} \|\xi_{u_1^{\mathrm{ap}}}\|_{\ell^2}.$$

Here, Lemma 5.22 provides a lower bound for the denominator $\sigma_2^2 - \sigma_1^2$.

In order to bound $\xi_{u_1^{\mathrm{ap}}}$, we will use Corollary 4.15 of Chapter 4 in [SS90] with $A = M_{\mathrm{stab}}^{\mathrm{tor}} M_{\mathrm{stab}}^{\mathrm{tor}\ *}$ and $X = u_1^{\mathrm{ap}}$. Denoting by $a := \|M_{\mathrm{stab}}^{\mathrm{tor}\ *} u_1^{\mathrm{ap}}\|_{\ell^2} \in \mathbb{R}$,[3] we have that

$$\begin{aligned}|a^2 - \sigma_1^2| &\leq \|M_{\mathrm{stab}}^{\mathrm{tor}} M_{\mathrm{stab}}^{\mathrm{tor}\ *} u_1^{\mathrm{ap}} - a^2 u_1^{\mathrm{ap}}\|_{\ell^2} \\ &\leq \|M_{\mathrm{stab}}^{\mathrm{thin}} M_{\mathrm{stab}}^{\mathrm{thin}*} u_1^{\mathrm{ap}} - a^2 u_1^{\mathrm{ap}}\|_{\ell^2} + \|(M_{\mathrm{stab}}^{\mathrm{thin}} R^* + R M_{\mathrm{stab}}^{\mathrm{thin}*} + R R^*) u_1^{\mathrm{ap}}\|_{\ell^2},\end{aligned}$$

where we have written the matrix $M_{\mathrm{stab}}^{\mathrm{tor}}$ as

$$M_{\mathrm{stab}}^{\mathrm{tor}} = M_{\mathrm{stab}}^{\mathrm{thin}} + R,$$

(this same decomposition, with $\lambda = 1.035i$ instead of λ^{ap} is found in the proof of Lemma 5.16.)

Hence, we find that

$$\begin{aligned}\|\xi_{u_1^{\mathrm{ap}}}\|_{\ell^2} = \|M_{\mathrm{stab}}^{\mathrm{tor}} M_{\mathrm{stab}}^{\mathrm{tor}\ *} u_1^{\mathrm{ap}} - \sigma_1^2 u_1^{\mathrm{ap}}\|_{\ell^2} &\leq \|M_{\mathrm{stab}}^{\mathrm{tor}} M_{\mathrm{stab}}^{\mathrm{tor}\ *} u_1^{\mathrm{ap}} - a^2 u_1^{\mathrm{ap}}\|_{\ell^2} + |a^2 - \sigma_1^2| \\ &\leq 2\|M_{\mathrm{stab}}^{\mathrm{thin}} M_{\mathrm{stab}}^{\mathrm{thin}*} u_1^{\mathrm{ap}} - a^2 u_1^{\mathrm{ap}}\|_{\ell^2} + 2\|(M_{\mathrm{stab}}^{\mathrm{thin}} R^* + R M_{\mathrm{stab}}^{\mathrm{thin}*} + R R^*) u_1^{\mathrm{ap}}\|_{\ell^2}.\end{aligned}$$

We bound $\rho := \|(M_{\mathrm{stab}}^{\mathrm{thin}} R^* + R M_{\mathrm{stab}}^{\mathrm{thin}*} + R R^*) u_1^{\mathrm{ap}}\|_{\ell^2}$ as follows. Let r_k and m_k denote the columns of R and $M_{\mathrm{stab}}^{\mathrm{thin}}$ respectively. Hence,

$$\rho \leq \|M_{\mathrm{stab}}^{\mathrm{thin}} R^*\|_2 + \|R M_{\mathrm{stab}}^{\mathrm{thin}*} u_1^{\mathrm{ap}}\|_{\ell^2} + \|R R^*\|_2 \leq \sum_{l=1}^{2N+2} \|m_l r_l^*\|_2 + \|r_l m_l^* u_1^{\mathrm{ap}}\|_{\ell^2} + \|r_l r_l^*\|_2$$

[3]The quantity a corresponds to M in [SS90]. Since we only have one column we also have that $\mu_1 = M$.

Since ab^* is a rank-one matrix for any vectors $a, b \in \mathbb{C}^{2N+2}$, we have that $\|ab^*\|_2 = \|ab^*\|_F \leq \|a\|_F \|b\|_F$, so we get that

$$\rho \leq \sum_{l=1}^{2N+2} \Big(\|m_l\|_{\ell^2} + |m_l^* u_1^{\mathrm{ap}}| + \|r_l\|_{\ell^2} \Big) \|r_l\|_{\ell^2}.$$

The computation of

$$\big| (\mathbf{b}(F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-}), u_1^{\mathrm{ap}})_{\ell^2} \big|,$$

is done using the approximations of $J^+ F_v^{\mathrm{ap}+}$ and $J^- F_v^{\mathrm{ap}-}$ described in Appendix III.D.2.1, namely h_+^{ap} and h_-^{ap}. Recall (5.33), so we have that

$$\mathbf{b}(F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-}) = \begin{pmatrix} \mathbf{b}^+(F_v^{\mathrm{ap}+}) \\ -\mathbf{b}^-(F_v^{\mathrm{ap}-}) \end{pmatrix}.$$

Consequently, applying Lemma D.10 and Parseval's identity, we obtain

$$\|\mathbf{b}(F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-}) - \widetilde{h^{\mathrm{ap}}}\|_{\ell^2}^2 \leq \frac{\rho_+ + \rho_-}{2\pi}.$$

Here the vector $\widetilde{h^{\mathrm{ap}}} \in \mathbb{C}^{2N+2}$ is given by

$$(\widetilde{h^{\mathrm{ap}}})_k = \begin{cases} (\widehat{h_+^{\mathrm{ap}}})_{k-2} & 1 \leq k \leq N+1, \\ (\widehat{h_-^{\mathrm{ap}}})_{k-N-3} & N+2 \leq k \leq 2N+2. \end{cases}$$

Then, by the Cauchy–Schwarz inequality, we have that

$$\big| (\mathbf{b}(F_v^{\mathrm{ap}+}, F_v^{\mathrm{ap}-}), u_1^{\mathrm{ap}})_{\ell^2} \big| \geq \Big| \sum_{j=-1}^{N-1} (\widehat{h_+^{\mathrm{ap}}})_j (\overline{u_1^{\mathrm{ap}}})_{j+2} - \sum_{j=-1}^{N-1} (\widehat{h_-^{\mathrm{ap}}})_j (\overline{u_1^{\mathrm{ap}}})_{j+N+3} \Big| - \frac{(\rho_+ + \rho_-)^{1/2}}{\sqrt{2\pi}}.$$

The full routine is done in `prod_fv_u1()` from `lemmas.py`. □

Proof of Lemma 5.25: For every $\lambda \in \mathbb{C}$, we have that

$$\lim_{z \to \alpha} i \frac{z-\alpha}{z} \Big((i \pm \lambda) \frac{z^N}{z^N P(z)} - i\theta^{\pm} \Big) = \lim_{z \to \alpha} i(z-\alpha)(i \pm \lambda) \frac{z^{N-1}}{z^N P(z)} = (\pm i\lambda - 1) \frac{\alpha^{N-1}}{\partial_z (z^N P(z))(\alpha)}.$$

Using the interval enclosures of α provided by Lemma C.3, this expression is evaluated with rigorously. The routine `proots_stab()` in `lemmas.py` verifies that, for every $\alpha \in Z$ and for both choices of sign, the resulting enclosure cannot contain an integer. □

Proof of Lemma 5.27: We know by Lemma 4.18 that β is a positive and bounded function. Hence,

$$0 < \int_0^{\pi} \beta(y)\, dy < \infty.$$

We then know that

$$I_\beta := e^{2\mathrm{Re}(\lambda^{\mathrm{ap}}) \int_0^{\pi} \beta(y)\, dy} > 1. \tag{E.15}$$

The quantity $\int_0^{\pi} \beta(y)\, dy$ is computed by the routine `integ_adaptive_1D()`.

Using (E.15) and κ_1 in Lemma 4.19, we obtain for every $x \in [0, \pi]$,

$$\begin{aligned} |\kappa_3^{\pm}(x)| &= e^{\pm 2\mathrm{Re}(\lambda^{\mathrm{ap}}) \int_0^x \beta(y)\, dy} \int_0^x \beta(y)^2 e^{\mp 2\mathrm{Re}(\lambda^{\mathrm{ap}}) \int_0^y \beta(s)\, ds}\, dy \\ &\leq e^{2\mathrm{Re}(\lambda^{\mathrm{ap}}) \int_0^{\pi} \beta(y)\, dy} \int_0^x \beta(y)^2\, dy = I_\beta \kappa_1(x). \end{aligned}$$

We finally apply Lemma 4.19. The whole computation is done in `kappa3()` from `lemmas.py`. □

Proof of Lemma 5.28: We construct the `Functions_1D` instances `integ_p` and `integ_m` with the routine `betamod_laap_sq_func_constructor()`. Here, `integ_p` represents $I^+(x) := (i + \lambda^{\mathrm{ap}})\beta(x) - i\theta$, while `integ_m` represents $I^-(x) := (i - \lambda^{\mathrm{ap}})\beta(x) + i\theta$.

We also construct an instance representing β^2 with `beta_sq_func_constructor()`.

Recall I_β from (E.15) and notice that

$$\|\kappa^+_{4,\kappa_3^\pm}\|_{L^1([0,\pi])} = \int_0^\pi |I^+(x)|^2 \kappa_3^\pm(x)\,dx \le I_\beta \int_0^\pi |I^+(x)|^2 \kappa_1(x)\,dx = I_\beta \int_0^\pi \int_0^x |I^+(x)|^2 \beta(y)^2\,dy\,dx.$$

The computation for $\kappa^-_{4,\kappa_3^\pm}$ follows in the same way.

Finally, the integrals

$$\int_0^\pi \int_y^\pi |I^+(x)|^2 \beta(y)^2\,dx\,dy \qquad \text{and} \qquad \int_0^\pi \int_y^\pi |I^-(x)|^2 \beta(y)^2\,dx\,dy$$

are evaluated with `triangular_integral()`. The whole computation is realized in `kappa4()` from `lemmas.py`. □

Proof of Lemma 5.31: We have to compute four different sums. We explain only how the third and fourth inequalities work, since the other two are analogous. Recall (D.18), that is $h_{4N+1} = \Pi^+$, and $h_{4N+2} = \Pi^-$ and

$$h_k = J^+ g^+_{k,\theta^+},\ h_{k+N} = J^+ g^-_{k,\theta^-},\ h_{k+2N} = J^- g^+_{k,\theta^-} \text{ and } h_{k+3N} = J^- g^-_{k,\theta^+} \text{ for all } k = 1,\dots,N.$$

Consequently, if we define $\mathfrak{S} := \{4N+1\} \cup \{k \in \mathbb{Z} : 1 \le k \le 2N\}$, the left member of the third inequality of the statement is

$$\sum_{k\in\mathfrak{S}} \|\partial_x h_k\|^2_{L^2([-\pi,\pi])} \le \sum_{k\in\mathfrak{S}} \left(\|\partial_x h_k^{\mathrm{ap}}\|_{L^2([-\pi,\pi])} + \|\partial_x(h_k - h_k^{\mathrm{ap}})\|_{L^2([-\pi,\pi])}\right)^2.$$

Now, we evaluate $\|\partial_x h_k^{\mathrm{ap}}\|_{L^2([-\pi,\pi])}$ with the local routine `_hk_norms_p()`, and use the Lemmas 5.29 and 5.30, and D.8 to get the upper bound ρ_k^1 for $\|\partial_x \tau_k\|_{L^2([-\pi,\pi])}$. Observe that in the proof of Lemma D.8 we enclose every residual quantity $\|\xi_k^{(s_J,s_g)}\|^2_{L^2([-\pi,\pi])}$, multiply it by the constant C_0 in Lemma 5.29, and compare it with the quantity ρ_k in `sing_resis.txt`:

$$\|h_k - h_k^{\mathrm{ap}}\|^2_{L^2([-\pi,\pi])} < C_0 \|\xi_k^{(s_J,s_g)}\|^2_{L^2([-\pi,\pi])} < \rho_k.$$

Now, let C_1^+ be the constant from the first inequality in Lemma 5.30, we obtain

$$\|\partial_x(h_k - h_k^{\mathrm{ap}})\|^2_{L^2([-\pi,\pi])} < C_1^+ \|\xi_k^{(s_J,s_g)}\|^2_{L^2([-\pi,\pi])} < \frac{C_1^+}{C_0}\rho_k.$$

The fourth inequality is analogous, if we define $\mathfrak{D} := \{4N+2\} \cup \{k \in \mathbb{Z} : 2N+1 \le k \le 4N\}$, we have to estimate now

$$\sum_{k\in\mathfrak{D}} \|\widetilde{\partial_x} h_k\|^2_{L^2([-\pi,\pi])} \le \sum_{k\in\mathfrak{D}} \left(\|\widetilde{\partial_x} h_k^{\mathrm{ap}}\|_{L^2([-\pi,\pi])} + \|\widetilde{\partial_x}\tau_k\|_{L^2([-\pi,\pi])}\right)^2.$$

The term $\|\widetilde{\partial_x} h_k^{\mathrm{ap}}\|_{L^2([-\pi,\pi])}$ is evaluated with the local routine `_hk_norms_m()`, while $\|\widetilde{\partial_x}\tau_k\|_{L^2([-\pi,\pi])}$ is bounded using again Lemmas 5.29, 5.30, and D.8. Notice that the constant we take now in Lemma 5.30 is the one in the second inequality. Denoting it by C_1^- we obtain

$$\|\widetilde{\partial_x}(h_k - h_k^{\mathrm{ap}})\|^2_{L^2([-\pi,\pi])} < C_1^- \|\xi_k^{(s_J,s_g)}\|^2_{L^2([-\pi,\pi])} < \frac{C_1^-}{C_0}\rho_k.$$

□

IV. Linear unstable de Gregorio steady states

This chapter is based on a preliminary version of the manuscript [Pas26]. The text has been adapted to the format and notation of this thesis. The version presented here proves the linear instability of all De Gregorio excited states.

IV.1 Introduction

The De Gregorio equation is the one-dimensional active scalar model

$$\begin{cases} \partial_t \omega + u\partial_x \omega = \omega \partial_x u, \quad (x,t) \in \mathbb{R} \times [0,\infty). \\ u = -\Lambda^{-1}\omega, \\ \omega(x,0) = \omega_0(x). \end{cases} \tag{1.1}$$

Here $\Lambda := (-\Delta)^{1/2}$.

Equations of this type were introduced as one-dimensional models for the vorticity formulation of the three-dimensional incompressible Euler equations, where the dynamics is governed by the competition between transport and vortex stretching. The model of Constantin, Lax and Majda [CLM85] keeps only the stretching term $\omega\partial_x u$ and develops singularities in finite time from smooth data. De Gregorio [De 90; De 96] added back the transport term $u\partial_x\omega$, whose regularizing effect can compete with the stretching and prevent blow-up. The two terms are interpolated by the one-parameter family of Okamoto, Sakajo and Wunsch [OSW08], of which (1.1) is the endpoint where transport is fully present.

The behavior of (1.1) depends strongly on both the regularity of the data and the underlying domain. On the real line, Chen, Hou and Huang [CHH21] proved finite-time blow-up from smooth odd initial data. Singularity formation for lower-regularity data was established in [EJ20b]. The periodic problem on $\mathbb{T}$ has a rather different character. Local well-posedness for $\omega_0 \in H^1(\mathbb{T})$ is proved in [OSW08]. It extends to $s > \frac{1}{2}$ by the methods of [Dan01; BCD11]. Jia, Stewart and Šverák [JSS19] outline this extension and prove that, for initial data C^2-close to an equilibrium $A\sin(x - x_0)$, the solution exists globally and converges exponentially to a nearby equilibrium in $H^s(\mathbb{T})$ for any $s < \frac{3}{2}$. Lei, Liu and Ren [LLR20] proved global well-posedness for initial data with nonnegative vorticity and stability near the ground state $-\sin x$. Chen [Che21] proved finite-time blow-up for the related gCLM equation on the circle when the transport coefficient is slightly below one, starting from smooth data with the same oddness and sign as the π-periodic class studied in [Che23]. For the De Gregorio equation, Chen [Che23] established a one-point blow-up criterion and global regularity for C^1 data in this class under an additional integral condition, while constructing finite-time blow-up solutions from $C^\alpha \cap H^s$ data for any $0 < \alpha < 1$ and $s < \frac{3}{2}$. The latter construction yields blow-up data arbitrarily close to $-\sin x$ in H^1 and, by spatial rescaling, to $-\sin(2x)$ within the π-periodic subspace of $\mathbb{T}$. Thus these equilibria admit blow-up data in arbitrarily small H^1 neighborhoods even though the corresponding sufficiently regular data in

Chen's symmetry class are global. Our instability result concerns H^2 perturbations in the full 2π-periodic space. The periodic stationary states

$$x \mapsto -\sin(nx), \qquad n \geq 1$$

form a natural hierarchy. The case $n = 1$ is the ground state, while the higher-frequency states $n \geq 2$ may be viewed as excited states. A natural question is then whether stability is a special feature of the lowest mode, or whether it persists along the whole family.

The purpose of this chapter is to show that it does not persist. We prove that every excited state $-\sin(nx)$, $n \geq 2$, has an unstable eigenvalue in the Sobolev space $H^2(\mathbb{T})$. Thus, within this family of steady states, the ground state is spectrally distinguished from all higher modes.

Instability of the first excited state $-\sin(2x)$ was recently obtained by Guo and Jiu [GJ25] through an energy method. Our spectral approach applies to every excited state and gives the following linear instability result.

Theorem 1.1 *For every $n \geq 2$, the stationary solution*

$$x \mapsto -\sin(nx)$$

of the De Gregorio equation is linearly unstable in $H^2(\mathbb{T})$.

Proof strategy. We use a decomposition by residue classes modulo n, which unfolds the linearization L_n into a one-parameter family of fiber operators L_θ, $\theta = \frac{k}{n}$ (Lemma 2.4). On each fiber the eigenvalue equation reduces to an ODE with regular singular points at $z = 0, z = 1$ and $z = -1$. Local solutions are constructed by the Frobenius method and matched at two of the three singular points, yielding a 4×4 matrix $\mathbf{M}_{\lambda,\theta}$ whose determinant encodes the eigenvalues. This Fuchsian approach was already used in [JSS19] for the linearization around the ground state. We also refer to our previous work [CGP26], where a matching condition of a Fuchsian system is used in another context.

For each n, we can choose k so that θ is convenient. A single computer-assisted certificate on these fibers yields *linear* instability of $-\sin(nx)$ for *every* n, with an explicit real eigenvalue $\lambda_\theta \in (0.56, 0.6)$.

IV.1.1 Computer-assisted methods

The verification is delicate because the entries of $\mathbf{M}_{\lambda,\theta}$ are computed from truncated power series depending on the parameters. We therefore use a computer-assisted proof. The matrix entries are enclosed by interval arithmetic and the truncation errors are controlled rigorously, as described in Appendix IV.B and Appendix IV.C. The code is available on GitHub at https://github.com/MiguelMGPascualCaballo/dgreg/tree/v1.0 .

There are by now many works in which computer-assisted arguments play a substantial role. We mention, for instance, Guo and Hadžić, Jang, Schrecker [Guo+22] for Euler–Poisson, Kobayashi [Kob10] for extreme Stokes waves, Chen, Hou, and Huang [CHH21] for the De Gregorio model, Castro, Córdoba, and Gómez-Serrano [CCG20a] for SQG, Elgindi and Pasqualotto [EP25] and Chen and Hou [CH22; CH25b; CH25a] for Boussinesq-type models and 3D Euler, Cadiot [Cad25] for Whitham-type equations, several works on the Kuramoto–Sivashinsky equation [AK10; FL17; GL17; Fig+17; Zgl04; ZM01], Buckmaster, Cao-Labora, and Gómez-Serrano [BCG25] for compressible Euler and Navier–Stokes, and Arioli, Gazzola, and Koch together with Bedrossian and Punshon-Smith [AGK21; BP24] for incompressible Navier–Stokes.

For general background on validated numerics we refer to Tucker's book [Tuc11]. For computer-assisted proofs in PDE, see also the survey [Góm19] and the book [NPW19].

IV.1.2 Functional setting and notation

We consider complex-valued functions. When needed, we will explicitly indicate that a function is real-valued.

In this chapter we use the following convention for the L^2 norm:

$$\|f\|^2_{L^2(\mathbb{T})} := \frac{1}{2\pi}\int_{-\pi}^{\pi} |f(x)|^2\,dx = \sum_{k\in\mathbb{Z}} |\widehat{f}_k|^2,$$

where

$$\widehat{f}_k := \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-ikx}\,dx.$$

For $s \geq 0$, $H^s(\mathbb{T})$ denotes the completion of $C^\infty(\mathbb{T})$ with respect to the norm

$$\|f\|^2_{H^s(\mathbb{T})} := \frac{1}{2\pi}\int_{-\pi}^{\pi} |f(x)|^2 + |\Lambda^s f(x)|^2\,dx = \sum_{k\in\mathbb{Z}} (1+|k|^{2s})|\widehat{f}_k|^2$$

Here $\Lambda^0 = I$, so the weight at $k = 0$ is also defined when $s = 0$. For $s > 0$, $H^{-s}(\mathbb{T})$ denotes the dual of $H^s(\mathbb{T})$ with respect to the L^2 pairing. In particular, $L^2(\mathbb{T}) \subset H^{-s}(\mathbb{T})$.

For $s \geq 0$, the operator $\partial_x^{-1} : H^s(\mathbb{T}) \to H^{s+1}(\mathbb{T})$ is the Fourier multiplier

$$\widehat{(\partial_x^{-1} g)}_k = \begin{cases} (ik)^{-1}\widehat{g}_k, & k \neq 0, \\ 0, & k = 0. \end{cases}$$

Thus $\partial_x\partial_x^{-1} g = g - \widehat{g}_0$. For smooth g, the equivalent integral formula is

$$\partial_x^{-1} g(x) = \int_0^x (g(y) - \widehat{g}_0)\,dy - \frac{1}{2\pi}\int_{-\pi}^{\pi}\left(\int_0^x (g(y) - \widehat{g}_0)\,dy\right)dx.$$

We use the same zero-mode convention for Λ^{-1}: its symbol is $|k|^{-1}$ for $k \neq 0$ and zero for $k = 0$.

The Hilbert transform H is the Fourier multiplier

$$\widehat{(Hf)}_k = -i\,\mathrm{sgn}(k)\widehat{f}_k.$$

With these conventions $\Lambda = H\partial_x$, and the velocity field in (1.1) satisfies $\partial_x u = H\omega$.

$\mathcal{H}$ denotes the space of holomorphic functions on the unit disk $\mathbb{D}$. If $F \in \mathcal{H}$, then F admits the expansion

$$F(z) = \sum_{k\geq 0} F_k z^k$$

with $F_k \in \mathbb{C}$.

For such F, write formally $F|_{\mathbb{T}}(x) := \sum_{k\geq 0} F_k e^{ikx}$. We also introduce the Hardy spaces

$$\mathcal{H}^s = \{F \in \mathcal{H} :\ F|_{\mathbb{T}} \in H^s(\mathbb{T})\},$$

endowed with the norm

$$\|F\|_{\mathcal{H}^s} := \|F|_{\mathbb{T}}\|_{H^s(\mathbb{T})}.$$

We will use the following notation

$$f^+ := \frac{1}{2}(f + iHf) + \frac{1}{2}\widehat{f}_0, \qquad f^- := \frac{1}{2}(f^\sharp + iHf^\sharp) + \frac{1}{2}\widehat{f}_{-1},$$

where $f^\sharp(x) := e^{-ix} f(-x)$.

It will be convenient to write

$$\mathcal{P}^{\pm} f := f^{\pm}$$

since sometimes we will need to apply $\mathcal{P}^{\pm}$ to equations rather than functions.

Notice that for $f \in H^s(\mathbb{T})$, with $s > \frac{1}{2}$, the Fourier series of f^+ and f^- are

$$f^+(x) = \sum_{k\geq 0} \widehat{f}_k \mathbf{e}^k, \qquad f^-(x) = \sum_{k\geq 0} \widehat{f}_{-k-1} \mathbf{e}^k.$$

Replacing $\mathbf{e}^1$ by z, we obtain holomorphic extensions $F^+, F^- \in \mathcal{H}^s$ of f^+ and f^- to the unit disk:

$$F^+(z) := \sum_{k\geq 0} \widehat{f}_k z^k, \qquad F^-(z) := \sum_{k\geq 0} \widehat{f}_{-k-1} z^k.$$

In other words, F^+ encodes the nonnegative Fourier modes of f, while F^- encodes the negative ones after reindexing.

For $n \geq 2$ and $l = 1, \ldots, n-1$, we also use the unfolding maps

$$\widehat{(\mathcal{P}_{l,n} f)}_k = \widehat{f}_{nk+l}.$$

Thus $\mathcal{P}_{l,n}$ selects the Fourier modes congruent to l modulo n and reindexes them by $nk + l \mapsto k$.

We denote by S_n and C_n the trigonometric functions given by

$$S_n(x) = \sin(nx), \qquad C_n(x) = \cos(nx).$$

Finally, we write $D_r(z_0) \subset \mathbb{C}$ for the open complex disk of radius $r > 0$ centered at z_0, i.e.

$$D_r(z_0) := \{z \in \mathbb{C} : |z - z_0| < r\}.$$

IV.2 The linearized equation

This section proves Theorem 2.1. We begin by linearizing (1.1) around $-S_n$.

Let ω and u be functions solving (1.1) and write

$$\omega = -S_n + \eta. \tag{2.1}$$

We can then write u as

$$u = \frac{1}{n} S_n + H\partial_x^{-1}\eta. \tag{2.2}$$

Linearizing (1.1) around the steady state $-S_n$, the equation for η reads

$$\partial_t \eta = L_n \eta, \tag{2.3}$$

where

$$L_n\eta := -\frac{1}{n} S_n \partial_x \eta + nC_n H\partial_x^{-1}\eta - S_n H\eta + C_n\eta. \tag{2.4}$$

Theorem 2.1 *For every $n \geq 2$, there exist $\lambda_n \in (0.56, 0.6)$ and $0 \neq \eta_n \in H^s(\mathbb{T})$ for every $s < \frac{3}{2} + \lambda_n$ such that*

$$e^{\lambda_n t}\eta_n(x)$$

solves (2.3).

Remark 2.2 *As we will see (Remark 3.8), the spectrum of L_n on $H^s(\mathbb{T})$ contains the band*

$$\left\{ z \in \mathbb{C} : |\mathrm{Re}(z)| < \frac{3}{2} - s \right\} \qquad \text{for all } s \in \left(0, \frac{3}{2}\right).$$

The spectral band prevents isolation when $s < \frac{3}{2} - \lambda_n$, where $\frac{3}{2} - \lambda_n \in (0.90, 0.94)$. For the selected fiber, Corollary 3.18 and the analyticity of the matching determinant (Remark 3.22) show that λ_n is an isolated eigenvalue of L_θ on $H^s(\mathbb{T})$ for every $s \in [1, 2]$.

The rest of this section is devoted to the proof of Theorem 2.1. Notice that Theorem 1.1 follows from Theorem 2.1 since $\lambda_n > 0.56 > \frac{1}{2}$, giving $2 < \frac{3}{2} + \lambda_n$ and hence $\eta_n \in H^2(\mathbb{T})$.

IV.2.1 Decoupling the modes

We begin by reformulating the problem in a way that works uniformly for all n. Denote by $\mathbf{e}^k : \mathbb{R} \to \mathbb{C}$ the function given by

$$\mathbf{e}^k(x) := e^{ikx}.$$

The key point is that L_n only couples neighboring modes inside each congruence class. In Fourier variables this takes the form

$$L_n \mathbf{e}^k = A_{k,n}\mathbf{e}^{k-n} + B_{k,n}\mathbf{e}^{k+n}, \tag{2.5}$$

where

$$A_{k,n} = \frac{(k+n)(k-n\mathrm{sgn}(k))}{2nk}, \qquad B_{k,n} = -\frac{(k-n)(k-n\mathrm{sgn}(k))}{2nk}, \tag{2.6}$$

for $k \neq 0$. Guided by this formula, we introduce the formal operator L_θ, with $\theta \in (0,1)$, defined by

$$L_\theta \mathbf{e}^k := \widetilde{A}_{k,\theta}\mathbf{e}^{k-1} + \widetilde{B}_{k,\theta}\mathbf{e}^{k+1}, \tag{2.7}$$

where

$$\widetilde{A}_{k,\theta} = \frac{(k+\theta+1)(k+\theta-\mathrm{sgn}(k+\theta))}{2(k+\theta)}, \qquad \widetilde{B}_{k,\theta} = -\frac{(k+\theta-1)(k+\theta-\mathrm{sgn}(k+\theta))}{2(k+\theta)}. \tag{2.8}$$

Remark 2.3 *Let $\theta \in (0,1)$. For every $k \in \mathbb{Z}$,*

$$\widetilde{A}_{-k-1,\theta} = \widetilde{B}_{k,1-\theta}.$$

This relation will be useful when proving Lemma 3.3.

The following lemma describes the precise relationship between L_n and L_θ.

Lemma 2.4 *Let $n \in \mathbb{N}$ with $n \geq 2$, and let $\eta : \mathbb{T} \to \mathbb{C}$ be an $H^1(\mathbb{T})$ function satisfying*

$$\widehat{\eta}_{nk} = 0 \quad \text{for all } k \in \mathbb{Z}.$$

Then η satisfies (2.3) if and only if, for every $1 \leq l \leq n-1$, the function $\eta^\theta := \mathcal{P}_{l,n}\eta$ satisfies

$$\partial_t \eta^\theta = L_\theta \eta^\theta, \quad \text{where } \theta := \frac{l}{n}. \tag{2.9}$$

Proof: Projecting (2.3) onto e^{ikx} for integers k that are not divisible by n gives

$$\partial_t \widehat{\eta}_k = \widehat{(L_n\eta)}_k = A_{k+n,n}\widehat{\eta}_{k+n} + B_{k-n,n}\widehat{\eta}_{k-n}.$$

Recall that the Fourier modes of $\mathcal{P}_{l,n}\eta$ are given by

$$\widehat{\mathcal{P}_{l,n}\eta}_k = \widehat{\eta}_{nk+l}.$$

Setting $\theta := \frac{l}{n}$ and $\eta^\theta := \mathcal{P}_{l,n}\eta$, we substitute $k \mapsto nk+l$ to obtain

$$\begin{aligned}\partial_t(\widehat{\eta^\theta})_k = \partial_t\widehat{\eta}_{nk+l} &= A_{n(k+1)+l,n}\widehat{\eta}_{n(k+1)+l} + B_{n(k-1)+l,n}\widehat{\eta}_{n(k-1)+l} \\ &= \widetilde{A}_{k+1,\theta}(\widehat{\eta^\theta})_{k+1} + \widetilde{B}_{k-1,\theta}(\widehat{\eta^\theta})_{k-1},\end{aligned} \tag{2.10}$$

which holds for all $k \in \mathbb{Z}$. Finally, (2.10) is precisely the projection of (2.9) onto e^{ikx}. □

This reduces the problem to the family of operators L_θ. We now show that it is enough to restrict attention to $\theta \in \left[\frac{1}{3}, \frac{1}{2}\right]$.

Lemma 2.5 *Let $n \in \mathbb{N}$ with $n \geq 2$. Then there exists $1 \leq h \leq n-1$ such that*

$$\frac{h}{n} \in \left[\frac{1}{3}, \frac{1}{2}\right].$$

Proof: For $n = 2, 3$ the choice $h = 1$ works. Assume $n \geq 4$ and set

$$h = \left\lceil \frac{n}{3} \right\rceil.$$

Then $h \geq \frac{n}{3}$ and $3h \leq n + 2$, which gives

$$h \leq \frac{n}{3} + \frac{2}{3} \leq \frac{n}{2},$$

where the last inequality uses $n \geq 4$. Hence $\frac{1}{3} \leq \frac{h}{n} \leq \frac{1}{2}$. □

By Lemma 2.5, for every n there exists a fraction $\theta = \frac{h}{n} \in \left[\frac{1}{3}, \frac{1}{2}\right]$. We now choose this fraction so that the divisibility argument can be applied. The verification in Section IV.3 is carried out at finitely many values of θ and on one short interval.

At this point it is enough to consider prime values of n. Lemma 2.4 shows that once instability is known for L_p, it follows for every L_n with $p \mid n$. The next lemma tells us which values of $\theta = \frac{h}{p} \in \left[\frac{1}{3}, \frac{1}{2}\right]$ need to be certified.

Lemma 2.6 *Let $N := 900$, let p^* be the smallest prime with $p^* \geq N$, and set*

$$J := \Big[\frac{5}{12} - \frac{1}{2p^*},\ \frac{5}{12} + \frac{1}{2p^*}\Big],$$

the interval of length $\frac{1}{p^*}$ centered at the midpoint of $\left[\frac{1}{3}, \frac{1}{2}\right]$. Then $J \subset \left[\frac{1}{3}, \frac{1}{2}\right]$, and:

1. *for every prime $p < N$ there is $h_p \in \{1, \ldots, p-1\}$ with $\theta_p := \frac{h_p}{p} \in \left[\frac{1}{3}, \frac{1}{2}\right]$.*
2. *for every prime $p \geq N$ there is $h_p \in \{1, \ldots, p-1\}$ with $\theta_p \in J$.*

Proof: Item 1 follows from Lemma 2.5, which shows that the interval $\left[\lceil \frac{p}{3} \rceil, \lfloor \frac{p}{2} \rfloor\right]$ is non-empty for every prime $p \geq 2$. For the certification in Lemma 3.20, we fix the explicit choice

$$h_p := \operatorname{round}\Big(p \cdot \frac{5}{12}\Big) \text{ clipped to } \left[\left\lceil \frac{p}{3} \right\rceil, \left\lfloor \frac{p}{2} \right\rfloor\right],$$

so that $\theta_p = \frac{h_p}{p} \in \left[\frac{1}{3}, \frac{1}{2}\right]$ (the clipping directly ensures this). This is the value certified in Lemma 3.20.

Since $p^* = 907 \geq 6$, we have $\frac{1}{2p^*} \leq \frac{1}{12}$. Hence

$$J \subset \left[\frac{5}{12} - \frac{1}{12}, \frac{5}{12} + \frac{1}{12}\right] = \left[\frac{1}{3}, \frac{1}{2}\right].$$

This proves that $J \subset \left[\frac{1}{3}, \frac{1}{2}\right]$.

It remains to prove item 2. Let $p \geq N$ be prime. Since p^* is the smallest prime greater than or equal to N, we have $p \geq p^*$, and hence

$$\frac{1}{p} \leq \frac{1}{p^*} = |J|.$$

Write $J = [\alpha, \beta]$ and choose

$$h := \lceil \alpha p \rceil.$$

Then

$$\alpha p \leq h < \alpha p + 1 \leq \beta p,$$

and therefore $\frac{h}{p} \in [\alpha, \beta) \subset J$. Since $J \subset \left[\frac{1}{3}, \frac{1}{2}\right] \subset (0,1)$, this integer satisfies $1 \leq h \leq p-1$. □

It remains to prove the following spectral statement.

Proposition 2.7 *Let $\theta = \theta_p$ for some prime $p < N$, or $\theta \in J$, with θ_p, J, N as in Lemma 2.6. Then there exist $\lambda_\theta \in (0.56, 0.6)$ and $0 \neq \eta_\theta \in H^s(\mathbb{T})$ for every $s < \frac{3}{2} + \lambda_\theta$ satisfying*

$$\lambda_\theta \eta_\theta = L_\theta \eta_\theta. \tag{2.11}$$

Proof: Section IV.3 is devoted to proving Proposition 2.7. □

We refer to Figure IV.1 for a numerical illustration of the eigenvalue λ_θ.

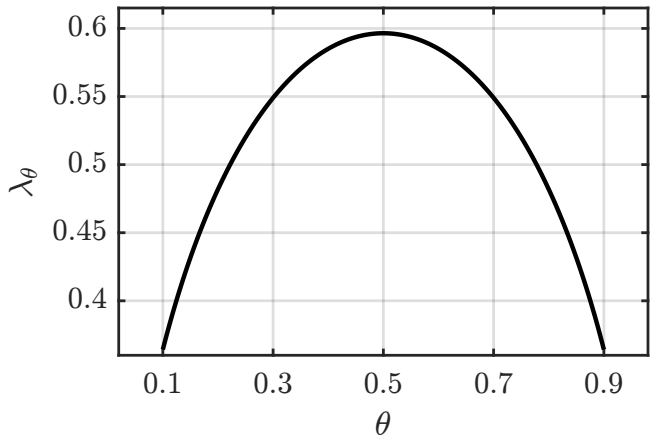


Figure IV.1: Numerical approximation of the eigenvalue λ_θ of L_θ.

IV.2.2 Conclusion of the proof of Theorem 2.1

Fix $n \geq 2$, and choose a prime divisor p of n. Write $n = pm$ with $m \in \mathbb{N}$. By Lemma 2.6, we can choose $h \in \{1, \dots, p-1\}$ so that $\theta := \frac{h}{p} \in \left[\frac{1}{3}, \frac{1}{2}\right]$. More precisely, $\theta = \theta_p$ if $p < N$, and $\theta \in J$ if $p \geq N$. We may then apply Proposition 2.7 and obtain $\lambda > 0.56$ and $0 \neq \eta_0 \in H^s(\mathbb{T})$ for every $s < \frac{3}{2} + \lambda$ satisfying (2.11).

Set $l := hm$. Since $1 \leq h \leq p-1$, we have $1 \leq l \leq n-1$, and also $\frac{l}{n} = \theta$. Therefore

$$\eta^\theta(x,t) = e^{\lambda t} \eta_0(x)$$

solves (2.9) with this choice of l.

We now define

$$\widetilde{\eta_0}(x) := e^{ilx} \eta_0(nx).$$

Since $(\widehat{\widetilde{\eta_0}})_{nk+l} = (\widehat{\eta_0})_k$ for all $k \in \mathbb{Z}$ and all other Fourier modes of $\widetilde{\eta_0}$ vanish, for $s \geq 0$ we have $\|\widetilde{\eta_0}\|^2_{H^s(\mathbb{T})} = \sum_k (1 + |nk+l|^{2s}) |(\widehat{\eta_0})_k|^2 \sim_{n,l,s} \|\eta_0\|^2_{H^s(\mathbb{T})}$, so $\widetilde{\eta_0} \in H^s(\mathbb{T})$ for every $0 \leq s < \frac{3}{2} + \lambda$. For $s < 0$, the claim follows from $\widetilde{\eta_0} \in H^0(\mathbb{T}) \subset H^s(\mathbb{T})$.

Then $\widetilde{\eta}_0$ satisfies

$$(\widehat{\widetilde{\eta}_0})_{nk} = 0 \qquad \text{for all } k \in \mathbb{Z}.$$

Moreover, for every $1 \leq l' \leq n-1$,

$$\mathcal{P}_{l',n}\widetilde{\eta}_0 = \delta_{l,l'}\eta_0.$$

By Lemma 2.4, the function $\widetilde{\eta}$ given by

$$\widetilde{\eta}(x,t) = e^{\lambda t}\widetilde{\eta}_0(x)$$

solves (2.3). This proves Theorem 2.1, with $\lambda_n := \lambda$ and $\eta_n := \widetilde{\eta}_0$.

IV.3 Proof of Proposition 2.7

To prove Proposition 2.7, we construct a real-valued continuous function ν of $\theta \in \left[\frac{1}{3}, \frac{1}{2}\right]$ and $\lambda \in (0.56, 0.6)$ whose zeros characterize the eigenvalues of L_θ. The function ν is defined from a real 4×4 matrix $\mathbf{M}_{\lambda,\theta}$ by

$$\nu(\lambda, \theta) := \det(\mathbf{M}_{\lambda,\theta}), \tag{3.1}$$

so that Proposition 2.7 reduces to showing that $\nu(\cdot, \theta)$ has a zero in $(0.56, 0.6)$ for $\theta = \theta_p$ at every prime $p < N$, and for every $\theta \in J$.

Section IV.3.1 reduces

$$L_\theta \eta = \lambda \eta \tag{3.2}$$

to a second-order singular ODE system. We study this system using Fuchsian theory in Section IV.3.2, where the matrix $\mathbf{M}_{\lambda,\theta}$ is also constructed in Definition 3.14.

Remark 3.1 *The function $\nu(\cdot, \theta)$ characterizes the eigenvalues of L_θ on a domain larger than $(0.56, 0.6)$. Indeed, we will work with $\lambda \in \mathbb{C}$ and specify any additional assumptions on λ within each lemma.*

IV.3.1 Reduction to finite dimension

Assume we have $\lambda \in \mathbb{C}$ and $0 \neq \eta \in L^2(\mathbb{T})$ solving (3.2). Taking the Fourier transform, we obtain

$$\lambda \widehat{\eta}_k = \widetilde{A}_{k+1,\theta}\widehat{\eta}_{k+1} + \widetilde{B}_{k-1,\theta}\widehat{\eta}_{k-1} \qquad \text{for all } k \in \mathbb{Z}, \tag{3.3}$$

where $\widetilde{A}_{k,\theta}$ and $\widetilde{B}_{k,\theta}$ were defined in (2.8).

Rather than solving the recurrence (3.3) directly, we follow [CGP26] and translate it into a holomorphic problem on the unit disk. To a solution η of (3.3) we associate holomorphic functions (F^+, F^-) whose generating series record the Fourier modes of η. These functions satisfy a second-order ODE system of the form (3.6), and its right-hand sides involve only the values $F^\pm(0)$. This reduces the eigenvalue problem to the existence of holomorphic solutions of (3.6).

For $\lambda \in \mathbb{C}$ and $\theta \in (0,1)$, define the operator $L^{\mathrm{hol}}_{\lambda,\theta} : \mathcal{H} \to \mathcal{H}$ by

$$L^{\mathrm{hol}}_{\lambda,\theta}F(z) := z^2(z^2-1)\partial_z^2 F(z) + zP_{\lambda,\theta}(z)\partial_z F(z) + Q_{\lambda,\theta}(z)F(z), \tag{3.4}$$

where

$$P_{\lambda,\theta}(z) := (2\theta-1)z^2 + 2\lambda z - (2\theta+1), \qquad Q_{\lambda,\theta}(z) := (\theta-1)^2 z^2 + 2\lambda\theta z - (\theta^2-1). \tag{3.5}$$

In terms of this operator, we seek pairs (F^+, F^-) of holomorphic functions on the open unit disk $\mathbb{D} := \{z \in \mathbb{C} : |z| < 1\}$ satisfying the system

$$\left.\begin{aligned} L^{\mathrm{hol}}_{\lambda,\theta^+} F^+(z) &= z\theta(2-\theta)F^-(0) + (1-\theta^2)F^+(0) \\ L^{\mathrm{hol}}_{\lambda,\theta^-} F^-(z) &= z(1-\theta^2)F^+(0) + \theta(2-\theta)F^-(0) \end{aligned}\right\} \tag{3.6}$$

Here $\theta^+ := \theta$ and $\theta^- := 1-\theta$. This is the equation the generating functions $F^\pm$ satisfy directly (Lemma 3.3 below).

As we will see, near $z = 1$, the kernel of $L^{\mathrm{hol}}_{\lambda,\theta}$ is locally spanned by a holomorphic function and a second independent solution of the form

$$(z-1)^{2-\lambda}$$

times a holomorphic function. Near $z = -1$, the identities

$$P_{\lambda,\theta}(z) = P_{-\lambda,\theta}(-z) \qquad \text{and} \qquad Q_{\lambda,\theta}(z) = Q_{-\lambda,\theta}(-z)$$

similarly yield a holomorphic solution and a singular solution of the form

$$(z+1)^{2+\lambda}$$

times a holomorphic function.

Our strategy is to construct $F^\pm$, holomorphic on $\mathbb{D}$, so that the coefficient of the singular term at $z = 1$ vanishes. The regularity of the eigenfunction is then dictated entirely by the behavior at $z = -1$, where the exponent $2 + \lambda$ implies that

$$\eta \in H^s(\mathbb{T}) \qquad \text{for all } s < \frac{3}{2} + \lambda.$$

Before showing the equivalence of both problems, we prove the following technical lemma.

Lemma 3.2 *Let $s \geq 0$ and $f \in H^s(\mathbb{T})$ with $\widehat{f}_k = 0$ for all $k < 0$. Then the power series $F(z) := \sum_{k\geq 0} \widehat{f}_k z^k$ converges absolutely for every $z \in \mathbb{D}$ and defines $F \in \mathcal{H}^s$.*

Proof: Since $f \in H^s(\mathbb{T})$, its Fourier coefficients satisfy

$$|\widehat{f}_k| \lesssim (1+k^2)^{|s|} \quad \text{for all } k \geq 0.$$

For any $r \in (0,1)$, the series $\sum_{k\geq 0} |\widehat{f}_k| r^k$ converges by comparison with $C\sum_{k\geq 0}(1+k^2)^{|s|}r^k < \infty$. Since this bound is uniform on $|z| \leq r$, the series converges locally uniformly in $\mathbb{D}$, so F is holomorphic there. By construction $F|_{\mathbb{T}} \in H^s(\mathbb{T})$, hence $F \in \mathcal{H}^s$. □

Lemma 3.3 *Let $\lambda \in \mathbb{C}$, $\theta \in (0,1)$, and $g \in H^s(\mathbb{T})$ for some $s \geq 0$. Set $\theta^+ := \theta$, $\theta^- := 1-\theta$. Define $g^\pm := \mathcal{P}^\pm g \in H^s(\mathbb{T})$ and let $(G^+, G^-) \in \mathcal{H}^s \oplus \mathcal{H}^s$ given by*

$$G^+(z) := 2i\sum_{k\geq 0} \widehat{g^+_k} z^k \qquad \text{and} \qquad G^-(z) := -2i\sum_{k\geq 0} \widehat{g^-_k} z^k.$$

Then there exists $\eta \in H^s(\mathbb{T})$ satisfying

$$(L_\theta - \lambda)\eta = g, \tag{3.7}$$

if and only if there exists $(F^+, F^-) \in \mathcal{H}^{s+1} \oplus \mathcal{H}^{s+1}$ solving

$$\left.\begin{aligned} L^{\mathrm{hol}}_{\lambda,\theta^+} F^+(z) &= z\theta(2-\theta)F^-(0) + (1-\theta^2)F^+(0) + zG^+(z) \\ L^{\mathrm{hol}}_{\lambda,\theta^-} F^-(z) &= z(1-\theta^2)F^+(0) + \theta(2-\theta)F^-(0) + zG^-(z) \end{aligned}\right\} \tag{3.8}$$

In particular, taking $g = 0$, the system (3.8) reduces to (3.6), recovering the eigenvalue equivalence with (2.11).

Proof: Let $\eta \in H^s(\mathbb{T})$ satisfy (3.7). Define $f := (\partial_x + i\theta)^{-1}\eta \in H^{s+1}(\mathbb{T})$. Applying $\mathcal{P}^+$ to (3.7), we obtain

$$\lambda(\partial_x + i\theta)f^+ + g^+ = \mathcal{P}^+ L_\theta \eta = \sum_{k\geq 0} (\widehat{\eta}_{k+1}\widetilde{A}_{k+1,\theta} + \widehat{\eta}_{k-1}\widetilde{B}_{k-1,\theta})\,\mathbf{e}^k. \tag{3.9}$$

Thus, for every $k \geq 0$,

$$\lambda(k+\theta)\widehat{f_k^+} = (k+1+\theta)\widehat{f}_{k+1}\widetilde{A}_{k+1,\theta} + (k-1+\theta)\widehat{f}_{k-1}\widetilde{B}_{k-1,\theta} + i\widehat{g_k^+}. \tag{3.10}$$

From (2.8), we get

$$(k-2+\theta)(k-2+2\delta_{k,0}+\theta)\widehat{f}_{k-1} + 2\lambda(k+\theta)\widehat{f_k^+} - (k+2+\theta)(k+\theta)\widehat{f}_{k+1} = 2i\widehat{g_k^+}. \tag{3.11}$$

The projection $f^+ := \mathcal{P}^+ f$ belongs to $H^{s+1}(\mathbb{T})$ and has only non-negative Fourier modes. By Lemma 3.2 applied to $f^+ \in H^{s+1}(\mathbb{T})$, the power series

$$F^+(z) := \sum_{k\geq 0} \widehat{f}_k^+ z^k$$

converges absolutely for every $z \in \mathbb{D}$ and defines $F^+ \in \mathcal{H}^{s+1}$.

Multiplying (3.11) by z^k and summing over $k \geq 0$, we obtain

$$I_1 + I_2 + I_3 = G^+ \tag{3.12}$$

where

$$\begin{aligned} I_1(z) &:= \sum_{k\geq 0}(k-2+\theta)(k-2(1-\delta_{k,0})+\theta)\widehat{f}_{k-1}z^k, \\ I_2(z) &:= \sum_{k\geq 0} 2\lambda(k+\theta)(\widehat{f^+})_k z^k, \qquad I_3(z) := -\sum_{k\geq 0}(k+2+\theta)(k+\theta)\widehat{f}_{k+1}z^k. \end{aligned}$$

We begin with I_1:

$$\begin{aligned} I_1(z) &= z\sum_{k\geq -1}(k-1+\theta)(k+1-2(1-\delta_{k+1,0})+\theta)\widehat{f}_k z^k \\ &= z\sum_{k\geq 0}(k-1+\theta)^2(\widehat{f^+})_k z^k - \theta(2-\theta)\widehat{f}_{-1} \\ &= z^3\partial_z^2 F^+(z) + (2\theta-1)z^2\partial_z F^+(z) + (1-\theta)^2 z F^+(z) - \theta(2-\theta)\widehat{f}_{-1}. \end{aligned}$$

Next,

$$I_2(z) = \sum_{k\geq 0} 2\lambda(k+\theta)(\widehat{f^+})_k z^k = 2\lambda z\partial_z F^+(z) + 2\lambda\theta F^+(z),$$

and

$$\begin{aligned} zI_3(z) &= z\sum_{k\geq 0} -(k+2+\theta)(k+\theta)\widehat{f}_{k+1}z^k \\ &= \sum_{k\geq 0} -(k+1+\theta)(k-1+\theta)(\widehat{f^+})_k z^k - (1-\theta)(1+\theta)\widehat{f}_0 \\ &= -z^2\partial_z^2 F^+(z) - (2\theta+1)z\partial_z F^+(z) + (1-\theta)(1+\theta)F^+(z) - (1-\theta)(1+\theta)\widehat{f}_0. \end{aligned}$$

Substituting the expressions for I_1, I_2, and I_3 into (3.12), we find that F^+ satisfies the first equation in (3.8).

By Remark 2.3 we have $\widetilde{A}_{-k,\theta^+} = \widetilde{B}_{k-1,\theta^-}$. Substituting $k \to -k-2$ and $\theta^+ \to \theta^-$ in the same identity also yields $\widetilde{B}_{-k-2,\theta^+} = \widetilde{A}_{k+1,\theta^-}$. These two identities allow us to find the recurrence that the Fourier modes of $\eta^\sharp$ satisfy, as the following computations show

$$\begin{aligned}\lambda\widehat{\eta^\sharp_k} &= \lambda\widehat{\eta}_{-k-1} = \widetilde{A}_{-k,\theta^+}\widehat{\eta}_{-k} + \widetilde{B}_{-k-2,\theta^+}\widehat{\eta}_{-k-2} - \widehat{g}_{-k-1} \\ &= \widetilde{B}_{k-1,\theta^-}\widehat{\eta}_{-k} + \widetilde{A}_{k+1,\theta^-}\widehat{\eta}_{-k-2} - \widehat{g}_{-k-1} = \widetilde{B}_{k-1,\theta^-}\widehat{\eta^\sharp_{k-1}} + \widetilde{A}_{k+1,\theta^-}\widehat{\eta^\sharp_{k+1}} - \widehat{g^\sharp_k}.\end{aligned}$$

For $f^- := \mathcal{P}^- f$, we have

$$\widehat{f^-_k} = \frac{i}{k+\theta^-}\widehat{\eta^\sharp_k}.$$

Hence, for all $k \geq 0$, it holds that

$$\begin{aligned}(k-2+\theta^-)(k-2(1-\delta_{k,0})+\theta^-)\widehat{f^\sharp_{k-1}} + 2\lambda(k+\theta^-)\widehat{f^-_k} \\ - (k+2+\theta^-)(k+\theta^-)\widehat{f^\sharp_{k+1}} = -2i\widehat{g^-_k},\end{aligned} \tag{3.13}$$

so that the extension F^- satisfies the second equation in (3.8).

Conversely, given $(F^+, F^-) \in \mathcal{H}^{s+1} \oplus \mathcal{H}^{s+1}$ solving (3.8), set $f^\pm := F^\pm|_{\mathbb{T}}$, and define f and $\eta := \partial_x f + i\theta f$ by

$$f(x) = f^+(x) + e^{-ix} f^-(-x).$$

The Fourier modes of $f^\pm$ satisfy (3.11) and (3.13), and reversing the computation gives (3.7). □

IV.3.2 Second-order ODE system

The analysis of (3.6) proceeds in two steps. In Section IV.3.2.1 we study the local behavior of solutions to (3.4), while in Section IV.3.2.2 we use these local solutions to construct global pairs of solutions to (3.6) with good boundary behavior on $\partial\mathbb{D} = \mathbb{T}$.

The main analytical problem is to understand, for a given holomorphic function G, whether there exists a holomorphic solution $F : \mathbb{D} \to \mathbb{C}$ to

$$L^{\mathrm{hol}}_{\lambda,\theta} F = G, \tag{3.14}$$

whether such a solution is unique, and what boundary regularity its trace $F|_{\mathbb{T}}$ possesses. Note that (3.6) is precisely a coupled system of two equations of the form (3.14), where the right-hand sides are linear in $F^\pm(0)$. The operator $L^{\mathrm{hol}}_{\lambda,\theta}$ is a second-order Fuchsian differential operator with regular singular points at $z = 0$ and $z = \pm 1$. We refer to [CL55; Inc56; WW96; Hil97; Tes12] for further details and background. This structure lets us describe the local behavior of solutions near each singularity explicitly, which is the content of Section IV.3.2.1.

IV.3.2.1 Local analysis

We begin with the homogeneous equation

$$z^2(z^2-1)\partial_z^2 F(z) + zP_{\lambda,\theta}(z)\partial_z F(z) + Q_{\lambda,\theta}(z)F(z) = 0, \tag{3.15}$$

where $P_{\lambda,\theta}$ and $Q_{\lambda,\theta}$ were given in (3.5).

The equation (3.15) has three regular singular points, namely

$$z = 0, \qquad z = 1, \qquad z = -1.$$

The next three lemmas describe the local form of the solutions near each singular point.

Lemma 3.4 *Let $\theta \in (0,1)$ and $\lambda \in \mathbb{C}$. Let $\Omega \subset \mathbb{C}$ be open and simply connected, with $0 \in \Omega$ and $1, -1 \notin \Omega$.*

Then there exists $r > 0$ such that, after fixing a branch of $\log(z)$ on any simply connected set $U \subset D_r(0) - \{0\}$, every solution of (3.15) on U is a linear combination of the functions $F^{h,0}_{\lambda,\theta,1}$ and $F^{h,0}_{\lambda,\theta,2}$, given by

$$F^{h,0}_{\lambda,\theta,1}(z) = z^{-\theta+1}\phi_1(z) \qquad \text{and} \qquad F^{h,0}_{\lambda,\theta,2}(z) = z^{-\theta-1}\phi_2(z) - F^{h,0}_{\lambda,\theta,1}(z)\log(z),$$

where $\phi_1, \phi_2 : D_r(0) \to \mathbb{C}$ are holomorphic functions with $\phi_1(0) = 1$ and $\phi_2(0) = -\frac{1}{2}$.

Proof: The statements follow from standard Frobenius theory for regular singular points after computing the corresponding indicial equations.

For completeness, and since explicit recursive formulas are useful for constructive computations, detailed derivations are provided in Appendix IV.A. □

Lemma 3.5 *Let $\theta \in (0,1)$ and $\lambda \in \mathbb{C} - \mathbb{Z}$. Let $\Omega \subset \mathbb{C}$ be open and simply connected, with $1 \in \Omega$ and $0, -1 \notin \Omega$.*

Then there exists $r > 0$ such that, after fixing a branch of $\log(z-1)$ on any simply connected set $U \subset D_r(1) - \{1\}$, every solution of (3.15) on U is a linear combination of the functions $F^{h,1}_{\lambda,\theta,1}$ and $F^{h,1}_{\lambda,\theta,2}$, given by

$$F^{h,1}_{\lambda,\theta,1}(z) = \phi_1(z) \qquad \text{and} \qquad F^{h,1}_{\lambda,\theta,2}(z) = (z-1)^{2-\lambda}\phi_2(z),$$

where ϕ_1 and ϕ_2 are holomorphic functions on $D_r(1)$, with $\phi_1(1) = \phi_2(1) = 1$.

Remark 3.6 *By Theorem 4.5 of [Tes12], in both Lemma 3.4 and Lemma 3.5 the functions ϕ_1 and ϕ_2 have radius of convergence at least 1 (the distance from each base point to the nearest other singularity).*

Proof: The statements follow from standard Frobenius theory for regular singular points after computing the corresponding indicial equations.

For completeness, and since explicit recursive formulas are useful for constructive computations, detailed derivations are provided in Appendix IV.A. □

Lemma 3.7 *Let $\theta \in (0,1)$ and $\lambda \in \mathbb{C} - \mathbb{Z}$. Let $\Omega \subset \mathbb{C}$ be open and simply connected, with $-1 \in \Omega$ and $0, 1 \notin \Omega$.*

Then for any simply connected set $U \subset \Omega - \{-1\}$, after fixing a branch of $\log(z+1)$ on U, every solution of (3.15) on U is a linear combination of $F^{h,-1}_{\lambda,\theta,1}$ and $F^{h,-1}_{\lambda,\theta,2}$, defined by

$$F^{h,-1}_{\lambda,\theta,1}(z) = \phi_1(z) \qquad \text{and} \qquad F^{h,-1}_{\lambda,\theta,2}(z) = (z+1)^{2+\lambda}\phi_2(z),$$

where ϕ_1 and ϕ_2 are holomorphic functions on Ω, with $\phi_1(-1) = \phi_2(-1) = 1$.

Proof: The statements follow from standard Frobenius theory for regular singular points after computing the corresponding indicial equations.

For completeness, and since explicit recursive formulas are useful for constructive computations, detailed derivations are provided in Appendix IV.A. □

Remark 3.8 *Fix $0 < s < \frac{3}{2}$ and a noninteger λ with $|\mathrm{Re}\lambda| < \frac{3}{2} - s$. For any $\theta \in (0,1)$, prescribe nonzero values (a^+, a^-) in (3.18). The resulting solutions are holomorphic throughout $\mathbb{D}$, and Lemma 3.13 gives $F^{\pm}(0) = a^{\pm}$. Near $z = \pm 1$, their only possible nonanalytic terms are the Frobenius branches with exponents $2 \mp \lambda$, respectively. The condition on λ places both components in $\mathcal{H}^{s+1}$, so Lemma 3.3 yields a nonzero eigenfunction of L_θ in $H^s(\mathbb{T})$. Taking $\theta = 1/n$ and lifting this eigenfunction η to $e^{ix}\eta(nx)$ gives an eigenfunction of L_n in $H^s(\mathbb{T})$, by the Fourier identity in Lemma 2.4. Since the spectrum is closed, the noninteger restriction on λ does not exclude the integer points of the band in Remark 2.2.*

In general, the local solutions from the previous lemmas are not holomorphic. However, the next lemmas show that one can still construct holomorphic solutions to (3.14) when we have a holomorphic affine term G.

Lemma 3.9 *Let $r > 0$ be as in Lemma 3.4 and let $G : D_r(0) \to \mathbb{C}$ be holomorphic. Then there exists a holomorphic function $F^{p,0}_{G,\lambda,\theta} : D_r(0) \to \mathbb{C}$ satisfying* (3.14).

Moreover, the Taylor coefficients of $F^{p,0}_{G,\lambda,\theta}$ can be computed recursively to arbitrary order.

Proof: The proof follows by variation of constants applied to the fundamental system given by Lemma 3.4. See Appendix IV.A. □

We now give conditions under which, for every holomorphic function G at 0, the function $F^{p,0}_{G,\lambda,\theta}$ from Lemma 3.9 is the unique holomorphic solution to (3.14). To prove this, we first need the following technical lemma.

Lemma 3.10 *Let $\theta \in (0,1)$ and $\lambda \in \mathbb{C}$. Let $F^{h,0}_{\lambda,\theta,1}, F^{h,0}_{\lambda,\theta,2}$ be as in Lemma 3.4. Then the unique linear combination of $F^{h,0}_{\lambda,\theta,1}$ and $F^{h,0}_{\lambda,\theta,2}$, which is holomorphic at 0, is the trivial zero function.*

Proof: Assume that there exists a holomorphic function F at 0 such that

$$F = C_1 F^{h,0}_{\lambda,\theta,1} + C_2 F^{h,0}_{\lambda,\theta,2}$$

for some $C_1, C_2 \in \mathbb{C}$.

Let $r > 0$ as in Lemma 3.4 and consider the set U given by

$$U := \{se^{ix} : 0 < s < r \text{ and } 0 < x < 2\pi\},$$

so that there is a choice of $\log(z)$ for which $F^{h,0}_{\lambda,\theta,1}$ and $F^{h,0}_{\lambda,\theta,2}$ are holomorphic on U.

Therefore, since F is holomorphic, we find for all $s \in (0,r)$ that

$$\begin{aligned}
0 &= \lim_{x\to 2\pi^-} F(se^{ix}) - \lim_{x\to 0} F(se^{ix}) \\
&= C_1\big(\lim_{x\to 2\pi^-} F^{h,0}_{\lambda,\theta,1}(se^{ix}) - \lim_{x\to 0} F^{h,0}_{\lambda,\theta,1}(se^{ix})\big) + C_2\big(\lim_{x\to 2\pi^-} F^{h,0}_{\lambda,\theta,2}(se^{ix}) - \lim_{x\to 0} F^{h,0}_{\lambda,\theta,2}(se^{ix})\big) \\
&= C_1\big(e^{(-\theta+1)2\pi i} - 1\big)s^{-\theta+1}\phi_1(s) - 2\pi i C_2 s^{-\theta+1} e^{(-\theta+1)(2\pi i)}\phi_1(s) \\
&\quad - C_2\big(e^{(-\theta+1)2\pi i} - 1\big)s^{-\theta+1}\log(s)\phi_1(s) + C_2\big(e^{(-\theta-1)2\pi i} - 1\big)s^{-\theta-1}\phi_2(s). \qquad (3.16)
\end{aligned}$$

Hence, multiplying (3.16) by $s^{\theta+1}$ yields

$$\begin{aligned}
0 &= C_1\big(e^{(-\theta+1)2\pi i} - 1\big)s^2\phi_1(s) - 2\pi i C_2 s^2 e^{(-\theta+1)(2\pi i)}\phi_1(s) \\
&\quad - C_2\big(e^{(-\theta+1)2\pi i} - 1\big)s^2\log(s)\phi_1(s) + C_2\big(e^{(-\theta-1)2\pi i} - 1\big)\phi_2(s). \qquad (3.17)
\end{aligned}$$

Since ϕ_1 and ϕ_2 are in particular smooth, the coefficient of $s^2\log(s)$ must vanish, so that

$$C_2\big(e^{(-\theta+1)2\pi i} - 1\big) = 0.$$

Since $\theta \notin \mathbb{Z}$, the factor $e^{(-\theta+1)2\pi i} - 1$ is non-zero, hence $C_2 = 0$. From (3.17) we then get

$$0 = C_1\big(e^{(-\theta+1)2\pi i} - 1\big)s^2\phi_1(s).$$

Finally, we have $C_1 = 0$ since $\theta \notin \mathbb{Z}$. □

Corollary 3.11 *Let G be a holomorphic function at 0. Then $F^{p,0}_{G,\lambda,\theta}$ is the unique holomorphic function solving* (3.14).

Proof: Let F be a holomorphic function at 0 solving (3.14). Then $\widetilde{F} := F - F^{p,0}_{G,\lambda,\theta}$ is holomorphic at 0 and solves (3.15). By Lemma 3.4, on a punctured neighbourhood of 0 we may write $\widetilde{F} = C_1 F^{h,0}_{\lambda,\theta,1} + C_2 F^{h,0}_{\lambda,\theta,2}$ for some $C_1, C_2 \in \mathbb{C}$. Since $\widetilde{F}$ is holomorphic at 0, Lemma 3.10 gives $C_1 = C_2 = 0$, so $\widetilde{F} = 0$ and $F = F^{p,0}_{G,\lambda,\theta}$. □

Lemma 3.12 *Let $r > 0$ and $F^{h,1}_{\lambda,\theta,1} : D_r(1) \to \mathbb{C}$ be as in Lemma 3.5, let $G : D_r(1) \to \mathbb{C}$ be holomorphic and assume $\lambda \notin \mathbb{Z}$. Then there exists a holomorphic function $F^{p,1}_{G,\lambda,\theta} : D_r(1) \to \mathbb{C}$ satisfying* (3.14).

Therefore, every holomorphic function F at 1 solving (3.14) *can be written as*

$$F = F^{p,1}_{G,\lambda,\theta} + C F^{h,1}_{\lambda,\theta,1},$$

for some $C \in \mathbb{C}$.

Moreover, the Taylor coefficients of $F^{p,1}_{G,\lambda,\theta}$ can be computed recursively to arbitrary order.

Proof: Existence of $F^{p,1}_{G,\lambda,\theta}$ follows by solving the Frobenius recurrence for the inhomogeneous equation at $z = 1$. For the characterization, note that among the two local solutions of (3.15) at 1, only $F^{h,1}_{\lambda,\theta,2} = (z-1)^{2-\lambda}\phi_2$ fails to be holomorphic there (as $\lambda \notin \mathbb{Z}$), so any holomorphic solution of (3.14) at 1 can only involve $F^{h,1}_{\lambda,\theta,1}$. Full details are in Appendix IV.A. □

IV.3.2.2 Global analysis

We now combine the local solutions of (3.6) near 0 and near 1 into global pairs (F^+, F^-) holomorphic on $\mathbb{D} \cup \{1\}$. The key step is analytic continuation along the segment $(0, 1)$: solutions of (3.6) near 0 and near 1 extend to overlap on this segment, and the requirement that they agree there reduces to a 4×4 homogeneous linear system whose matrix we denote by $\mathbf{M}_{\lambda,\theta}$. Proposition 2.7 then reduces to finding a zero of $\det \mathbf{M}_{\cdot,\theta}$ in $(0.56, 0.6)$ for the finitely many $\theta_p = \frac{h_p}{p}$ with $p < N = 900$ prime and for the interval J of Lemma 2.6, which is verified with computer assistance in Lemma 3.20.

The following lemma shows that, in (3.6), the values $F^+(0)$ and $F^-(0)$ may be replaced by arbitrary constants, provided we only require F^+ and F^- to be holomorphic at 0.

Lemma 3.13 *Let $\theta^+ \in (0, 1)$, $\lambda \in \mathbb{C}$ and $\theta^- := 1 - \theta^+$. Let $a^+, a^- \in \mathbb{C}$. Let $F^+, F^-, G^+, G^- \in \mathcal{H}$.*

Assume that the following identities hold

$$L^{\mathrm{hol}}_{\lambda,\theta^+} F^+(z) = z\theta^+(2-\theta^+)a^- + (1-\theta^+)(1+\theta^+)a^+ + zG^+(z)$$
$$L^{\mathrm{hol}}_{\lambda,\theta^-} F^-(z) = z\theta^-(2-\theta^-)a^+ + (1-\theta^-)(1+\theta^-)a^- + zG^-(z).$$

Then we have

$$F^+(0) = a^+ \qquad \text{and} \qquad F^-(0) = a^-.$$

Proof: We evaluate the system at $z = 0$, yielding

$$Q_{\lambda,\theta^\pm}(0) F^\pm(0) = (1-\theta^\pm)(1+\theta^\pm)a^\pm.$$

Since $\theta^\pm \notin \mathbb{Z}$, we find that $F^\pm(0) = a^\pm$. □

We now present the criterion we use to find eigenvalues. This result is purely theoretical, but its application will still require computer assistance. We introduce the affine terms

$$G^\theta_1(z) := \theta(2-\theta)z \qquad \text{and} \qquad G^\theta_2(z) := (1-\theta)(1+\theta).$$

The following definition introduces the notation needed for the next proposition.

Definition 3.14 *Let $\theta^+ \in (0,1)$, $\lambda \in \mathbb{C}$, $\theta^- := 1-\theta^+$, and set the midpoint $x_0 := \frac{1}{2}$. For $a^+, a^- \in \mathbb{C}$, consider the system*

$$\left.\begin{aligned} L^{\mathrm{hol}}_{\lambda,\theta^+} F^+(z) &= a^- G_1^{\theta^+}(z) + a^+ G_2^{\theta^+}(z), \\ L^{\mathrm{hol}}_{\lambda,\theta^-} F^-(z) &= a^+ G_1^{\theta^-}(z) + a^- G_2^{\theta^-}(z). \end{aligned}\right\} \tag{3.18}$$

By Corollary 3.11, the holomorphic solutions of (3.18) *near* 0 *are*

$$\mathcal{F}^{+,0}_{\lambda,\theta^+} = a^- F^{p,0}_{\lambda,\theta^+,1} + a^+ F^{p,0}_{\lambda,\theta^+,2} \qquad \text{and} \qquad \mathcal{F}^{-,0}_{\lambda,\theta^-} = a^+ F^{p,0}_{\lambda,\theta^-,1} + a^- F^{p,0}_{\lambda,\theta^-,2}.$$

By Lemma 3.12, the holomorphic solutions of (3.18) *near* 1 *depend on two additional constants $C^+, C^- \in \mathbb{C}$ and read*

$$\begin{aligned} \mathcal{F}^{+,1}_{\lambda,\theta^+} &= a^- F^{p,1}_{\lambda,\theta^+,1} + a^+ F^{p,1}_{\lambda,\theta^+,2} + C^+ F^{h,1}_{\lambda,\theta^+,1}, \\ \mathcal{F}^{-,1}_{\lambda,\theta^-} &= a^+ F^{p,1}_{\lambda,\theta^-,1} + a^- F^{p,1}_{\lambda,\theta^-,2} + C^- F^{h,1}_{\lambda,\theta^-,1}. \end{aligned}$$

By analytic continuation along the segment $(0,1)$, all four functions extend holomorphically across this segment.

Define the function-valued 2×2 blocks $M_\pm$ and column vectors $h_\pm$ by

$$M_+ := \begin{pmatrix} F^{p,-1}_{\lambda,\theta^+,1} - F^{p,0}_{\lambda,\theta^+,1} & F^{p,-1}_{\lambda,\theta^+,2} - F^{p,0}_{\lambda,\theta^+,2} \\ \partial_z(F^{p,-1}_{\lambda,\theta^+,1} - F^{p,0}_{\lambda,\theta^+,1}) & \partial_z(F^{p,-1}_{\lambda,\theta^+,2} - F^{p,0}_{\lambda,\theta^+,2}) \end{pmatrix}, \qquad h_+ := \begin{pmatrix} F^{h,1}_{\lambda,\theta^+,1} \\ \partial_z F^{h,1}_{\lambda,\theta^+,1} \end{pmatrix},$$

$$M_- := \begin{pmatrix} F^{p,-1}_{\lambda,\theta^-,2} - F^{p,0}_{\lambda,\theta^-,2} & F^{p,-1}_{\lambda,\theta^-,1} - F^{p,0}_{\lambda,\theta^-,1} \\ \partial_z(F^{p,-1}_{\lambda,\theta^-,2} - F^{p,0}_{\lambda,\theta^-,2}) & \partial_z(F^{p,-1}_{\lambda,\theta^-,1} - F^{p,0}_{\lambda,\theta^-,1}) \end{pmatrix}, \qquad h_- := \begin{pmatrix} F^{h,1}_{\lambda,\theta^-,1} \\ \partial_z F^{h,1}_{\lambda,\theta^-,1} \end{pmatrix}.$$

Finally, set

$$\mathbf{M}_{\lambda,\theta} := \begin{pmatrix} h_+ & 0 & M_+ \\ 0 & h_- & M_- \end{pmatrix}(x_0) \in \mathbb{C}^{4\times 4},$$

where the 0's denote 2×1 zero blocks.

The function ν is the determinant of $\mathbf{M}_{\lambda,\theta}$. The following Proposition proves that studying the zeros of $\det \mathbf{M}_{\lambda,\theta}$ is indeed a good criterion to see if $\lambda \notin \mathbb{Z}$ is an eigenvalue. In addition, requiring $\mathrm{Re}\lambda > \frac{1}{2}$ also allows us to characterize the resolvent set by the nonvanishing of ν. In our application, the zero of $\nu(\cdot,\theta)$ will moreover be real (Lemma 3.19).

Proposition 3.15 *Let $\theta \in (0,1)$ and $\lambda \in \mathbb{C} - \mathbb{Z}$.*

Homogeneous problem. *There exists a non-trivial pair $(F^+, F^-) \in \mathcal{H}^s \oplus \mathcal{H}^s$ for all $s < \frac{5}{2} + \mathrm{Re}\lambda$ of holomorphic functions on $\mathbb{D}\cup\{1\}$ solving* (3.6) *if and only if* $\det \mathbf{M}_{\lambda,\theta} = 0$. *The space of such pairs has dimension* $\dim\ker \mathbf{M}_{\lambda,\theta}$.

Forced problem. *Assume additionally* $\mathrm{Re}(\lambda) > \frac{1}{2}$ *and let $s \in [1,2]$. For every $(G^+, G^-) \in \mathcal{H}^s \oplus \mathcal{H}^s$, there exists a unique pair $(F^+, F^-) \in \mathcal{H}^{s+1} \oplus \mathcal{H}^{s+1}$ solving* (3.8) *if and only if* $\det \mathbf{M}_{\lambda,\theta} \neq 0$. *Moreover,* $\|(F^+,F^-)\|_{\mathcal{H}^{s+1}\oplus\mathcal{H}^{s+1}} \lesssim \|(G^+,G^-)\|_{\mathcal{H}^s\oplus\mathcal{H}^s}$.

Proof: Both $\mathcal{F}^{\pm,1}_{\lambda,\theta^\pm}$ and $\mathcal{F}^{\pm,0}_{\lambda,\theta^\pm}$ solve the same second-order ODE on $(0,1)$. By uniqueness of the Cauchy problem at a regular point, $\mathcal{F}^{+,1}_{\lambda,\theta^+}$ and $\mathcal{F}^{+,0}_{\lambda,\theta^+}$ coincide on $(0,1)$ if and only if

$$\mathcal{F}^{+,1}_{\lambda,\theta^+}(x_0) = \mathcal{F}^{+,0}_{\lambda,\theta^+}(x_0) \qquad \text{and} \qquad \partial_z\mathcal{F}^{+,1}_{\lambda,\theta^+}(x_0) = \partial_z\mathcal{F}^{+,0}_{\lambda,\theta^+}(x_0),$$

and analogously for $\mathcal{F}^{-,1}_{\lambda,\theta^-}$ and $\mathcal{F}^{-,0}_{\lambda,\theta^-}$. These four scalar equations form a homogeneous linear system in $(C^+, C^-, a^-, a^+) \in \mathbb{C}^4$ whose matrix is precisely

$$\mathbf{M}_{\lambda,\theta},$$

and a non-trivial solution exists if and only if

$$\det \mathbf{M}_{\lambda,\theta} = 0.$$

Assume now that $\det \mathbf{M}_{\lambda,\theta} = 0$, and let (C^+, C^-, a^-, a^+) be a non-trivial vector in the kernel. The functions $\mathcal{F}^{\pm,0}_{\lambda,\theta^\pm}$ and $\mathcal{F}^{\pm,1}_{\lambda,\theta^\pm}$ then agree on $(0,1)$ and define a single holomorphic function $F^\pm$ on a neighborhood of $[0,1]$. Since $\mathbb{D} \cup \{1\}$ is simply connected and the only interior singular point of (3.4), namely $z = 0$, is one where $F^\pm$ is holomorphic by Corollary 3.11, $F^\pm$ extends uniquely by analytic continuation to all of $\mathbb{D} \cup \{1\}$. Moreover, (F^+, F^-) is non-trivial: if $(F^+, F^-) = 0$, then applying $L^{\mathrm{hol}}_{\lambda,\theta^+}$ to $\mathcal{F}^{+,0}_{\lambda,\theta^+} = 0$ yields

$$a^- G_1^{\theta^+} + a^+ G_2^{\theta^+} = 0.$$

Since $G_1^{\theta^+}$ and $G_2^{\theta^+}(z)$ are linearly independent and nonzero for $\theta^+ \in (0,1)$, this forces $a^\pm = 0$. Therefore,

$$\mathcal{F}^{\pm,1}_{\lambda,\theta^\pm} = C^\pm F^{h,1}_{\lambda,\theta^\pm,1} = 0,$$

which forces $C^\pm = 0$, contradicting non-triviality of (C^+, C^-, a^-, a^+).

Conversely, given a non-trivial pair (F^+, F^-) holomorphic on $\mathbb{D} \cup \{1\}$ solving (3.6), set $a^\pm := F^\pm(0)$ (consistent with (3.18) by Lemma 3.13). By Corollary 3.11, $F^\pm$ coincides with $\mathcal{F}^{\pm,0}_{\lambda,\theta^\pm}$ near 0. By Lemma 3.12, it coincides with $\mathcal{F}^{\pm,1}_{\lambda,\theta^\pm}$ near 1 for some $C^\pm \in \mathbb{C}$. The matching at x_0 yields

$$\mathbf{M}_{\lambda,\theta} \begin{pmatrix} C^+ \\ C^- \\ a^- \\ a^+ \end{pmatrix} = 0,$$

and the vector is non-trivial since otherwise $F^\pm$ would vanish near 0 and hence everywhere by analytic continuation. Therefore $\det \mathbf{M}_{\lambda,\theta} = 0$.

The forced part ($\Leftarrow$) is Lemma 3.17 below and uses the estimate in Lemma 3.16, proven in the appendix. For the converse ($\Rightarrow$): if $\det \mathbf{M}_{\lambda,\theta} = 0$, the homogeneous part gives a non-trivial pair $(F_0^+, F_0^-) \neq (0,0)$ solving (3.8) with $G^+ = G^- = 0$. Then both $(0,0)$ and (F_0^+, F_0^-) solve the forced system for $G = 0$, so uniqueness fails. □

Lemma 3.16 *Let $\theta \in (0,1)$, $\lambda \in \mathbb{C} - \mathbb{Z}$ with $\mathrm{Re}\lambda > \frac{1}{2}$, $s \in [1,2]$, and $\epsilon \in (0,1)$. There exists $C = C_{\lambda,\theta,s,\epsilon} > 0$ such that for every $F \in \mathcal{H}^{s+1}$ satisfying $L^{\mathrm{hol}}_{\lambda,\theta} F = G$ with $G \in \mathcal{H}^s$,*

$$\|F\|_{\mathcal{H}^{s+1}} \leq C\Big(\|G\|_{\mathcal{H}^s} + \|F\|_{\mathcal{H}^{s+1-\epsilon}}\Big).$$

Proof: See Appendix IV.A.3. □

Lemma 3.17 *Let $\theta \in (0,1)$, $\lambda \in \mathbb{C} - \mathbb{Z}$ with $\mathrm{Re}\lambda > \frac{1}{2}$, $s \in [1,2]$, and $\det \mathbf{M}_{\lambda,\theta} \neq 0$. For every $(G^+, G^-) \in \mathcal{H}^s \oplus \mathcal{H}^s$, there exists a unique pair $(F^+, F^-) \in \mathcal{H}^{s+1} \oplus \mathcal{H}^{s+1}$ solving* (3.8).

Moreover, $\|(F^+, F^-)\|_{\mathcal{H}^{s+1} \oplus \mathcal{H}^{s+1}} \lesssim \|(G^+, G^-)\|_{\mathcal{H}^s \oplus \mathcal{H}^s}$.

Proof: A priori estimate. Apply Lemma 3.16 (with the given $s \in [1,2]$) to each scalar equation with $\epsilon \in (0,1)$ fixed. The coupling terms $F^\pm(0)$ are bounded by point evaluation, continuous on $\mathcal{H}^{s+1-\epsilon}$ since $s + 1 - \epsilon > \frac{1}{2}$. Adding the two inequalities gives

$$\|(F^+, F^-)\|_{\mathcal{H}^{s+1} \oplus \mathcal{H}^{s+1}} \leq C\big(\|(G^+, G^-)\|_{\mathcal{H}^s \oplus \mathcal{H}^s} + \|(F^+, F^-)\|_{\mathcal{H}^{s+1-\epsilon} \oplus \mathcal{H}^{s+1-\epsilon}}\big). \tag{3.19}$$

Suppose the stated estimate $\|(F^+,F^-)\|_{\mathcal{H}^{s+1}\oplus\mathcal{H}^{s+1}} \lesssim \|(G^+,G^-)\|_{\mathcal{H}^s\oplus\mathcal{H}^s}$ fails. Then there exist solutions (F_j^+,F_j^-) with

$$\|(F_j^+,F_j^-)\|_{\mathcal{H}^{s+1}\oplus\mathcal{H}^{s+1}} = 1 \qquad \text{and} \qquad \|(G_j^+,G_j^-)\|_{\mathcal{H}^s\oplus\mathcal{H}^s} \to 0.$$

The compact embedding $\mathcal{H}^{s+1}\oplus\mathcal{H}^{s+1} \Subset \mathcal{H}^{s+1-\epsilon}\oplus\mathcal{H}^{s+1-\epsilon}$ yields, after passing to a subsequence, strong convergence $(F_j^+,F_j^-)\to(F^+,F^-)$ in $\mathcal{H}^{s+1-\epsilon}\oplus\mathcal{H}^{s+1-\epsilon}$. The limit lies in $\mathcal{H}^{s+1}\oplus\mathcal{H}^{s+1}$ and solves the homogeneous system (3.6).

Since $\mathrm{Re}\lambda > \frac{1}{2} \geq \frac{3}{2} - s$ (as $s \geq 1$), the branch $(z-1)^{2-\lambda}$ lies in $\mathcal{H}^t$ only for $t < \frac{5}{2} - \mathrm{Re}\lambda < 2$, so $(z-1)^{2-\lambda} \notin \mathcal{H}^{s+1}$. Hence $(F^\pm)$ carry no contribution from this mode and extend holomorphically to $z=1$. By the homogeneous part of Proposition 3.15, the condition $\det \mathbf{M}_{\lambda,\theta} \neq 0$ forces $(F^+,F^-)=0$. Therefore $(F_j^+,F_j^-)\to 0$ in $\mathcal{H}^{s+1-\epsilon}\oplus\mathcal{H}^{s+1-\epsilon}$, and (3.19) gives a contradiction.

Existence. If (G^+,G^-) is polynomial then $zG^\pm$ is analytic near $z=1$. Lemmas 3.9 and 3.12 then provide holomorphic local solutions near $z=0$ and $z=1$, and the matching system at x_0 is uniquely solvable since $\det \mathbf{M}_{\lambda,\theta} \neq 0$, giving a global pair in $\mathcal{H}^{s+1}\oplus\mathcal{H}^{s+1}$. For general (G^+,G^-), approximate in $\mathcal{H}^s\oplus\mathcal{H}^s$ by polynomials. The a priori estimate shows the corresponding solutions are Cauchy in $\mathcal{H}^{s+1}\oplus\mathcal{H}^{s+1}$ so their limit solves the system.

Uniqueness follows from the a priori estimate with $G^+=G^-=0$. □

Corollary 3.18 *Let $\theta\in(0,1)$, $\lambda\in\mathbb{C}-\mathbb{Z}$ with $\mathrm{Re}\lambda > \frac{1}{2}$.*

1. *If $\det \mathbf{M}_{\lambda,\theta} \neq 0$, then $\lambda \in \rho(L_\theta)$ on $H^s(\mathbb{T})$ for every $s\in[1,2]$, that is, the operator $L_\theta - \lambda : \mathcal{D}_s \to H^s(\mathbb{T})$ is bijective and its inverse is bounded on $H^s(\mathbb{T})$, where $\mathcal{D}_s := \{\eta \in H^s(\mathbb{T}) : L_\theta\eta \in H^s(\mathbb{T})\}$.*

2. *If $\det \mathbf{M}_{\lambda,\theta} = 0$, then λ is an eigenvalue of L_θ on $H^s(\mathbb{T})$ for every $s < \frac{3}{2} + \mathrm{Re}\lambda$. For $s\in[1,2]$, its geometric multiplicity is $\dim\ker \mathbf{M}_{\lambda,\theta}$.*

Proof: *Part 1.* Fix any $s\in[1,2]$. For any $g\in H^s(\mathbb{T})$, Lemma 3.3 reduces solving $(L_\theta-\lambda)\eta = g$ in $H^s(\mathbb{T})$ to finding $(F^+,F^-)\in\mathcal{H}^{s+1}\oplus\mathcal{H}^{s+1}$ solving (3.8). Under $\mathrm{Re}\lambda > \frac{1}{2}$ and $\det \mathbf{M}_{\lambda,\theta}\neq 0$, Lemma 3.17 provides a unique such pair with $\|(F^+,F^-)\|_{\mathcal{H}^{s+1}\oplus\mathcal{H}^{s+1}} \lesssim \|(G^+,G^-)\|_{\mathcal{H}^s\oplus\mathcal{H}^s} \lesssim \|g\|_{H^s(\mathbb{T})}$. Translating back via Lemma 3.3 gives a unique $\eta\in H^s(\mathbb{T})$ satisfying $(L_\theta-\lambda)\eta = g$ and $\|\eta\|_{H^s(\mathbb{T})} \lesssim \|g\|_{H^s(\mathbb{T})}$.

Since $L_\theta\eta = \lambda\eta + g \in H^s(\mathbb{T})$, we have $\eta\in\mathcal{D}_s$ so $(L_\theta-\lambda)^{-1}$ is bounded on $H^s(\mathbb{T})$.

Part 2. If $\det \mathbf{M}_{\lambda,\theta}=0$, the homogeneous part of Proposition 3.15 gives a nonzero space of solution pairs of dimension $\dim\ker \mathbf{M}_{\lambda,\theta}$, whose members belong to $\mathcal{H}^r\oplus\mathcal{H}^r$ for every $r < \frac{5}{2} + \mathrm{Re}\lambda$. By Lemma 3.3, the reconstruction $\eta := \partial_x f + i\theta^+ f$ satisfies $(L_\theta-\lambda)\eta = 0$ and belongs to $H^s(\mathbb{T})$ for every $s < \frac{3}{2} + \mathrm{Re}\lambda$. For $s<0$, the regularity follows from $\eta\in H^0(\mathbb{T}) \subset H^s(\mathbb{T})$.

For $s\in[1,2]$, Lemma 3.3 associates every eigenfunction with a pair in $\mathcal{H}^{s+1}\oplus\mathcal{H}^{s+1}$. Since $s+1 \geq 2 > \frac{5}{2} - \mathrm{Re}\lambda$, this regularity excludes the branch $(z-1)^{2-\lambda}$. The pair is therefore holomorphic at $z=1$, and Proposition 3.15 identifies the eigenspace with $\ker \mathbf{M}_{\lambda,\theta}$. □

The following lemma reduces the analysis of $\det \mathbf{M}_{\lambda,\theta}$ to real values of λ.

Lemma 3.19 *Let $\theta\in(0,1)$ and $\lambda\in\mathbb{R}-\mathbb{Z}$. Then $\det \mathbf{M}_{\lambda,\theta}\in\mathbb{R}$.*

Proof: Fix $\theta\in(0,1)$ and $\lambda\in\mathbb{R}-\mathbb{Z}$. The polynomial coefficients $P_{\lambda,\theta}, Q_{\lambda,\theta}$ of (3.5) and the affine right-hand sides $G_1^{\theta^\pm}, G_2^{\theta^\pm}$ have real coefficients in z. Therefore the Frobenius recurrences at the regular singular points $z=0$ and $z=1$ (Lemmas 3.4 and 3.5) are real, and the local series ϕ_1,ϕ_2 together with the particular solutions $F^{p,k}_{\lambda,\theta^\pm,i}$ supplied by Lemmas 3.9 and 3.12 all have real Taylor coefficients at their respective base points. In particular each takes real values on a real interval around its base point.

Analytic continuation along the real segment $(0,1)$ preserves real-valuedness, and the same holds for first derivatives. Hence every entry of $\mathbf{M}_{\lambda,\theta}$ is real, and $\det \mathbf{M}_{\lambda,\theta}\in\mathbb{R}$. □

The next lemma provides the sign change needed for the intermediate value theorem, and its verification is computer-assisted.

Lemma 3.20 *For $\theta = \theta_p$, for every prime $p < N$, and for every $\theta \in J$,*

$$\det \mathbf{M}_{\lambda,\theta} < 0 \quad \textit{when } \lambda = 0.56, \qquad \det \mathbf{M}_{\lambda,\theta} > 0 \quad \textit{when } \lambda = 0.6.$$

Proof: The proof is computer-assisted. The code is available on GitHub at https://github.com/MiguelMGPascualCaballo/dgreg/tree/v1.0. We refer to Appendix IV.C for details about the implementation.

□

Corollary 3.21 *For $\theta = \theta_p$, for every prime $p < N$, and for every $\theta \in J$, there exists $\lambda_\theta \in (0.56, 0.6)$ such that $\det \mathbf{M}_{\lambda,\theta} = 0$.*

Proof: Fix one such θ. The map $\lambda \mapsto \det \mathbf{M}_{\lambda,\theta}$ is continuous on $[0.56, 0.6]$, since by standard Frobenius theory [Tes12, Theorem 4.2] the entries of $\mathbf{M}_{\lambda,\theta}$ depend analytically on $(\lambda, \theta) \in (\mathbb{C} - \mathbb{Z}) \times (0, 1)$.

By Lemma 3.19 this map is real-valued, and by Lemma 3.20 it changes sign between the endpoints of $[0.56, 0.6]$, so the intermediate value theorem yields a zero in the interior. □

Remark 3.22 *Lemma 3.20 also certifies that $\nu(\cdot, \theta) \neq 0$: at $\lambda = 0.6$ one has $\nu > 0$, so ν is not the zero function. Since $\nu(\cdot, \theta)$ is analytic on $\mathbb{C} - \mathbb{Z}$ by the Frobenius theory cited above, the Identity Theorem implies that its zero set is discrete in $\mathbb{C} - \mathbb{Z}$. For each $s \in [1, 2]$, Corollary 3.18 identifies the zeros in $\{\lambda \in \mathbb{C} - \mathbb{Z} : \mathrm{Re}\lambda > \frac{1}{2}\}$ with the eigenvalues of L_θ on $H^s(\mathbb{T})$ in that set. All these eigenvalues are therefore isolated on $H^s(\mathbb{T})$. See Figure IV.2 for a numerical plot of $\nu(\cdot, \theta)$ at the two endpoint values $\theta = \frac{1}{3}$ and $\theta = \frac{1}{2}$. For the endpoint values $\theta = \frac{1}{3}$ and $\theta = \frac{1}{2}$, the corresponding numerical eigenfunctions are shown in Figure IV.3.*

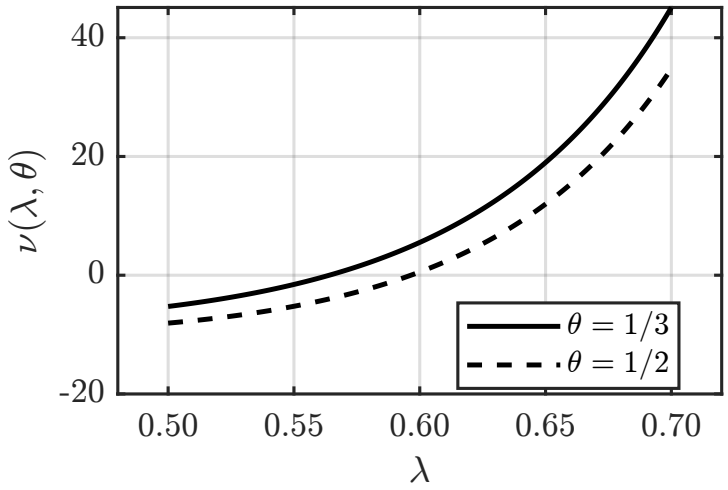


Figure IV.2: Numerical evaluation of $\nu(\cdot, \theta)$ for $\theta = \frac{1}{3}$ and $\theta = \frac{1}{2}$. The zeros correspond to eigenvalues of L_θ.

IV.3.3 Conclusion of the proof

Fix $\theta = \theta_p$ for some prime $p < N$, or $\theta \in J$. Corollary 3.21 gives $\lambda_\theta \in (0.56, 0.6)$ such that $\det \mathbf{M}_{\lambda,\theta} = 0$ at (λ_θ, θ). By Proposition 3.15, there is then a non-trivial pair (F_θ^+, F_θ^-) of holomorphic functions on $\mathbb{D} \cup \{1\}$ solving (3.6) at (λ_θ, θ).

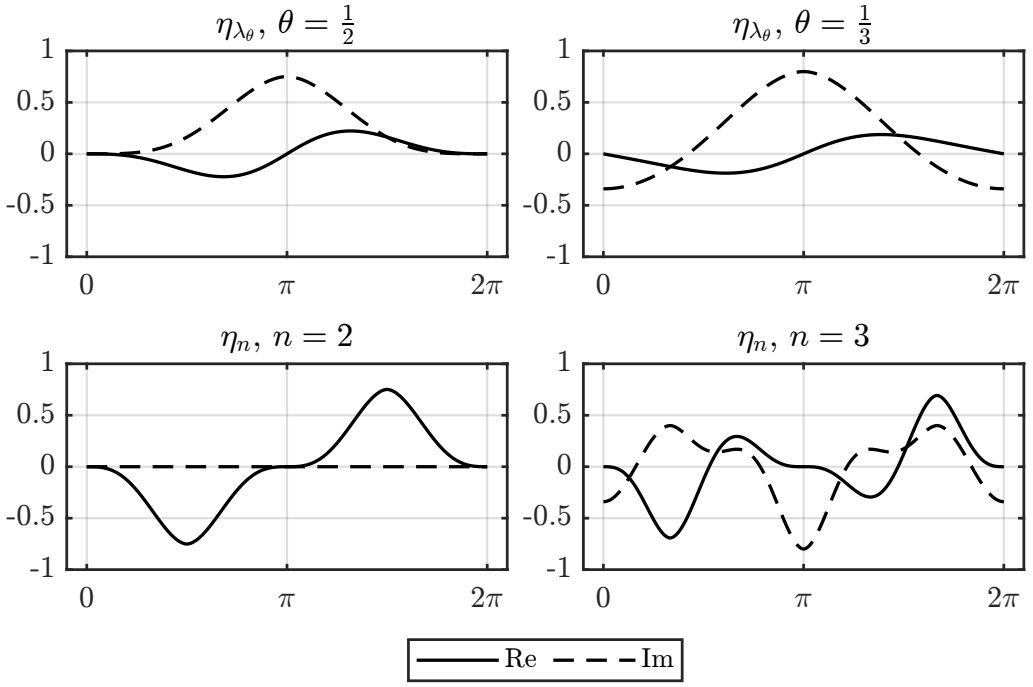


Figure IV.3: Numerical plots of the eigenfunctions obtained from the matching construction for $\theta = \frac{1}{2}$ and $\theta = \frac{1}{3}$, together with their lifted counterparts for $n = 2$ and $n = 3$.

The only point of $\overline{\mathbb{D}}$ where $F_\theta^\pm$ may fail to be holomorphic is $z = -1$. Indeed, the right-hand sides of (3.6) are analytic at $z = -1$, so local Frobenius theory shows that, near $z = -1$, $F_\theta^\pm$ is of the form

$$F_\theta^\pm(z) = \psi^\pm(z) + \beta^\pm(z+1)^{2+\lambda_\theta}\phi_2^\pm(z),$$

where $\psi^\pm$ and $\phi_2^\pm$ are analytic at -1 and $\beta^\pm \in \mathbb{C}$.

Restricting to $\mathbb{T}$ via $z = e^{ix}$, the branch $(e^{ix}+1)^{2+\lambda_\theta}\phi_2^\pm(e^{ix})$ behaves like $|x-\pi|^{2+\lambda_\theta}$ near $x = \pi$. Its Fourier coefficients therefore decay like $|k|^{-3-\lambda_\theta}$. Hence the traces $F_\theta^\pm|_{\mathbb{T}}$ lie in $H^s(\mathbb{T})$ for every $s < \frac{5}{2} + \lambda_\theta$, equivalently $(F_\theta^+, F_\theta^-) \in \mathcal{H}^s \oplus \mathcal{H}^s$ for the same range of s.

By Lemma 3.3, this yields an eigenfunction η_θ of L_θ with eigenvalue λ_θ such that $\eta_\theta \in H^s(\mathbb{T})$ for every $0 \le s < \frac{3}{2} + \lambda_\theta$. The same holds for $s < 0$ by the inclusion $H^0(\mathbb{T}) \subset H^s(\mathbb{T})$. Since $\lambda_\theta > 0.56 > \frac{1}{2}$, it follows in particular that $\eta_\theta \in H^2(\mathbb{T})$. This completes the proof of Proposition 2.7.

IV.A Fuchsian theory

In this appendix we derive explicit local representations of the solutions to (3.15). The qualitative statements used in the main text follow from standard Frobenius theory for regular singular points, but the recursive formulas below are useful both for constructive computations and for the rigorous numerical part of the proof.

Although the computations in Lemmas 3.4 and 3.5 are essentially standard and closely follow [Tes12], we include them here since the order of presentation is adapted to the supplementary code and may help the reader compare the argument with the implementation.

IV.A.1 Homogeneous solutions

This section provides an overview of the Frobenius theory used in our argument.

Let P and Q be analytic functions on a simply connected neighborhood V^{z_0} of $z_0 \in \mathbb{C}$, and consider the equation

$$(z-z_0)^2\partial_z^2 F(z) + (z-z_0)P(z)\partial_z F(z) + Q(z)F(z) = 0. \tag{A.1}$$

The local structure of solutions on any simply connected set $U \subset V^{z_0}$ with $z_0 \notin U$ is well known. After choosing a branch of $\log(z-z_0)$ on U, every solution is a linear combination of two explicit local branches involving analytic functions at z_0 and, possibly, $\log(z-z_0)$.

Proof of Lemma 3.4: Throughout this proof, p_k and q_k denote the Taylor coefficients of $P^0_{\lambda,\theta}$ and $Q^0_{\lambda,\theta}$ at $z=0$. Let

$$P^0_{\lambda,\theta}(z) := \frac{(2\theta-1)z^2 + 2\lambda z - (2\theta+1)}{z^2-1} \qquad \text{and} \qquad Q^0_{\lambda,\theta}(z) := \frac{(\theta-1)^2 z^2 + 2\lambda\theta z - (\theta^2-1)}{z^2-1}.$$

Then (3.15) takes the form of (A.1) with $P = P^0_{\lambda,\theta}$, $Q = Q^0_{\lambda,\theta}$, and $z_0 = 0$:

$$z^2\partial_z^2 F(z) + zP^0_{\lambda,\theta}(z)\partial_z F(z) + Q^0_{\lambda,\theta}(z)F(z) = 0. \tag{A.2}$$

The indicial equation at $z=0$ has roots $1-\theta$ and $-1-\theta$. Since the difference between the roots equals 2, the second local $F^{h,0}_{\lambda,\theta,2}$ branch may contain a logarithmic term. We now derive the corresponding constructive formulas explicitly. The first local $F^{h,0}_{\lambda,\theta,1}$ branch has the form

$$F^{h,0}_{\lambda,\theta,1} = z^{-\theta+1}\phi_1(z),$$

where ϕ_1 is analytic at 0. We choose $\phi_1(0) = 1$ and write

$$\phi_1(z) =: 1 + \sum_{k\geq 1} a_k z^k, \qquad P^0_{\lambda,\theta}(z) =: \sum_{k\geq 0} p_k z^k, \qquad Q^0_{\lambda,\theta}(z) =: \sum_{k\geq 0} q_k z^k,$$

Hence the coefficients a_k are determined for all $k \geq 1$ by the following recurrence

$$\mathcal{I}(k+r)a_k + \sum_{j=0}^{k-1} a_j\big((j+r)p_{k-j} + q_{k-j}\big) = 0, \tag{A.3}$$

where $r = 1-\theta$ and $\mathcal{I}(x) := x(x-1) + xp_0 + q_0$.

The Taylor expansions of $P^0_{\lambda,\theta}$ and $Q^0_{\lambda,\theta}$ are given by

$$\begin{aligned} P^0_{\lambda,\theta}(z) &= (1+2\theta) + (1-\lambda)\frac{z}{1-z} - (1+\lambda)\frac{z}{1+z} \\ &= (1+2\theta) + \sum_{k\geq 1}(1-\lambda)z^k + (1+\lambda)(-z)^k \end{aligned} \tag{A.4}$$

$$\begin{aligned} Q^0_{\lambda,\theta}(z) &= -2\lambda\theta z - 1 + \theta^2 - (1-\theta+\lambda\theta)\frac{z^2}{1-z} - (1-\theta-\lambda\theta)\frac{z^2}{1+z} \\ &= -2\lambda\theta z - 1 + \theta^2 - \sum_{k\geq 2}\big((1-\theta+\lambda\theta)z^k + (1-\theta-\lambda\theta)(-z)^k\big). \end{aligned} \tag{A.5}$$

In particular, we write the first Taylor coefficients centered at 0 of p and q

$$\begin{aligned} &p_0 = 1+2\theta, && p_1 = -2\lambda, && p_2 = 2, \\ &q_0 = -1+\theta^2, && q_1 = -2\lambda\theta, && q_2 = -2(1-\theta). \end{aligned}$$

Using (A.3), we have $a_0 = 1$ and

$$\begin{aligned} a_1 &= -\frac{ra_0p_1 + q_1}{r(r+1) + (r+1)p_0 + q_0} = \frac{2\lambda}{3}, \\ a_2 &= -\frac{ra_0p_2 + (r+1)a_1p_1 + a_0q_2 + a_1q_1}{(r+2)(r+1) + (r+2)p_0 + q_0} = \frac{\lambda^2}{3}. \end{aligned}$$

To construct $F^{h,0}_{\lambda,\theta,2}$, we use reduction of order and write it in the form $CF^{h,0}_{\lambda,\theta,1}$. Substituting $F(z) = C(z)F^{h,0}_1(z)$ into (A.2) yields

$$z^2\partial_z^2 C(z)F^{h,0}_{\lambda,\theta,1}(z) + 2z^2\partial_z C(z)\partial_z F^{h,0}_{\lambda,\theta,1}(z) + z\partial_z C(z)F^{h,0}_{\lambda,\theta,1}(z)P^0_{\lambda,\theta}(z) = 0. \tag{A.6}$$

Since $p_0 = 1 + 2\theta$, the function $\widetilde{P}^0$ given by

$$\widetilde{P}^0(z) := \frac{1}{z}(P^0_{\lambda,\theta}(z) - (1+2\theta)) = \sum_{k\geq 0} p_{k+1}z^k$$

is analytic at 0. From (A.6) we obtain

$$\begin{aligned} z\frac{\partial_z^2 C(z)}{\partial_z C(z)}F^{h,0}_{\lambda,\theta,1}(z) &= -2z\partial_z F^{h,0}_{\lambda,\theta,1}(z) - F^{h,0}_{\lambda,\theta,1}(z)P^0_{\lambda,\theta}(z) \\ &= \big(-2z\partial_z\phi_1(z) + 2(\theta-1)\phi_1(z) - \phi_1(z)P^0_{\lambda,\theta}(z)\big)z^{-\theta+1}. \end{aligned}$$

Dividing by $zF^{h,0}_{\lambda,\theta,1}(z) = z^{2-\theta}\phi_1(z)$ yields

$$\frac{\partial_z^2 C(z)}{\partial_z C(z)} = -2\frac{\partial_z\phi_1(z)}{\phi_1(z)} - \frac{1}{z}(2 - 2\theta + P^0_{\lambda,\theta}(z)) = -2\frac{\partial_z\phi_1(z)}{\phi_1(z)} - \frac{3}{z} - \widetilde{P}^0(z). \tag{A.7}$$

Choose $r > 0$ so that $\phi_1 \neq 0$ on $D_r(0)$. Integrating (A.7) we obtain

$$\partial_z C(z) = \frac{1}{z^3}\psi(z), \qquad \psi(z) := \frac{1}{\phi_1(z)^2}\exp\Big(-\int^z \widetilde{P}^0(y)\,dy\Big),$$

where the antiderivative is defined by

$$\int^z \widetilde{P}^0(y)\,dy := \sum_{k\geq 0}\frac{1}{k+1}p_{k+1}z^{k+1} = \sum_{k\geq 1}\frac{1}{k}p_k z^k.$$

This primitive is holomorphic at 0. We have the same for ψ since ϕ_1 does not vanish at 0.

Hence, integrating once more, we obtain

$$C(z) = \psi_2\log(z) + \sum_{\substack{k\geq 0\\ k\neq 2}}\frac{1}{k-2}\psi_k z^{k-2}.$$

Here $\psi(z) = \sum_{k\geq 0}\psi_k z^k$.

Define $\phi_2 : D_r(0) \to \mathbb{C}$ by

$$\phi_2(z) = \phi_1(z)\sum_{\substack{k\geq 0\\ k\neq 2}}\frac{1}{k-2}\psi_k z^k,$$

so that $F^{h,0}_{\lambda,\theta,2}$ is given by

$$F^{h,0}_{\lambda,\theta,2}(z) = \psi_2 F^{h,0}_{\lambda,\theta,1}(z)\log(z) + z^{-\theta-1}\phi_2(z).$$

Finally, we compute the exact value of ψ_2. Notice that

$$\partial_z\psi = -2\frac{\partial_z\phi_1}{\phi_1}\psi - \widetilde{P}^0\psi \qquad \text{and} \qquad \partial_z^2\psi = -2\frac{\partial_z^2\phi_1}{\phi_1}\psi + 2\Big(\frac{\partial_z\phi_1}{\phi_1}\Big)^2\psi + \frac{(\partial_z\psi)^2}{\psi} - \partial_z\widetilde{P}^0\psi.$$

Since $\phi_1(0) = \psi(0) = 1$, we have

$$\psi_2 = \frac{1}{2}\left.\frac{d^2\psi}{dz^2}\right|_{z=0} = -2a_2 + a_1^2 + \frac{\psi_1^2}{2} - \frac{p_2}{2} = -\frac{2\lambda^2}{3} + \frac{4\lambda^2}{9} + \frac{2\lambda^2}{9} - 1 = -1.$$

where we have used that $\psi_1 = -2a_1 - p_1 = \frac{2\lambda}{3}$. □

Proof of Lemma 3.5: Throughout this proof, p_k and q_k denote the Taylor coefficients of $P^1_{\lambda,\theta}$ and $Q^1_{\lambda,\theta}$ at $z = 1$. Dividing (3.15) by $z^2(z+1)$ and multiplying by $(z-1)$ casts it in the Fuchsian standard form of (A.1) with $z_0 = 1$:

$$(z-1)^2\partial_z^2 F(z) + (z-1)P^1_{\lambda,\theta}(z)\partial_z F(z) + Q^1_{\lambda,\theta}(z)F(z) = 0,$$

where

$$P^1_{\lambda,\theta}(z) := \frac{P_{\lambda,\theta}(z)}{z(z+1)}, \qquad Q^1_{\lambda,\theta}(z) := \frac{(z-1)Q_{\lambda,\theta}(z)}{z^2(z+1)}.$$

Both $P^1_{\lambda,\theta}$ and $Q^1_{\lambda,\theta}$ are analytic on a neighborhood of $z = 1$. Setting $p_0 := P^1_{\lambda,\theta}(1)$ and $q_0 := Q^1_{\lambda,\theta}(1)$, direct computation yields

$$p_0 = \frac{P_{\lambda,\theta}(1)}{2} = -(1-\lambda), \qquad q_0 = 0,$$

so the indicial equation $r(r-1) + rp_0 + q_0 = 0$ becomes $r^2 - (2-\lambda)r = 0$, with roots $r_1 = 0$ and $r_2 = 2-\lambda$. Since $\lambda \in \mathbb{C} - \mathbb{Z}$, the difference $r_2 - r_1 = 2 - \lambda \notin \mathbb{Z}$, and both local solutions are pure power series:

$$F^{h,1}_{\lambda,\theta,1}(z) = \phi_1(z), \qquad F^{h,1}_{\lambda,\theta,2}(z) = (z-1)^{2-\lambda}\phi_2(z),$$

with ϕ_1, ϕ_2 analytic at 1 and normalized so that $\phi_1(1) = \phi_2(1) = 1$.

Expanding

$$\phi_j(z) = \sum_{k\geq 0} a_k^{(j)}(z-1)^k, \qquad P^1_{\lambda,\theta}(z) = \sum_{k\geq 0} p_k(z-1)^k, \qquad Q^1_{\lambda,\theta}(z) = \sum_{k\geq 0} q_k(z-1)^k,$$

the Taylor coefficients of ϕ_j are determined for $k \geq 1$ by the Frobenius recurrence

$$\big((k+r_j)(k+r_j-1) + (k+r_j)p_0 + q_0\big)a_k^{(j)} + \sum_{l=0}^{k-1}\big((l+r_j)p_{k-l} + q_{k-l}\big)a_l^{(j)} = 0, \qquad \text{(A.8)}$$

together with $a_0^{(j)} = 1$. The leading factor equals $k(k-2+\lambda)$ for $r_1 = 0$ and $k(k+2-\lambda)$ for $r_2 = 2-\lambda$. Both are nonzero for every $k \geq 1$ since $\lambda \notin \mathbb{Z}$, so (A.8) determines ϕ_1 and ϕ_2 uniquely. □

Proof of Lemma 3.7: Throughout this proof, p_k and q_k denote the Taylor coefficients of $P^{-1}_{\lambda,\theta}$ and $Q^{-1}_{\lambda,\theta}$ at $z = -1$. Dividing (3.15) by $z^2(z-1)$ and multiplying by $(z+1)$ casts it in the Fuchsian standard form of (A.1) with $z_0 = -1$:

$$(z+1)^2\partial_z^2 F(z) + (z+1)P^{-1}_{\lambda,\theta}(z)\partial_z F(z) + Q^{-1}_{\lambda,\theta}(z)F(z) = 0,$$

where

$$P^{-1}_{\lambda,\theta}(z) := \frac{P_{\lambda,\theta}(z)}{z(z-1)}, \qquad Q^{-1}_{\lambda,\theta}(z) := \frac{(z+1)Q_{\lambda,\theta}(z)}{z^2(z-1)}.$$

Both $P^{-1}_{\lambda,\theta}$ and $Q^{-1}_{\lambda,\theta}$ are analytic on a neighborhood of $z = -1$. Setting $p_0 := P^{-1}_{\lambda,\theta}(-1)$ and $q_0 := Q^{-1}_{\lambda,\theta}(-1)$, direct computation yields

$$p_0 = \frac{P_{\lambda,\theta}(-1)}{2} = -(1+\lambda), \qquad q_0 = 0,$$

so the indicial equation $r(r-1) + rp_0 + q_0 = 0$ becomes $r^2 - (2+\lambda)r = 0$, with roots $r_1 = 0$ and $r_2 = 2+\lambda$. Since $\lambda \in \mathbb{C} - \mathbb{Z}$, the difference $r_2 - r_1 = 2 + \lambda \notin \mathbb{Z}$, and both local solutions are pure power series:

$$F^{h,-1}_{\lambda,\theta,1}(z) = \phi_1(z), \qquad F^{h,-1}_{\lambda,\theta,2}(z) = (z+1)^{2+\lambda}\phi_2(z),$$

with ϕ_1, ϕ_2 analytic at -1 and normalized so that $\phi_1(-1) = \phi_2(-1) = 1$. The Taylor coefficients of ϕ_j are determined for $k \geq 1$ by the Frobenius recurrence (A.8), now with $(z-1)$ replaced by $(z+1)$ and with the coefficients p_k, q_k of $P^{-1}_{\lambda,\theta}, Q^{-1}_{\lambda,\theta}$. The leading factor equals $k(k-2-\lambda)$ for $r_1 = 0$ and $k(k+2+\lambda)$ for $r_2 = 2+\lambda$, both nonzero for $k \geq 1$ since $\lambda \notin \mathbb{Z}$. □

IV.A.2 Analytic particular solutions

This section proves Lemmas 3.9 and 3.12. Rather than constructing the particular solutions by variation of constants, we obtain them directly: we look for a power series solving the inhomogeneous equation, read off the resulting Frobenius recurrence, and prove its convergence by a majorant. This is both shorter and closer to what the computer-assisted part actually evaluates, since the same recurrence produces the Taylor coefficients used in Appendix IV.B.

IV.A.2.1 A holomorphic solution from the Frobenius recurrence

Let $z_0 \in \mathbb{C}$, let P, Q, G be holomorphic on a disk $D_R(z_0)$, $R > 0$, and consider the inhomogeneous Fuchsian equation in normal form

$$(z - z_0)^2 \partial_z^2 F(z) + (z - z_0) P(z) \partial_z F(z) + Q(z) F(z) = G(z). \tag{A.9}$$

Write the local variable $x := z - z_0$ and expand

$$P(z) = \sum_{k \geq 0} p_k x^k, \qquad Q(z) = \sum_{k \geq 0} q_k x^k, \qquad G(z) = \sum_{k \geq 0} g_k x^k.$$

We look for a holomorphic solution $F(z) = \sum_{k \geq 0} f_k x^k$. Substituting into (A.9) and collecting the coefficient of x^k gives, for every $k \geq 0$,

$$\mathcal{I}(k) f_k + \sum_{j=0}^{k-1} \big(j p_{k-j} + q_{k-j} \big) f_j = g_k, \qquad \mathcal{I}(k) := k(k-1) + k p_0 + q_0, \tag{A.10}$$

where $\mathcal{I}$ is the indicial polynomial of (A.9), with roots r_1, r_2 so that $\mathcal{I}(k) = (k - r_1)(k - r_2)$.

Lemma A.1 *With the notation above, assume*

(i) $\mathcal{I}(k) \neq 0$ for every integer $k \geq 1$, and

(ii) either $\mathcal{I}(0) \neq 0$, or else $\mathcal{I}(0) = 0$ and $g_0 = 0$.

Then (A.9) *has a holomorphic solution f on $D_R(z_0)$ whose Taylor coefficients are determined by the recurrence* (A.10)*. In the resonant case $\mathcal{I}(0) = 0$ the coefficient f_0 is free and we fix $f_0 := 0$. In particular f has radius of convergence at least R.*

Proof: *Coefficients.* For $k \geq 1$, hypothesis (i) lets us solve (A.10) for f_k in terms of $f_0, \dots, f_{k-1}$. For $k = 0$, (A.10) reads $\mathcal{I}(0) f_0 = g_0$: if $\mathcal{I}(0) \neq 0$ this fixes f_0, while if $\mathcal{I}(0) = 0$ then $g_0 = 0$ by (ii), the equation $0 = 0$ holds automatically, and we are free to set $f_0 := 0$. Thus all f_k are well defined.

Convergence. Fix $\rho \in (0, R)$. Since P, Q, G are holomorphic on $D_R(z_0)$, Cauchy's estimates provide $M \geq 1$ with

$$|p_k| \leq M \rho^{-k}, \qquad |q_k| \leq M \rho^{-k}, \qquad |g_k| \leq M \rho^{-k} \qquad (k \geq 0).$$

Because $\mathcal{I}(k) = (k - r_1)(k - r_2) \sim k^2$ as $k \to \infty$ and $\mathcal{I}(k) \neq 0$ for every $k \geq 1$, the ratios $\frac{|\mathcal{I}(k)|}{k^2}$ are bounded away from 0, so there is $c > 0$ with

$$|\mathcal{I}(k)| \geq c k^2 \qquad (k \geq 1). \tag{A.11}$$

From (A.10), using $|j p_{k-j} + q_{k-j}| \leq M(j+1)\rho^{-(k-j)}$, we get for $k \geq 1$

$$|f_k| \leq \frac{1}{c k^2} \Big(M \rho^{-k} + \sum_{j=0}^{k-1} M (j+1) \rho^{-(k-j)} |f_j| \Big).$$

Set $b_k := |f_k|\rho^k$ and $\Xi_k := \sum_{j=0}^{k}(j+1)b_j$. Multiplying the previous bound by ρ^k gives

$$b_k \le \frac{M}{ck^2}(1+\Xi_{k-1}) \qquad (k \ge 1).$$

Since $\frac{k+1}{k^2} \le \frac{2}{k}$, it follows that for every $k \ge 1$

$$\Xi_k + 1 = (\Xi_{k-1}+1) + (k+1)b_k \le (\Xi_{k-1}+1)\Big(1+\frac{2M}{ck}\Big).$$

Iterating and using

$$\prod_{j=1}^{k}\Big(1+\frac{2M}{cj}\Big) \le \exp\Big(\frac{2M}{c}\sum_{j=1}^{k}\frac{1}{j}\Big) \le C_1 k^{2M/c},$$

we obtain $\Xi_k \le C_1(\Xi_0+1)k^{2M/c}$, which grows only polynomially in k. Hence

$$b_k \le \frac{M}{ck^2}(1+\Xi_{k-1}) \le C_2 k^{2M/c}$$

is polynomially bounded as well, so

$$|f_k| = b_k\rho^{-k} \le C_2\frac{k^{2M/c}}{\rho^k}, \qquad \text{whence} \qquad \limsup_{k\to\infty}|f_k|^{1/k} \le \frac{1}{\rho}.$$

As $\rho \in (0,R)$ was arbitrary, $\sum_{k\ge0} f_k x^k$ has radius of convergence at least R. Its sum f is holomorphic on $D_R(z_0)$ and, since its coefficients satisfy (A.10), which is equivalent to (A.9), f solves the equation. □

We now apply Lemma A.1 at the two singular points $z_0 = 0$ and $z_0 = 1$. In each case the right-hand side G of (3.14) is holomorphic, and putting (3.14) into the normal form (A.9) divides G by exactly the factor used for the homogeneous equation in the proofs of Lemmas 3.4 and 3.5.

Proof of Lemma 3.9: At $z_0 = 0$, dividing (3.14) by z^2-1 casts it in the normal form (A.9) with $P = P^0_{\lambda,\theta}$ and $Q = Q^0_{\lambda,\theta}$ as in the proof of Lemma 3.4, and right-hand side

$$G^0(z) := \frac{G(z)}{z^2-1},$$

which is holomorphic on $D_r(0)$ since $r < 1$. The indicial polynomial has roots $1-\theta$ and $-1-\theta$, so

$$\mathcal{I}(n) = (n-1+\theta)(n+1+\theta) \ne 0 \qquad \text{for every integer } n \ge 0,$$

because $\theta \in (0,1)$. Hence hypotheses (i)–(ii) of Lemma A.1 hold, with $\mathcal{I}(0) \ne 0$, and Lemma A.1 produces a holomorphic solution $F^{p,0}_{G,\lambda,\theta} : D_r(0) \to \mathbb{C}$ of (3.14) whose Taylor coefficients are computed recursively by (A.10). □

Proof of Lemma 3.12: At $z_0 = 1$, dividing (3.14) by $z^2(z+1)$ and multiplying by $(z-1)$ casts it in the normal form (A.9) with $P = P^1_{\lambda,\theta}$ and $Q = Q^1_{\lambda,\theta}$ as in the proof of Lemma 3.5, and right-hand side

$$G^1(z) := \frac{(z-1)G(z)}{z^2(z+1)},$$

which is holomorphic on $D_r(1)$. The indicial polynomial has roots $r_1 = 0$ and $r_2 = 2-\lambda$, so

$$\mathcal{I}(n) = n(n-2+\lambda),$$

which vanishes at $n = 0$ and, since $\lambda \notin \mathbb{Z}$, is nonzero for every $n \geq 1$. The root $r_1 = 0$ is resonant. However, the normalization above carries an explicit factor $(z-1)$, so that

$$g_0 = G^1(1) = 0$$

for *every* holomorphic G. Hence hypotheses (i)–(ii) of Lemma A.1 hold in the resonant case with $g_0 = 0$, and Lemma A.1 produces a holomorphic solution $F^{p,1}_{G,\lambda,\theta} : D_r(1) \to \mathbb{C}$ of (3.14), normalized by $F^{p,1}_{G,\lambda,\theta}(1) = f_0 = 0$, whose Taylor coefficients are computed recursively by (A.10).

Finally, let F be any holomorphic solution of (3.14) at 1. Then $\widetilde{F} := F - F^{p,1}_{G,\lambda,\theta}$ is holomorphic at 1 and solves (3.15), so by Lemma 3.5 we have $\widetilde{F} = C_1 F^{h,1}_{\lambda,\theta,1} + C_2 F^{h,1}_{\lambda,\theta,2}$ for some $C_1, C_2 \in \mathbb{C}$. Since $\widetilde{F}$ and $F^{h,1}_{\lambda,\theta,1} = \phi_1$ are holomorphic at 1 while $F^{h,1}_{\lambda,\theta,2} = (z-1)^{2-\lambda}\phi_2$ is not (as $\lambda \notin \mathbb{Z}$), we must have $C_2 = 0$, and therefore $F = F^{p,1}_{G,\lambda,\theta} + C_1 F^{h,1}_{\lambda,\theta,1}$. □

IV.A.3 A priori estimate at the boundary singular points

This appendix contains the proof of Lemma 3.16. The argument rests on a pair of weighted Hardy inequalities that we state and prove first.

IV.A.3.1 Weighted Hardy operators

For $\sigma \geq 0$, we use on $\mathbb{R}$ the same Sobolev convention as on $\mathbb{T}$:

$$\|f\|^2_{H^\sigma(\mathbb{R})} := \int_{\mathbb{R}} (1 + |\xi|^{2\sigma}) |\widehat{f}(\xi)|^2 \, d\xi, \qquad \widehat{f}(\xi) := \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(x) e^{-ix\xi} \, dx.$$

At $\sigma = 0$ the multiplier $|\xi|^0$ means 1, also at $\xi = 0$. For $\delta > 0$, $\beta \in \mathbb{C}$, and w supported in $(-\delta, \delta)$, define for $x \in (-\delta, \delta) - \{0\}$,

$$(\mathsf{H}^0_\beta w)(x) := |x|^{-\beta} \int_0^{|x|} \tau^{\beta-1} w(\operatorname{sgn}(x)\tau) \, d\tau, \qquad (\mathsf{H}^\delta_\beta w)(x) := |x|^{\beta} \int_{|x|}^{\delta} \tau^{-\beta-1} w(\operatorname{sgn}(x)\tau) \, d\tau.$$

For $|x| \geq \delta$ use the same formula for $\mathsf{H}^0_\beta w$ (with w extended by zero) and set $\mathsf{H}^\delta_\beta w = 0$. Writing $D_x := x\partial_x$, these operators satisfy

$$D_x(\mathsf{H}^0_\beta w) + \beta \mathsf{H}^0_\beta w = w, \qquad D_x(\mathsf{H}^\delta_\beta w) - \beta \mathsf{H}^\delta_\beta w = -w. \tag{A.12}$$

Lemma A.2 *Let $\sigma \geq 0$ and $w \in H^\sigma(\mathbb{R})$ be supported in $(-\delta, \delta)$.*

(i) *If $\operatorname{Re}\beta > \frac{1}{2}$, then*

$$\|\mathsf{H}^0_\beta w\|_{H^\sigma(\mathbb{R})} \leq \frac{\|w\|_{H^\sigma(\mathbb{R})}}{\operatorname{Re}\beta - \frac{1}{2}}.$$

(ii) *If $\operatorname{Re}\beta > \sigma - \frac{1}{2}$, then*

$$\|\mathsf{H}^\delta_\beta w\|_{H^\sigma(\mathbb{R})} \leq \frac{\|w\|_{H^\sigma(\mathbb{R})}}{\operatorname{Re}\beta - \sigma + \frac{1}{2}}.$$

Proof: Let w be as in the statement and put $T_t w(x) := w(tx)$ for $t > 0$. Changing variables gives

$$\|T_t w\|^2_{H^\sigma(\mathbb{R})} = t^{-1} \int_{\mathbb{R}} (1 + t^{2\sigma}|\xi|^{2\sigma}) |\widehat{w}(\xi)|^2 \, d\xi. \tag{A.13}$$

For $0 < t \le 1$ the weight in (A.13) is at most $1 + |\xi|^{2\sigma}$. For $t \ge 1$ it is at most $t^{2\sigma}(1 + |\xi|^{2\sigma})$. Thus

$$\|T_t w\|_{H^\sigma(\mathbb{R})} \le \begin{cases} t^{-1/2}\|w\|_{H^\sigma(\mathbb{R})}, & 0 < t \le 1, \\ t^{\sigma-1/2}\|w\|_{H^\sigma(\mathbb{R})}, & t \ge 1. \end{cases} \tag{A.14}$$

Under the respective hypotheses of the two parts, the substitution $\tau = t|x|$ and the support of w give, for $x \neq 0$,

$$(\mathsf{H}^0_\beta w)(x) = \int_0^1 t^{\beta-1} T_t w(x)\, dt, \qquad (\mathsf{H}^\delta_\beta w)(x) = \int_1^\infty t^{-\beta-1} T_t w(x)\, dt. \tag{A.15}$$

For part (i), assume $\mathrm{Re}\beta > \frac{1}{2}$. The first case of (A.14) gives

$$\int_0^1 \|t^{\beta-1} T_t w\|_{H^\sigma(\mathbb{R})}\, dt \le \|w\|_{H^\sigma(\mathbb{R})} \int_0^1 t^{\mathrm{Re}\beta - 3/2}\, dt = \frac{\|w\|_{H^\sigma(\mathbb{R})}}{\mathrm{Re}\beta - \frac{1}{2}}. \tag{A.16}$$

For part (ii), assume instead $\mathrm{Re}\beta > \sigma - \frac{1}{2}$. The second case of (A.14) gives

$$\int_1^\infty \|t^{-\beta-1} T_t w\|_{H^\sigma(\mathbb{R})}\, dt \le \|w\|_{H^\sigma(\mathbb{R})} \int_1^\infty t^{-\mathrm{Re}\beta + \sigma - 3/2}\, dt = \frac{\|w\|_{H^\sigma(\mathbb{R})}}{\mathrm{Re}\beta - \sigma + \frac{1}{2}}. \tag{A.17}$$

The finite bounds in (A.16) and (A.17) show that the integrals in (A.15) converge in $H^\sigma(\mathbb{R})$. The triangle inequality then yields

$$\|\mathsf{H}^0_\beta w\|_{H^\sigma(\mathbb{R})} \le \frac{\|w\|_{H^\sigma(\mathbb{R})}}{\mathrm{Re}\beta - \frac{1}{2}} \qquad \text{and} \qquad \|\mathsf{H}^\delta_\beta w\|_{H^\sigma(\mathbb{R})} \le \frac{\|w\|_{H^\sigma(\mathbb{R})}}{\mathrm{Re}\beta - \sigma + \frac{1}{2}}.$$

□

Proof of Lemma 3.16: Throughout, $A \lesssim B$ means $A \le C_{\lambda,\theta,\epsilon} B$. Write $z = e^{ix}$, so $\partial_x = iz\partial_z$ and $z^2\partial_z^2 = (z\partial_z)^2 - z\partial_z = -\partial_x^2 + i\partial_x$. Hence

$$z^2(z^2-1)\partial_z^2 F = (z^2-1)(-\partial_x^2 + i\partial_x)F \qquad \text{and} \qquad zP_{\lambda,\theta}\partial_z F = -iP_{\lambda,\theta}\partial_x F.$$

The trace $f := F|_{\mathbb{T}}$ therefore satisfies

$$(1-z^2)\partial_x^2 f + i(z^2 - 1 - P_{\lambda,\theta})\partial_x f + Q_{\lambda,\theta} f = g, \qquad g := G|_{\mathbb{T}}. \tag{A.18}$$

Since $F \in \mathcal{H}^{s+1}$ and $G \in \mathcal{H}^s$, the functions $f, \partial_x f, g$ are continuous on $\mathbb{T}$.

Write $\sigma := s - 1 \in [0,1]$. Since $(1+k^2)^{s+1} \lesssim k^4(1+k^2)^{s-1} + 1$,

$$\|f\|^2_{H^{s+1}(\mathbb{T})} \lesssim \|\partial_x^2 f\|^2_{H^{s-1}(\mathbb{T})} + \|f\|^2_{L^2(\mathbb{T})}, \tag{A.19}$$

and $\|f\|_{L^2(\mathbb{T})} \lesssim \|F\|_{\mathcal{H}^{s+1-\epsilon}}$. It therefore suffices to prove

$$\|\partial_x^2 f\|_{H^{s-1}(\mathbb{T})} \lesssim \|G\|_{\mathcal{H}^s} + \|F\|_{\mathcal{H}^{s+1-\epsilon}}. \tag{A.20}$$

Fix $0 < \delta < \frac{\pi}{2}$ and a smooth partition $1 = \chi_0^2 + \chi_+^2 + \chi_-^2$ on $\mathbb{T}$ where

$$\chi_+ = 1 \text{ on } \left\{|x| \le \frac{1}{2}\delta\right\}, \quad \operatorname{supp}\chi_+ \subset \left\{|x| < \frac{3}{4}\delta\right\},$$
$$\chi_- = 1 \text{ on } \left\{|x-\pi| \le \frac{1}{2}\delta\right\}, \quad \operatorname{supp}\chi_- \subset \left\{|x-\pi| < \frac{3}{4}\delta\right\},$$

and χ_0 is supported away from $z = \pm 1$. Then

$$\|v\|_{H^{s-1}(\mathbb{T})} \lesssim \|\chi_0 v\|_{H^{s-1}(\mathbb{T})} + \|\chi_+ v\|_{H^{s-1}(\mathbb{T})} + \|\chi_- v\|_{H^{s-1}(\mathbb{T})}. \tag{A.21}$$

Interior piece. On $\operatorname{supp}\chi_0$ one has $|1-z^2|\geq c(\delta)>0$, so (A.18) gives

$$\chi_0\partial_x^2 f=\frac{\chi_0}{1-z^2}(g-i(z^2-1-P_{\lambda,\theta})\partial_x f-Q_{\lambda,\theta}f).$$

The factor $\frac{\chi_0}{1-z^2}$ is smooth, so multiplication by it is bounded on $H^{s-1}(\mathbb{T})$, giving

$$\|\chi_0\partial_x^2 f\|_{H^{s-1}(\mathbb{T})}\lesssim\|g\|_{H^{s-1}(\mathbb{T})}+\|\partial_x f\|_{H^{s-1}(\mathbb{T})}+\|f\|_{H^{s-1}(\mathbb{T})}\lesssim\|G\|_{\mathcal{H}^s}+\|F\|_{\mathcal{H}^{s+1-\epsilon}}.$$

For functions supported in either singular chart, the $H^\sigma(\mathbb{T})$ norm is equivalent to the $H^\sigma(\mathbb{R})$ norm of the zero extension in the local coordinate.

Near $z=-1$ (admissible branch). Here the local exponent is $2+\lambda$, the admissible branch. Writing x for the local coordinate at this singular point (so that $z^2-1=e^{2ix}-1$ vanishes at $x=0$), set

$$h(x):=\frac{e^{2ix}-1}{x}=2i\int_0^1 e^{2isx}\,ds.$$

Notice that h is smooth at 0, $h(0)=2i$, $h\neq 0$ for $|x|<\delta$, after choosing $\delta>0$ sufficiently small. Dividing (A.18) by $-h$ gives on $\{|x|<\delta\}$:

$$D_x\partial_x f=\widetilde{g}+\widetilde{p}\partial_x f+\widetilde{q}f,\qquad \widetilde{p}:=\frac{c_1}{h},\quad \widetilde{q}:=\frac{Q_{\lambda,\theta}}{h},\quad \widetilde{g}:=-\frac{g}{h},\tag{A.22}$$

where $c_1=i(z^2-1-P_{\lambda,\theta})$ and $D_x f(x):=x\partial_x f(x)$.

Differentiating (A.22) and setting $v=\partial_x^2 f$:

$$D_x v+(1-\widetilde{p})v=\kappa_-,\qquad \kappa_-:=\partial_x\widetilde{g}+(\partial_x\widetilde{p}+\widetilde{q})\partial_x f+f\partial_x\widetilde{q}.\tag{A.23}$$

Multiplying (A.23) by χ_- and writing $V:=\chi_- v$,

$$D_x V+(1-\widetilde{p})V=\chi_-\kappa_-+vD_x\chi_-=:K_-.\tag{A.24}$$

The term $vD_x\chi_-$ is supported in $\{\frac{1}{2}\delta<|x|<\frac{3}{4}\delta\}$, where the interior estimate applies, giving

$$\|K_-\|_{H^{s-1}(\mathbb{T})}\lesssim\|G\|_{\mathcal{H}^s}+\|F\|_{\mathcal{H}^{s+1-\epsilon}}.\tag{A.25}$$

Extend K_- by zero to $\mathbb{R}$ in the local coordinate.

Write $1-\widetilde{p}=-\lambda+x\rho$ and $R(x):=\int_0^x\rho(y)\,dy$. Extend $e^R K_-$ by zero to $\mathbb{R}$. On either side of zero, the difference between V and $-e^{-R}\mathsf{H}_\lambda^\delta[e^R K_-]$ is a multiple of $|x|^\lambda e^{-R(x)}$. Both vanish near $\pm\delta$, so the multiples are zero and

$$V=-e^{-R}\mathsf{H}_\lambda^\delta[e^R K_-].\tag{A.26}$$

Since $\mathrm{Re}\lambda>\frac{1}{2}\geq\sigma-\frac{1}{2}$, Lemma A.2(ii) applies. A smooth cutoff supported in $(-\delta,\delta)$ and equal to one on $\operatorname{supp}\chi_-$ gives

$$\|\chi_- v\|_{H^\sigma(\mathbb{R})}\lesssim\|\mathsf{H}_\lambda^\delta[e^R K_-]\|_{H^\sigma(\mathbb{R})}\lesssim\|K_-\|_{H^\sigma(\mathbb{R})}\lesssim\|G\|_{\mathcal{H}^s}+\|F\|_{\mathcal{H}^{s+1-\epsilon}}.$$

Near $z=+1$ (excluded branch). The same local computation, in the coordinate y centered at this point, gives

$$y\partial_y v+(1-\widetilde{p}_+)v=\kappa_+,\qquad 1-\widetilde{p}_+(0)=\lambda.$$

Set $V_+:=\chi_+ v$ and $K_+:=\chi_+\kappa_++vD_y\chi_+$, where $D_y:=y\partial_y$. The interior estimate on $\operatorname{supp}D_y\chi_+$ gives, after extension by zero to $\mathbb{R}$,

$$\|K_+\|_{H^\sigma(\mathbb{R})}\lesssim\|G\|_{\mathcal{H}^s}+\|F\|_{\mathcal{H}^{s+1-\epsilon}}.$$

Write $1 - \widetilde{p}_+(y) = \lambda + y\rho_+(y)$ and $R_+(y) := \int_0^y \rho_+(r)\,dr$. Extend $e^{R_+}K_+$ by zero to $\mathbb{R}$. By Lemma A.2(i), $e^{-R_+}\mathsf{H}^0_\lambda[e^{R_+}K_+]$ belongs to $H^\sigma(-\delta,\delta)$. On either side of zero, its difference from V_+ is a multiple of $|y|^{-\lambda}e^{-R_+(y)}$. Both V_+ and the particular solution belong to $L^2(-\delta,\delta)$. Since $\mathrm{Re}\lambda > \frac{1}{2}$, the multiple must vanish. Thus

$$V_+ = e^{-R_+}\mathsf{H}^0_\lambda[e^{R_+}K_+] \qquad \text{on } (-\delta,\delta).$$

A smooth cutoff supported in $(-\delta,\delta)$ and equal to one on $\operatorname{supp}\chi_+$ gives

$$\|\chi_+ v\|_{H^\sigma(\mathbb{R})} \lesssim \|K_+\|_{H^\sigma(\mathbb{R})} \lesssim \|G\|_{\mathcal{H}^s} + \|F\|_{\mathcal{H}^{s+1-\epsilon}}.$$

Conclusion. Inserting all three bounds into (A.21) gives (A.20). Combined with (A.19) this yields

$$\|F\|_{\mathcal{H}^{s+1}} = \|f\|_{H^{s+1}(\mathbb{T})} \lesssim \|\partial_x^2 f\|_{H^{s-1}(\mathbb{T})} + \|f\|_{L^2(\mathbb{T})} \lesssim \|G\|_{\mathcal{H}^s} + \|F\|_{\mathcal{H}^{s+1-\epsilon}},$$

completing the proof. □

IV.B Error bounds for Taylor approximations

The computer-assisted verification of Lemma 3.20 requires rigorous enclosures for the entries of $\mathbf{M}_{\lambda,\theta}$, which are values at $x_0 = \frac{1}{2}$ of analytic functions associated with the Fuchsian operator $L^{\mathrm{hol}}_{\lambda,\theta}$. Each such function is approximated in practice by a polynomial Taylor truncation matching the first coefficients of the exact solution at the relevant singular point. The aim of this appendix is to develop the residual estimate that allows us to control the distance between an analytic function f and its truncated Taylor approximation f^{ap}.

IV.B.1 A residual estimate for inhomogeneous Fuchsian equations

Throughout this section let $b > 0$, let $p, q, g : [0,b] \to \mathbb{C}$ be analytic, and consider the Fuchsian operator

$$\mathcal{L}f(x) := x^2\partial_x^2 f(x) + xp(x)\partial_x f(x) + q(x)f(x)$$

together with an analytic solution f of

$$\mathcal{L}f = g \quad \text{on } [0,b]. \tag{B.1}$$

Given an analytic approximation $f^{\mathrm{ap}} : [0,b] \to \mathbb{C}$ matching the first N_0 derivatives of f at 0, we measure its accuracy by the residual

$$\xi(x) := \int_0^x \int_0^y \frac{g(s) - \mathcal{L}f^{\mathrm{ap}}(s)}{s^2}\,ds\,dy, \tag{B.2}$$

which is well defined and analytic on $[0,b]$ provided $g - \mathcal{L}f^{\mathrm{ap}}$ vanishes to order at least 2 at 0.

For $N_0 \geq 2$, write

$$\mathcal{A}_{N_0}([0,b]) := \{h \text{ analytic on } [0,b] : \partial_x^k h(0) = 0 \text{ for every } 0 \leq k < N_0\},$$

and, for $h \in \mathcal{A}_{N_0}([0,b])$ written as $h = x^{N_0}u$, define

$$\|h\|_{\mathcal{A}_{N_0}} := \|u\|_{L^\infty([0,b])}.$$

Lemma B.1 *Let $N_0 \geq 2$ and assume the contraction condition*

$$\ell := \frac{\|p\|_{L^\infty([0,b])}}{N_0} + \frac{\|D_x p - p - q\|_{L^\infty([0,b])}}{N_0(N_0-1)} < 1, \tag{B.3}$$

where $D_x = x\partial_x$.

Let f be an analytic solution of (B.1)*, let $f^{\mathrm{ap}} : [0,b] \to \mathbb{C}$ be analytic with*

$$f^{\mathrm{ap}(k)}(0) = f^{(k)}(0) \qquad \text{for every } 0 \leq k \leq N_0 - 1,$$

and let $\tau := f - f^{\mathrm{ap}}$ and ξ as in (B.2)*. Then $\tau \in \mathcal{A}_{N_0}([0,b])$, and*

$$\|\tau\|_{\mathcal{A}_{N_0}([0,b])} \leq \frac{1}{1-\ell}\|\xi\|_{\mathcal{A}_{N_0}([0,b])}. \tag{B.4}$$

In particular,

$$\|\tau\|_{L^\infty([0,b])} \leq \frac{b^{N_0}}{1-\ell}\|\xi\|_{\mathcal{A}_{N_0}([0,b])}, \tag{B.5a}$$

$$\|\partial_x \tau\|_{L^\infty([0,b])} \leq \frac{N_0 \ell b^{N_0-1}}{1-\ell}\|\xi\|_{\mathcal{A}_{N_0}([0,b])} + \|\partial_x \xi\|_{L^\infty([0,b])}. \tag{B.5b}$$

Proof: The matching conditions imply that $\tau = f - f^{\mathrm{ap}}$ belongs to $\mathcal{A}_{N_0}([0,b])$. Define T on this space by

$$Th(x) := -\int_0^x \int_0^y \frac{sp(s)\partial_x h(s) + q(s)h(s)}{s^2}\, ds\, dy.$$

For $h \in \mathcal{A}_{N_0}([0,b])$, the numerator vanishes to order at least N_0, so the integrals converge, Th is analytic, and $Th \in \mathcal{A}_{N_0}([0,b])$. Since $h(0) = \partial_x h(0) = 0$, Taylor's integral formula gives

$$(I - T)h(x) = \int_0^x \int_0^y \frac{\mathcal{L}h(s)}{s^2}\, ds\, dy.$$

Apply this identity directly to τ. The equation $\mathcal{L}\tau = g - \mathcal{L}f^{\mathrm{ap}}$ and the definition of ξ yield

$$(I - T)\tau = \xi. \tag{B.6}$$

Writing $\tau(x) = x^{N_0}u(x)$, a direct computation gives

$$sp(s)\partial_x \tau(s) + q(s)\tau(s) = s^{N_0}\big(N_0 p(s) + q(s)\big)u(s) + s^{N_0+1}p(s)\partial_x u(s).$$

Hence, we have

$$\begin{aligned} T\tau(x) &= -\int_0^x \int_0^y \frac{sp(s)\partial_x \tau(s) + q(s)\tau(s)}{s^2}\, ds\, dy \\ &= -\int_0^x \int_0^y s^{N_0-2}\big(N_0 p(s) + q(s)\big)u(s) + s^{N_0-1}p(s)\partial_x u(s)\, ds\, dy \\ &= -\int_0^x y^{N_0-1}p(y)u(y)\, dy + \int_0^x \int_0^y s^{N_0-2}\big(s\partial_x p(s) - p(s) - q(s)\big)u(s)\, ds\, dy, \end{aligned}$$

where we have integrated by parts.

Bounding each integrand by its L^∞ norm and using

$$\int_0^x y^{N_0-1}\, dy = \frac{x^{N_0}}{N_0}, \qquad \int_0^x \int_0^y s^{N_0-2}\, ds\, dy = \frac{x^{N_0}}{N_0(N_0-1)},$$

we obtain

$$\frac{|T\tau(x)|}{x^{N_0}} \le \ell \|u\|_{L^\infty([0,b])} \qquad \text{for all } x \in (0,b],$$

hence

$$\|T\tau\|_{\mathcal{A}_{N_0}([0,b])} \le \ell \|\tau\|_{\mathcal{A}_{N_0}([0,b])}.$$

By (B.6) and the preceding estimate,

$$\|\tau\|_{\mathcal{A}_{N_0}} \le \|T\tau\|_{\mathcal{A}_{N_0}} + \|\xi\|_{\mathcal{A}_{N_0}} \le \ell \|\tau\|_{\mathcal{A}_{N_0}} + \|\xi\|_{\mathcal{A}_{N_0}}.$$

Rearranging and using $\ell < 1$ proves (B.4). The bound (B.5a) follows from $\|\tau\|_{L^\infty([0,b])} \le b^{N_0} \|\tau\|_{\mathcal{A}_{N_0}}$.

For the derivative bound (B.5b), differentiating (B.6) yields

$$\partial_x \tau(x) = -x^{N_0-1} p(x) u(x) + \int_0^x s^{N_0-2} \big(s \partial_x p(s) - p(s) - q(s)\big) u(s)\, ds + \partial_x \xi(x).$$

Bounding each integrand by its L^∞ norm and using

$$x^{N_0-1} \le b^{N_0-1}, \qquad \int_0^x s^{N_0-2}\, ds = \frac{x^{N_0-1}}{N_0-1} \le \frac{b^{N_0-1}}{N_0-1},$$

together with the identity $\|p\|_{L^\infty([0,b])} + \frac{\|D_x p - p - q\|_{L^\infty([0,b])}}{N_0-1} = N_0 \ell$, gives

$$\|\partial_x \tau\|_{L^\infty([0,b])} \le N_0 \ell b^{N_0-1} \|u\|_{L^\infty([0,b])} + \|\partial_x \xi\|_{L^\infty([0,b])},$$

and the bound (B.5b) follows from $\|u\|_{L^\infty([0,b])} = \|\tau\|_{\mathcal{A}_{N_0}([0,b])} \le (1-\ell)^{-1} \|\xi\|_{\mathcal{A}_{N_0}([0,b])}$. □

IV.B.2 Application to the entries of $\mathbf{M}_{\lambda,\theta}$

The matrix $\mathbf{M}_{\lambda,\theta}$ defined in Definition 3.14 is assembled from the values at $x_0 = \frac{1}{2}$ of the functions

$$F^{h,1}_{\lambda,\theta^\pm,1} \qquad \text{and} \qquad F^{p,k}_{\lambda,\theta^\pm,i}, \quad i \in \{1,2\}, \quad k \in \{0,1\},$$

together with their first derivatives ∂_z. For each such global function F we pass to the local coordinate x at the relevant singular point. Near $z = 1$ set $f(x) := F(1-x)$ (so $x = 1 - z$), and near $z = 0$ set $f(x) := F(x)$ (so $x = z$). Each f is analytic near $x = 0$ and satisfies a local Fuchsian equation $\mathcal{L}_{z_0} f = g$ identified below. Writing $b_0 := \frac{1}{2}$, the global midpoint $z = x_0 = \frac{1}{2}$ corresponds to $x = b_0$ in both charts, and the matrix entries of Definition 3.14 relate to f via

$$F(x_0) = f(x_*), \qquad \partial_z F(x_0) = \sigma \partial_x f(x_*), \qquad x_* := b_0, \qquad \sigma := \begin{cases} -1 & \text{near } z = 1, \\ +1 & \text{near } z = 0. \end{cases} \tag{B.7}$$

Local equations at $z = 0$. Set $x := z$ and write the local operator at $z = 0$ as

$$\mathcal{L}_0 f(x) := x^2 \partial_x^2 f(x) + x p^0_{\lambda,\theta}(x) \partial_x f(x) + q^0_{\lambda,\theta}(x) f(x), \tag{B.8}$$

where $p^0_{\lambda,\theta}$ and $q^0_{\lambda,\theta}$ are the coefficients of the Frobenius normal form at $z_0 = 0$, written in the local variable $x = z$. From $P^0_{\lambda,\theta}, Q^0_{\lambda,\theta}$ in the proof of Lemma 3.4 one obtains the rational functions

$$p^0_{\lambda,\theta}(x) = P^0_{\lambda,\theta}(x) = \frac{P_{\lambda,\theta}(x)}{x^2 - 1}, \qquad q^0_{\lambda,\theta}(x) = Q^0_{\lambda,\theta}(x) = \frac{Q_{\lambda,\theta}(x)}{x^2 - 1},$$

which are analytic on $|x| < 1$, with nearest poles at $x = \pm 1$.

The norms $\|p\|_{L^\infty}$ and $\|D_x p - p - q\|_{L^\infty}$ entering the contraction condition (B.3) are bounded by hand in the following lemma.

Lemma B.2 *It holds for* $\lambda \in \{0.56, 0.6\}$ *and* $\theta \in (0,1)$ *that*

$$\|p^0_{\lambda,\theta}\|_{L^\infty([0,\frac{1}{2}])} < 4, \qquad \text{and} \qquad \|D_x p^0_{\lambda,\theta} - p^0_{\lambda,\theta} - q^0_{\lambda,\theta}\|_{L^\infty([0,\frac{1}{2}])} < 8.$$

Consequently, the contraction condition (B.3) *is satisfied for all* $N_0 \geq 6$.

Proof: We have

$$p^0_{\lambda,\theta}(x) = 1 + 2\theta - (1+\lambda)\frac{x}{1+x} + (1-\lambda)\frac{x}{1-x} = 1 + 2\theta + 2x\frac{x-\lambda}{1-x^2}.$$

Taking the supremum yields

$$\|p^0_{\lambda,\theta}\|_{L^\infty([0,\frac{1}{2}])} \leq 1 + 2\theta + \frac{|x-\lambda|}{1-\frac{1}{4}} \leq 3 + \frac{4}{5} < 4,$$

Here we have used that $\lambda \in \{0.56, 0.6\}$ so that $|x-\lambda| \leq \frac{3}{5}$ for all $x \in [0, \frac{1}{2}]$.

The second bound in the statement is derived in the same way. A direct computation yields

$$x\partial_x p^0_{\lambda,\theta}(x) - p^0_{\lambda,\theta}(x) - q^0_{\lambda,\theta}(x) = \frac{r_{\lambda,\theta}(x)}{(x^2-1)^2}. \tag{B.9}$$

where the function $r_{\lambda,\theta}$ is given by

$$r_{\lambda,\theta}(x) := -x^4\theta^2 - (2\lambda\theta + 4\lambda)x^3 + (2\theta^2 + 2\theta + 4)x^2 + 2x\lambda\theta - \theta^2 - 2\theta.$$

Notice that each monomial of $r_{\lambda,\theta}(x)$ has constant sign for $\lambda, \theta \geq 0$ and $x \geq 0$. We then have for all $x \in [0, \frac{1}{2}]$, $\theta \in (0,1)$ and $\lambda \in \{0.56, 0.6\}$ that

$$(2\theta^2 + 2\theta + 4)x^2 + 2x\lambda\theta \in [0,3] \qquad \text{and} \qquad -x^4\theta^2 - (2\lambda\theta + 4\lambda)x^3 - \theta^2 - 2\theta \in [-4, 0].$$

Therefore, we have

$$\|D_x p^0_{\lambda,\theta} - p^0_{\lambda,\theta} - q^0_{\lambda,\theta}\|_{L^\infty([0,\frac{1}{2}])} \leq \frac{16}{9} \sup_{x\in[0,\frac{1}{2}]} |r_{\lambda,\theta}(x)| \leq 8.$$

Finally, we check that the contraction condition (B.3) is satisfied for all $N_0 \geq 6$:

$$\ell \leq \frac{4}{N_0} + \frac{8}{N_0(N_0-1)} \leq \frac{28}{30} < 1.$$

□

Local equations at $z = 1$. Set $x := 1 - z$ and write the local operator at $z = 1$ as

$$\mathcal{L}_1 f(x) := x^2\partial_x^2 f(x) + x p^1_{\lambda,\theta}(x)\partial_x f(x) + q^1_{\lambda,\theta}(x) f(x), \tag{B.10}$$

where $p^1_{\lambda,\theta}$ and $q^1_{\lambda,\theta}$ are the coefficients of the Frobenius normal form (A.1) at $z_0 = 1$, written in the local variable x. Constructing them from (3.5) as in the proof of Lemma 3.5 gives the rational functions

$$p^1_{\lambda,\theta}(x) = \frac{P_{\lambda,\theta}(1-x)}{(x-1)(x-2)}, \qquad q^1_{\lambda,\theta}(x) = \frac{xQ_{\lambda,\theta}(1-x)}{(x-1)^2(x-2)},$$

which are analytic on $|x| < 1$, with nearest poles at $x = 1$ and $x = 2$.

The corresponding bounds at $z = 1$ are as follows.

Lemma B.3 *It holds for* $\lambda \in \{0.56, 0.6\}$ *and* $\theta \in (0,1)$ *that*

$$\|p^1_{\lambda,\theta}\|_{L^\infty([0,\frac{1}{2}])} \le 8 \qquad \text{and} \qquad \|D_x p^1_{\lambda,\theta} - p^1_{\lambda,\theta} - q^1_{\lambda,\theta}\|_{L^\infty([0,\frac{1}{2}])} \le 16.$$

Consequently, the contraction condition (B.3) *is satisfied for all* $N_0 \ge 10$.

Proof: The first estimate is derived as follows

$$\begin{aligned}\|p^1_{\lambda,\theta}\|_{L^\infty([0,\frac{1}{2}])} &= \sup_{x\in[0,\frac{1}{2}]} \left| \frac{(2\theta-1)(x-1)^2 - 2\lambda(x-1) - (2\theta+1)}{(x-1)(x-2)} \right| \\ &\le 2 \sup_{x\in[0,\frac{1}{2}]} \left|(2\theta-1)(x-1)^2 - 2\lambda(x-1) - (2\theta+1)\right|.\end{aligned}$$

As in the proof of Lemma B.2, we enclose each monomial separately. Let $x \in [0, \frac{1}{2}]$. Then

$$(2\theta-1)(x-1)^2 \in [-1,1], \qquad -2\lambda(x-1) \in \left[\tfrac{1}{2}, \tfrac{6}{5}\right], \qquad -(2\theta+1) \in [-3,-1].$$

Hence, we obtain

$$\|p^1_{\lambda,\theta}\|_{L^\infty([0,\frac{1}{2}])} \le 2\left|[-1,1] + [\tfrac{1}{2}, \tfrac{6}{5}] + [-3,-1]\right| \le 2|[-4,2]| \le 8.$$

In order to compute the second bound, we write

$$x\partial_x p^1_{\lambda,\theta}(x) - p^1_{\lambda,\theta}(x) - q^1_{\lambda,\theta}(x) = \frac{r_{\lambda,\theta}(x)}{(x-2)^2(x-1)^2},$$

where $r_{\lambda,\theta}$ is given by

$$\begin{aligned}r_{\lambda,\theta}(x) := &-x^4\theta^2 + (4\theta^2 + 2\theta\lambda + 4\lambda)x^3 \\ &+ (-4\theta^2 - 6\theta\lambda + 2\theta - 12\lambda + 4)x^2 + (4\theta\lambda - 4\theta + 12\lambda - 8)x - 4\lambda + 4.\end{aligned}$$

Hence, for every $x \in [0, \frac{1}{2}]$ we have

$$-x^4\theta^2 \in \left[-\tfrac{1}{16}, 0\right], \qquad (4\theta^2 + 2\theta\lambda + 4\lambda)x^3 \in [0,1], \qquad (-4\theta^2 - 6\theta\lambda + 2\theta - 12\lambda + 4)x^2 \in [-3,0],$$
$$(4\theta\lambda - 4\theta + 12\lambda - 8)x \in [-2,0], \qquad -4\lambda + 4 \in \left[\tfrac{8}{5}, 2\right].$$

The fourth bound holds because $4\theta\lambda - 4\theta + 12\lambda - 8 = 4\theta(\lambda-1) + 4(3\lambda-2) \in [-4,0]$ for $\theta \in [0,1]$, $\lambda \in \{0.56, 0.6\}$ (both summands are non-positive since $\lambda < 1$ and $3\lambda < 2$), so the product with $x \le \frac{1}{2}$ lies in $[-2,0]$. Adding all the intervals yields

$$r_{\lambda,\theta}(x) \in [-4,3] \qquad \text{for all } x \in [0, \tfrac{1}{2}].$$

Since $|x-2| \ge 1$ and $|x-1| \ge \frac{1}{2}$ on $[0, \frac{1}{2}]$, we obtain

$$\|D_x p^1_{\lambda,\theta} - p^1_{\lambda,\theta} - q^1_{\lambda,\theta}\|_{L^\infty([0,\frac{1}{2}])} \le \frac{4}{1^2(1/2)^2} = 16.$$

The contraction condition then follows from

$$\ell \le \frac{8}{N_0} + \frac{16}{N_0(N_0-1)} \le \frac{8}{10} + \frac{16}{90} = \frac{44}{45} < 1$$

for every $N_0 \ge 10$. □

IV.B.3 Rigorous control of the tails

The contraction constant in (B.3) is controlled by the hand estimates of Lemmas B.2 and B.3. The code does not recompute those suprema from Taylor coefficients. What remains to be computed for each matrix function is the residual size $\|\xi\|_{\mathcal{A}_{N_0}}$ and the derivative size $\|\partial_x \xi\|_{L^\infty}$. These are infinite sums, and this subsection records the Cauchy estimates used by the code to bound their tails after the exact coefficients have been summed up to order N_{res}.

We fix throughout $N_0 \geq 2$, $N_{\mathrm{res}} \geq N_0$, $b_0 = \frac{1}{2}$, and a Cauchy radius $\rho \in (b_0, 1)$. We set $r := \frac{b_0}{\rho} \in (0,1)$, which is the geometric ratio governing all tail bounds below.

We first record explicit sup-bounds $M_p(\rho)$ and $M_q(\rho)$ at each singular point. The constants c_j^p and c_j^q in Lemmas B.4 and B.5 below are the partial fraction coefficients of

$$\frac{P_{\lambda,\theta}(z)}{z(z^2-1)} \qquad \text{and} \qquad \frac{Q_{\lambda,\theta}(z)}{z^2(z^2-1)}.$$

Explicitly, c_0^p, c_1^p, c_2^p are the coefficients of $\frac{1}{z}, \frac{1}{z-1}, \frac{1}{z+1}$, respectively, in the first expansion, and $c_0^q, c_1^q, c_2^q, c_3^q$ are the coefficients of $\frac{1}{z}, \frac{1}{z^2}, \frac{1}{z-1}, \frac{1}{z+1}$, respectively, in the second. See (C.1). The local formulas use $z = x$ near $z = 0$ and $z = 1 - x$ near $z = 1$.

In each case, with $z_0 \in \{0, 1\}$ the corresponding singular point, $M_p(\rho)$ and $M_q(\rho)$ denote explicit upper bounds for

$$\max_{|x|\leq\rho} |p^{z_0}_{\lambda,\theta}(x)| = \|P^{z_0}_{\lambda,\theta}\|_{L^\infty(D_\rho(z_0))} \qquad \text{and} \qquad \max_{|x|\leq\rho} |q^{z_0}_{\lambda,\theta}(x)| = \|Q^{z_0}_{\lambda,\theta}\|_{L^\infty(D_\rho(z_0))}.$$

Lemma B.4 *Fix $\rho \in (0,1)$. In the local coordinate $x = z$, the coefficients of $\mathcal{L}_0$ can be written as*

$$p^0_{\lambda,\theta}(x) = c_0^p + x\Big(\frac{c_1^p}{x-1} + \frac{c_2^p}{x+1}\Big) \qquad \textit{and} \qquad q^0_{\lambda,\theta}(x) = c_1^q + c_0^q x + x^2\Big(\frac{c_2^q}{x-1} + \frac{c_3^q}{x+1}\Big).$$

Since $|x \pm 1| \geq 1 - \rho$ for $|x| \leq \rho < 1$,

$$M_p(\rho) = |c_0^p| + \frac{\rho(|c_1^p| + |c_2^p|)}{1-\rho} \qquad \textit{and} \qquad M_q(\rho) = |c_1^q| + |c_0^q|\rho + \frac{\rho^2(|c_2^q| + |c_3^q|)}{1-\rho}$$

are upper bounds for $\max_{|x|\leq\rho} |p^0_{\lambda,\theta}(x)|$ and $\max_{|x|\leq\rho} |q^0_{\lambda,\theta}(x)|$, respectively.

Proof: For $|x| \leq \rho < 1$ we have

$$|x \pm 1| \geq 1 - \rho.$$

Applying the triangle inequality to the decompositions

$$p^0_{\lambda,\theta}(x) = c_0^p + x\Big(\frac{c_1^p}{x-1} + \frac{c_2^p}{x+1}\Big) \qquad \text{and} \qquad q^0_{\lambda,\theta}(x) = c_1^q + c_0^q x + x^2\Big(\frac{c_2^q}{x-1} + \frac{c_3^q}{x+1}\Big)$$

gives

$$|p^0_{\lambda,\theta}(x)| \leq |c_0^p| + \frac{|x|(|c_1^p| + |c_2^p|)}{1-\rho} \leq |c_0^p| + \frac{\rho(|c_1^p| + |c_2^p|)}{1-\rho},$$
$$|q^0_{\lambda,\theta}(x)| \leq |c_1^q| + |c_0^q||x| + \frac{|x|^2(|c_2^q| + |c_3^q|)}{1-\rho} \leq |c_1^q| + |c_0^q|\rho + \frac{\rho^2(|c_2^q| + |c_3^q|)}{1-\rho}.$$

Taking the maximum over $|x| \leq \rho$ gives the claimed bounds. □

Lemma B.5 *In the local coordinate $x = 1 - z$ of* (B.10)*, the coefficients of $\mathcal{L}_1$ are*

$$p^1_{\lambda,\theta}(x) = c^p_1 + x\Big(\frac{c^p_0}{x-1} + \frac{c^p_2}{x-2}\Big),$$
$$q^1_{\lambda,\theta}(x) = -c^q_2 x + x^2\Big(-\frac{c^q_0}{x-1} + \frac{c^q_1}{(x-1)^2} - \frac{c^q_3}{x-2}\Big).$$

For $|x| \le \rho < 1$, the bounds $|x-1| \ge 1-\rho$ and $|x-2| \ge 2-\rho$ give

$$M_p(\rho) = |c^p_1| + \rho\Big(\frac{|c^p_0|}{1-\rho} + \frac{|c^p_2|}{2-\rho}\Big),$$
$$M_q(\rho) = |c^q_2|\rho + \rho^2\Big(\frac{|c^q_0|}{1-\rho} + \frac{|c^q_1|}{(1-\rho)^2} + \frac{|c^q_3|}{2-\rho}\Big)$$

as upper bounds for $\max_{|x|\le\rho} |p^1_{\lambda,\theta}(x)|$ and $\max_{|x|\le\rho} |q^1_{\lambda,\theta}(x)|$, respectively.

Proof: Apply the triangle inequality to the displayed partial fraction representations and use the lower bounds for $|x-1|$ and $|x-2|$. □

Lemma B.6 *Let $\theta \in (0,1)$ and $\rho \in (0,1)$. For $j = 1, 2$, the affine term G^θ_j in* (3.18) *gives the following forcing terms in the local equations at $z = 0$ and $z = 1$, respectively:*

$$g_{0,j}(x) = \frac{G^\theta_j(x)}{x^2-1}, \qquad g_{1,j}(x) = -\frac{xG^\theta_j(1-x)}{(1-x)^2(2-x)}.$$

Define

$$M_{g_{0,1}}(\rho) := \frac{\theta(2-\theta)\rho}{1-\rho^2}, \qquad M_{g_{1,1}}(\rho) := \frac{\theta(2-\theta)\rho}{(1-\rho)(2-\rho)}$$
$$M_{g_{0,2}}(\rho) := \frac{(1-\theta)(1+\theta)}{1-\rho^2}, \qquad M_{g_{1,2}}(\rho) := \frac{(1-\theta)(1+\theta)\rho}{(1-\rho)^2(2-\rho)}.$$

For each $k \in \{0,1\}$ and $j \in \{1,2\}$,

$$\sup_{|x|\le\rho} |g_{k,j}(x)| \le M_{g_{k,j}}(\rho).$$

For the homogeneous solution, take $M_g = 0$.

Proof: The formulas follow from (3.18) by setting $z = x$ or $z = 1 - x$ and dividing by the leading coefficient. The bounds follow by the triangle inequality. □

We now bound the residual tail, including the normalized forcing.

Lemma B.7 *Let $f^{\mathrm{ap}}(x) = \sum_{m=0}^{N_0-1} f^{\mathrm{ap}}_m x^m$, with $N_0 \ge 2$, $N_{\mathrm{res}} \ge N_0$, $b_0 = \frac{1}{2}$, and $\rho \in (b_0, 1)$. Let M_p, M_q be the bounds at radius ρ from Lemma B.4 or Lemma B.5, according to the local chart, and let M_g be the corresponding bound from Lemma B.6. Define*

$$C_{\mathrm{val}} := \sum_{m=0}^{N_0-1} |f^{\mathrm{ap}}_m|\rho^m, \qquad C_{\mathrm{der}} := \sum_{m=1}^{N_0-1} m|f^{\mathrm{ap}}_m|\rho^m.$$

Set $r := \frac{b_0}{\rho}$. Then, for $k > N_{\mathrm{res}}$,

$$|\xi_k| \le \frac{M_pC_{\mathrm{der}} + M_qC_{\mathrm{val}} + M_g}{k(k-1)}\rho^{-k}, \qquad k|\xi_k| \le \frac{M_pC_{\mathrm{der}} + M_qC_{\mathrm{val}} + M_g}{k-1}\rho^{-k},$$

and consequently

$$\sum_{k>N_{\mathrm{res}}} |\xi_k| b_0^{k-N_0} \le \frac{M_p C_{\mathrm{der}} + M_q C_{\mathrm{val}} + M_g}{N_{\mathrm{res}}(N_{\mathrm{res}}+1) b_0^{N_0}} \frac{r^{N_{\mathrm{res}}+1}}{1-r},$$
$$\sum_{k>N_{\mathrm{res}}} k|\xi_k| b_0^{k-1} \le \frac{M_p C_{\mathrm{der}} + M_q C_{\mathrm{val}} + M_g}{N_{\mathrm{res}} b_0} \frac{r^{N_{\mathrm{res}}+1}}{1-r}.$$

Proof: For $k > N_{\mathrm{res}} \ge N_0$, the coefficient of $x^2\partial_x^2 f^{\mathrm{ap}}$ vanishes. Set

$$d_k := \left(g - pD_x f^{\mathrm{ap}} - q f^{\mathrm{ap}}\right)_k.$$

Cauchy's estimates give

$$|d_k| \le (M_p C_{\mathrm{der}} + M_q C_{\mathrm{val}} + M_g)\rho^{-k}.$$

Since $\xi_k = \frac{d_k}{k(k-1)}$ by (B.2),

$$\sum_{k>N_{\mathrm{res}}} |\xi_k| b_0^{k-N_0} \le \sum_{k>N_{\mathrm{res}}} \frac{M_p C_{\mathrm{der}} + M_q C_{\mathrm{val}} + M_g}{k(k-1)} \frac{r^k}{b_0^{N_0}} \le \frac{M_p C_{\mathrm{der}} + M_q C_{\mathrm{val}} + M_g}{N_{\mathrm{res}}(N_{\mathrm{res}}+1) b_0^{N_0}} \frac{r^{N_{\mathrm{res}}+1}}{1-r},$$
$$\sum_{k>N_{\mathrm{res}}} k|\xi_k| b_0^{k-1} \le \sum_{k>N_{\mathrm{res}}} \frac{M_p C_{\mathrm{der}} + M_q C_{\mathrm{val}} + M_g}{k-1} \frac{r^k}{b_0} \le \frac{M_p C_{\mathrm{der}} + M_q C_{\mathrm{val}} + M_g}{N_{\mathrm{res}} b_0} \frac{r^{N_{\mathrm{res}}+1}}{1-r}.$$

□

IV.B.4 Certified bounds for the matrix entries

We now combine the residual estimate with the inputs established above. The contraction constants come from the hand estimates of Lemmas B.2 and B.3. The residual norms are computed from the exact coefficients up to N_{res}, plus the Cauchy tail of Lemma B.7.

Lemma B.8 *Fix $N_0 \ge 2$ and assume that the corresponding contraction constant L_f satisfies $L_f < 1$. For every local representative f in the sense of* (B.7) *of each function among $F^{h,1}_{\lambda,\theta^\pm,1}$ and $F^{p,k}_{\lambda,\theta^\pm,j}$, $j \in \{1,2\}$, $k \in \{0,1\}$, let f^{ap} be its order-N_0 Taylor approximation and let ξ_f be the residual defined by* (B.2)*. Then there are explicitly computable bounds $\delta_f, \delta'_f > 0$, valid for each fixed $\lambda \in \{0.56, 0.6\}$ and for θ in the certified interval, such that*

$$\left|f(b_0) - f^{\mathrm{ap}}(b_0)\right| \le \delta_f, \qquad \left|\partial_x f(b_0) - \partial_x f^{\mathrm{ap}}(b_0)\right| \le \delta'_f, \tag{B.11}$$

with $b_0 = \frac{1}{2}$. Explicitly,

$$\delta_f = \frac{b_0^{N_0}}{1-L_f}\left\|\xi_f\right\|_{\mathcal{A}_{N_0}}, \qquad \delta'_f = \frac{N_0 L_f b_0^{N_0-1}}{1-L_f}\left\|\xi_f\right\|_{\mathcal{A}_{N_0}} + \left\|\partial_x \xi_f\right\|_{L^\infty([0,b_0])}, \tag{B.12}$$

where L_f is the contraction constant (B.3) *for the corresponding local operator.*

Proof: Each local representative f is analytic near $x = 0$ and satisfies $\mathcal{L}_{z_0} f = g$ with $\mathcal{L}_{z_0} \in \{\mathcal{L}_1, \mathcal{L}_0\}$, as described in Section IV.B.2. The approximation f^{ap} matches the first N_0 Taylor coefficients of f by construction, so Lemma B.1 applies. Its value and derivative estimates give (B.12). The condition $L_f < 1$ is guaranteed by Lemmas B.2 and B.3 for the values of N_0 used in the proof. The quantities $\left\|\xi_f\right\|_{\mathcal{A}_{N_0}}$ and $\left\|\partial_x \xi_f\right\|_{L^\infty}$ are then obtained from the exact finite sums and the Cauchy tail of Lemma B.7. □

Corollary B.9 *Let the local functions defining the entries of $\mathbf{M}_{\lambda,\theta}$ be ordered as in Definition 3.14. For each entry of $\mathbf{M}_{\lambda,\theta}$, let ε_{jk} be the sum of the corresponding bounds δ_f or δ'_f from Lemma B.8, according to whether the entry involves values or derivatives and whether contributions from both singular points are present. Then the true matrix $\mathbf{M}_{\lambda,\theta}$ is enclosed entrywise by the truncated matrix $\mathbf{M}^{\mathrm{ap}}_{\lambda,\theta}$ enlarged by these radii:*

$$\left|(\mathbf{M}_{\lambda,\theta})_{jk} - (\mathbf{M}^{\mathrm{ap}}_{\lambda,\theta})_{jk}\right| \le \varepsilon_{jk}.$$

IV.C Computer-assisted implementation

As promised in the proof of Lemma 3.20, this appendix describes the computer-assisted proof of that lemma, which is the only computer-assisted ingredient in the proof of Theorem 2.1. We explain how the program represents the quantities appearing in the lemma, describe the main routines, and give the verification procedure. The analytic estimates used to control the errors are established in Appendices IV.A and IV.B.

The source code is available at https://github.com/MiguelMGPascualCaballo/dgreg/tree/v1.0 . The programs are written in Python for SageMath 10.9. Symbolic computations use exact rational arithmetic. Numerical evaluations use Arb ball arithmetic through `RealBallField` at 200 bits of precision. A real ball represents an interval containing the exact quantity, and arithmetic operations preserve this inclusion. A numerical inequality is accepted only when the resulting enclosure proves it on the entire parameter interval under consideration.

IV.C.1 Supplementary code and computational data

The supplementary files implement the verification of Lemma 3.20. The data needed for the proof are constructed by the programs from the equations and rational parameters specified below.

Files for the determinant certificate. The construction of the matrix $\mathbf{M}_{\lambda,\theta}$ and its residuals is implemented in `local_fuchs.py` and `cert_residuals.py`. The modules `cert_eval.py` and `cert_det_post.py` enclose the truncation errors and the determinant. The covering of the required parameter set is implemented in `cert_proof.py`. Finally, `run_proof.py` runs the symbolic checks in `verify_residuals.py` and then the determinant certificate.

For this certificate, the parameters are

$$N = 900, \qquad N_0 = 31, \qquad N_{\mathrm{res}} = 62,$$
$$b_0 = \frac{1}{2}, \qquad \rho = \frac{3}{4}, \qquad \texttt{center} = \frac{5}{12}.$$

Here N is the prime threshold in Lemma 2.6, N_0 is the number of Taylor coefficients retained in each local approximation, and N_{res} is the largest residual index computed explicitly. The point b_0 is the local coordinate of the matching point, and ρ is the radius used for the Cauchy estimates. The point `center` is the centre of the interval J of Lemma 2.6 and the target used to choose the fractions θ_p. The two values of λ are 0.56 and 0.6.

Local solution data. For each fixed rational value of λ, the local Taylor coefficients are elements of

$$K = \mathrm{Frac}(\mathbb{Q}[\theta]).$$

The program uses the same global coordinate z as this chapter. The keys `p1` and `00` correspond to $z = 1$ and $z = 0$, with local coordinates $x = 1 - z$ and $x = z$, respectively. Both charts reach the matching point at $x = \frac{1}{2}$. The keys `the` and `eht` respectively represent the branches θ and $1 - \theta$.

For each chart and branch, `all_res_sym()` returns a block containing the coefficients of the approximations f^{ap}, the coefficients ξ_k of their residuals (B.2), and the partial fraction coefficients `term_pp` and `term_qq`. The function keys `phi1`, `fp1`, and `fp2` identify the normalized homogeneous solution and the two particular solutions. Only `fp1` and `fp2` are needed at $z = 0$.

Each approximation has the form

$$f^{\mathrm{ap}}(x) = \sum_{k=0}^{N_0-1} f_k x^k.$$

For each such function, the program computes the finite sequence $(\xi_0, \dots, \xi_{N_{\mathrm{res}}})$ exactly over K. Its entries with $k < N_0$ vanish by construction. The full residual generally has infinitely many nonzero coefficients. The contribution of the coefficients with $k > N_{\mathrm{res}}$ is bounded separately by Lemma B.7.

IV.C.2 Objects, methods and routines

Construction of the local solutions. The routines in `local_fuchs.py` construct the normalized local equations and solve their coefficient recurrences. The global partial fraction identities are

$$\frac{P_{\lambda,\theta}(z)}{z(z^2-1)} = \frac{1+2\theta}{z} + \frac{\lambda-1}{z-1} - \frac{\lambda+1}{z+1}, \tag{C.1a}$$

$$\frac{Q_{\lambda,\theta}(z)}{z^2(z^2-1)} = -\frac{2\lambda\theta}{z} - \frac{1-\theta^2}{z^2} + \frac{1-\theta+\lambda\theta}{z-1} - \frac{1-\theta-\lambda\theta}{z+1}. \tag{C.1b}$$

The affine terms satisfy

$$\frac{G_1^\theta(z)}{z^2(z^2-1)} = -\frac{\theta(2-\theta)}{2}\left(\frac{2}{z} - \frac{1}{z-1} - \frac{1}{z+1}\right),$$

$$\frac{G_2^\theta(z)}{z^2(z^2-1)} = -\frac{1-\theta^2}{2}\left(\frac{2}{z^2} - \frac{1}{z-1} + \frac{1}{z+1}\right).$$

The routines `default_parameter_matrices()` and `specialize_terms()` produce these coefficients for each branch. The routines `local_coeffs_at_p1()` and `local_coeffs_at_0()` then construct the corresponding normalized local equations. The signs of the affine terms agree with Definition 3.14.

The homogeneous approximation is computed by `hom_sol_fro_const()` using the Frobenius recurrence. At $z = 1$ the particular approximations are computed directly by the routine `particular_sol_r1_is_0_direct()`, and at $z = 0$ by `particular_sol_ordinary_direct()`. These routines use the coefficient recurrence of the local equation, with constant coefficient zero for the particular solutions at $z = 1$. The routine `symbolic_matrix_A()` assembles the values and derivatives at the matching point in the order prescribed by Definition 3.14. Global derivatives are obtained as $-\partial_x$ in the first chart and ∂_x in the second chart.

Residual bounds. The module `cert_eval.py` implements the estimates of Appendix IV.B for inputs enclosed in real balls. Its main routines are as follows.

- `L_bound()` computes the rational upper bound for the contraction constant in (B.3), using Lemmas B.2 and B.3.
- `rational_M_00()` and `rational_M_p1()` compute $M_p(\rho)$ and $M_q(\rho)$ from Lemmas B.4 and B.5. The routine `forcing_M()` computes the forcing bound M_g of Lemma B.6.
- `fapprox_norms()` computes C_{val} and C_{der} from the coefficients of f^{ap}. These are combined with M_p, M_q, and M_g by `xi_tails()` to bound the residual tails in Lemma B.7.

- `norm_AN0()` and `norm_deriv()` compute the finite sums involving ξ_k for $N_0 \le k \le N_{\mathrm{res}}$.
- `delta_block()` adds the finite and tail contributions, checks that the contraction bound is less than one, and applies Lemma B.8. Its output is the pair (δ_f, δ_f') bounding the errors in the value and derivative of f.

Taylor models for rational functions. The entries of the truncated matching matrix are rational functions of θ. To enclose them on $[c-r, c+r]$, `_entry_tm()` in `cert_det_post.py` constructs a Taylor model of order one. Write an entry as $\frac{u}{v}$, where $u, v \in \mathbb{Q}[\theta]$, and set $t = \theta - c$. Exact polynomial arithmetic gives

$$u(c+t) - (a_0 + a_1 t)v(c+t) = t^2 h(t),$$

where a_0 and a_1 are the first two Taylor coefficients of $\frac{u}{v}$. If $v(c+t) = \sum_j v_j t^j$ and $h(t) = \sum_j h_j t^j$, the program computes

$$d = |v_0| - \sum_{j \ge 1} |v_j| r^j.$$

When $d > 0$, the remainder is bounded by

$$\left| \frac{u(c+t)}{v(c+t)} - a_0 - a_1 t \right| \le \frac{r^2}{d} \sum_j |h_j| r^j, \qquad |t| \le r.$$

If the denominator bound is not positive, the interval must be subdivided. At a single rational parameter value, the entry is evaluated directly.

The class `TM` represents the polynomial part and an interval remainder, and implements addition and multiplication with certified remainders. The exact rational coefficients are enclosed in real balls before the models are combined. The routine `_tm_det()` computes a determinant by cofactor expansion in this arithmetic, and `enclose()` returns an interval containing all its values. The routine `eval_series_tight()` also uses shifted numerators and denominators to enclose the coefficients needed in the residual estimates.

Adaptive subdivision. The covering routine `verify_det_sign()` starts with a list containing the entire rational parameter interval. It removes one interval, encloses the determinant over it, and accepts the interval if the desired sign is certified. Otherwise it bisects the interval and returns both halves to the list. The accepted intervals together with the pending intervals always cover the original domain. A successful return requires that the pending list be empty. Bounds on the subdivision depth or the number of intervals cause an error if the verification cannot be completed.

IV.C.3 Details of the proof

Proof of Lemma 3.20: Fix one of the two rational endpoints $\lambda \in \{0.56, 0.6\}$. The routine `build_post()` constructs

$$\mathbf{M}^{\mathrm{ap}}_{\lambda,\theta} = \texttt{symbolic_matrix_A}(N_0 - 1, \lambda),$$

obtained by replacing every local solution in Definition 3.14 by its polynomial approximation. The routine `all_res_sym()` constructs the corresponding residual data. Both computations are exact over K and are performed once for each value of λ.

Let I be a parameter interval. The Taylor model determinant gives an enclosure $\nu^{\mathrm{ap}}(I)$ containing $\det \mathbf{M}^{\mathrm{ap}}_{\lambda,\theta}$ for all $\theta \in I$. Next, `delta_per_function()` encloses the residual coefficients, approximation coefficients, and partial fraction coefficients on I. It passes these to `delta_block()`

to obtain the bounds (δ_f, δ'_f) from (B.12). The routine `_radius_matrix()` arranges them into the entrywise radii ε_{jk} of Corollary B.9. Thus

$$\mathbf{M}_{\lambda,\theta} = \mathbf{M}^{\mathrm{ap}}_{\lambda,\theta} + E, \qquad |E_{jk}| \le \varepsilon_{jk}$$

uniformly for $\theta \in I$.

Choose certified upper bounds S_j for the row sums $\sum_k |(\mathbf{M}^{\mathrm{ap}}_{\lambda,\theta})_{jk}|$, and set $\varepsilon_j = \sum_k \varepsilon_{jk}$. Multilinearity in the rows and the bound of a determinant by the product of the row ℓ^1 norms give

$$|\det \mathbf{M}_{\lambda,\theta} - \det \mathbf{M}^{\mathrm{ap}}_{\lambda,\theta}| \le \Delta, \qquad \Delta = \prod_j (S_j + \varepsilon_j) - \prod_j S_j.$$

Indeed, the difference of the determinants is the sum of all terms containing at least one row of E. Bounding each such term gives precisely the expansion on the right. The routine `certified_det_post()` therefore returns an enclosure of

$$\nu^{\mathrm{ap}}(I) + [-\Delta, \Delta].$$

A strictly negative enclosure proves the required sign at $\lambda = 0.56$, and a strictly positive enclosure proves it at $\lambda = 0.6$.

It remains to cover the parameter values in Lemma 2.6. The routine `prove()` first checks each prime $p < N$ at the single point

$$\theta_p = \frac{h_p}{p},$$

returned by `theta_of()`: h_p is the integer nearest to $p \cdot \texttt{center}$, with $\texttt{center} = \frac{5}{12}$, clipped to $\left[\lceil \frac{p}{3} \rceil, \lfloor \frac{p}{2} \rfloor\right]$, so that $\theta_p \in \left[\frac{1}{3}, \frac{1}{2}\right]$, as in the proof of Lemma 2.6. At such a point the enclosure is computed on a ball of radius zero. The remaining primes $p \ge N$ are covered by the interval

$$J = \left[\frac{5}{12} - \frac{1}{2p^*}, \frac{5}{12} + \frac{1}{2p^*}\right], \qquad p^* = \texttt{next_prime}(N-1) = 907,$$

of length $\frac{1}{p^*}$. The routine `verify_det_sign()` applies adaptive bisection to J, accepting an interval only when the determinant enclosure has the required strict sign. It subdivides intervals for which the sign is undecided or a coefficient denominator has not been bounded away from zero. A certified opposite sign or an unresolved interval below the prescribed minimum width causes an error.

Before the covering begins, the program checks the contraction bounds for both local charts. It also checks finitely many parameter values as a diagnostic. An undecided point enclosure causes an error reporting the truncation order and arithmetic precision. These point checks accompany the subsequent certification over the whole interval.

The driver `run_proof.py` performs the symbolic consistency checks and then both sign coverings with the parameters given above. The residual estimates and determinant enclosures on the accepted intervals provide the computer-assisted verification of the lemma. □

Reproducing the computation. From the directory containing the supplementary code, the certificate is run with

```
sage -python run_proof.py
```

This command includes the exact symbolic consistency checks and the determinant sign verification.

V. Conclusions and future work

In Chapter II, Theorem 1.1 shows the existence of nontrivial homogeneous stationary solutions for the SQG equation. Theorem 3.2 gives the analogous result for the generalized De Gregorio equation when $\alpha > \frac{1}{2}$.

In Chapter III, Theorem 2.1 proves the existence of a Burgers–Hilbert traveling wave. Moreover, Theorem 3.2 establishes its linear instability.

Finally, in Chapter IV, linear instability of the excited states of the De Gregorio equation is proved in Theorem 1.1.

These results leave several problems open. The linear instability of the Burgers–Hilbert traveling wave and of the excited states of De Gregorio does not by itself imply nonlinear instability. Proving nonlinear instability in either setting remains open.

Independently and at the same time, the existence of homogeneous steady states for the generalized SQG equation was proved in [AGJ25]. This result includes Theorem 1.1. The question of their stability remains open both for generalized SQG and for the generalized De Gregorio equation.

Several questions remain for the Burgers–Hilbert equation. One is to continue the branch of traveling waves constructed in [CCZ23] up to the unstable wave found in Chapter III. Another is to show that the cusped traveling wave of [DG23] lies in the closure of this branch. A further open problem is to prove linear instability for the corresponding $\frac{2\pi}{n}$-periodic wave with $n \geq 2$, beyond the case $n = 3$. A more ambitious goal is to obtain counterparts of the preceding instability results for the V-states.

More generally, the reduction of the spectral problems to a Fuchsian equation on the disk used in Chapters III and IV could also be applied to other spectral problems.

Conclusiones y trabajo futuro

En el Capítulo II, el Teorema 1.1 muestra la existencia de soluciones estacionarias homogéneas no triviales de la ecuación SQG. El Teorema 3.2 proporciona el resultado análogo para la ecuación de De Gregorio generalizada cuando $\alpha > \frac{1}{2}$.

En el Capítulo III, el Teorema 2.1 demuestra la existencia de una onda viajera de la ecuación de Burgers–Hilbert. Además, el Teorema 3.2 establece su inestabilidad lineal.

Finalmente, en el Capítulo IV, se demuestra en el Teorema 1.1 la inestabilidad lineal de los estados excitados de la ecuación de De Gregorio.

Estos resultados dejan abiertos varios problemas. La inestabilidad lineal de la onda viajera de Burgers–Hilbert y de los estados excitados de De Gregorio no implica por sí sola inestabilidad no lineal. Demostrar la inestabilidad no lineal en cualquiera de los dos casos sigue siendo un problema abierto.

De manera independiente y al mismo tiempo, en [AGJ25] se demostró la existencia de estados estacionarios homogéneos para la ecuación SQG generalizada. Este resultado incluye el Teorema 1.1. La cuestión de su estabilidad sigue abierta tanto para SQG generalizado como De Gregorio generalizado.

Quedan varias cuestiones abiertas para la ecuación de Burgers–Hilbert. Una de ellas es continuar la rama de ondas viajeras construida en [CCZ23] hasta la onda inestable encontrada en el Capítulo III. Otra es demostrar que la onda viajera con cúspide de [DG23] pertenece a la clausura de esta rama. Otro problema abierto es demostrar la inestabilidad lineal de la onda $\frac{2\pi}{n}$-periódica correspondiente para $n \geq 2$, más allá del caso $n = 3$. Un objetivo más ambicioso es extender a los V-states los resultados de inestabilidad anteriores.

De manera más general, la reducción de los problemas espectrales a una ecuación fuchsiana en el disco utilizada en los Capítulos III y IV podría aplicarse también a otros problemas espectrales.

Bibliography

[AGJ25] K. Abe, J. Gómez-Serrano, and I.-J. Jeong. *Homogeneous steady states for the generalized surface quasi-geostrophic equations*. 2025. arXiv: 2510.03009.

[AGK21] G. Arioli, F. Gazzola, and H. Koch. "Uniqueness and bifurcation branches for planar steady Navier–Stokes equations under Navier boundary conditions". In: *Journal of Mathematical Fluid Mechanics* 23.3 (2021), p. 49. DOI: 10.1007/s00021-021-00572-4.

[Ahl79] L. V. Ahlfors. *Complex analysis*. 3rd ed. McGraw-Hill New York, 1979.

[AK10] G. Arioli and H. Koch. "Computer-assisted methods for the study of stationary solutions in dissipative systems, applied to the Kuramoto-Sivashinski equation". In: *Arch. Ration. Mech. Anal.* 197.3 (2010), pp. 1033–1051. DOI: 10.1007/s00205-010-0309-7.

[Ao+21] W. Ao, J. Dávila, M. del Pino, M. Musso, and J. Wei. "Travelling and rotating solutions to the generalized inviscid surface quasi-geostrophic equation". In: *Trans. Amer. Math. Soc.* 374.9 (2021), pp. 6665–6689. DOI: 10.1090/tran/8406.

[BC93] A. L. Bertozzi and P. Constantin. "Global regularity for vortex patches". In: *Communications in Mathematical Physics* 152.1 (1993), pp. 19–28. DOI: 10.1007/BF02097055.

[BCD11] H. Bahouri, J.-Y. Chemin, and R. Danchin. *Fourier Analysis and Nonlinear Partial Differential Equations*. Vol. 343. Grundlehren der Mathematischen Wissenschaften. Heidelberg: Springer, 2011. DOI: 10.1007/978-3-642-16830-7.

[BCG25] T. Buckmaster, G. Cao-Labora, and J. Gómez-Serrano. "Smooth imploding solutions for 3d compressible fluids". In: *Forum of Mathematics, Pi* 13 (2025), e6. DOI: 10.1017/fmp.2024.12.

[Ber+21] J. B. van den Berg, M. Breden, J.-P. Lessard, and L. van Veen. "Spontaneous periodic orbits in the Navier–Stokes flow". In: *Journal of Nonlinear Science* 31.2 (2021), p. 41. DOI: 10.1007/s00332-021-09695-4.

[BH10] J. Biello and J. K. Hunter. "Nonlinear Hamiltonian waves with constant frequency and surface waves on vorticity discontinuities". In: *Comm. Pure Appl. Math.* 63.3 (2010), pp. 303–336. DOI: 10.1002/cpa.20304.

[Bha97] R. Bhatia. *Matrix analysis*. Vol. 169. Graduate Texts in Mathematics. Springer Science & Business Media, 1997.

[BHM23] M. Berti, Z. Hassainia, and N. Masmoudi. "Time quasi-periodic vortex patches of Euler equation in the plane". In: *Invent. Math.* 233.3 (2023), pp. 1279–1391. DOI: 10.1007/s00222-023-01195-4.

[BHP23] A. Bulut, M. K. Huynh, and S. Palasek. *Convex integration above the Onsager exponent for the forced Euler equations*. 2023. arXiv: 2301.00804.

[BL08] J. B. van den Berg and J.-P. Lessard. "Chaotic braided solutions via rigorous numerics: chaos in the Swift-Hohenberg equation". In: *SIAM Journal on Applied Dynamical Systems* 7.3 (2008), pp. 988–1031. DOI: 10.1137/070709128.

[BM21] P. Baldi and R. Montalto. "Quasi-periodic incompressible Euler flows in 3D". In: *Adv. Math.* 384 (2021), Paper No. 107730. DOI: 10.1016/j.aim.2021.107730.

[BN14] A. Bressan and K. T. Nguyen. "Global existence of weak solutions for the Burgers–Hilbert equation". In: *SIAM Journal on Mathematical Analysis* 46.4 (2014), pp. 2884–2904. DOI: 10.1137/140957536.

[BP24] J. Bedrossian and S. Punshon-Smith. "Chaos in stochastic 2d Galerkin-Navier-Stokes". In: *Communications in Mathematical Physics* 405.4 (2024), p. 107. DOI: 10.1007/s00220-024-04949-0.

[Bur82] J. Burbea. "Motions of vortex patches". In: *Letters in Mathematical Physics* 6.1 (1982), pp. 1–16. DOI: 10.1007/BF02281165.

[BV19] T. Buckmaster and V. Vicol. "Nonuniqueness of weak solutions to the Navier-Stokes equation". In: *Ann. of Math. (2)* 189.1 (2019), pp. 101–144. DOI: 10.4007/annals.2019.189.1.3.

[BZ17] A. Bressan and T. Zhang. "Piecewise smooth solutions to the Burgers–Hilbert equation". In: *Communications in Mathematical Sciences* 15.1 (2017), pp. 165–184. DOI: 10.4310/CMS.2017.v15.n1.a7.

[Cad25] M. Cadiot. "Constructive proofs of existence and stability of solitary waves in the Whitham and capillary–gravity Whitham equations". In: *Nonlinearity* 38.3 (2025), p. 035021. DOI: 10.1088/1361-6544/adb5e8.

[Cao+21] D. Cao, J. Wan, G. Wang, and W. Zhan. "Rotating vortex patches for the planar Euler equations in a disk". In: *J. Differential Equations* 275 (2021), pp. 509–532. DOI: 10.1016/j.jde.2020.11.027.

[Cas+25a] Á. Castro, D. Faraco, F. Mengual, and M. Solera-Diana. "A proof of Vishik's nonuniqueness theorem for the forced 2D Euler equation". In: *Journal für die reine und angewandte Mathematik (Crelles Journal)* 824 (Apr. 2025), pp. 253–288. DOI: 10.1515/crelle-2025-0025.

[Cas+25b] Á. Castro, D. Faraco, F. Mengual, and M. Solera-Diana. *Unstable vortices, sharp non-uniqueness with forcing, and global smooth solutions for the SQG equation.* 2025. arXiv: 2502.10274.

[CCF05] A. Córdoba, D. Córdoba, and M. A. Fontelos. "Formation of singularities for a transport equation with nonlocal velocity". In: *Annals of Mathematics* 162.3 (2005), pp. 1377–1389. DOI: 10.4007/annals.2005.162.1377.

[CCG10] Á. Castro, D. Córdoba, and F. Gancedo. "Singularity formations for a surface wave model". In: *Nonlinearity* 23 (2010), pp. 2835–2847. DOI: 10.1088/0951-7715/23/11/006.

[CCG16] Á. Castro, D. Córdoba, and J. Gómez-Serrano. "Uniformly rotating analytic global patch solutions for active scalars". In: *Annals of PDE* 2.1 (2016), pp. 1–34. DOI: 10.1007/s40818-016-0007-3.

[CCG20a] A. Castro, D. Córdoba, and J. Gómez-Serrano. "Global smooth solutions for the inviscid SQG equation". In: *Memoirs of the American Mathematical Society* 266.1292 (2020). DOI: 10.1090/memo/1292.

[CCG20b] Á. Castro, D. Córdoba, and J. Gómez-Serrano. "Global smooth solutions for the inviscid SQG equation". In: *Mem. Amer. Math. Soc.* 266.1292 (2020), pp. v+89. DOI: 10.1090/memo/1292.

[CCZ21] Á. Castro, D. Córdoba, and F. Zheng. "The lifespan of classical solutions for the inviscid Surface Quasi-geostrophic equation". In: *Annales de l'Institut Henri Poincaré C, Analyse non linéaire* 38.5 (2021), pp. 1583–1603. DOI: 10.1016/j.anihpc.2020.12.005.

[CCZ23] Á. Castro, D. Córdoba, and F. Zheng. "Stability of traveling waves for the Burgers-Hilbert equation". In: *Anal. PDE* 16.9 (2023), pp. 2109–2145. DOI: 10.2140/apde.2023.16.2109.

[CF13] N. Crouseilles and E. Faou. "Quasi-periodic solutions of the 2D Euler equation". In: *Asymptot. Anal.* 81.1 (2013), pp. 31–34. DOI: 10.3233/ASY-2012-1117.

[CG25] G. Castro-López and J. Gómez-Serrano. "Existence of analytic non-convex V-states of the 2D Euler equation". In: *Communications in Mathematical Physics* 406.9 (2025), Paper No. 217. DOI: 10.1007/s00220-025-05382-7.

[CGL18] R. Castelli, M. Gameiro, and J.-P. Lessard. "Rigorous numerics for ill-posed PDEs: periodic orbits in the Boussinesq equation". In: *Arch. Ration. Mech. Anal.* 228.1 (2018), pp. 129–157. DOI: 10.1007/s00205-017-1186-0.

[CGP26] Á. Castro, J. Gómez-Serrano, and M. M. G. Pascual-Caballo. *Linear instability of a Burgers–Hilbert traveling wave*. 2026. arXiv: 2605.03920.

[CH21] J. Chen and T. Y. Hou. "Finite time blowup of 2D Boussinesq and 3D Euler equations with $C^{1,\alpha}$ velocity and boundary". In: *Comm. Math. Phys.* 383.3 (2021), pp. 1559–1667. DOI: 10.1007/s00220-021-04067-1.

[CH22] J. Chen and T. Y. Hou. *Stable nearly self-similar blowup of the 2D Boussinesq and 3D Euler equations with smooth data I: Analysis*. 2022. arXiv: 2210.07191.

[CH25a] J. Chen and T. Y. Hou. "Singularity formation in 3D Euler equations with smooth initial data and boundary". In: *Proc. Natl. Acad. Sci. USA* 122.27 (2025), e2500940122. DOI: 10.1073/pnas.2500940122.

[CH25b] J. Chen and T. Y. Hou. "Stable nearly self-similar blowup of the 2D Boussinesq and 3D Euler equations with smooth data II: Rigorous numerics". In: *Multiscale Model. Simul.* 23.1 (2025), pp. 25–130. DOI: 10.1137/23M1580395.

[Cha03] D. Chae. "The quasi-geostrophic equation in the Triebel-Lizorkin spaces". In: *Nonlinearity* 16.2 (2003), pp. 479–495. DOI: 10.1088/0951-7715/16/2/307.

[Che21] J. Chen. "On the slightly perturbed De Gregorio model on S^1". In: *Arch. Ration. Mech. Anal.* 241.3 (2021), pp. 1843–1869. DOI: 10.1007/s00205-021-01685-w.

[Che23] J. Chen. "On the regularity of the De Gregorio model for the 3D Euler equations". In: *Journal of the European Mathematical Society* 27.4 (2023), pp. 1619–1677.

[Che93] J.-Y. Chemin. "Persistance de structures géométriques dans les fluides incompressibles bidimensionnels". In: *Annales de l'École Normale Supérieure* 26 (1993), pp. 517–542. DOI: 10.24033/asens.1679.

[Che95] J.-Y. Chemin. *Fluides parfaits incompressibles*. Vol. 230. Astérisque. Société Mathématique de France, 1995.

[CHH21] J. Chen, T. Y. Hou, and D. Huang. "On the Finite Time Blowup of the De Gregorio Model for the 3D Euler Equations". In: *Communications on Pure and Applied Mathematics* 74.6 (2021), pp. 1282–1350. DOI: 10.1002/cpa.21991.

[CIN25] P. Constantin, M. Ignatova, and Q.-H. Nguyen. "Global regularity for critical SQG in bounded domains". In: *Communications on Pure and Applied Mathematics* 78.1 (2025), pp. 3–59. DOI: 10.1002/cpa.22221.

[CJ22] K. Choi and I.-J. Jeong. "Stability and instability of Kelvin waves". In: *Calc. Var. Partial Differential Equations* 61.6 (2022). DOI: 10.1007/s00526-022-02334-0.

[CL55] A. Coddington and N. Levinson. *Theory of Ordinary Differential Equations*. International series in pure and applied mathematics. McGraw-Hill, 1955. ISBN: 9780070992566.

[CLM85] P. Constantin, P. D. Lax, and A. J. Majda. "A simple one-dimensional model for the three-dimensional vorticity equation". In: *Communications on Pure and Applied Mathematics* 38.6 (1985), pp. 715–724. DOI: 10.1002/cpa.3160380605.

[CLZ21] D. Cao, S. Lai, and W. Zhan. "Traveling vortex pairs for 2D incompressible Euler equations". In: *Calc. Var. Partial Differential Equations* 60.5 (2021), Paper No. 190. DOI: 10.1007/s00526-021-02068-5.

[CM22] D. Córdoba and L. Martínez-Zoroa. "Non existence and strong ill-posedness in C^k and Sobolev spaces for SQG". In: *Adv. Math.* 407 (2022), Paper No. 108570, 74. DOI: 10.1016/j.aim.2022.108570.

[CMO24] D. Córdoba, L. Martínez-Zoroa, and W. S. Ożański. "Instantaneous gap loss of Sobolev regularity for the 2D incompressible Euler equations". In: *Duke Math. J.* 173.10 (2024), pp. 1931–1971. DOI: 10.1215/00127094-2023-0052.

[CMT94a] P. Constantin, A. J. Majda, and E. Tabak. "Formation of strong fronts in the 2-D quasigeostrophic thermal active scalar". In: *Nonlinearity* 7.6 (1994), pp. 1495–1533. DOI: 10.1088/0951-7715/7/6/001.

[CMT94b] P. Constantin, A. J. Majda, and E. G. Tabak. "Singular front formation in a model for quasigeostrophic flow". In: *Physics of Fluids* 6.1 (Jan. 1994), pp. 9–11. DOI: 10.1063/1.868050.

[CN18a] P. Constantin and H. Q. Nguyen. "Global Weak Solutions for SQG in Bounded Domains". In: *Communications on Pure and Applied Mathematics* 71.11 (2018), pp. 2323–2333. DOI: 10.1002/cpa.21720.

[CN18b] P. Constantin and H. Q. Nguyen. "Local and global strong solutions for SQG in bounded domains". In: *Phys. D* 376/377 (2018), pp. 195–203. DOI: 10.1016/j.physd.2017.08.008.

[Dan01] R. Danchin. "A few remarks on the Camassa–Holm equation". In: *Differential Integral Equations* 14.8 (2001), pp. 953–988. DOI: 10.57262/die/1356123175.

[De 90] S. De Gregorio. "On a one-dimensional model for the three-dimensional vorticity equation". In: *Journal of Statistical Physics* 59.5 (1990), pp. 1251–1263. DOI: 10.1007/BF01334750.

[De 96] S. De Gregorio. "A Partial Differential Equation Arising in a 1D Model for the 3D Vorticity Equation". In: *Mathematical Methods in the Applied Sciences* 19.15 (1996), pp. 1233–1255. DOI: 10.1002/(SICI)1099-1476(199610)19:15⟨1233::AID-MMA828⟩3.0.CO;2-W.

[DG23] J. Dahne and J. Gómez-Serrano. "Highest Cusped Waves for the Burgers–Hilbert Equation". In: *Archive for Rational Mechanics and Analysis* 247.5 (Aug. 2023). DOI: 10.1007/s00205-023-01904-6.

[DGR24] M. Dai, V. Giri, and R.-O. Radu. *An Onsager-type theorem for SQG*. 2024. arXiv: 2407.02582.

[DLM07] S. Day, J.-P. Lessard, and K. Mischaikow. "Validated continuation for equilibria of PDEs". In: *SIAM Journal on Numerical Analysis* 45.4 (2007), pp. 1398–1424. DOI: 10.1137/050645968.

[DP23] M. Dai and Q. Peng. *Non-unique weak solutions of forced SQG*. 2023. arXiv: 2310.13537.

[DZ78] G. S. Deem and N. J. Zabusky. "Vortex waves: Stationary "V-states", interactions, recurrence, and breaking". In: *Physical Review Letters* 40.13 (1978), pp. 859–862. DOI: 10.1103/PhysRevLett.40.859.

[EJ20a] T. M. Elgindi and I.-J. Jeong. "On singular vortex patches, II: long-time dynamics". In: *Trans. Amer. Math. Soc.* 373.9 (2020), pp. 6757–6775. DOI: 10.1090/tran/8134.

[EJ20b] T. M. Elgindi and I.-J. Jeong. "On the effects of advection and vortex stretching". In: *Arch. Ration. Mech. Anal.* 235.3 (2020), pp. 1763–1817. DOI: 10.1007/s00205-019-01455-9.

[EJ20c] T. M. Elgindi and I.-J. Jeong. "Symmetries and Critical Phenomena in Fluids". In: *Communications on Pure and Applied Mathematics* 73 (2020), pp. 257–316. DOI: 10.1002/cpa.21829.

[EJ23] T. M. Elgindi and I.-J. Jeong. "On singular vortex patches, I: Well-posedness issues". In: *Mem. Amer. Math. Soc.* 283.1400 (2023), pp. 1–102. DOI: 10.1090/memo/1400.

[EJ25] T. M. Elgindi and M. J. Jo. *Cusp formation in vortex patches.* 2025. arXiv: 2504.02705.

[Elg21] T. M. Elgindi. "Finite-time singularity formation for $C^{1,\alpha}$ solutions to the incompressible Euler equations on $\mathbb{R}^3$". In: *Ann. of Math. (2)* 194.3 (2021), pp. 647–727. DOI: 10.4007/annals.2021.194.3.2.

[EM20] T. M. Elgindi and N. Masmoudi. "L^∞ ill-posedness for a class of equations arising in hydrodynamics". In: *Arch. Ration. Mech. Anal.* 235.3 (2020), pp. 1979–2025. DOI: 10.1007/s00205-019-01457-7.

[EP25] T. M. Elgindi and F. Pasqualotto. "Invertibility of a Linearized Boussinesq Flow: A Symbolic Approach". In: *Communications in Mathematical Physics* 406.11 (2025), p. 261. DOI: 10.1007/s00220-025-05367-6.

[EPT23] A. Enciso, D. Peralta-Salas, and F. Torres de Lizaur. "Quasi-periodic solutions to the incompressible Euler equations in dimensions two and higher". In: *J. Differential Equations* 354 (2023), pp. 170–182. DOI: 10.1016/j.jde.2023.01.013.

[Fig+17] J.-L. Figueras, M. Gameiro, J.-P. Lessard, and R. de la Llave. "A framework for the numerical computation and a posteriori verification of invariant objects of evolution equations". In: *SIAM Journal on Applied Dynamical Systems* 16.2 (2017), pp. 1070–1088. DOI: 10.1137/16M1073777.

[FL17] J.-L. Figueras and R. de la Llave. "Numerical computations and computer assisted proofs of periodic orbits of the Kuramoto-Sivashinsky equation". In: *SIAM Journal on Applied Dynamical Systems* 16.2 (2017), pp. 834–852. DOI: 10.1137/16M1073790.

[Fra00] L. E. Fraenkel. *An introduction to maximum principles and symmetry in elliptic problems.* Vol. 128. Cambridge Tracts in Mathematics. Cambridge: Cambridge University Press, 2000, pp. x+340. DOI: 10.1017/CBO9780511569203.

[FS05] S. Friedlander and R. Shvydkoy. "The unstable spectrum of the surface quasi-geostrophic equation". In: *J. Math. Fluid Mech.* 7 (2005), S81–S93. DOI: 10.1007/s00021-004-0129-3.

[Gar20] C. García. "Kármán vortex street in incompressible fluid models". In: *Nonlinearity* 33.4 (2020), pp. 1625–1676. DOI: 10.1088/1361-6544/ab6309.

[Gar21] C. García. "Vortex patches choreography for active scalar equations". In: *J. Nonlinear Sci.* 31.5 (2021), Paper No. 75, 31. DOI: 10.1007/s00332-021-09729-x.

[Ger31] S. A. Gerschgorin. "Über die Abgrenzung der Eigenwerte einer Matrix". In: *Bulletin de l'Académie des Sciences de l'URSS. Classe des sciences mathématiques et na* 6 (6 1931), pp. 749–754.

[GH23] C. García and S. V. Haziot. "Global bifurcation for corotating and counter-rotating vortex pairs". In: *Comm. Math. Phys.* 402.2 (2023), pp. 1167–1204. DOI: 10.1007/s00220-023-04741-6.

[GHH26] C. García, Z. Hassainia, and T. Hmidi. *Time-periodic leapfrogging vortex rings in the 3D Euler equations*. 2026. arXiv: 2603.21644.

[GHM24] C. García, T. Hmidi, and J. Mateu. "Time periodic solutions close to localized radial monotone profiles for the 2D Euler equations". In: *Ann. PDE* 10.1 (2024), Paper No. 1, 75. DOI: 10.1007/s40818-023-00166-5.

[GHR25] C. García, Z. Hassainia, and E. Roulley. "Dynamics of vortex cap solutions on the rotating unit sphere". In: *Journal of Differential Equations* 417 (2025), pp. 1–63. DOI: 10.1016/j.jde.2024.11.012.

[GHS04] Y. Guo, C. Hallstrom, and D. Spirn. "Dynamics near an unstable Kirchhoff ellipse". In: *Comm. Math. Phys.* 245.2 (2004), pp. 297–354. DOI: 10.1007/s00220-003-1017-z.

[GHS20] C. García, T. Hmidi, and J. Soler. "Non uniform rotating vortices and periodic orbits for the two-dimensional Euler equations". In: *Arch. Ration. Mech. Anal.* 238.2 (2020), pp. 929–1085. DOI: 10.1007/s00205-020-01561-z.

[GJ25] J. Guo and Q. Jiu. *Stability and Instability on the De Gregorio Modification of the Constantin-Lax-Majda model*. 2025. arXiv: 2506.02800.

[GL17] M. Gameiro and J.-P. Lessard. "A posteriori verification of invariant objects of evolution equations: periodic orbits in the Kuramoto-Sivashinsky PDE". In: *SIAM Journal on Applied Dynamical Systems* 16.1 (2017), pp. 687–728. DOI: 10.1137/16M1073789.

[GLM08] M. Gameiro, J.-P. Lessard, and K. Mischaikow. "Validated continuation over large parameter ranges for equilibria of PDEs". In: *Mathematics and Computers in Simulation* 79.4 (2008), pp. 1368–1382. DOI: 10.1016/j.matcom.2008.03.014.

[GO21] J. Gómez-Serrano and G. Orriols. "Any three eigenvalues do not determine a triangle". In: *Journal of Differential Equations* 275 (2021), pp. 920–938. DOI: 10.1016/j.jde.2020.11.002.

[God21] L. Godard-Cadillac. "Smooth traveling-wave solutions to the inviscid surface quasi-geostrophic equations". In: *C. R. Math. Acad. Sci. Paris* 359 (2021), pp. 85–98. DOI: 10.5802/crmath.159.

[Góm+21] J. Gómez-Serrano, J. Park, J. Shi, and Y. Yao. "Symmetry in stationary and uniformly-rotating solutions of active scalar equations". In: *Duke Math. J.* 170.13 (2021), pp. 2957–3038. DOI: 10.1215/00127094-2021-0002.

[Góm19] J. Gómez-Serrano. "Computer-assisted proofs in PDE: a survey". In: *SeMA Journal* 76.2 (2019), pp. 459–484. DOI: 10.1007/s40324-019-00186-x.

[GPS25] J. Gómez-Serrano, J. Park, and J. Shi. "Existence of non-trivial non-concentrated compactly supported stationary solutions of the 2D Euler equation with finite energy". In: *Mem. Amer. Math. Soc.* 311.1577 (2025), p. 82. DOI: 10.1090/memo/1577.

[GS19] P. Gravejat and D. Smets. "Smooth travelling-wave solutions to the inviscid surface quasi-geostrophic equation". In: *Int. Math. Res. Not. IMRN* 6 (2019), pp. 1744–1757. DOI: 10.1093/imrn/rnx177.

[Guo+22] Y. Guo, M. Hadžić, J. Jang, and M. Schrecker. "Gravitational collapse for polytropic gaseous stars: self-similar solutions". In: *Arch. Ration. Mech. Anal.* 246.2 (2022), pp. 957–1066. DOI: 10.1007/s00205-022-01827-8.

[Hel+95] I. M. Held, R. T. Pierrehumbert, S. T. Garner, and K. L. Swanson. "Surface quasi-geostrophic dynamics". In: *J. Fluid Mech.* 282 (1995), pp. 1–20. DOI: 10.1017/S0022112095000012.

[HH21] Z. Hassainia and T. Hmidi. "Steady asymmetric vortex pairs for Euler equations". In: *Discrete Contin. Dyn. Syst.* 41.4 (2021), pp. 1939–1969. DOI: 10.3934/dcds.2020348.

[HHM25] Z. Hassainia, T. Hmidi, and N. Masmoudi. "Rigorous derivation of the leapfrogging motion for planar Euler equations". In: *Invent. Math.* 242.3 (2025), pp. 725–825. DOI: 10.1007/s00222-025-01368-3.

[HHR24a] Z. Hassainia, T. Hmidi, and E. Roulley. *Desingularization of time-periodic vortex motion in bounded domains via KAM tools*. 2024. arXiv: 2408.16671.

[HHR24b] Z. Hassainia, T. Hmidi, and E. Roulley. "Invariant KAM tori around annular vortex patches for 2D Euler equations". In: *Comm. Math. Phys.* 405.11 (2024), Paper No. 270. DOI: 10.1007/s00220-024-05141-0.

[HI12] J. K. Hunter and M. Ifrim. "Enhanced life span of smooth solutions of a Burgers–Hilbert equation". In: *SIAM Journal on Mathematical Analysis* 44 (2012), pp. 2039–2052. DOI: 10.1137/110849791.

[Hil97] E. Hille. *Ordinary differential equations in the complex domain*. Courier Corporation, 1997.

[HJ15a] V. M. Hur and M. A. Johnson. "Modulational instability in the Whitham equation for water waves". In: *Studies in Applied Mathematics* 134.1 (2015), pp. 120–143. DOI: 10.1111/sapm.12061.

[HJ15b] V. M. Hur and M. A. Johnson. "Modulational instability in the Whitham equation with surface tension and vorticity". In: *Nonlinear Analysis: Theory, Methods & Applications* 129 (2015), pp. 104–118. DOI: 10.1016/j.na.2015.08.019.

[HK21] S. He and A. Kiselev. "Small-scale creation for solutions of the SQG equation". In: *Duke Math. J.* 170.5 (2021), pp. 1027–1041. DOI: 10.1215/00127094-2020-0064.

[HM16a] T. Hmidi and J. Mateu. "Bifurcation of rotating patches from Kirchhoff vortices". In: *Discrete and Continuous Dynamical Systems* 36.10 (2016), pp. 5401–5422. DOI: 10.3934/dcds.2016038.

[HM16b] T. Hmidi and J. Mateu. "Degenerate bifurcation of the rotating patches". In: *Adv. Math.* 302 (2016), pp. 799–850. DOI: 10.1016/j.aim.2016.07.022.

[HM17] T. Hmidi and J. Mateu. "Existence of corotating and counter-rotating vortex pairs for active scalar equations". In: *Comm. Math. Phys.* 350.2 (2017), pp. 699–747. DOI: 10.1007/s00220-016-2784-7.

[Hmi15] T. Hmidi. "On the trivial solutions for the rotating patch model". In: *J. Evol. Equ.* 15.4 (2015), pp. 801–816. DOI: 10.1007/s00028-015-0281-7.

[HMV13] T. Hmidi, J. Mateu, and J. Verdera. "Boundary regularity of rotating vortex patches". In: *Archive for Rational Mechanics and Analysis* 209.1 (2013), pp. 171–208. DOI: 10.1007/s00205-013-0618-8.

[HMW20] Z. Hassainia, N. Masmoudi, and M. H. Wheeler. "Global bifurcation of rotating vortex patches". In: *Comm. Pure Appl. Math.* 73.9 (2020), pp. 1933–1980. DOI: 10.1002/cpa.21855.

[Hoz+16a] F. de la Hoz, Z. Hassainia, T. Hmidi, and J. Mateu. "An analytical and numerical study of steady patches in the disc". In: *Anal. PDE* 9.7 (2016), pp. 1609–1670. DOI: 10.2140/apde.2016.9.1609.

[Hoz+16b] F. de la Hoz, T. Hmidi, J. Mateu, and J. Verdera. "Doubly connected V-states for the planar Euler equations". In: *SIAM J. Math. Anal.* 48.3 (2016), pp. 1892–1928. DOI: 10.1137/140992801.

[HP16] V. M. Hur and A. K. Pandey. "Modulational instability in nonlinear nonlocal equations of regularized long wave type". In: *Physica D: Nonlinear Phenomena* 325 (2016), pp. 98–112. DOI: 10.1016/j.physd.2016.03.005.

[HR25] Z. Hassainia and E. Roulley. “Boundary effects on the emergence of quasi-periodic solutions for Euler equations”. In: *Nonlinearity* 38.1 (2025), p. 015016. DOI: 10.1088/1361-6544/ad9ba7.

[Hun+15] J. K. Hunter, M. Ifrim, D. Tataru, and T. K. Wong. “Long time solutions for a Burgers–Hilbert equation via a modified energy method”. In: *Proceedings of the American Mathematical Society* 143 (2015), pp. 3407–3412. DOI: 10.1090/proc/12215.

[Hun+22] J. K. Hunter, R. C. Moreno-Vasquez, J. Shu, and Q. Zhang. “On the approximation of vorticity fronts by the Burgers-Hilbert equation”. In: *Asymptot. Anal.* 129.2 (2022), pp. 141–177. DOI: 10.3233/asy-211724.

[Hun18] J. K. Hunter. “The Burgers–Hilbert equation”. In: *Theory, Numerics and Applications of Hyperbolic Problems II*. Vol. 237. Springer Proceedings in Mathematics and Statistics. Cham: Springer, 2018, pp. 41–57. DOI: 10.1007/978-3-319-91548-7_3.

[HW22] Z. Hassainia and M. H. Wheeler. “Multipole vortex patch equilibria for active scalar equations”. In: *SIAM J. Math. Anal.* 54.6 (2022), pp. 6054–6095. DOI: 10.1137/21M1415339.

[Inc56] E. L. Ince. *Ordinary Differential Equations*. Reprint of the 1944 edition. New York: Dover Publications, 1956, p. 558. ISBN: 0486603490.

[JK24] I.-J. Jeong and J. Kim. “Strong ill-posedness for SQG in critical Sobolev spaces”. In: *Anal. PDE* 17.1 (2024), pp. 133–170. DOI: 10.2140/apde.2024.17.133.

[Joh13] M. A. Johnson. “Stability of small periodic waves in fractional KdV-type equations”. In: *SIAM Journal on Mathematical Analysis* 45.5 (2013), pp. 3168–3193. DOI: 10.1137/120894397.

[JSS19] H. Jia, S. Stewart, and V. Sverak. “On the De Gregorio Modification of the Constantin–Lax–Majda Model”. In: *Archive for Rational Mechanics and Analysis* 231.2 (2019), pp. 1269–1304. DOI: 10.1007/s00205-018-1298-1.

[Kat80] T. Kato. *Perturbation Theory for Linear Operators*. 2nd. Berlin, Heidelberg, New York: Springer-Verlag, 1980. ISBN: 3-540-58661-X.

[Kir74] G. Kirchhoff. *Vorlesungen über mathematische Physik*. Leipzig: Teubner, 1874.

[KN12] A. Kiselev and F. Nazarov. “A simple energy pump for the surface quasi-geostrophic equation”. In: *Nonlinear partial differential equations*. Vol. 7. Abel Symp. Springer, Heidelberg, 2012, pp. 175–179. ISBN: 978-3-642-25360-7. DOI: 10.1007/978-3-642-25361-4_9.

[Kob10] K. Kobayashi. “On the global uniqueness of Stokes’ wave of extreme form”. In: *IMA Journal of Applied Mathematics* 75.5 (2010), pp. 647–675. DOI: 10.1093/imamat/hxq037.

[KS15] C. Klein and J.-C. Saut. “A numerical approach to blow-up issues for dispersive perturbations of Burgers’ equation”. In: *Physica D: Nonlinear Phenomena* 295–296 (2015), pp. 46–65. DOI: 10.1016/j.physd.2014.12.004.

[KV20] S. G. Krupa and A. F. Vasseur. “Stability and uniqueness for piecewise smooth solutions to a nonlocal scalar conservation law with applications to Burgers–Hilbert equation”. In: *SIAM Journal on Mathematical Analysis* 52.3 (2020), pp. 2491–2530. DOI: 10.1137/19M1257883.

[LI24] S.-Z. Looi and P. Isett. *A proof of Onsager’s Conjecture for the SQG equation*. 2024. arXiv: 2407.02578.

[LLR20] Z. Lei, J. Liu, and X. Ren. “On the Constantin–Lax–Majda model with convection”. In: *Comm. Math. Phys.* 375.1 (2020), pp. 765–783. DOI: 10.1007/s00220-019-03584-4.

[Mar08] F. Marchand. "Existence and regularity of weak solutions to the quasi-geostrophic equations in the spaces L^p or $\dot{H}^{-1/2}$". In: *Comm. Math. Phys.* 277.1 (2008), pp. 45–67. DOI: 10.1007/s00220-007-0356-6.

[MB02] A. J. Majda and A. L. Bertozzi. *Vorticity and incompressible flow.* Vol. 27. Cambridge Texts in Applied Mathematics. Cambridge University Press, Cambridge, 2002, pp. xii+545. ISBN: 0-521-63057-6.

[MR25] A. Maspero and A. M. Radakovic. "Full description of Benjamin–Feir instability for generalized Korteweg–de Vries equations". In: *SIAM J. Math. Anal.* 57.3 (2025), pp. 3030–3070. DOI: 10.1137/24M1652076.

[MW83] J. Marsden and A. Weinstein. "Coadjoint orbits, vortices, and Clebsch variables for incompressible fluids". In: vol. 7. 1-3. Order in chaos (Los Alamos, N.M., 1982). 1983, pp. 305–323. DOI: 10.1016/0167-2789(83)90134-3.

[NPW19] M. T. Nakao, M. Plum, and Y. Watanabe. *Numerical Verification Methods and Computer-Assisted Proofs for Partial Differential Equations.* Vol. 53. Springer Series in Computational Mathematics. Singapore: Springer, 2019. ISBN: 9789811376689. DOI: 10.1007/978-981-13-7669-6.

[OSW08] H. Okamoto, T. Sakajo, and M. Wunsch. "On a generalization of the Constantin–Lax–Majda equation". In: *Nonlinearity* 21.10 (Sept. 2008), pp. 2447–2461. DOI: 10.1088/0951-7715/21/10/013.

[OSW14] H. Okamoto, T. Sakajo, and M. Wunsch. "Steady-states and traveling-wave solutions of the generalized Constantin-Lax-Majda equation". In: *Discrete and Continuous Dynamical Systems* 34.8 (Aug. 2014), pp. 3155–3170. DOI: 10.3934/dcds.2014.34.3155.

[Par22] J. Park. "Quantitative estimates for uniformly-rotating vortex patches". In: *Adv. Math.* 411 (2022), Paper No. 108779, 44. DOI: 10.1016/j.aim.2022.108779.

[Pas25] M. M. G. Pascual-Caballo. *Stationary homogeneous solutions for the inviscid SQG equation.* 2025. arXiv: 2510.03108.

[Pas26] M. M. G. Pascual-Caballo. *On the instability of De Gregorio excited states.* Manuscript in preparation. 2026.

[PCN17] J. A. Pava, E. J. Cardoso, and F. Natali. "Stability properties of periodic traveling waves for the intermediate long wave equation". In: *Revista Matemática Iberoamericana* 33.2 (2017), pp. 417–448. DOI: 10.4171/RMI/943.

[Ped79] J. Pedlosky. *Geophysical fluid dynamics.* Springer, 1979. ISBN: 9780387903682.

[Res95] S. G. Resnick. *Dynamical problems in non-linear advective partial differential equations.* Thesis (Ph.D.)–The University of Chicago. ProQuest LLC, Ann Arbor, MI, 1995, p. 76.

[RS25] R.-O. Radu and N. Stevenson. *Desingularization of nondegenerate rotating vortex patches.* 2025. arXiv: 2511.18592.

[Sch30] J. Schauder. "Der Fixpunktsatz in Funktionalraümen". ger. In: *Studia Mathematica* 2.1 (1930), pp. 171–180. DOI: 10.4064/sm-2-1-171-180.

[SS90] G. W. Stewart and J.-g. Sun. *Matrix Perturbation Theory.* Computer Science and Scientific Computing. Boston: Academic, 1990. ISBN: 0126702306.

[SW22] J.-C. Saut and Y. Wang. "The wave breaking for Whitham-type equations revisited". In: *SIAM J. Math. Anal.* 54.2 (2022), pp. 2295–2319. DOI: 10.1137/20M1345207.

[Tes12] G. Teschl. *Ordinary Differential Equations and Dynamical Systems.* Vol. 140. Graduate Studies in Mathematics. American Mathematical Society, 2012. DOI: 10.1090/gsm/140.

[Tuc11] W. Tucker. *Validated Numerics: A Short Introduction to Rigorous Computations*. A short introduction to rigorous computations. Princeton, NJ: Princeton University Press, 2011. ISBN: 9780691147819.

[Tur85] B. Turkington. "Corotating steady vortex flows with n-fold symmetry". In: *Nonlinear Anal.* 9.4 (1985), pp. 351–369. DOI: 10.1016/0362-546X(85)90059-8.

[Wu05] J. Wu. "Solutions of the 2D quasi-geostrophic equation in Hölder spaces". In: *Nonlinear Anal.* 62.4 (2005), pp. 579–594. DOI: 10.1016/j.na.2005.03.053.

[WW96] E. T. Whittaker and G. N. Watson. *A Course of Modern Analysis*. Cambridge Mathematical Library. Reprint of the 4th edition (1927). Cambridge: Cambridge University Press, 1996, p. 608. ISBN: 9780521588072.

[WXZ22] Y. Wang, X. Xu, and M. Zhou. *Degenerate bifurcations of two-fold doubly-connected uniformly rotating vortex patches*. 2022. arXiv: 2212.01869.

[Yan21] R. Yang. "Shock Formation of the Burgers–Hilbert Equation". In: *SIAM Journal on Mathematical Analysis* 53.5 (2021), pp. 5756–5802. DOI: 10.1137/21M1399348.

[Zgl04] P. Zgliczyński. "Rigorous numerics for dissipative partial differential equations. II. Periodic orbit for the Kuramoto-Sivashinsky PDE—a computer-assisted proof". In: *Foundations of Computational Mathematics* 4.2 (2004), pp. 157–185. DOI: 10.1007/s10208-002-0080-8.

[Zha25] X. Zhao. *An Onsager-type Theorem for General 2D Active Scalar Equations*. 2025. arXiv: 2412.11094.

[ZM01] P. Zgliczyński and K. Mischaikow. "Rigorous numerics for partial differential equations: the Kuramoto-Sivashinsky equation". In: *Foundations of Computational Mathematics* 1.3 (2001), pp. 255–288. DOI: 10.1007/s002080010010.